

LG-X350

Service Manual



LG Electronics

Table Of Contents

1. INTRODUCTION.....	3	4.13 FM Radio Trouble.....	85
1.1 Purpose.....	3	4.14 RX & PAM Trouble.....	87
1.2 Regulatory Information.....	3	4.15 Bluetooth Trouble.....	96
2. PERFORMANCE.....	5	4.16 Camera Trouble.....	98
2.1 H/W Features.....	5	4.17 ISDB-T(NMI326) Trouble.....	101
2.2 S/W Features.....	6	5. DOWNLOAD.....	104
3. TECHNICAL BRIEF.....	20	5.1 Download setup.....	104
3.1 Digital Main Processor(MT6236).....	20	5.2 Download Process.....	105
3.2 Power Amplifier Module(SKY77547).....	31	6. BLOCK DIAGRAM.....	113
3.3 Transceiver Module(AD6548).....	33	7. CIRCUIT DIAGRAM.....	114
3.4 Memory Module (K521H12ACB-B060).....	36	8. BGA IC PIN Check.....	125
3.5 FM Radio Module (AR1200).....	40	9. PCB LAYOUT.....	128
3.6 TV Module(NMI326).....	43	10. ENGINEERING MODE.....	130
3.7 LCD Interface.....	45	11. CALIBRATION.....	143
3.8 SIM & Micro SD Card Interface.....	48	11.1 Test Equipment Setup.....	143
3.9 KEYPAD Interface.....	49	11.2 Calibration Steps.....	144
3.10 Battery Charging Block Interface.....	50	12. STAND ALONE TEST.....	183
3.11 Audio Interface.....	51	12.1 Test Configuration.....	183
3.12 Vibrator Interface.....	53	12.2 META tool Install process.....	184
3.13 Camera Interface.....	54	12.3 Rx Test.....	187
3.14 Key LED Interface.....	56	12.4 Tx Test.....	193
4. TROUBLE SHOOTING.....	57	13. #EV#.....	199
4.1 Power On Trouble.....	57	14. Rear Cover Disassemble	
4.2 SIM Card Trouble.....	62	SOP.....	201
4.3 Vibrator Trouble.....	64		
4.4 Keypad Trouble.....	66		
4.5 Key Backlight Trouble.....	69		
4.6 LCM Backlight Trouble.....	70		
4.7 LCM Trouble.....	72		
4.8 Microphone Trouble.....	74		
4.9 Receiver Trouble.....	76		
4.10 Speaker Trouble.....	78		
4.11 Headphone Trouble.....	80		
4.12 Charging Trouble.....	83		

1. INTRODUCTION

1.1 Purpose

This manual provides information necessary to repair, description and download the features of this model.

1.2 Regulatory Information

A. Security

Toll fraud, the unauthorized use of telecommunications system by an unauthorized part (for example, persons other than your company's employees, agents, subcontractors, or person working on your company's behalf) can result in substantial additional charges for your telecommunications services.

System users are responsible for the security of own system. There are may be risks of toll fraud associated with your telecommunications system. System users are responsible for programming and configuring the equipment to prevent unauthorized use. The manufacturer does not warrant that this product is immune from the above case but will prevent unauthorized use of common-carrier telecommunications service of facilities accessed through or connected to it.

The manufacturer will not be responsible for any charges that result from such unauthorized use.

B. Incidence of Harm

If a telephone company determines that the equipment provided to customer is faulty and possibly causing harm or interruption in service to the telephone network, it should disconnect telephone service until repair can be done. A telephone company may temporarily disconnect service as long as repair is not done.

C. Changes in Service

A local telephone company may make changes in its communications facilities or procedure. If these changes could reasonably be expected to affect the use of the this phone or compatibility with the network, the telephone company is required to give advanced written notice to the user, allowing the user to take appropriate steps to maintain telephone service.

D. Maintenance Limitations

Maintenance limitations on this model must be performed only by the manufacturer or its authorized agent. The user may not make any changes and/or repairs except as specifically noted in this manual.

Therefore, note that authorized alternations or repair may affect the regulatory status of the system and may void any remaining warranty.

E. Notice of Radiated Emissions

This model complies with rules regarding radiation and radio frequency emission as defined by local regulatory agencies. In accordance with these agencies, you may be required to provide information such as the following to the end user.

F. Pictures

The pictures in this manual are for illustrative purposes only; your actual hardware may look slightly different.

G. Interference and Attenuation

Phone may interfere with sensitive laboratory equipment, medical equipment, etc. Interference from unsuppressed engines or electric motors may cause problems.

H. Electrostatic Sensitive Devices



Boards, which contain Electrostatic Sensitive Devices(ESD),are indicated by the sign .



Following information is ESD handing:

- . Service personnel should ground themselves by using a wrist strap when exchange system boards.
- . When repairs are made to a system board , they should spread the floor with anti-static mat which is also grounded .
- . Use a suitable, grounded soldering iron .
- . Keep sensitive parts in these protective packages until these are used.
- . When returning system boards or parts like EEPROM to the factory, use the protective packages as described.

2. PERFORMANCE

2.1 H/W Features

Solution	MT6236A/B-L A/D	Media Tek
Type	Bar type	
Antenna Type	Internal (Quad-Band)	850/900/1800/1900
Main Display	2.3" 320x240 QVGA	Tianma
GPRS	Class 10	EDGE Rx
ISDB-T	NM326AAA	NMI
MMS	Yes, 1.2	
Camera	2.0M FF	Hansung (Toshiba 1/5")
Battery	1500mAh Li-ion inner pack	LGC BL-44JN
Audio player	Yes	MP3/AAC/WAV
FM Receiver	Yes	(87.5~108MHz)
MPEG4/H.263	Yes (support 3GP)	
H.264	Yes	
AAC	Yes	
AAC+	Yes	
FM alarm	Yes	
Scheduled FM recording	Yes	
MP4 for incoming call/ power on off animation and screen saver	Yes	
Loud Speaker	Yes	
Audio player--real resuming	Yes	
Video recording	Yes	
Memory Size	<u>1Gb</u> + 512Mb	NAND Boot. Samsung 1.8V
Internal NAND	Yes	(Maximum is 44M for user memory)
Memory Card	Micro SD	Up to 16GB
Bluetooth	Yes, version 2.1 3.0	W/ EDR.
USB	Yes, USB 2.0 full speed	
WAP	Yes, 2.0	Opera Mobile
Java	Yes	
EMAIL	Yes	
DRM	Yes	DRM 1.0 FL Only
MPEG4 caller ID	Yes	
OTA	Yes	
Music Equalizer	Yes	
In flight mode	Yes	

2.2 S/W Features

2.1.1 System Specification

Item	Target Specification
Form Factor	Bar Type
Size	111 x 61 x 13.5 mm
Weight	115g with battery 85g without battery
Battery	3.7V, 1500mAh Li-Ion
Talk Time (TBC)	Up to 8 hrs@1500mAh (PCL:10)
Standby Time (TBC)	Up to 800 hrs @ 1500mAh (SIM1 P.P.: 9)
Antenna	Embedded type
LCD	2.3" 320x240 QVGA
FM	Yes,
Camera	2.0M pixel FF Auto mode by default setting
Back Light	White LED
Keypad Backlight Color	White
Vibrator	Yes
Loud Speaker	Yes, 18*12 , out put : 0.8W over
Microphone	Yes
Earphone Jack	Yes 3.5Φ
SIM Socket	Yes, 1.8/3.0V
Volume Key	Side key (up/down)
Basic Accessory	Travel Adaptor(700MA)
	Standard Battery (1500mA, Li-Ion)
	Stereo Headset with button (FM)
	Micro USB Data Cable

2.1.2 General Features

Function	Target Specification																									
Basic Display	RSSI (7 Level, 'no service, 0, 1, 2, 4, 5, 7')																									
	Battery Indicator (4 Level, 0~3)																									
	Battery Indicator	Battery Indicator	Voltage																							
		3 -> 2	3.72 ± 0.03V																							
		2 -> 1	3.62 ± 0.03 V																							
		1 -> 0	3.55 ± 0.03 V																							
	Antenna display (7 level settings and the corresponding RSSI)																									
	<table><tr><th>Model</th><th>Antenna Bar</th><th>Display</th><th>Range (dBm)</th><th>Remarks</th></tr><tr><td rowspan="7">CDMA GSM 3G</td><td rowspan="7">7 ea</td><td></td><td>-92 or higher</td><td rowspan="7">If there's a specific requirement by Operator, must follow it.</td></tr><tr><td></td><td>-93 ~ -97</td></tr><tr><td></td><td>-98 ~ -100</td></tr><tr><td></td><td>-101 ~ -103</td></tr><tr><td></td><td>-104 ~ -105</td></tr><tr><td></td><td>-106 or lower</td></tr><tr><td>OFF</td><td>No service</td></tr></table>				Model	Antenna Bar	Display	Range (dBm)	Remarks	CDMA GSM 3G	7 ea		-92 or higher	If there's a specific requirement by Operator, must follow it.		-93 ~ -97		-98 ~ -100		-101 ~ -103		-104 ~ -105		-106 or lower	OFF	No service
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			-106 or lower																							
OFF			No service																							
Icons Indicator																										
Others reference to "Phone Personalization Setting"																										
Speech Codec	FR/EFR/HR/AMR																									
Keypad	Number of Keys: 48 Key (include 37 QWERTY keys, 4 function keys, 5 way navigation keys, 2 side keys)																									
	Soft Function Keys : 2																									
	International Access (+)																									
User Profile (Audio Settings)	User Selectable and Customizable Profiles (4 profiles: Normal, Outdoor, Silent, Flight mode)																									
	Key Tone																									
	Key Tone Volume (7 Level - 0 ~ 6, 0 for Mute)																									
	Key tone setting (DTMF, English)																									
	Ring Tone																									
	Ring Tone Volume (7 Level - 0 ~ 6, 0 for Mute)																									
	Built-in Ring Tone Pattern: 20																									
	Customizable Ring Tone Link: 1																									
	Alert Type																									
	5 Types - Ring, Vibration Only, Vibration and Ring, Ring after vibration, Silent																									

	Power on/off tones
	Built-in Ring Tone Pattern: 3 (include Silent) Built-in Ring Tone Pattern: 4
	Power Off Tone
	Built-in Ring Tone Pattern: 3 (include Silent)
	Message Tone
	Built-in Ring Tone Pattern: 10
	Warning Tone
	Built-in Ring Tone Pattern: 1
	Error Tone
	Built-in Ring Tone Pattern: 1
	Camp On Tone
	Built-in Ring Tone Pattern: 1
	Connect Tone
	Built-in Ring Tone Pattern: 1
	Status LED
	Built-in Lighting Pattern: No (None, Pattern 1)
	Charger-in Status LED
	Built-in Lighting Pattern: No (None, Pattern 1)
	Answer Mode
	Any Key Answer, Send Key only
	Auto (Only available for headset mode while headset plugged in) TBD
Personal Information Management	Calendar - Month view only
	Schedule - 6 fields (Date, time, Expired Date, Subject, Alarm, Repeat)
Tools and Utilities	Alarm
	5 sets of Alarm
	4 major fields for each set - On/Off, Time, Repeat type, Audio option
	World Clock
	Cities list: TBD cities
	Daylight saving time support: activated by user selection
	Home city set
	Calculator
	Addition, Subtraction, Multiplication, Division
	Unit Converter
	Weight, Length
	Memo
	To do

Phone Personalization Setting	Greeting Text
	Shortcuts
	Flight Mode
	Time and Date Setting
	Wallpaper
	Screen Saver
	Power On Animation
	Power Off Animation
	LCD Backlight
	PLMN/Service Indicator (Display of PLMN Name/Service Provider Name from SIM)
	Date Time Display
	Own Number Display
	Restore Factory Default Setting
Security	Phone Lock
	Key Lock
Input Method	Engine
	T9 (except Chinese)
	Support Language
	Depends on customer and market requirement.
	Total supported languages will be limited to memory condition.
	Predictive word input
Game	Java Games, provided by LGE.
Anti-theft Mobile Tracker (ATMT)	Provide this feature by following LG spec. (GSM_VVLT 0 5_LMT_20071117_1.ppt)

2.1.3 GSM/GPRS Features

Function	Target Specification
GPRS	GPRS Multi slot Class 10
Data Service	BS 24 - 26 (2400-9600 bit/s), asynchronous, non-transparent, UDI. CSD rate up to 9.6K bit/s
Call History (Dual SIM)	Last Dialed Number: 80 (single SIM:40)
	Last Received Number: 80 (single SIM:40)
	Last Missed Number: 80 (single SIM:40)
	Scratch Pad Memory (Save an input number in call): 1
Call Cost	Last Call Time
	Total Dialed Call Time
	Total Received Call Time
	Last Call Cost
	Total Cost
	Max Cost
	Price Per Unit
GPRS Counter	Last Sent (unit in Byte)
	Last Received (unit in Byte)
	All Sent (unit in Byte)
	All Received (unit in Byte)
Call Management	Call Swap
	Call Retrieve
	Automatic Redial
	Speed Dialing
	Last Number Redial
	Support 40-digits Dialing Number from Idle, Phonebook in handset, and Call Log
	1. Idle dial screen: OK 2. Phonebook (Phone): OK 3. Phonebook (SIM): Have limitation, depend on SIM card. 4. Call Log: OK 5. SMS: support 20-digit 6. MMS: support 40-digit
Call Related Supplementary Services	Call Hold
	Call Waiting
	Calling Line Identity Presentation
	Calling Line Identity Restriction
	Connected Identification Restriction
	Call Divert All voice Calls

	Call Divert if unreachable
	Call Divert if no answer
	Call Divert if busy
	Call Divert all data calls
	Cancel all divert
	Call Barring All Outgoing Calls
	Call Barring All Outgoing International Calls
	Call Barring All outgoing International except home
	Call Barring All incoming Calls
	Call Barring All incoming Calls when roaming
	Multi-party Call (up to 7 calls, 5 in conference, 1 on held, 1 waiting)
	Miniute minder (On, Off)
Phone Book	Quick Search (Notice: Quick search function only works in Phonebook, SMS and MMS. In other application, this phone supports regular search.)
	Alpha Store and Recall
	Access Phone Book in call
	Copy & Move
	Fixed Dial Number
	Service Dial Number
	Speed Dial Number
	SOS Number
	Phone Entry: 1000 names 12 fields – Name, Number, Mobile, Home, Office number, Fax number, Email Address, Company Name, Associate Picture, Associate Video, Associate Sound, Birthday
	Caller Group-5 caller group- Friends, Family, VIP, Colleagues, School
	Own Numbers: User can change the own numbers of handset. (Sets of own numbers depends on SIM)
	vCard:
	Note: This phone doesn't support phone number search.
Message	SMS
	Standard SMS
	SMS Reply Path
	SMS Delivery Report
	Valid period (1 hour/12 hours/1 day/1 week/Maximum) Message Type (Text, Fax, Page, Email) Message Indication Type refer to GSM 03.40

	Write Message
	Read Message
	Edit Message (For MMS, Edit only conformance messages, unknown media not supported, unknown SMIL not supported)
	Reply Message
	Send Message
	Delete Message
	Forward Message
	Use Sender's Number
	Message Templates
	Extract media from Message (MMS/EMS)
	Store Media (MMS/EMS)
	Delete Media (MMS/EMS)
Cell Broadcast	Read Cell Broadcast
	Cell Broadcast Mode: Receive On/Off
	Cell Broadcast Message Language
	Channel Setting
Network	Automatic Network Selection
	Manual Network Selection
	Network Service Status
	Preferred Network (User definition)
	GPRS connection mode selection: Always, When Needed
SIM	Common Operation
	SIM Application Toolkit (Release 98 Class 2 certified)
	Prepaid SIM operation,
	Security
	PIN
	Personalization (Service provider lock, Network lock)
DTMF	DTMF Signaling
	DTMF Enable & Disable

2.1.4 Multimedia Features

Function	Target Specification
Camera	Image size: 320X240, 640X480, 1280x960, 1600x1200 .
	Zoom: 1x ~ 4x, (320x240:4x, 640x480:2x, 1280x960:1x, 1600x1200:1x)
	Image Quality: Super fine, Fine, Normal
	White Balance: Auto, Daylight, Tungsten, Fluorescent, Cloud, Incandescence
	Shot: Three Shot Sounds
	EV: -4 ~+4
	Screen Mode: Auto, Night
	Banding: 60Hz/50Hz
	Effect settings: (Total 14 6 types) Normal, Grayscale, Sepia, Color Invert, Blackboard , Embossment ,
	Storage Selection: Phone, Memory card (Only available when external memory card supported)
	Delay timer: Off/ 3 / 5/ 10 Sec
Image Viewer	Thumbnail supported
	Browse Style: List, Matrix
	View
	Use as: To Wallpaper, Caller ID, Send via: MMS, Bluetooth, Email
	Rename
	Delete
	Delete All
	Sort: By Name, Type, Time, Size, None
	Storage Selection: Get list from Phone, Memory card (Only available when external memory card supported)
	Image Format Support

	JPEG Baseline
	GIF87a
	GIF89a
	WBMP
	BMP
Music Player	Play
	Pause
	Resume
	Stop
	Next
	Previous
	Fast forward
	Rewind
	Storage Selection: Get list from Phone, Memory card (Only available when external memory card supported)
	Auto-Generate Playlist
	Skin: 1 skins
	Repeat Mode: Off, One Song, All Songs ,All off
	Shuffle Play
	Background Play
	Equalizer Setting: 8 sets Normal, Bass, Dance, Classical, Treble, Party, Pop, Rock
	Volume Control: 21 level (0~20, 0 for Mute)
	Playlist Edit: Add, Remove, Remove All
	Sound Format Support
	MP3
	AMR
	MIDI
	WAV
	AAC
Video Player	Play
	Pause
	Stop

	Fast forward
	Rewind
	Forward:
	To MMS, Bluetooth
	Rename
	Delete
	Delete All
	Sort:
	By Name, Type, Date
	Storage Selection:
	Get list from Phone, Memory card
	Volume Control:
	21 level (0~20, 0 for Mute)
Video Recorder	White Balance:
	Auto, Daylight, Tungsten, Fluorescent, Cloud, Incandescence
	EV: -4 ~+4
	Night Mode:
	On/Off
	Banding: 60Hz/50Hz
	Video Quality:
	Good Super fine, Fine, Normal
	Record Audio:
	On/Off
	Encode Format:
	MPEG4, H.263
	Effect settings: (Total 44 6 types)
	Normal, Grayscale, Sepia, Color Invert, Blackboard, Embossment
	Storage Selection:
	Phone, Memory card (Only available when external memory card supported)
	Record
	Pause
	Resume Recording
	Stop

Sound Recorder	Storage Selection: Phone, Memory card (Only available when external memory card supported)
	Encode Format: WAV, AMR, AWB
	Record
	Pause
	Resume Recording
	Stop
Melody Compose	Edit
	Play
	Save
	Instrument Selection: 10 types Piano, Guitar, Violin, Saxophone, Steel Drums, Flute, Harmonica, Trumpet, Music Box, Xylophone
	Play Speed: Fast, Normal, Slow
	[Notice] Melody composer only support one instrument in one melody file, so the last chosen instrument will be used to play this melody file
FM Radio	Frequencies: 87.5 ~ 108.0
	User definable Preset Channel List
	Channel Auto Search
	Background Play
	Record
	Record Format: AMR, WAV, AWB
	Record Storage: Phone, Memory Card (Only available when external memory card supported)
	Preset Channel List generated by auto search
JAVA	MIDP 2.0
	CLDC 1.1
	Memory Limit 2MB
	Support JSR 139,118,120,135,185,75

2.1.5 Connectivity Features

Function	Target Specification
WAP	WAP 2.0 Spec.
	WAP Push OTA/Message
	WAP Provisioning Service
	CSD/GPRS data connection
	Bookmark
	Wireless Telephony Application (WTA) support: Only Public WTA support, supported functions listing below -
	* Make a telephone call * Send a string of DTMF tones over an established voice connection * Add an entry to the telephone book of the device
	Support OTA push and push message
	OTA Provisioning & OTA download
	Supports WML, WCSS, XHTML mp
Bluetooth	Version 3.0 (w/ EDR)
	Profile: GAP,SDAP,DUN,SPP,HSP,HFP,OPP,FTP,A2DP, AVRCP
USB	Mass Storage Device <u>2.0</u>
	Virtual COM

3. TECHNICAL BRIEF

3.1 Digital Main Processor

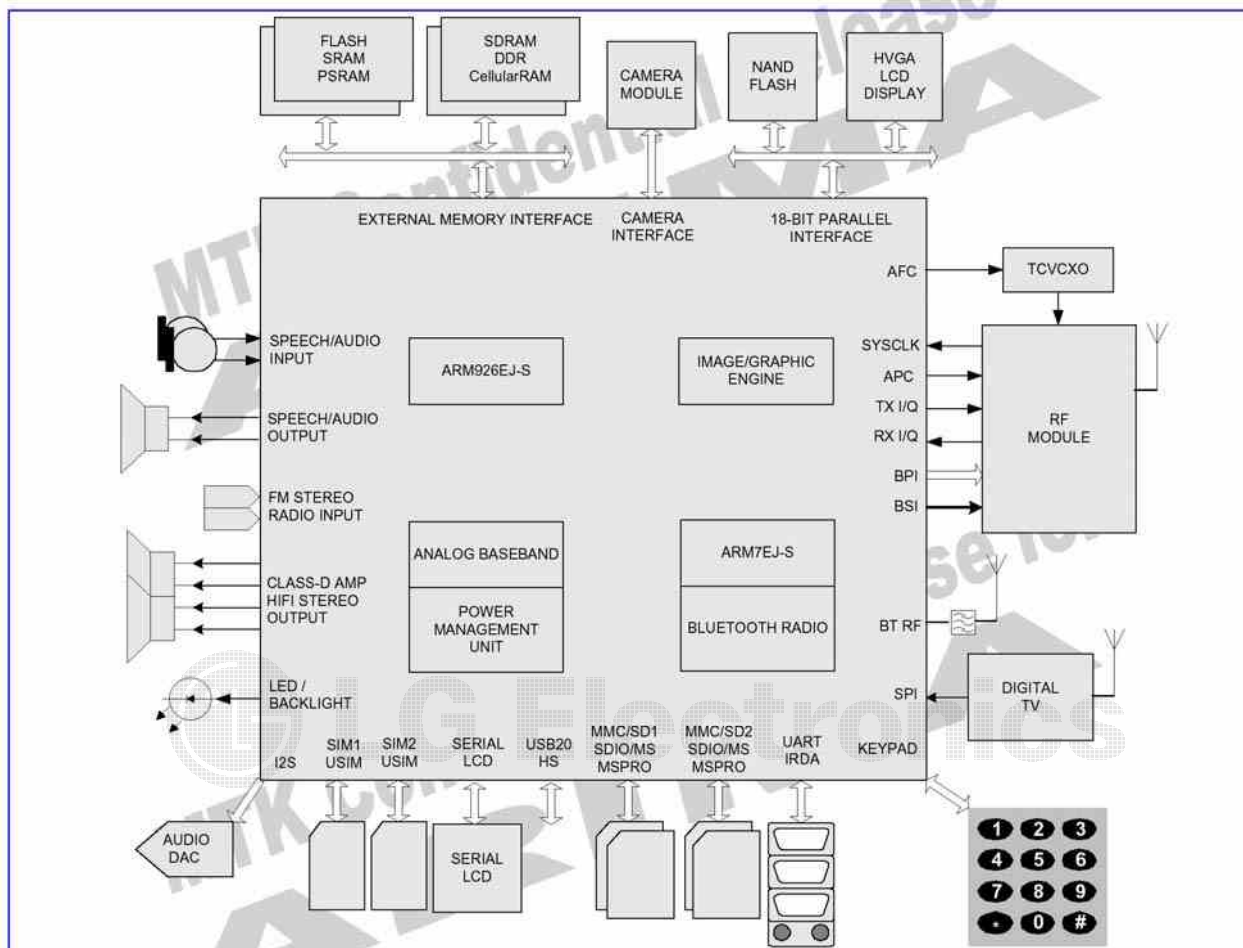


Figure.3-1-1 MT6236 FUNCTIONAL BLOCK DIAGRAM

3.1.1 System Overview

MT6236 is a monolithic chip that integrates leading edge power management unit, analog baseband, and radio circuitry based on 65nm low power process.

MT6236 is a feature-rich and extremely powerful single-chip solution for high-end GSM/GPRS and EDGE capability. Based on the 32-bit ARM926EJ-STM RISC processor, MT6236's superb processing power, along with high bandwidth architecture and dedicated hardware support, provides a platform for high performance GPRS/EDGE Class 12 MODEM application and leading-edge multimedia applications.

Besides, MT6236 also features a highly integrated Bluetooth transceiver, which is fully compliance with Bluetooth specification v2.1+EDR.

Platform

MT6236 is capable of running the ARM926EJ-STM RISC processor at up to 312 MHz, thus providing fast data processing capabilities. In addition to the high clock frequency, separate CODE and DATA caches are also added to further improve the overall system efficiency.

For large amounts of data transfer, high performance DMA (Direct Memory Access) with hardware flow control is implemented, which greatly enhances the data movement speed while reducing MCU processing load.

Memory

To provide the greatest capacity for expansion and maximum bandwidth for data intensive applications such as multimedia features, MT6236 supports up to 3 external memory devices through 16-bit host interface. High performance devices such as Mobile DDR SDRAM and Cellular RAM are supported for maximum bandwidth. Traditional devices such as burst flash and Pseudo SRAM are also supported. For greatest compatibility, the memory interface can also be used to connect to legacy devices such as Color/Parallel LCD, and multimedia companion chips are all supported through this interface. To minimize power consumption and ensure low noise, this interface is designed for flexible I/O voltage and allows lowering of the supply voltage down to 1.8V. The driving strength is configurable for signal integrity adjustment. The data bus also employs retention technology to prevent the bus from floating during a turn over.

Multi-media

The MT6236 multimedia subsystem provides both conventional parallel interface and MIPI CSI-2 interface for CMOS sensors. The camera resolution is up to 5Mpixels. With its advanced image signal processing technology, MT6236 offers high quality image and video contents. Built-in JPEG encode hardware enables real-time capture of high-resolution images.

The MT6236 also provides fancy UI capabilities through its hardware 2D accelerator. The 2D accelerator performs high speed linear transforms with filtering. SW based video codecs can be used to process various video types such as MPEG-4, H264, etc. To take advantage of the high MCU performance, GIF and PNG decoders are implemented by software as well.

In addition, MT6236 is implemented with a high performance audio synthesis technology, as well as a high quality audio amplifier to provide superior audio experiences.

Connectivity and Storage

To take advantage of its incredible multimedia strengths, MT6236 incorporates myriads of advanced connectivity and storage options for data storage and communication. MT6236 supports UART, Fast IrDA, USB 2.0 High Speed, SDIO, WIFI Interface, and MMC/SD/MS/MS Pro storage systems. These interfaces provide MT6236 users with the highest level of flexibility in implementing high-end solutions.

To achieve a complete user interface, MT6236 also brings together all the necessary peripheral blocks for a multimedia GSM/GPRS phone. The peripheral blocks include the Keypad Scanner with the capability to detect multiple key presses, SIM Controller, Real Time Clock, PWM, Serial LCD Controller, and General Purpose Programmable I/Os.

Furthermore, to provide better configurability and bandwidth for multimedia products, an additional 18-bit parallel interface is incorporated. This interface enables connection to LCD panels and NAND flash devices on the shared interface for additional multimedia data storage.

Audio

Using a highly integrated mixed-signal Audio Front-End, the MT6236 architecture provides easy audio interfacing with direct connection to the audio transducers. The audio interface integrates D/A and A/D Converters for Voice band, as well as high resolution Stereo D/A Converters for Audio band. In addition, MT6236 also provides Stereo Input and Analog MUX.

MT6236 supports AMR codec to adaptively optimize speech and audio quality. Moreover, HE-AAC codec is implemented to deliver CD-quality audio at low bit rates.

In addition, an 900mW class-D amplifier is also embedded to save the BOM cost of adopting external amplifier.

Radio

MT6236 integrates a mixed-signal baseband front-end in order to provide a well-organized radio interface with flexibility for efficient customization. The front-end contains gain and offset calibration mechanisms, and filters with programmable coefficients for comprehensive compatibility control on RF modules. This approach provides the usage of a high resolution D/A Converter for controlling VCXO or crystal, reducing the need for an expensive TCVCXO. MT6236 achieves great MODEM performance by utilizing a 14-bit high resolution A/D Converter in the RF downlink path. Furthermore, to reduce the need for extra external current-driving component, the driving strength of some BPI outputs is designed to be configurable.

Bluetooth radio

MT6236 offers a highly integrated Bluetooth radio and baseband processor. Only a minimum of external components are required. MT6236 provides superior sensitivity and class 1 output power, thus ensures the quality of the connection with a wide range of Bluetooth devices.

MT6236 is fully compliant with Bluetooth v2.1 and offers the enhanced data rates of up to 3Mbps. It also provides the coexistence protocol with 802.11 system, with Mediatek's proprietary interface.

MT6236 has rich support of Bluetooth profiles. It enables diversified applications that are widely used on the handset with excellent interoperability.

Debug Function

The JTAG interface enables in-circuit debugging of the software program with the ARM926EJ-STM core. With this standardized debugging interface, MT6236 provides developers with a wide set of options in choosing ARM development kits from different third party vendors.

Power Management

An power management is embedded in MT6236 to provide the rich features that an high-end feature phone supports, including Li-ion battery charger, high performance and low quiescent current LDOs, power efficient switching regulator, and drivers for LED and backlight.

The MT6236 offers various low-power features to help reduce system power consumption. MT6236 is also fabricated in an advanced low power 65nm CMOS process, hence providing an overall ultra low leakage solution.

Package

The MT6236 device is offered in a 12.5mm×12.5mm, 382-ball, 0.5 mm pitch, TFBGA package.

3.1.2 Platform Features

General

Integrated voice-band, audio-band and base-band analog front end
Integrated full-featured power management unit.

MCU Subsystem

ARM926EJ-STM 32-bit RISC processor
Java hardware acceleration for fast Java-based games and applets
Operating frequency: 13/78/156/312 MHz
64K Byte Instruction TCM and 176K Byte Data TCM
16K Byte Instruction Cache and 16K Byte Data Cache
High performance multi-layer AMBA bus
Dedicated DMA bus with 18 DMA channels
On-chip boot ROM for Factory Flash Programming
Watchdog timer for system crash recovery
3 sets of General Purpose Timer
Circuit Switch Data coprocessor
Division coprocessor
PPP Framer coprocessor

External Memory Interface

Supports up to 3 external devices
Supports 16-bit memory components with maximum size of up to 64M Byte each
Supports Mobile DDR SDRAM and Cellular RAM
Supports Flash and SRAM/PSRAM with burst mode
Industry standard Parallel LCD interface
Supports multimedia companion chips with 8/16 bits data width
Flexible I/O voltage of 1.8V ~ 2.8V for memory interface
Configurable driving strength for memory interface

User Interfaces

8-row × 8-column keypad controller with hardware scanner
Supports multiple key presses for gaming
Dual SIM/USIM controller with hardware T=0/T=1 protocol control
Real Time Clock (RTC) operating with a low-quiescent-current power supply
General Purpose I/Os (GPIOs) available for auxiliary application
4 sets of Pulse Width Modulation (PWM) output
7 external interrupt lines
4-channel auxiliary 10-bit A/D Converter

Security

Supports security key and 128 bit chip random ID

Connectivity

3 UARTs with hardware flow control and supports of baud rate up to 921600bps
IrDA modulator/demodulator with hardware framer. Supports SIR/MIR/FIR operating speeds.

USB 2.0 high speed capability.

Multi Media Card, Secure Digital Memory Card, Memory Stick, Memory Stick Pro host controller with flexible I/O

voltage power

Supports SDIO interface for SDIO peripherals as well as WIFI connectivity

DAI/PCM and I2S interface for Audio application

I2C master interface for peripheral management including image sensor.

SPI master interface for peripheral management including digital TV chip.

Power Management

Li-ion battery charger

High efficiency bulk converter for core power and memory power supply with programmable voltage scaling

13 LDOs for the power supply of memory card, camera, Bluetooth, RF, SIM card, and other diversified usage

One boost regulator and Four Open-Drain Output Switches to Supply/Control the LED LDO type Vibrator

One NMOS switch to control R(GB) LED

Thermal Overload Protection

Under Voltage Lock-out Protection

Over Voltage Protection

Different level of power down modes with sophisticated software control enables excellent power saving

performance.

Test and Debug

Built-in digital and analog loop back modes for both Audio and Baseband Front-End

DAI port complying with GSM Rec.11.10

JTAG port for debugging embedded MCU

3.1.3 MODEM Features

Radio Interface and Baseband Front End

- GMSK modulator with analog I and Q channel outputs
- 10-bit D/A Converter for uplink baseband I and Q signals
- 14-bit high resolution A/D Converter for downlink baseband I and Q signals
- Calibration mechanism of offset and gain mismatch for baseband A/D Converter and

D/A Converter

- 10-bit D/A Converter for Automatic Power Control
- 13-bit high resolution D/A Converter for Automatic Frequency Control
- Programmable Radio RX filter
- 2 channels Baseband Serial Interface (BSI) with 3-wire control
- Bi-directional BSI interface. RF chip register read access with 3-wire or 4-wire interface.
- 10-Pin Baseband Parallel Interface (BPI) with programmable driving strength
- Multi-band support

Voice and Modem CODEC

- Dial tone generation
- Voice memo
- Noise reduction
- Echo suppression
- Advanced sidetone Oscillation Reduction
- Digital sidetone generator with programmable gain
- Two programmable acoustic compensation filters

GSM/GPRS quad vocoders for adaptive multirate (AMR), enhanced full rate (EFR), full rate (FR) and half rate (HR)

- GSM channel coding, equalization and A5/1, A5/2, and A5/3 ciphering
- GPRS GEA1, GEA2 and GEA3 ciphering
- Programmable GSM/GPRS modem
- Packet Switched Data with CS1/CS2/CS3/CS4 coding schemes
- GSM Circuit Switch Data
- GPRS Class 12

Voice Interface and Voice Front End

Two microphone inputs sharing one low noise amplifier with programmable gain and automatic gain control (AGC) mechanisms

- Voice power amplifier with programmable gain
- 2nd order Sigma-Delta A/D Converter for voice uplink path
- D/A Converter for voice downlink path
- Supports half-duplex hands-free operation
- Compliant with GSM 03.50

3.1.4 Multi-Media Features

LCD/NAND Flash Interface

Dedicated Parallel Interface supports 3 external devices with 8-/16-bit NAND flash interface, 8-/9-/16-/18-bit

Parallel interface, and Serial interface for LCM

Built-in NAND Flash Controller with 1-bit ECC for mass storage

LCD Controller

Supports simultaneous connection to up to 3 parallel LCD and 2 serial LCD modules

Supports LCM format: RGB332, RGB444, RGB565, RGB666, RGB888

Supports LCD module with maximum resolution up to 320x480 at 24bpp

Per pixel alpha channel

True color engine

Supports hardware display rotation

Capable of combining display memories with up to 4 blending layers

Image Signal Processor

8/10 bit Bayer format image input

RGB565/YUV422 format image input

Capable of processing image of size up to 5M pixels

Color correction matrix

Gamma correction

Automatic exposure (AE) control

Automatic focus control

Automatic white balance (AWB) control

Programmable AE/AWB windows

Edge enhancement support

Shading compensation

Defect Pixel compensation

JPEG Decoder

SW based JPEG decode

Supports various YUV format, DC/AC Huffman tables, and quantization tables.

JPEG Encoder

ISO/IEC 10918-1 JPEG baseline mode

ISO/IEC 10918-2 compliance

Supports YUV422 and YUV420 and grayscale formats

Supports EXIF/JFIF

Standard DC and AC Huffman tables

Provides 14 levels of encode quality

Supports continuous shooting

Image Data Processing

- Supports Digital Zoom

- Supports RGB and YCbCr image processing

- High throughput hardware scaler. Capable of tailoring an image to an arbitrary size.

- Horizontal scaling with bicubic interpolation

- Vertical scaling with bicubic interpolation

- Simultaneous scaling for video buffer and display buffer

- YUV and RGB color space conversion

- Boundary padding

- Pixel processing: hue/saturation/intensity/color adjustment, Gamma correction and grayscale/invert/sepia-tone effects

- Hardware accelerated image editing

- Photo frame capability

- RGB/YCbCr format thumbnail output

MPEG-4/H.263 CODEC

- Software Video CODEC

- ISO/IEC 14496-2 simple profile:

- decode @ level 0/1/2/3

- encode @ level 0

- ITU-T H.263 profile 0 @ level 10

- Encoder resync marker and HEC

- Supported visual tools for decoder: I-VOP, P-VOP, AC/DC Prediction, 4-MV, Unrestricted MV, Error Resilience, Short Header

- Error Resilience for decoder: Slice Resynchronization, Data Partitioning, Reversible VLC

- Supported visual tools for encoder: I-VOP, P-VOP, Half-Pel, DC Prediction, Unrestricted MV, Reversible VLC, Short Header

- Supports encoding motion vector of range up to -64/+63.5 pixels

- HE-AAC decode support

- AAC/AMR/WB-AMR audio decode support

- AMR/WB-AMR audio encode support

H.264 Decode

- Software Decoder

- ISO/IEC 14496-10 baseline profile : decode @ level 3

2D Accelerator

Supports 32-bpp ARGB8888, 24-bpp RGB888, 16-bpp RGB565, and 8-bpp index color modes

Rectangle fill with constant or gradient color

BitBlt: capable with 7 rotation types, ROP4,, Mask

Alpha blending with 7 rotation types, per-pixel alpha and pre-multiplied alpha

Font caching: normal and italic font

Linear transform : Support perspective transform, truncate/nearest/bi-linear sample filter.

Command queue with max depth of 1024

Audio CODEC

Supports HE-AAC codec decode

Supports AAC codec decode

Wavetable synthesis with up to 64 tones

3.1.5 Bluetooth Features

Radio features

Fully compliant with Bluetooth specification 2.1 + EDR

Low out-of-band spurious emissions supports simultaneous operation with GPS, GSM/GPRS worldwide radio systems

Low-IF architecture with high degree of linearity and high order channel filter

Integrated T/R switch and Balun.

Fully integrated PA provides 10dBm output power

-90dBm sensitivity with excellent interference rejection performance

Hardware AGC dynamically adjusts receiver performance in changing environments

Baseband features

Up to 7 simultaneous active ACL links

Up to 3 simultaneous SCO and eSCO links with CVSD coding

eSCO support

Scatternet support: Up to 4 piconets simultaneously with background inquiry/page scan

Sniff mode, hold mode, and park mode support

AFH and PTA collaborative support for WLAN/BT coexistence

Idle mode and sleep mode enables ultra low power consumption

PCM interface and built-in transcoders for A-law, μ -law and linear voice with re-transmission support

Built-in hardware modem engine for access code correlation, header error correction, forward error correction, CRC, whitening, and encryption

Channel quality driven data rate adaptation

Channel assessment for AFH

Platform features

Embedded ARM7 processor for Bluetooth protocol stack with built-in memory system.

Fully verified ROM based system with code patch for feature enhancement.

3.1.6 General Description

Figure 3-1-2 depicts the block diagram of MT6236. Based on a multi-processor architecture, MT6236 integrates an ARM926EJ-STM core, the main processor running high-level GSM/EDGE protocol software as well as multimedia applications, two digital signal processor core, which manages the low-level MODEM as well as advanced audio functions, and an ARM7EJ-STM core, the dedicated processor running Bluetooth baseband and link control protocol, as well as the Bluetooth radio control.

MT6236 consists of the following subsystems:

Microcontroller Unit (MCU) Subsystem: includes an ARM926EJ-STM RISC processor and its accompanying memory management and interrupt handling logics;

Digital Signal Processor (DSP) Subsystem: includes a DSP and its accompanying memory, memory controller, and interrupt controller;

MCU/DSP Interface: the junction at which the MCU and the DSP exchange hardware and software information;

Microcontroller Peripherals: includes all user interface modules and RF control interface modules;

Microcontroller Coprocessors: runs computing-intensive processes in place of the Microcontroller;

DSP Peripherals: hardware accelerators for GSM/GPRS channel codec;

Multimedia Subsystem: integrates several advanced accelerators to support multimedia applications;

Voice Front End: the data path for converting analog speech to and from digital speech;

Audio Front End: the data path for converting stereo audio from an audio source;

Baseband Front End: the data path for converting a digital signal to and from an analog signal from the RF modules;

Timing Generator: generates the control signals related to the TDMA frame timing; and,

Power, Reset and Clock Subsystem: manages the power, reset, and clock distribution inside MT6236.

Bluetooth subsystem: includes an ARM7EJ-STM RISC processor with embedded ROM/RAM system, baseband processor, and a high-performance radio block.

Power management unit: a self-contained power supply source which also controls the charging and system startup

Circuitry.

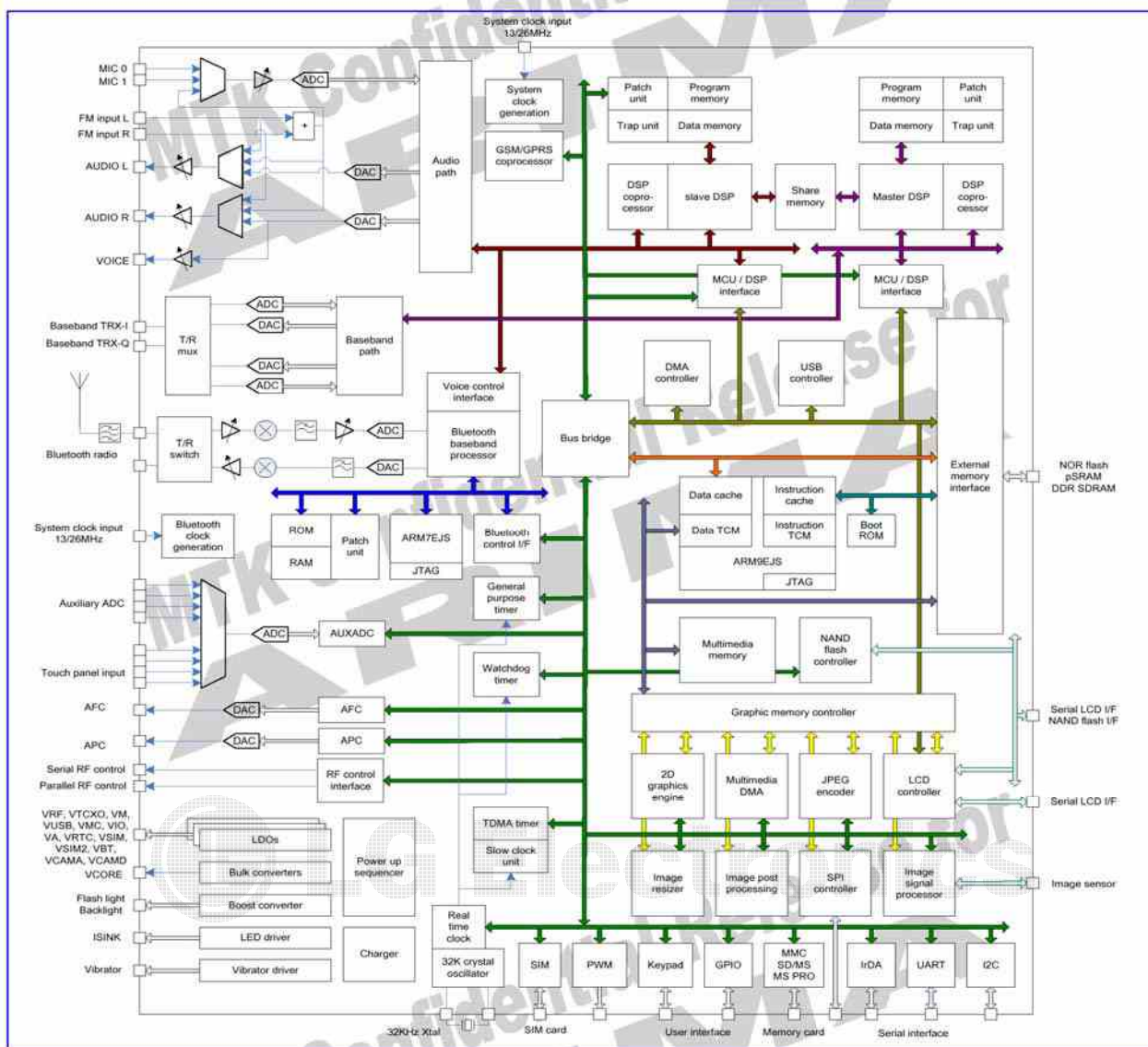


Figure.3-1-2 MT6236 BLOCK DIAGRAM

3.2 Power Amplifier Module (SKY77547)

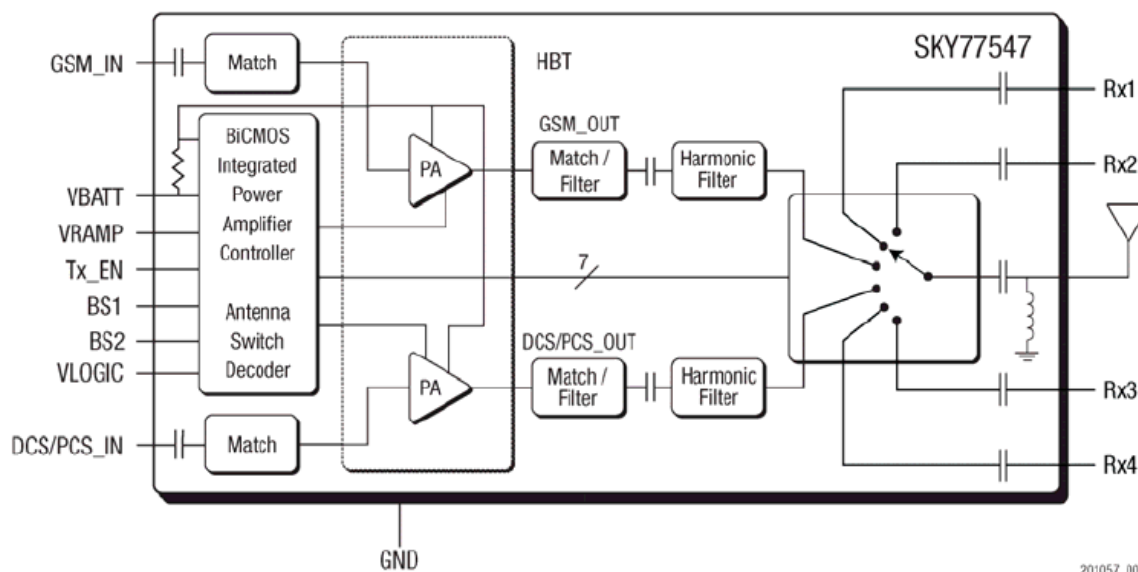


Figure.3-2-1 SKY77547 FUNCTIONAL BLOCK DIAGRAM

The SKY77547 is a transmit and receive front-end module (FEM) with Integrated Power Amplifier Control (iPAC™) for quad-band cellular handsets comprising GSM850/900 and DCS1800/PCS1900 operation. Designed in a low profile, compact form factor, the SKY77547 offers a complete Transmit VCO-to-Antenna and Antenna-to-Receive SAW filter solution. The FEM also supports Class 12 General Packet Radio Service (GPRS) multi-slot operation.

The module consists of a GSM850/900 PA block and a DCS1800/PCS1900 PA block, impedancematching circuitry for 50 Ω input and output impedances, Tx harmonics filtering, high linearity and a low insertion loss PHEMT RF switch, and a Power Amplifier Control (PAC) block with internal current sense resistor. A custom BiCMOS integrated circuit provides the internal PAC function and decoder circuitry to control the RF switches. The two Heterojunction Bipolar Transistor (HBT) PA blocks are fabricated onto a single Gallium Arsenide (GaAs) die. One PA block supports the GSM850/900 bands and the other PA block supports the DCS1800/PCS1900 bands. Both PA blocks share common power supply pads to distribute current. The output of each PA block and the outputs to the four receive pads are connected to the antenna pad through a PHEMT RF switch. The GaAs die, PHEMT die, Silicon (Si) die and passive components are mounted on a multi-layer laminate substrate. The assembly is encapsulated with plastic overmold.

Band selection and control of transmit and receive are performed using four external control pads. Refer to the block diagram in [Figure 3-2-1](#) below. The band select pads, BS1 and BS2, select GSM850, GSM900, DCS, and PCS modes of operation. Transmit enable Tx_EN controls receive or transmit mode of the RF switch (Tx = logic 1). Proper timing

between transmit enable Tx_EN and Analog Power Control VRAMP allows for high isolation between the antenna and Tx-VCO while the VCO is being tuned prior to the transmit burst.

The SKY77547 is compatible with logic levels from 1.2 V to VCC for BS1, BS2, and Tx_EN pads, depending on the level applied to the VLOGIC pad. This feature provides additional flexibility for the designer in the selection of FEM interface control logic.

Table 3. SKY77547 Mode Control Logic

Mode	VLOGIC	Input Control Bits		
		Tx_EN	BS1	BS2
STANDBY	0	X ¹	X ¹	X ¹
Rx1	1	0	0	0
Rx2	1	0	0	1
Rx3	1	0	1	1
Rx4	1	0	1	0
LB_Tx	1	1	0	X ¹
HB_Tx	1	1	1	X ¹

¹ X = DON'T CARE

² Rx1, Rx2, Rx3, and Rx4 are broadband receive ports and each supports the GSM850, GSM900, DCS, and PCS bands.



extremely low noise removing the need for external SAW filters prior to the PA.

The AD6548/9 uses a single integrated LO VCO for both the receive and the transmit circuits. The synthesizer lock times are optimized for GPRS applications up to and including class 12.

To dramatically reduce the BOM both TX Translational loop and main PLL Loop Filters are fully integrated into the device.

AD6548 incorporates a complete reference crystal calibration system. This allows the external VCTCXO to be replaced with a low cost crystal. No other external components are required. The AD6549 uses the traditional VCTCXO reference source.

The AD6548/9 also contains on-chip low dropout voltage regulators (LDOs) to deliver regulated supply voltages to the functions on chip, with a battery input voltage of between 2.9V and 5.5V. Comprehensive power down options are included to minimize power consumption in normal use.

A standard 3 wire serial interface is used to program the IC. The interface features low-voltage digital interface buffers compatible with logic levels from 1.6V to 3.0V.

The AD6548/9 is packaged in a 5mm × 5mm , 32-lead LFCSP package.

ORDERING GUIDE	Model TemperatureRange	Package
AD6548BCPZ	-20°C to +85°C	LFCSP-32
AD6549BCPZ	-20°C to +85°C	LFCSP-32

3.3.2 Features

Fully Integrated GSM Transceiver including

Direct Conversion Receiver

- 4 Differential LNAs
- Integrated Active RX Channel Select Filters
- Programmable Gain Baseband Amplifiers

Translation Loop Direct VCO Modulator

- Integrated TX VCO and tank
- External TX filters eliminated
- Integrated Loop filter components

High performance multi band PLL system

- Fast Fractional-N Synthesizer
- Integrated Local Oscillator VCO
- Fully Integrated Loop filters
- Crystal Reference Oscillator & Tuning System (AD6548)

Power Management

- Integrated LDOs allow direct battery supply connection

Small footprint

32-Lead 5 X 5 mm ChipScale Package

APPLICATIONS

Dual, Triple and Quad Band Radios

- GSM850, E-GSM 900, DCS1800 and PCS1900
- GPRS to Class 12- EDGE RX

3.3.3 Pin Descriptions

No	Name	Description	No	Name	Description
1	VCC_FE	Front end supply (IP) ³	17	VCC_REF	Reference Oscillator Supply (IP)
2	I	I baseband input/output	18	VAFC	AD6548 Crystal Freq control (IP) AD6549: Connect to VCC_REF
3	IB	I baseband input/output	19	REFINB	Crystal / VCTCXO Connection
4	VCC_BBI	Baseband I, TX path supply (IP) ³	20	REFIN	Crystal Connection
5	SDATA	Serial port data	21	REF_OP	Reference Frequency Output
6	SCLK	Serial port clock	22	QB	Q baseband input/output
7	SEN	Serial port enable	23	Q	Q baseband input/output
8	N/C	Not connected	24	VCC_BBQ	Baseband Q supply (IP) ³
9	VLDO3	TX LDO Output ¹	25	RX1900B	PCS 1900 LNA input
10	TXOP_LO	Transmit O/P (850/900MHz)	26	RX1900	PCS 1900 LNA input
11	TXOP_HI	Transmit O/P (1800/1900MHz)	27	RX1800B	DCS 1800 LNA input
12	VCC_TXVCO	TX VCO supply (1)	28	RX1800	DCS 1800 LNA input
13	VDD	Serial interface supply	29	RX900B	E-GSM 900 LNA input
14	VBAT	Battery I/P for LDO reg's	30	RX900	E-GSM 900 LNA input
15	VLDO1	LDO regulator Output ²	31	RX850B	GSM 850 LNA input
16	VLDO2	LO VCO Supply ¹	32	RX850	GSM 850 LNA input

3.4 Memory Module (K521H12ACB-B060)

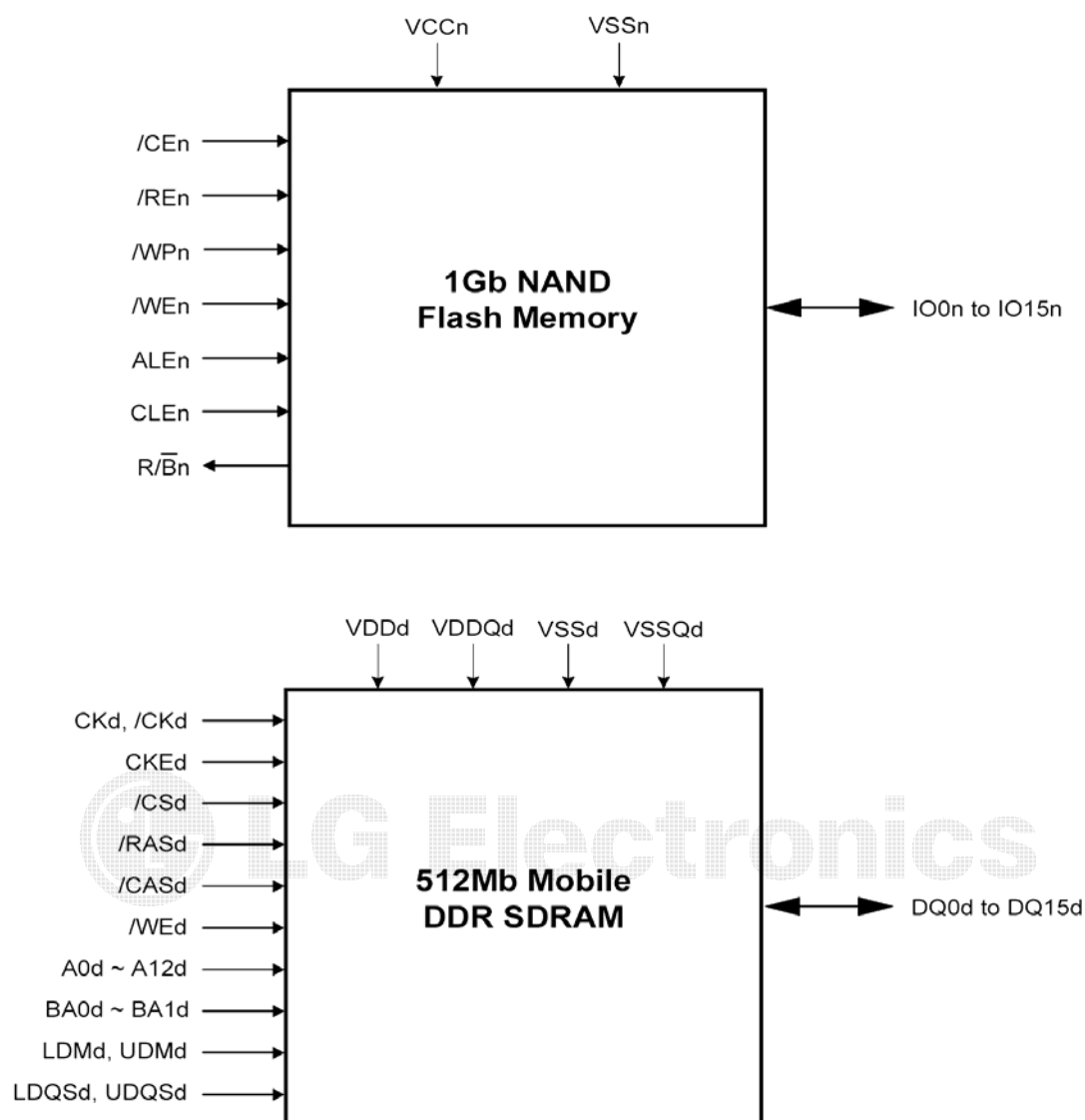


Figure.3-4-1 K521H12ACB-B060 FUNCTIONAL BLOCK DIAGRAM

3.4.1 General Description

The K521H12ACB is a Multi Chip Package Memory which combines 1Gbit NAND Flash Memory and 512Mbit DDR synchronous high data rate Dynamic RAM.

NAND cell provides the most cost-effective solution for the solid state application market. A program operation can be performed in typical 250 μ s on the (1K+32)Word page and an erase operation can be performed in typical 2ms on a (64K+2K)Word block. Data in the data register can be read out at 42ns cycle time per Word. The I/O pins serve as the ports for address and data input/output as well as command input. The on-chip write controller automates all program and erase functions including pulse repetition, where required, and

internal verification and margining of data. Even the write-intensive systems can take advantage of the device's extended reliability of 100K program/erase cycles by providing ECC(Error Correcting Code) with real time mapping-out algorithm. The device is an optimum solution for large nonvolatile storage applications such as solid state file storage and other portable applications requiring non-volatility.

In 512Mb Mobile DDR, Synchronous design make a device controlled precisely with the use of system clock. Range of operating frequencies, programmable burst length and programmable latencies allow the same device to be useful for a variety of high bandwidth, high performance memory system applications.

The K521H12ACB is suitable for use in data memory of mobile communication system to reduce not only mount area but also power consumption. This device is available in 107-ball FBGA Type.

3.4.2 FEATURES

<Common>

- Operating Temperature : -25°C ~ 85°C
- Package : 107ball FBGA Type - 10.5 x 13 x 1.2mmt, 0.8mm pitch

<NAND Flash>

- Voltage Supply : 1.7V ~ 1.95V
- Organization
 - Memory Cell Array : (64M + 2M) x 16bit for 1Gb
 - Data Register : (1K + 32) x 16bit
- Automatic Program and Erase
 - Page Program : (1K + 32)Word
 - Block Erase : (64K + 2K)Word
- Page Read Operation
 - Page Size : (1K + 32)Word
 - Random Read : 40μs(Max.)
 - Serial Access : 42ns(Min.)
- Fast Write Cycle Time
 - Page Program time : 250μs(Typ.)
 - Block Erase Time : 2ms(Typ.)
- Command/Address/Data Multiplexed I/O Port
- Hardware Data
 - Program/Erase Lockout During Power Transitions
- Reliable CMOS Floating-Gate Technology
 - Endurance : 100K Program/Erase Cycles with 1bit/256Word ECC for x16
- Command Driven Operation

Unique ID for Copyright Protection

<Mobile DDR>

- VDD/VDDQ = 1.8V/1.8V
- Double-data-rate architecture; two data transfers per clock cycle
- Bidirectional data strobe(DQS)
- Four banks operation
- Differential clock inputs(CK and CK)
- MRS cycle with address key programs
 - CAS Latency (3)
 - Burst Length (2, 4, 8, 16)
 - Burst Type (Sequential & Interleave)
- EMRS cycle with address key programs
 - Partial Array Self Refresh (Full, 1/2, 1/4 Array)
 - Output Driver Strength Control (Full, 1/2, 1/4, 1/8, 3/4, 3/8, 5/8, 7/8)
- Internal Temperature Compensated Self Refresh
- All inputs except data & DM are sampled at the positive going edge of the system clock(CK).
- Data I/O transactions on both edges of data strobe, DM for masking.
- Edge aligned data output, center aligned data input.
- No DLL; CK to DQS is not synchronized.
- LDM - UDM for write masking only.
- Clock Stop capability.
- Auto refresh duty cycle
 - 7.8us

Operating Frequency

	DDR333
Speed @CL3 ¹⁾	166MHz

NOTE :

1) CAS Latency

Address configuration

Organization	Bank	Row	Column
32Mx16	BA0,BA1	A0 - A12	A0 - A9

- DM is internally loaded to match DQ and DQS

3.4.3 PIN CONFIGURATION

	1	2	3	4	5	6	7	8	9	10
A		DNU							DNU	DNU
B	DNU	NC	DQ0d	VDDd	VSSn	VCCn	NC	A3d	NC	DNU
C		VSSd	DQ2d	DQ1d	CLEn	/CEn	A0d	A1d	A2d	
D		VDDQd	DQ4d	DQ3d	ALEn	/WEEn	BA0d	BA1d	A10d	
E		VSSQd	DQ6d	DQ5d	/REn	R/Bn	/RASd	NC	/CSd	
F		VDDQd	LDQsd	DQ7d	/WPn	NC	/CASd	/WEd	VSSd	
G		VSSd	LDMd	/CKd	NC	NC	A12d	CKEd	VDDd	
H		VDDd	UDMd	CKd	NC	NC	A8d	A9d	A11d	
J		VSSQd	UDQsd	DQ8d	IO0n	IO2n	IO4n	IO6n	A7d	
K		VDDQd	DQ9d	DQ10d	IO8n	IO10n	IO12n	IO14n	A6d	
L		VSSQd	DQ11d	DQ12d	IO1n	IO3n	IO5n	IO7n	A5d	
M		VDDd	DQ13d	DQ14d	IO9n	IO11n	IO13n	IO15n	A4d	
N	DNU	NC	DQ15d	VSSd	VSSn	VCCQn	VCCn	VSSn	NC	DNU
P	DNU	DNU							DNU	DNU

107 FBGA: Top View (Ball Down)

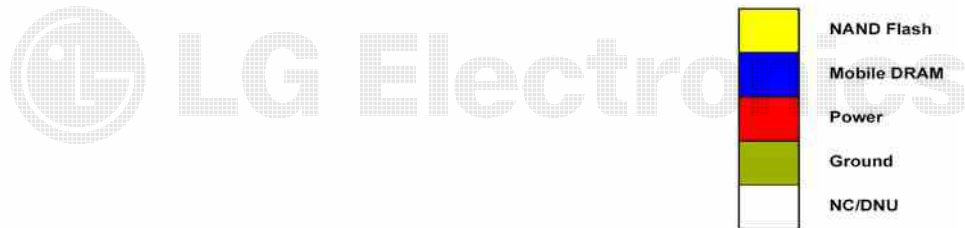


Figure.3-4-2 107 Balls FBGA Package (TOP View)

3.4.4 PIN DESCRIPTION

Pin Name	Pin Function(Mobile DRAM)
CKd, /CKd	Differential System Clock
CKEd	Clock Enable
/CSd	Chip Select
/RASd	Row Address Strobe
/CASd	Column Address Strobe
/WEd	Write Enable
A0d ~ A12d	Address Input
BA0d ~ BA1d	Bank Address Input
LDMd,UDMd	Lower / Upper Input Data Mask
LDQsd , UDQsd	Lower / Upper Data Strobe
DQ0d ~ DQ15d	Data Input/Output
VDDd	Power Supply
VDDQd	Data Out Power
VSSd	Ground
VSSQd	DQ Ground

Pin Name	Pin Function(NAND Flash)
/CEn	Chip Enable
/REn	Read Enable
/WPn	Write Protection
/WEEn	Write Enable
ALEn	Address Latch Enable
CLEn	Command Latch Enable
R/Bn	Ready/Busy Output
IO0n ~ IO15n	Data Input/Output
VCCn	Power Supply
VSSn	Ground

Pin Name	Pin Function
DNU	Do Not Use
NC	No Connection

3.5 FM Radio Module (AR1200)

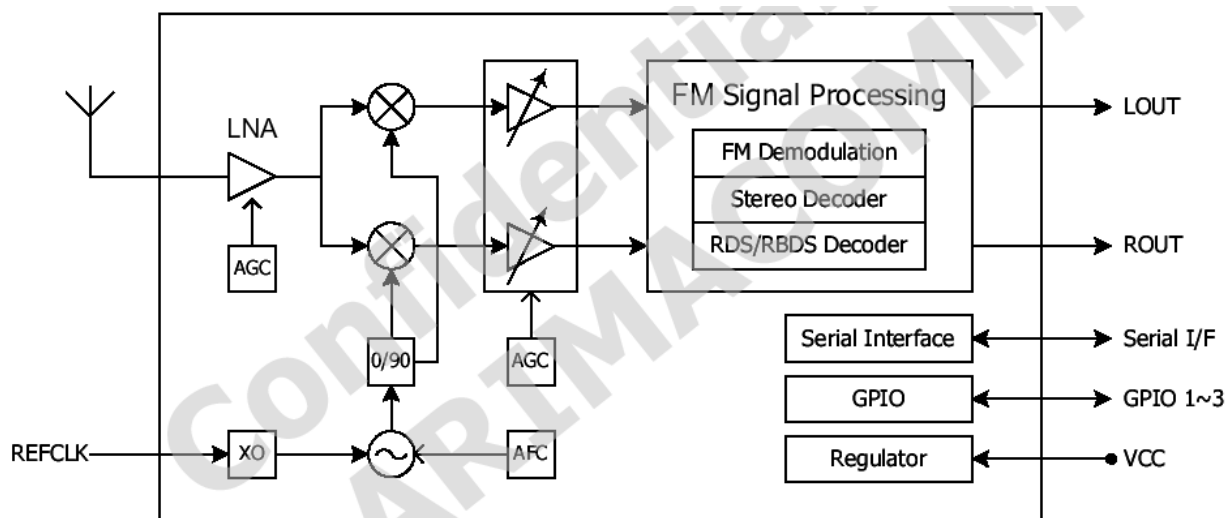


Figure. 3-5-1 AR1200 FM Receiver Block Diagram

3.5.1 Features

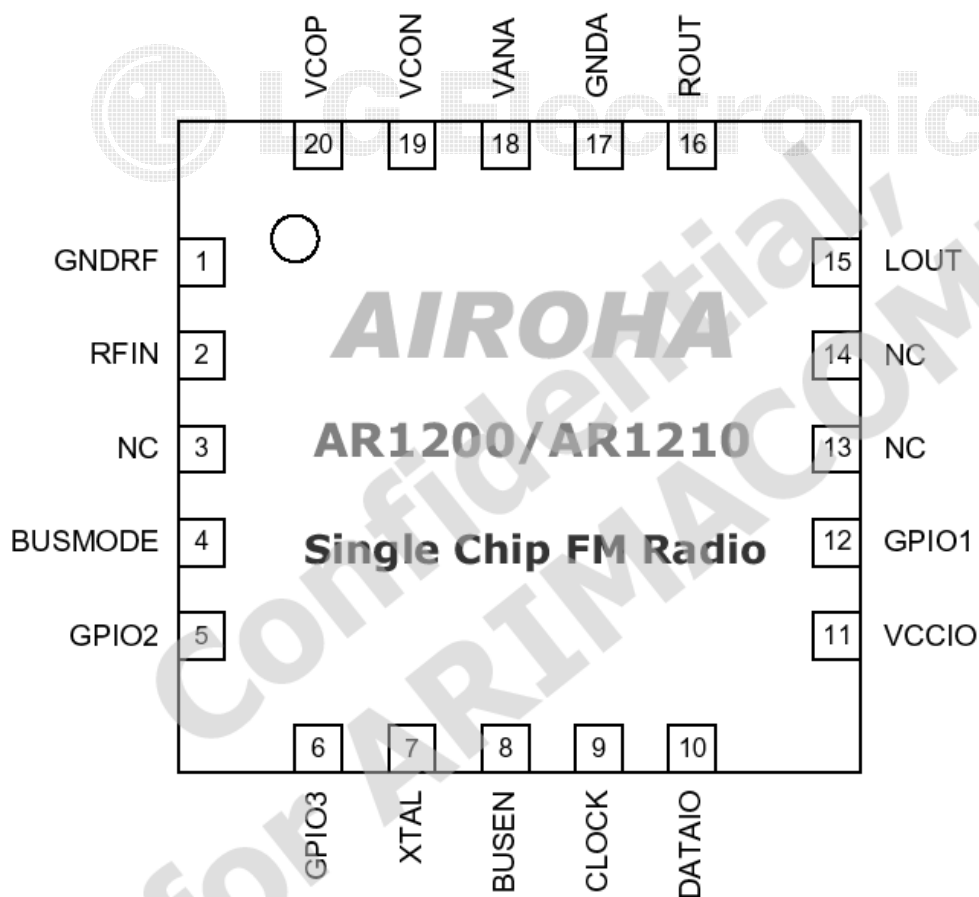
- Worldwide FM band support: 76~108MHz
- Highest integration level with minimized external BOM cost
- Frequency synthesizer with integrated VCO and Automatic Frequency Control (AFC)
- Integrated XO with external reference clock input or external 32.768KHz crystal
- Automatic Gain Control (AGC) on LNA/VGA amplifiers
- Signal strength measurement
- Programmable de-emphasis time constant (50/75us)
- Adaptive noise suppression
- Analog output with volume control and Line-level outputs
- Serial control interface for 2-wire and 3-wire modes
- Support both RDS & RBDS (AR1200 only)
- Embedded Seek and Tuning function
- Integrated LDO regulators support 2.7 to 3.6 V supply voltage
- Support 32Ω output load resistance
- QFN 3x3x0.55mm 20-pin package

3.5.2 Description

AR1200/AR1210 is a highly integrated single chip stereo FM radio receiver for all kinds of applications. AR1200/AR1210 supports worldwide FM bands from 76 to 108MHz. It integrates LNA, Mixer, Oscillator and LDO regulator to minimize the external BOM cost. The built-in FM signal processing unit with noise reduction mechanism provides optimum sound quality. A simple 2-wire/3-wire interface allows easy control from the host. For AR1200, Radio Data System (RDS) and Radio Broadcast Data System (RBDS) demodulator and decoder are also supported.

AR1200 and AR1210 are pin-to-pin compatible and are the same in all FM radio receiver functions, specs and packages except the RDS/RBDS function. AR1200 supports RDS/RBDS receiver function, while AR1210 do not.

3.5.3 Pin Assignment



3.5.4 Pin Name Description

PIN	SIGANL	TYPE	DESCRIPTION
1	GNDRF	GND	RF Ground
2	RFIN	Input, Analog	RF Input
3	NC		Not Connected
4	BUSMODE	Input, Digital Control	2-wire/3-wire Bus Mode Selection
5	GPIO2	Input/Output, Digital	General I/O Port 2
6	GPIO3	Input/Output, Digital	General I/O Port 3
7	XTAL	Analog	XTAL Oscillator Input
8	BUSEN	Input, Digital Control	Serial Interface
9	CLOCK	Input, Digital Control	Serial Interface
10	DATAIO	Input/Output, Digital	Serial Interface
11	VCCIO	VCC Supply	Supply Voltage for I/O Ports
12	GPIO1	Input/Output, Digital	General I/O Port 1
13	NC		Not Connected
14	NC		Not Connected
15	LOUT	Output, Analog	Left Audio Output
16	ROUT	Output, Analog	Right Audio Output
17	GNDA	GND	Analog Ground
18	VANA	VCC Supply	Supply Voltage for Analog Circuits
19	VCON	Input, Analog	VCO Tank Input
20	VCOP	Input, Analog	VCO Tank Input

3.6 TV Module(NMI326)

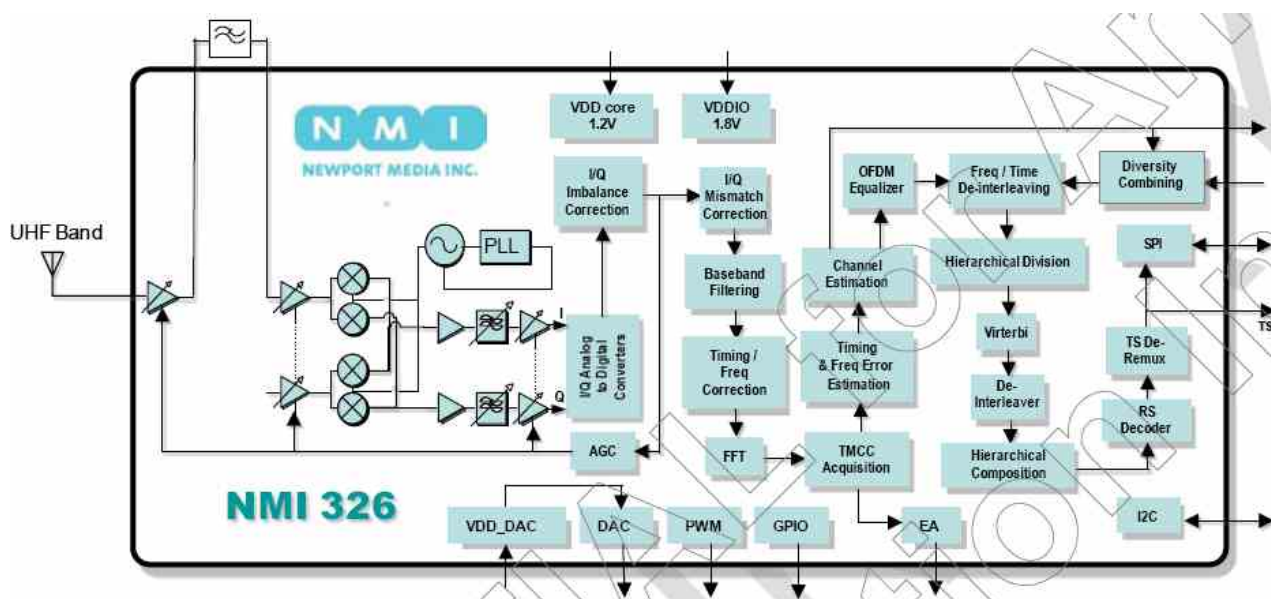


Figure.3-6-1 NMI326 Block Diagram

3.6.1 General Descriptions

The NMI326 is a System-on-Chip (SoC) Integrated Circuit (IC) that implements the Integrated Services Digital Broadcasting for Terrestrial Television Broadcasting (ISDB-T) specifications of the Association of Radio Industries and Businesses (ARIB) Standard STD-B29 and Brazil Standard NBR 15604. Specifically, the NMI326 supports 1-segment ISDB-T reception. Figure 1 shows that the NMI326 is a SoC that implements the RF tuner, demodulator, decoder/de-interleaver, and peripherals on a single die.

3.6.2 Key Features

The key features of the NMI326 are:

- Power consumption: approximately 50mW
- Requires only two supply voltages, namely, 1.2V and I/O voltage
- Ultra-low sensitivity: -100dBm for QPSK mode rate 2/3 (13-segment power)
- Superior linearity: +10dBm maximum input signal
- Up to 60 dB Desired to Undesired Ratio towards ISDB-T blockers
- Doppler >235Hz for QPSK Rate 2/3
- General purpose DAC and PWM for antenna control
- Fully integrated VCOs, PLL loop filter and crystal oscillator
- Fully integrated wideband LNAs and matching networks
- Integrated ADC
- Dual automatic and programmable AGC for RF and IF gain saves power and reduces host processor overhead
- Automatic mode and guard detection
- Fully compliant 1-segment ISDB-T reception according to ARIB Standard STD-B29 and Brazil Standard NBR 15604.
- Automatic re-acquisition upon loss of input signal
- Common phase error correction
- Serial transport stream interface
- SPI interface
- AES-128 ECB encrypted transport stream
- Includes device drivers and middleware support for all popular processors and operating systems

3.6.3 Description

The NMI326 is a fully compliant ISDB-T 1-segment receiver that implements the RF tuner and digital demodulator on the same die. NMI326 receives an input single-ended RF signal from the antenna through the UHF band RF input. The output of NMI326 is an MPEG-2 transport stream.

The tuner is a highly integrated ISDB-T receiver that covers UHF band from 470 MHz to 807 MHz. The tuner integrates the LNA/matching networks, VCO, PLL/loop filter, crystal oscillator, baseband filters and VGA.

The demodulator receives a baseband input from the tuner and provides an Automatic Gain Control (AGC) signal to control the tuner gain. The demodulator implements the full acquisition, synchronization, demodulation, and channel decoding as required by the ISDB specifications STD-B29 and NBR 15604. The demodulator outputs an MPEG-2 transport stream.

Figure 1 shows that the NMI326 communicates with the host via the transport stream (TS) serial interface, the I²C interface, and the SPI interface. Two NM326 (6X6 BGA) devices can be used to support diversity or picture-in-picture.

The NMI326 supports the following transmission parameters:



3.7 LCD Interface

LCM

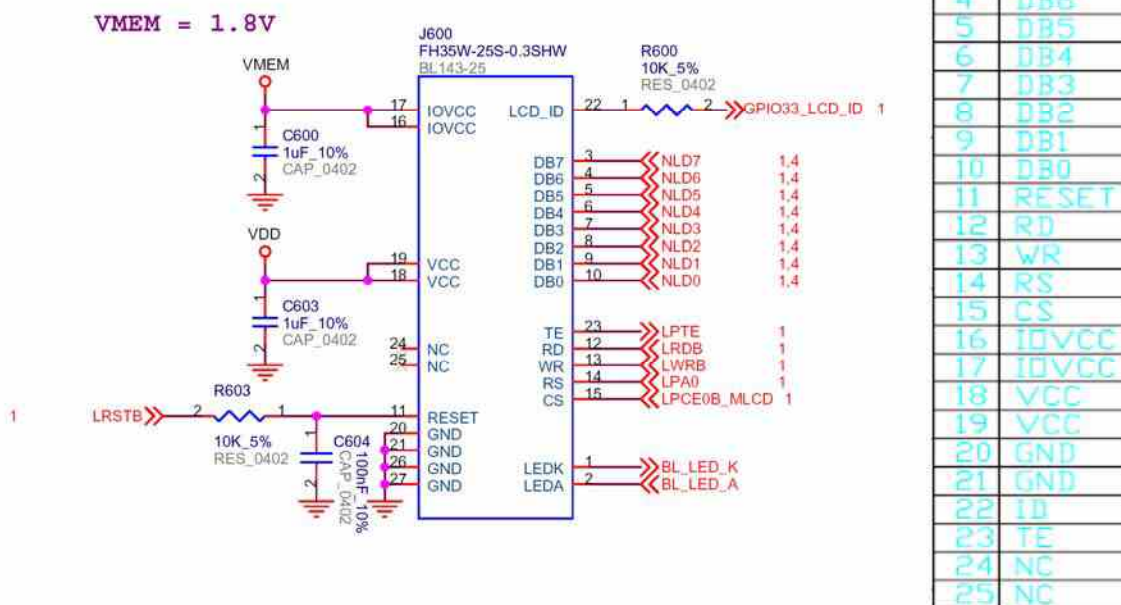


Figure.3-7-1 LCD Interface

3.7.1 General Specifications

Feature		Spec
Display Spec	Size	2.3 inch
	Resolution	320(RGB) x 240
	Interface	CPU 8 bit
	Color Depth	65K
	Technology Type	a-Si
	Pixel Pitch (mm)	0.1461 x 0.1461
	Pixel Configuration	R.G.B. Vertical Stripe
	Display Mode	TM with Normally White
	Surface Treatment	Clear type (3H)
	Viewing Direction	6 o'clock
	Gray Scale Inversion Direction	12 o'clock
Mechanical Characteristics	LCM (W x H x D) (mm)	51.95x46.15x2.1
	Active Area(mm)	46.752x35.064
	With /Without TSP	Without TSP
	Weight (g)	TBD
	LED Numbers	4 LEDs
Electronic	Driver IC	ILI9342 (Polished)

3.7.2 Block Diagram

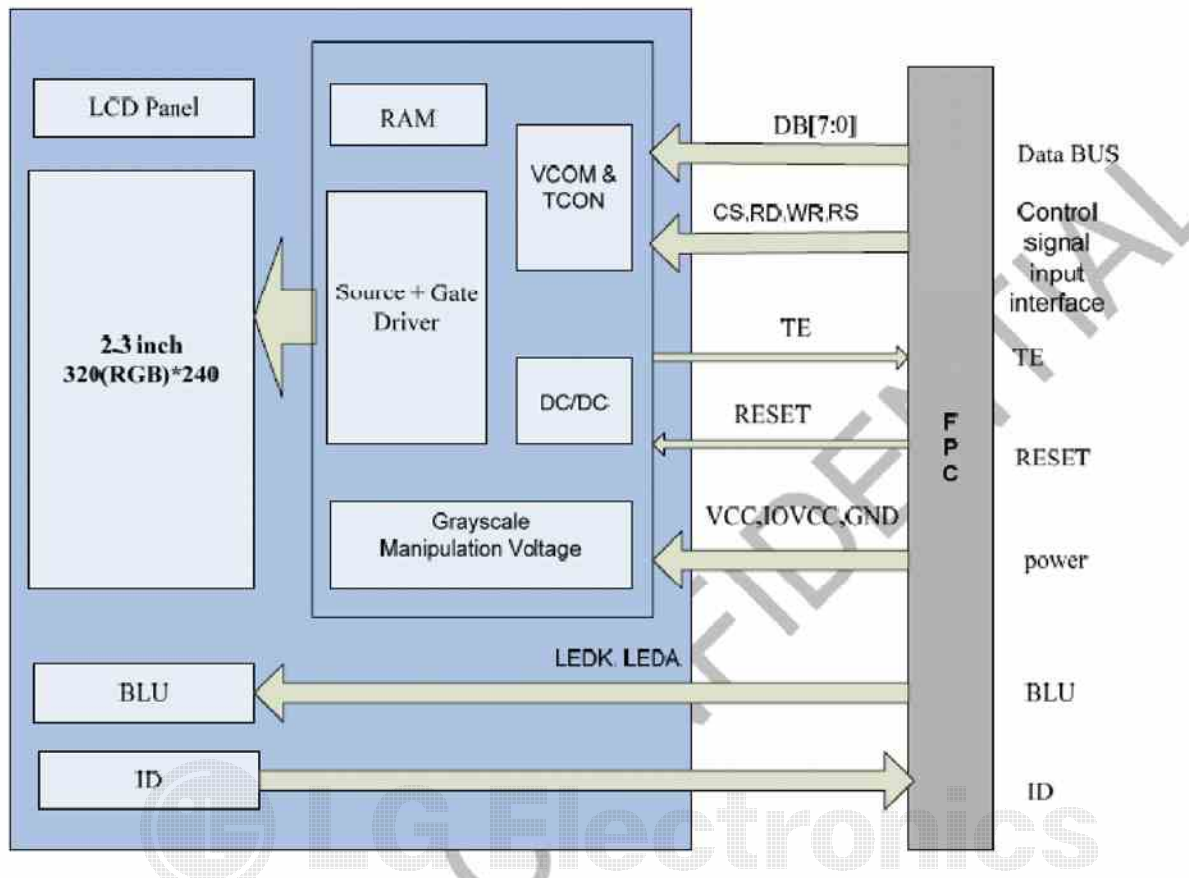


Figure. 3-7-2 TM023KDH41 Block Diagram

3.7.3 Pin description

No	SYMBOL	I/O	Description	Remark
1	LEDK	P	LED Cathode	
2	LEDA	P	LED Anode	
3	DB7	I	Data input/output	
4	DB6	I	Data input/output	
5	DB5	I	Data input/output	
6	DB4	I	Data input/output	
7	DB3	I	Data input/output	
8	DB2	I	Data input/output	
9	DB1	I	Data input/output	
10	DB0	I	Data input/output	
11	RESET	I	Reset signal	
12	RD	I	A read strobe signal and enables an operation to read out data when the signal is low	
13	WR	I	A write strobe signal and enables an operation to write data when the signal is low	
14	RS	I	A register select signal	
15	CS	I	A chip select signal	
16	IOVCC	P	I/O pad and digital power supply	
17	IOVCC	P	I/O pad and digital power supply	
18	VCC	P	Analog power supply	
19	VCC	P	Analog power supply	
20	GND	P	Power Ground	
21	GND	P	Power Ground	
22	ID	O	Customer Identification Pin	
23	TE	O	Tearing effect output, If not used, please open this pin	
24	NC	---	No connection	
25	NC	---	No connection	

Note 1: I/O definition:

I---Input O---Output P---Power(Ground)

3.8 SIM & Micro SD Card Interface

The MT6236 contains a dedicated smart card interface to allow the MCU access to the SIM card. It can operate via 4 terminals, using VSIM, SIMI/O, SIMRST, SIMCLK

The VSIM is used to control the external voltage supply to the SIM card. SIMRST is used as the SIM card reset signal. SIMI/O and SIMCLK are used for data exchange purpose.

The SIM interface acts as a half duplex asynchronous communication port and its data format is composed of ten consecutive bits: a start bit in state Low, eight information bits, and a tenth bit used for parity checking.

The SD Card control signal and data signal is controlled by MT6236.

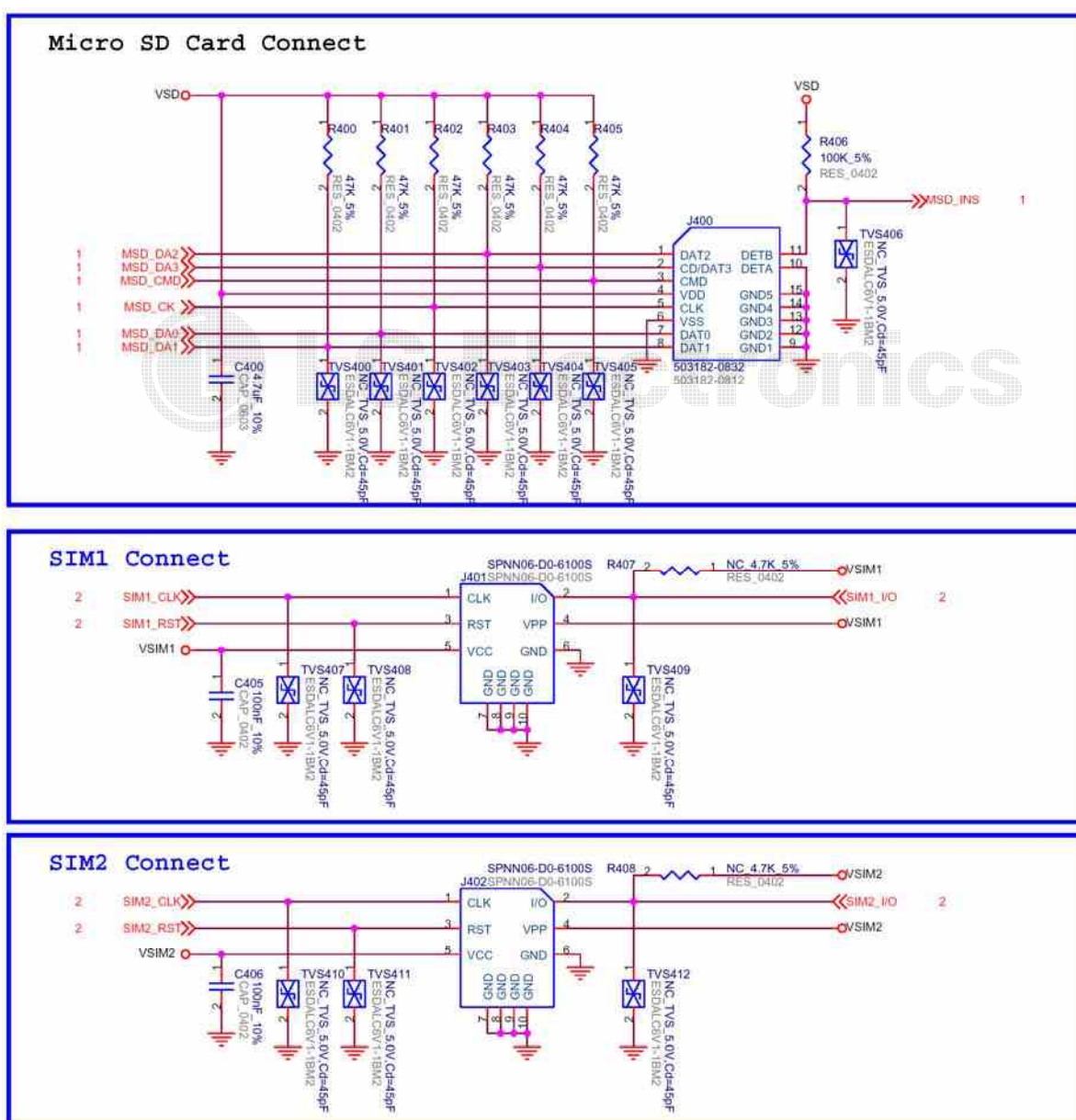


Figure.3-8-1. SIM & Micro SD Card Interface

3.9 KEYPAD Interface

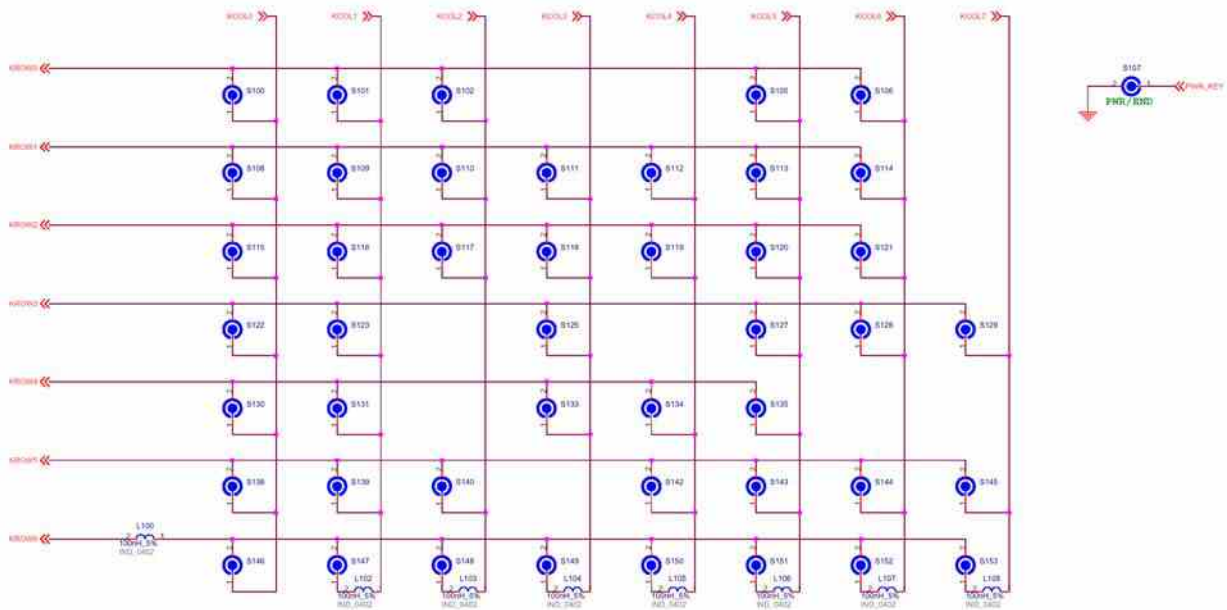


Figure.3-9-1. KEYPAD Interface

The keypad can be divided into two parts: one is the keypad interface including 8 columns and 8 rows with one dedicated power-key; the other is the key detection block which provides key pressed, key released and de-bounce mechanisms. Each time the key is pressed or released, i.e. something different in the 8 x 8 matrix or power-key, the key detection block senses the change and recognizes if a key has been pressed or released. Whenever the key status changes and is stable, a KEYPAD IRQ is issued.

The MCU can then read the key(s) pressed directly in KP_HI_KEY, KP_MID_KEY and KP_LOW_KEY registers. To ensure that the key pressed information is not missed, the status register in keypad is not read-cleared by APB read command. The status register can only be changed by the key-pressed detection FSM.

3.10 Battery Charging Block Interface

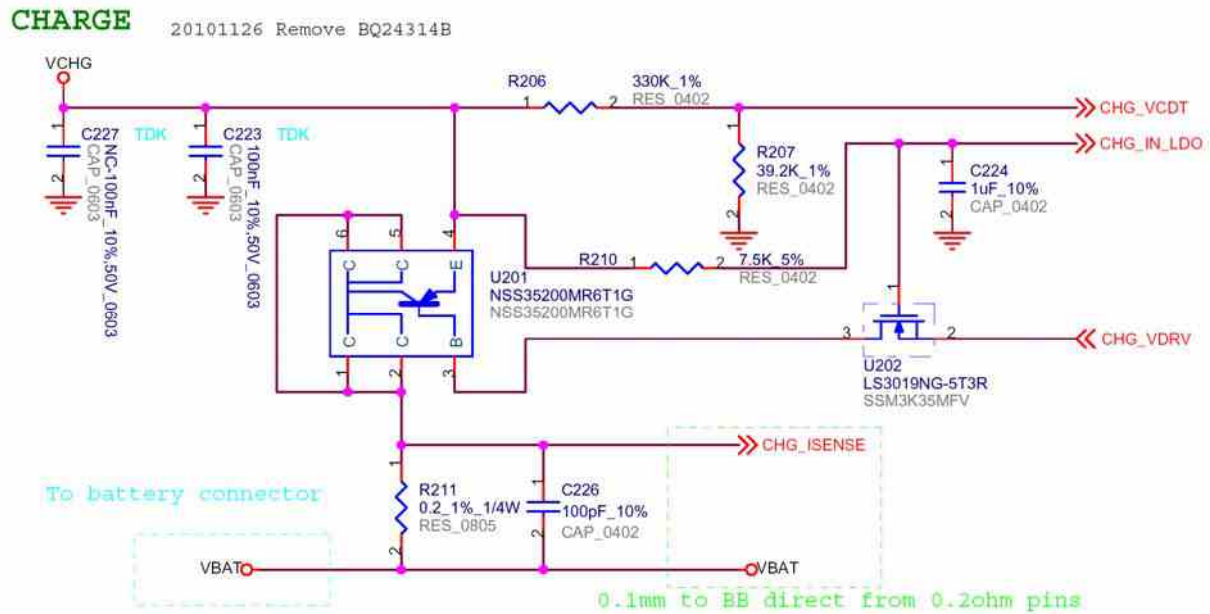


Figure.3-10-1 Charging IC Interface

The NSS35200MR6T1G is controlled by MT6236.



3.11 Audio Interface

3.11.1 Speaker

AUDIO AMP

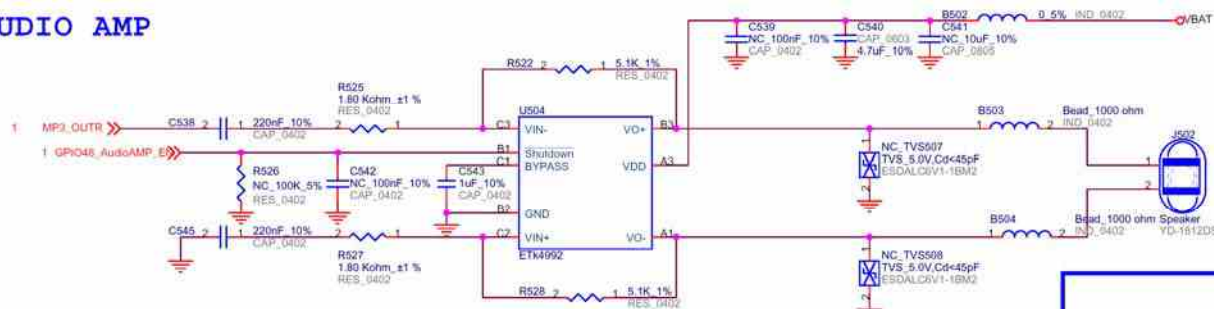


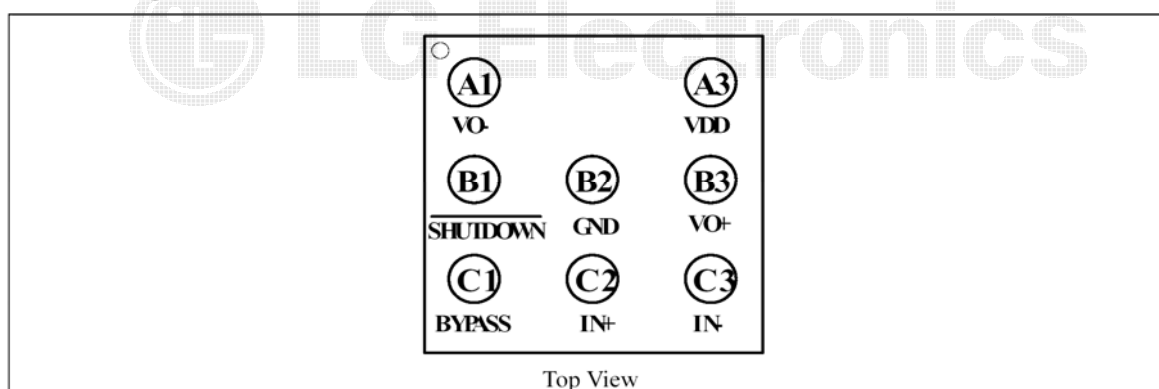
Figure.3-11-1 Audio Amp IC and Speaker Interface

The ETK4992 is a fully differential audio power amplifier designed for portable communication device applications. It is capable of delivering 1.25 watt of continuous average power to an 8. BTL load with less than 1% distortion (THD+N) from a 5V battery voltage. It operates from 2.2 to 5.5V.

Features like 90dB PSRR at 217Hz, improved RF-rectification immunity, the space-saving 8-bump WLCSP package, the advanced pop & click circuitry, a minimal count of external components and low-power shutdown mode make ETK4992 ideal for wireless handsets.

The ETK4992 is unity-gain stable, and the gain can be configured by external resistors.

Pin Configuration



Pin Function

Pin No.	Name	I/O	Function
A1	VO-	O	Negative differential output.
B1	SHUTDOWN	I	Shutdown Pin, active low.
C1	BYPASS	I	Common mode voltage. Connect a bypass capacitor to GND for common mode voltage filtering. The bypass capacitor is optional.
B2	GND	POWER	Ground.
C2	IN+	I	Positive differential input.
A3	VDD	POWER	Power supply.
B3	VO+	O	Positive differential output.
C3	IN-	I	Negative differential input.

3.11.2 Microphone & receiver

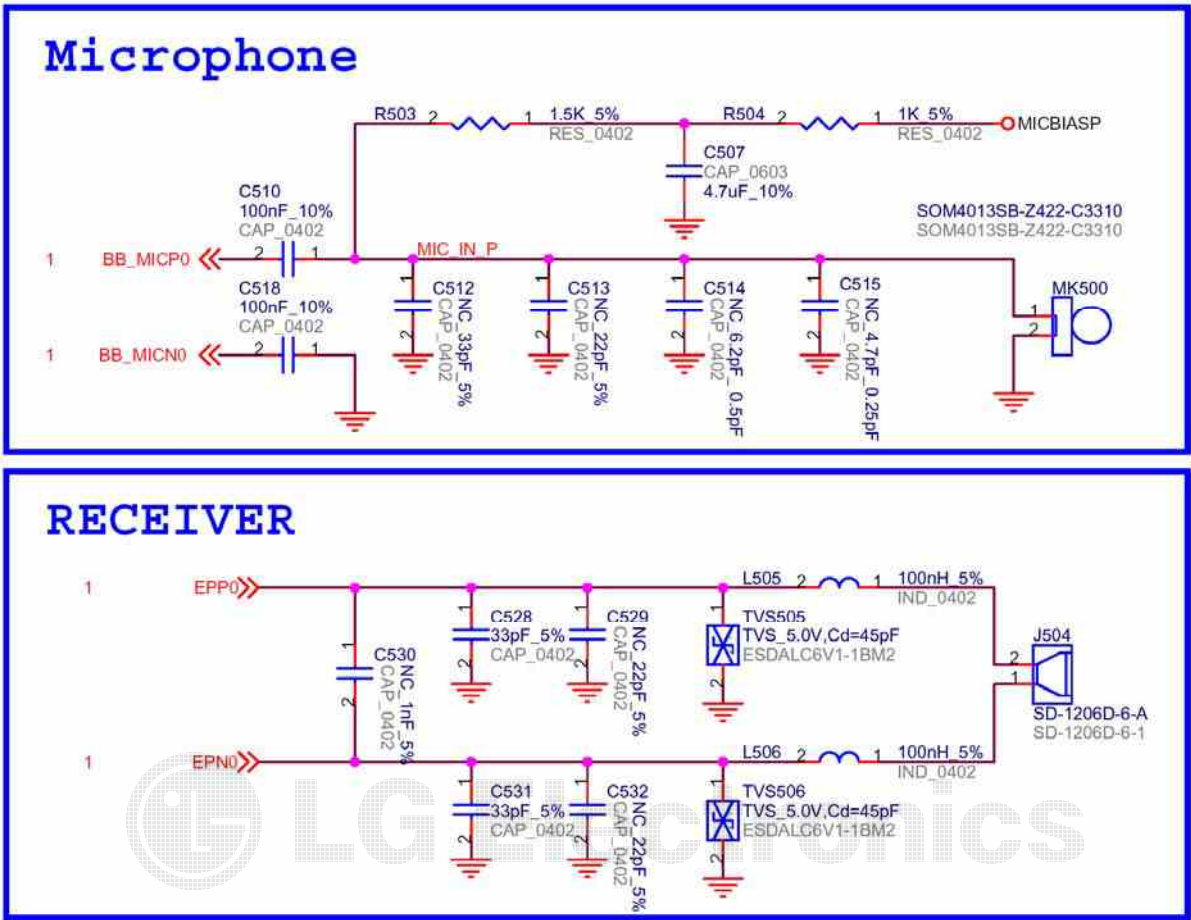


Figure.3-11-2 Microphone & Receiver Interface

3.12 Vibrator Interface

VIBRATOR

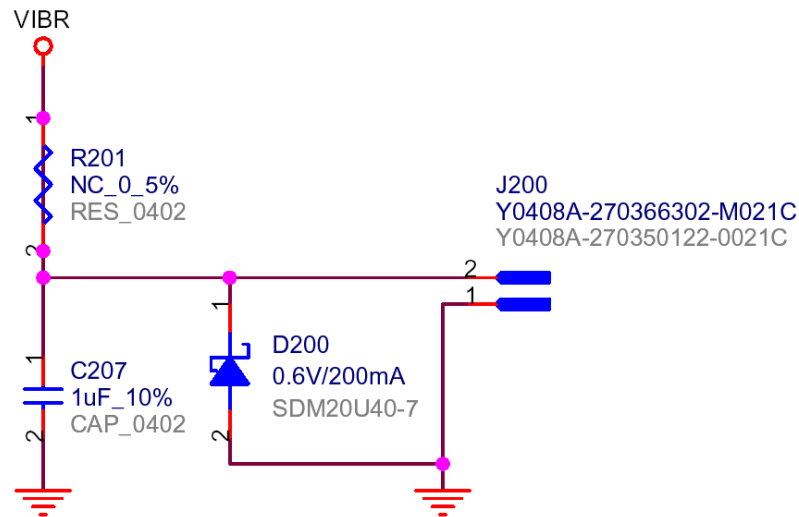


Figure.3-12-1 Vibrator Interface

This handset has Vibrator operation. Control signal is controlled by MT6236



3.13 Camera Interface

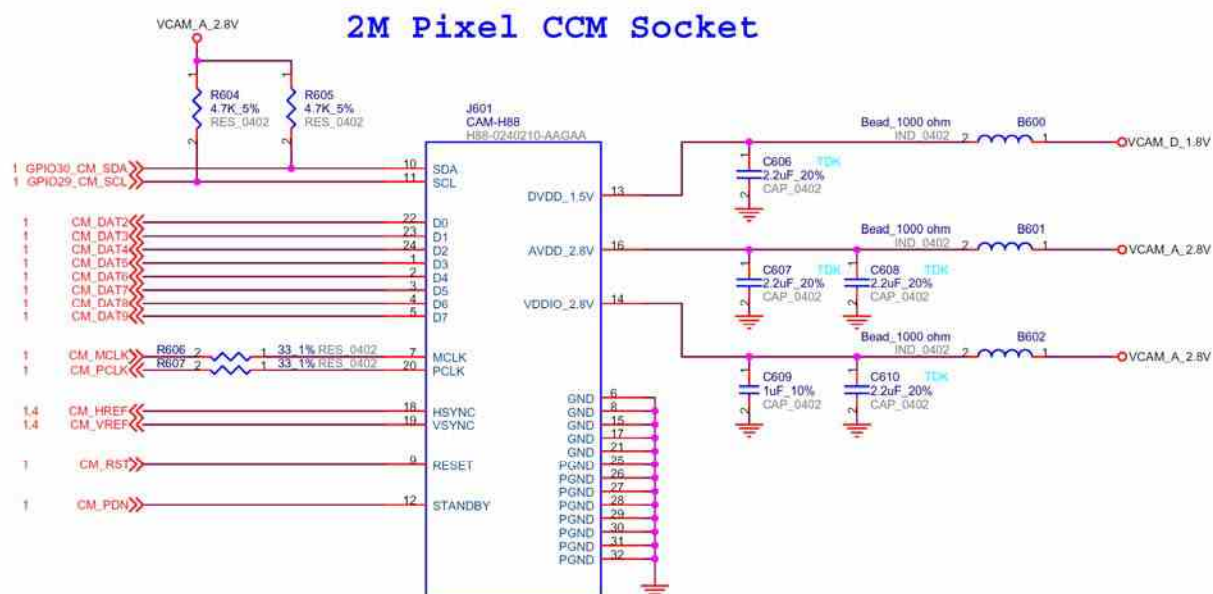


Figure.3-13-1 Camera Interface

3.13.1 Pin Description

Camera Module Connector Pin Description

No.	SYMBOL	DESCRIPTION	No.	SYMBOL	DESCRIPTION
1	Data3	Pixel data output 3	13	VDDD	Digital Power(1.5V)
2	Data4	Pixel data output 4	14	VDDIO	I/O Power (1.8V or 2.8V)
3	Data5	Pixel data output 5	15	GND	Ground
4	Data6	Pixel data output 6	16	VDDA	Analog Power(2.8V)
5	Data7	Pixel data output 7	17	GND	Ground
6	GND	Ground	18	HSYNC	Horizontal Sync Output
7	MCLK	System Main clock	19	VSYNC	Vertical Sync output
8	GND	Ground	20	PCLK	Pixel clock output
9	RESET	Sensor Reset (Active "L")	21	GND	Ground
10	SDA	I2C standard data I/O port	22	Data0	Parallel pixel data0
11	SCL	I2C standard clock input	23	Data1	Parallel pixel data1
12	STANDBY	Sensor sleep mode. (Active"L")	24	Data2	Parallel pixel data2

3.13.2 Camera Features

	Item	Feature		Remark
Module Feature	Dimension	7.0 (W) × 7.0 (L) × 4.1 (H) : mm		
	Lens Construction	3 Aspheric Plastic Lens		
	Focusing range	50 cm to ∞		
	F Number	F2.8 (±5%)		
	Focal Length	2.86 mm		
	TV Distortion	-0.23% under		
	Angle of view	64.0 Deg (±5%)		Diagonal
	Focusing target	110 cm		
Electro-Optical Feature	Pixel Array	1616 x 1216 (Number of effective pixels : 1.96M)		
	Pixel Size	1.75μm square		
	Array Pattern	RGB Bayer		
	Optical Format	1/5"		
	Power Consumption	Normal	233mW max. @ 15fps, Full resolution AVDD=2.8V, DVDD=1.8V, IOVDD=1.8V	
		Standby	4.01mW max. @ 15fps AVDD=2.8V, DVDD=1.8V, IOVDD=1.8V	
	Operation Voltage	IOVDD:1.7-3.0V DVDD:1.7-1.9V AVDD: 2.6-3.0V		
	Clock Speed	6 ~ 27MHz		
	Frame rate	15 fps @ UXGA 30 fps @ VGA		
	Operation Temperature	-20~+60°C		
	Optical filter	Infrared rays cutting filter		
Features	Video Mode	UXGA / SXGA / Quad VGA / XGA / SVGA / VGA and etc		
	Output Format	YUV422, Parallel output		
		Window of interest (Windowing to any arbitrary size with panning) Image sizing (programmable decimation for image down sizing) Digital zoom Picture flip (Horizontal flip and Vertical flip) Picture effects (monochrome, negative, sepia, antique, sketch, emboss) Auto luminance control (=Auto exposure) Auto white balance Bad pixel correction Auto flicker detection and correction Gamma correction (programmable gamma table) Lens shading correction Color manager Standby mode Power down mode for low power consumption		

3.14 Key LED Interface

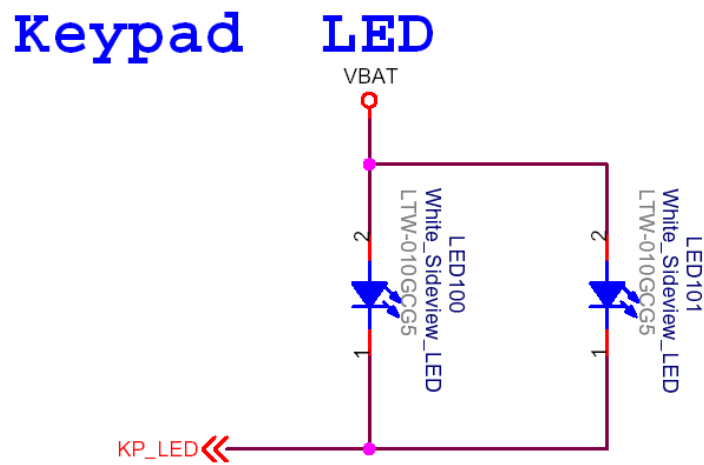


Figure.3-14-1 Key LED Interface

This handset has 2 LEDs that illuminates white color.

Control signal is controlled by MT6236 and handset has 3 methods, ON, OFF, Dimming



4. Trouble Shooting

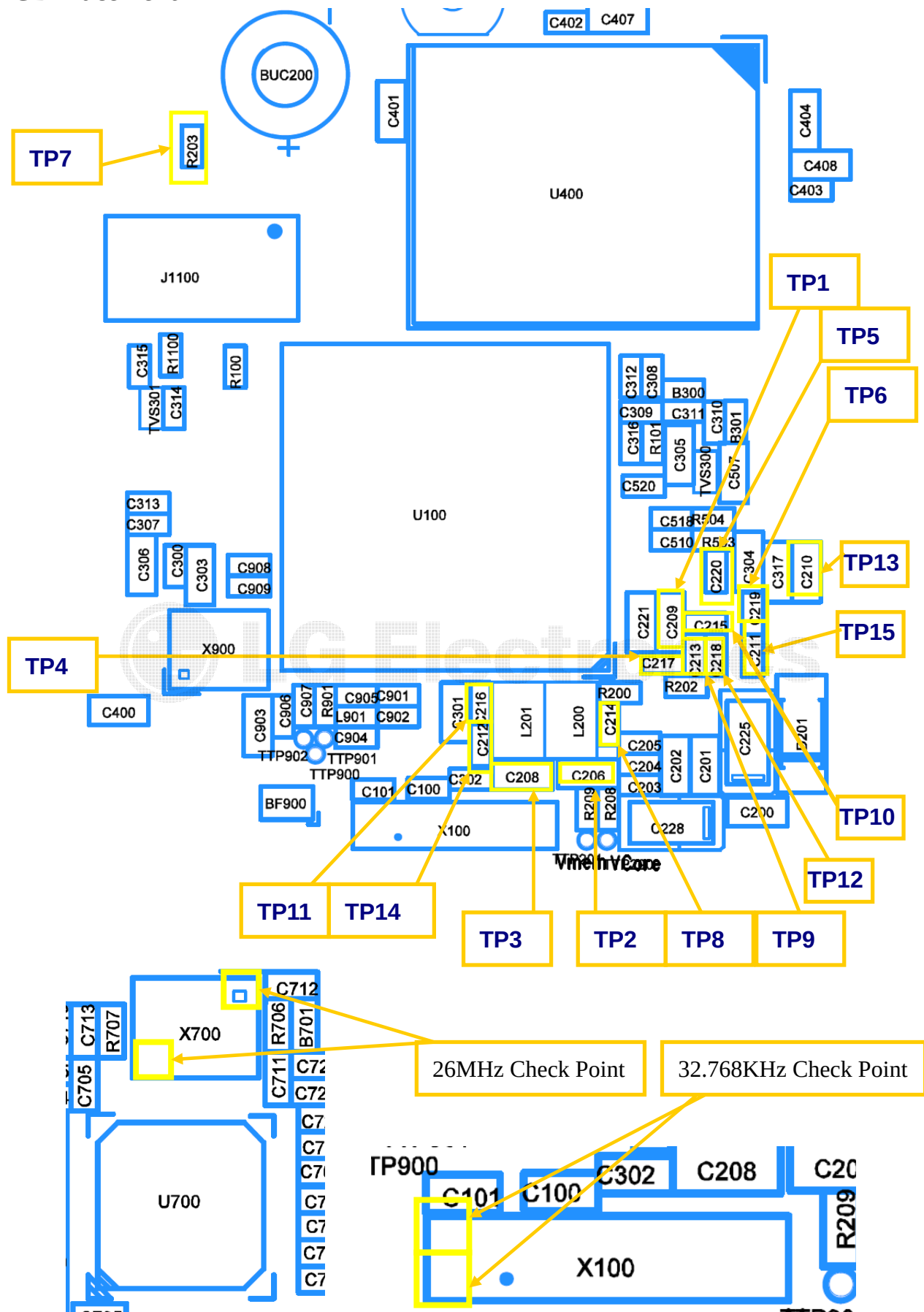
4.1 Power On Trouble

Test Point

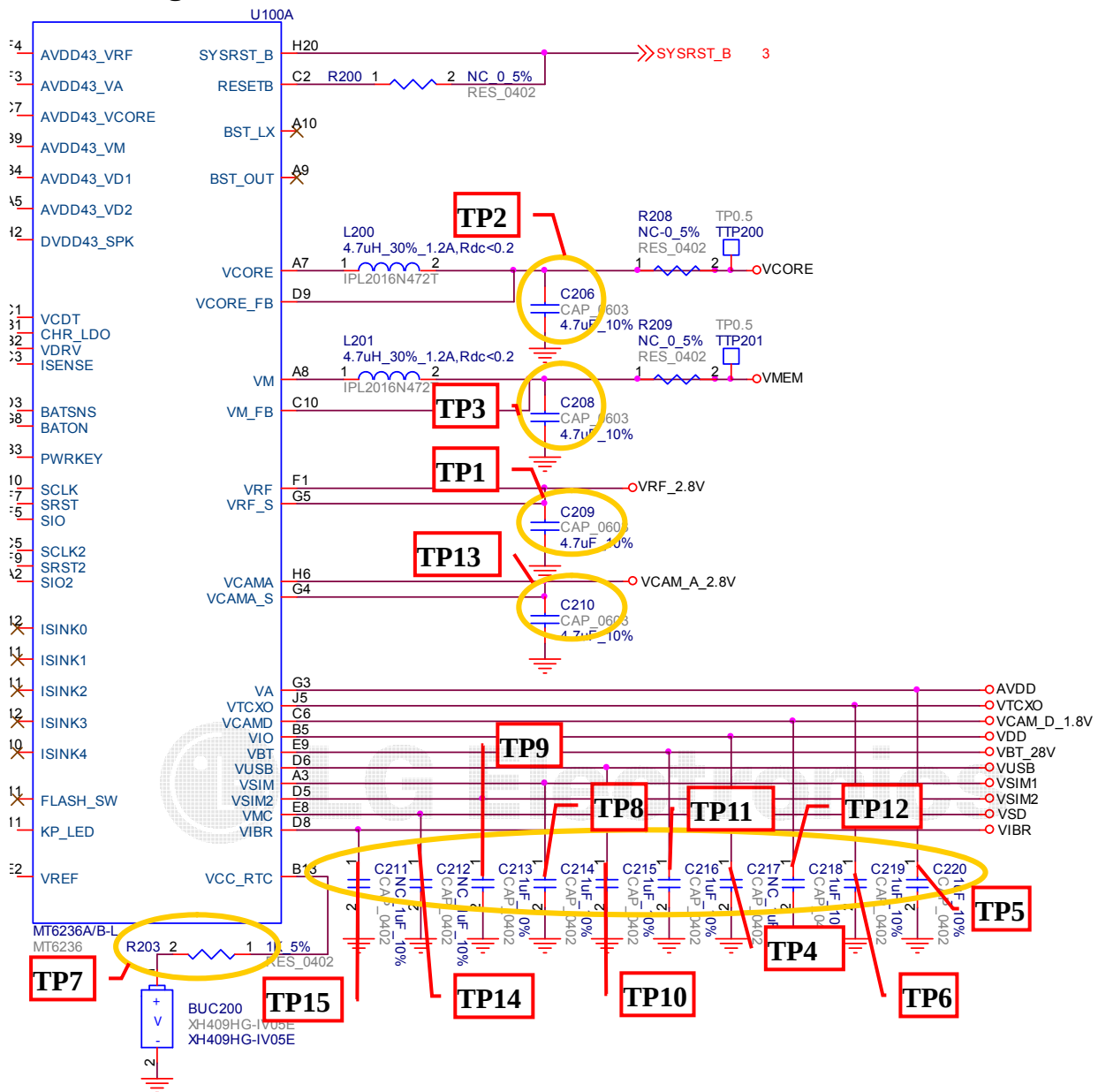
- Battery Voltage(Need to over 3.34V)
- Power-On key detection(PWRON signal)
- Outputs LDO by U100

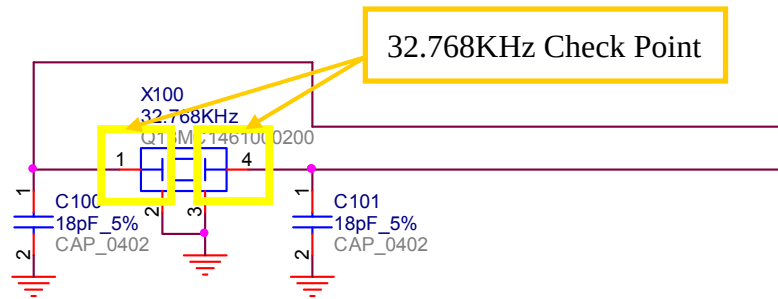
Name	Output	TP	Component	Describe
VRF	2.8V	TP1	C209	Power on check
VCORE	0.8~1.3V	TP2	C206	Power on check
VMEM	1.8V	TP3	C208	Power on check
VDD	2.8V	TP4	C217	Power on check
AVDD	2.8V	TP5	C220	Power on check
VTCXO	2.8V	TP6	C219	Power on check
VRTC	2.8V	TP7	R203	RTC voltage check point
VSIM1	1.8V/3V	TP8	C214	SIM1 card fail check point
VSIM2	1.8V/3V	TP9	C213	SIM1 card fail check point
VUSB	3.3V	TP10	C215	USB fail check point
VBT	2.8V	TP11	C216	Bluetooth fail check point
VCAMD	2.8V	TP12	C218	Camera fail check point
VCAMA	1.8V	TP13	C210	Camera fail check point
VSD	2.8V	TP14	C212	Micro-SD Card fail check point
VIBR	3V	TP15	C211	Vibrate fail check point

PCB Placement

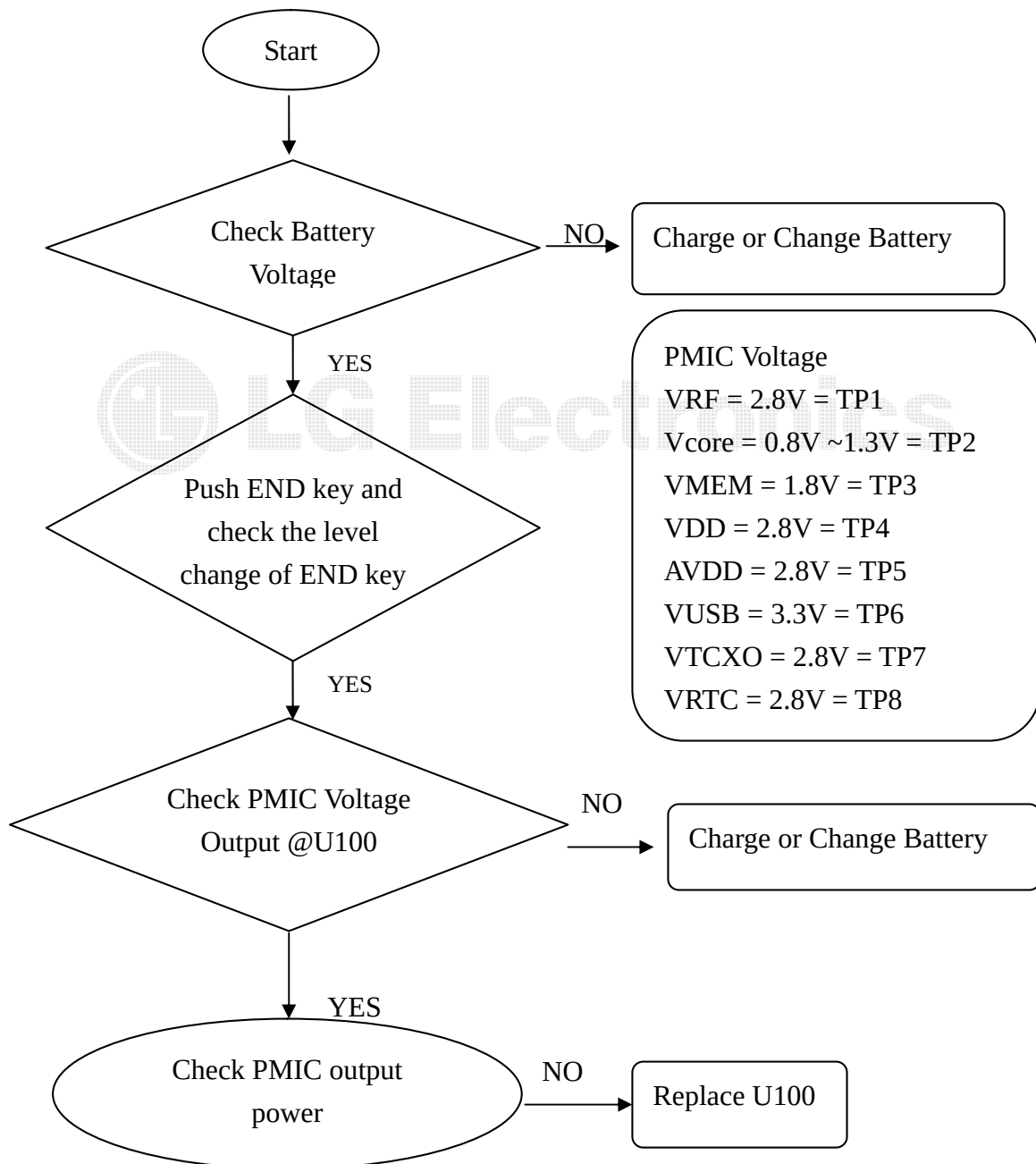


Circuit Diagram

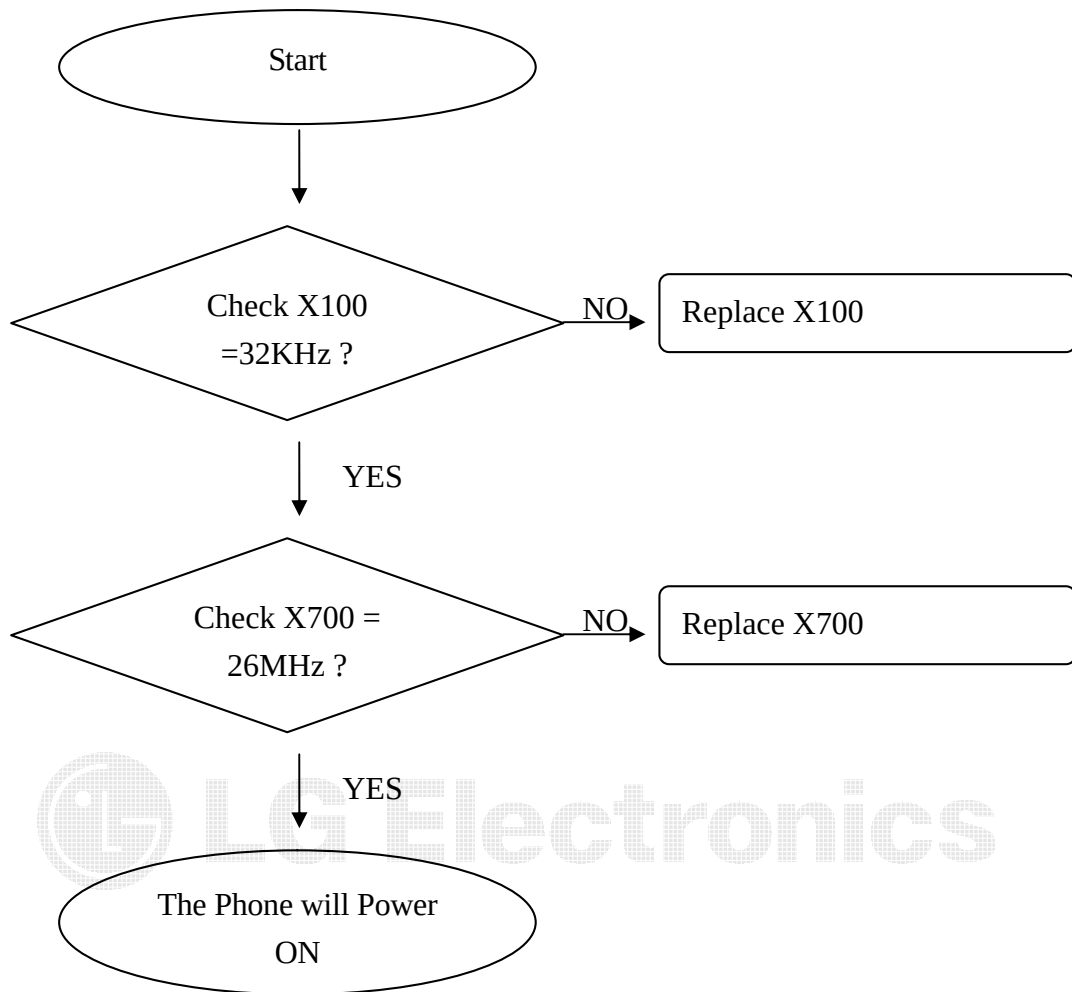




Checking Flow - Power Unit

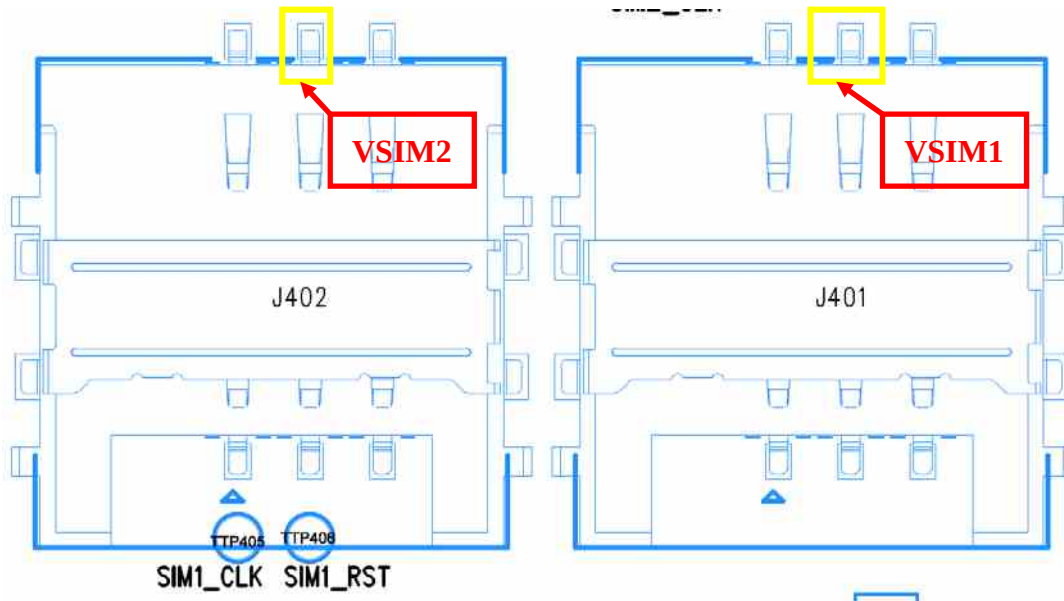


Checking Flow - Clock Unit



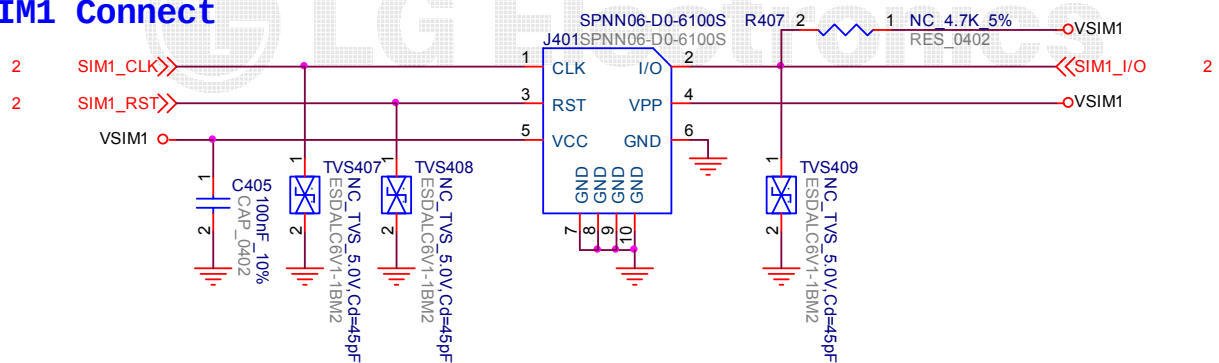
4.2 SIM Card Trouble

Test Point

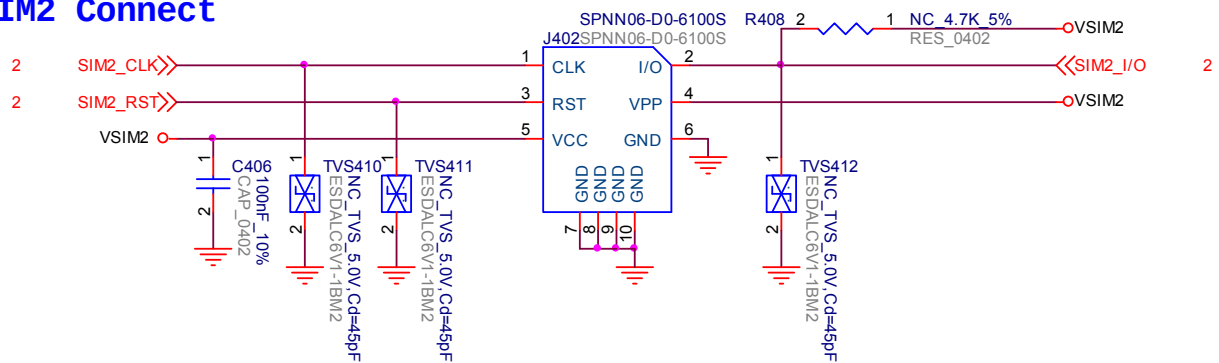


Circuit Diagram

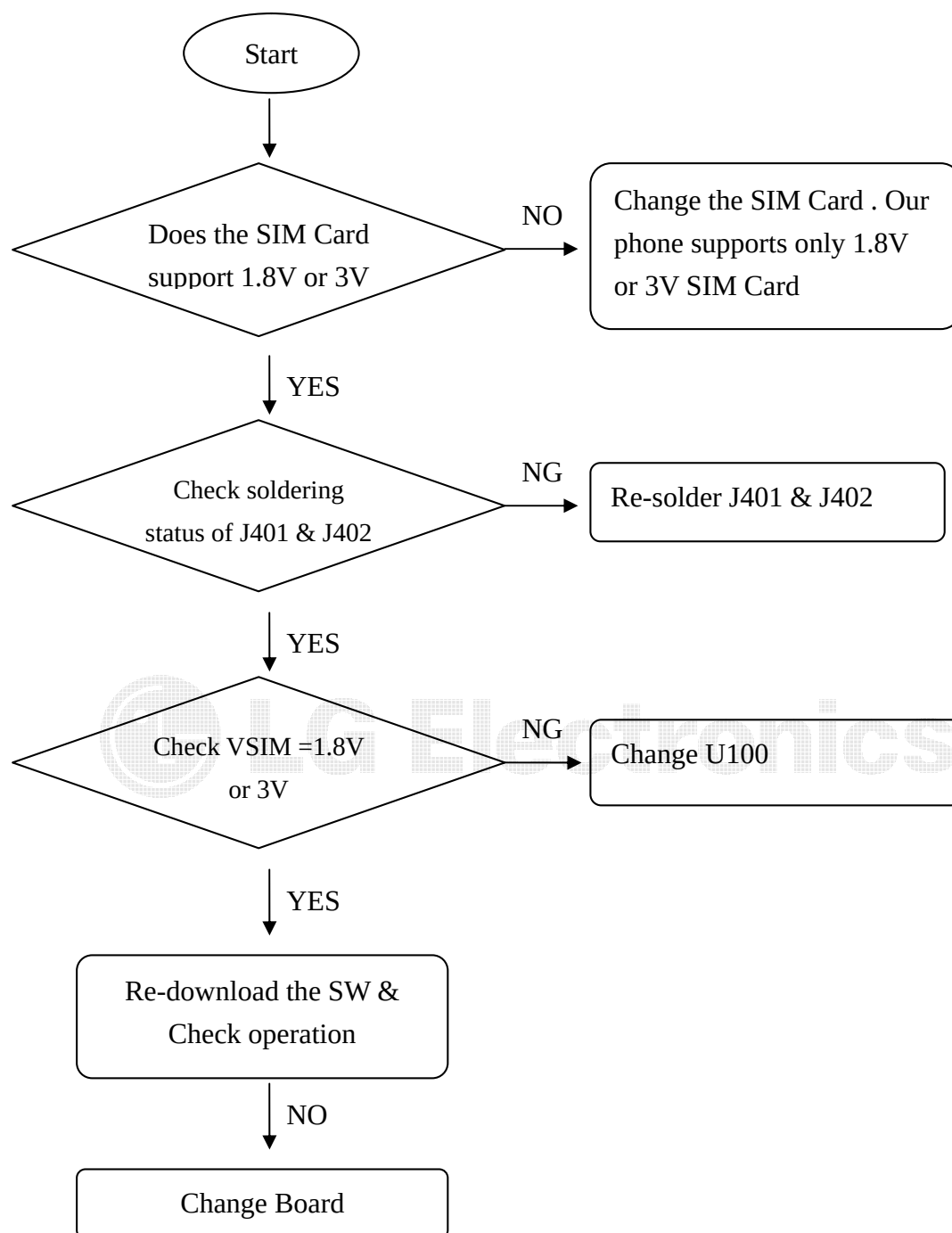
SIM1 Connect



SIM2 Connect

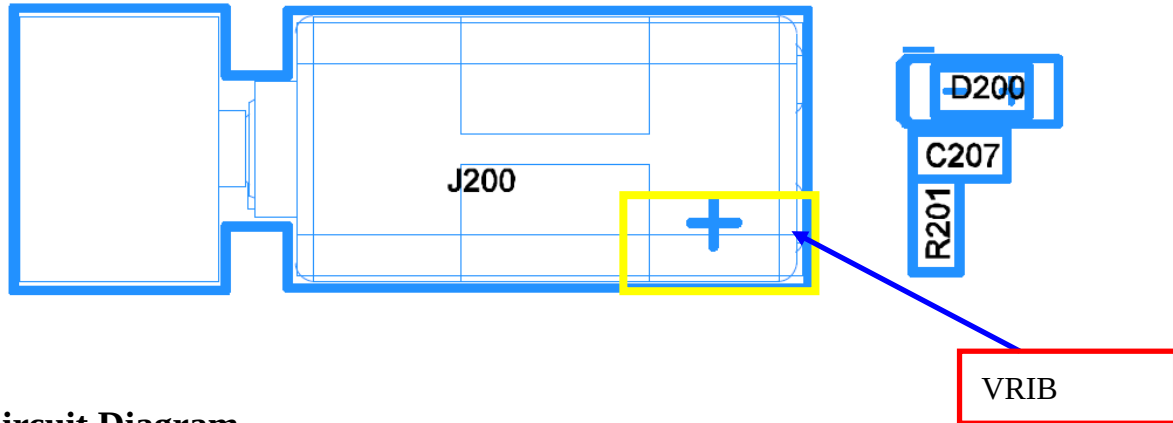


Checking Flow



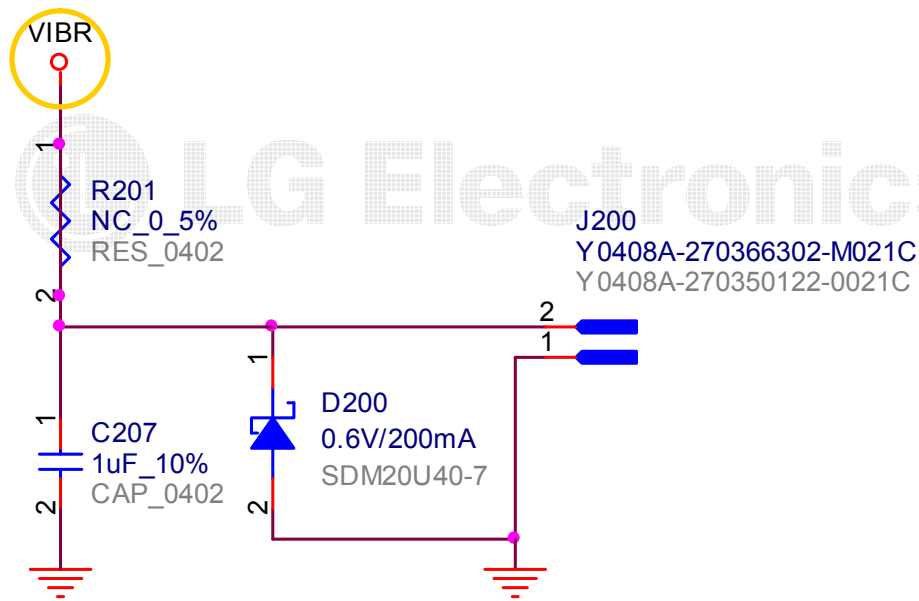
4.3 Vibrator Trouble

Test Point

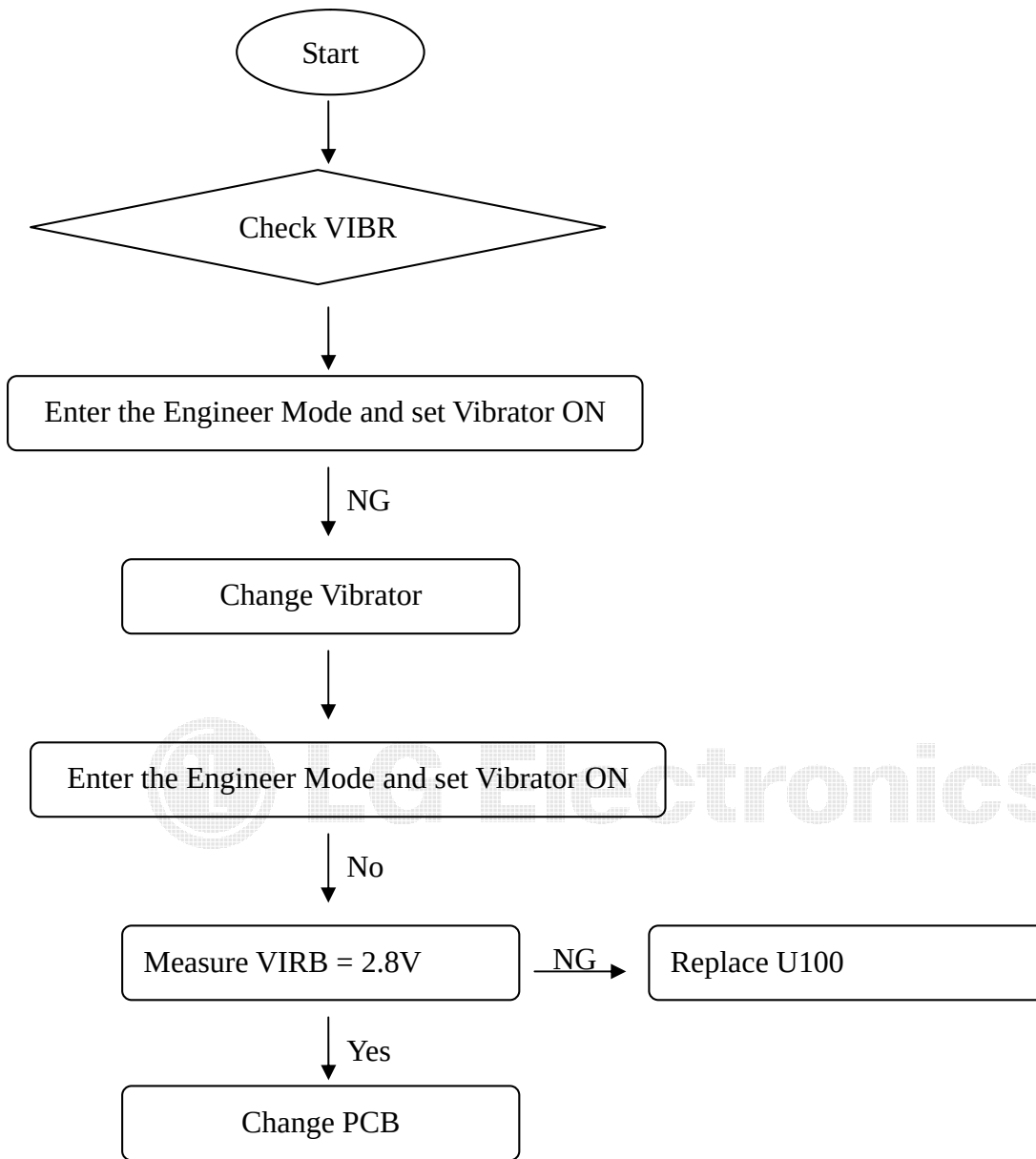


Circuit Diagram

VIBRATOR



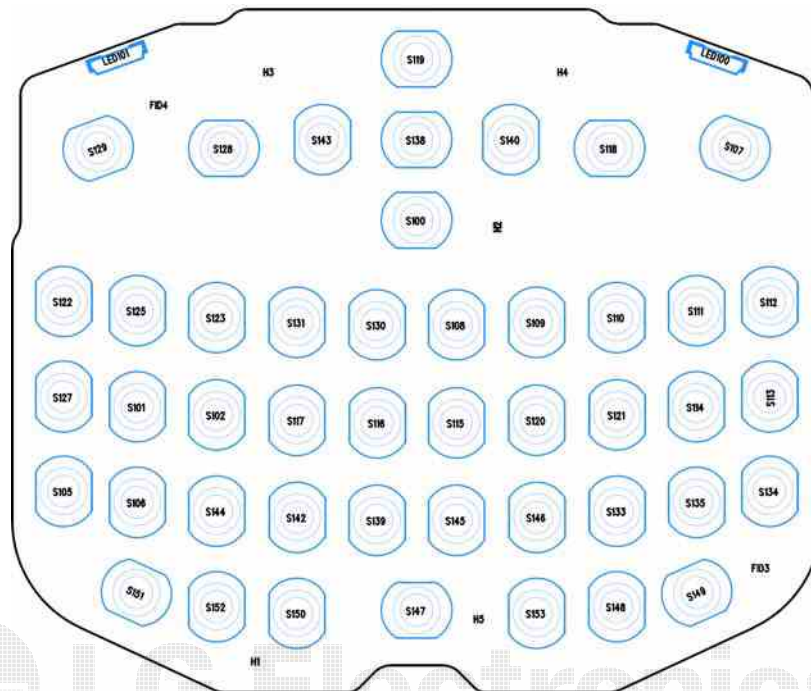
Checking Flow



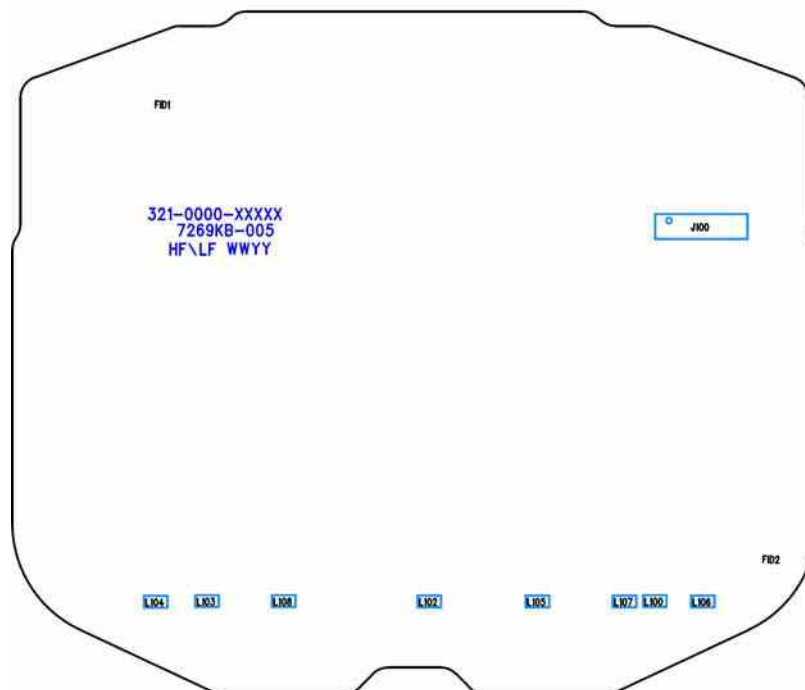
4.4 Keypad Trouble

Test Point

Key pad Top View

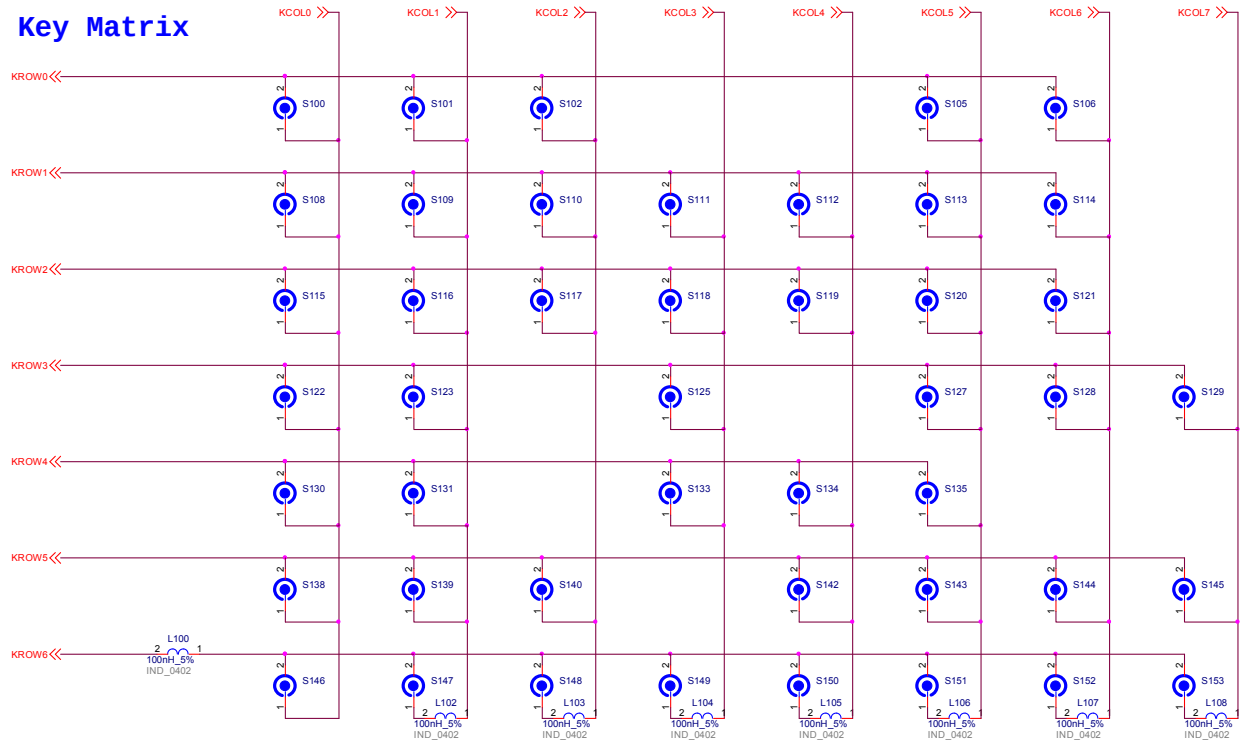


Key pad BOTTOM View

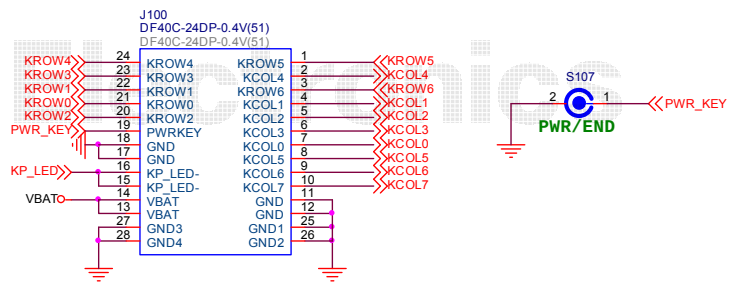
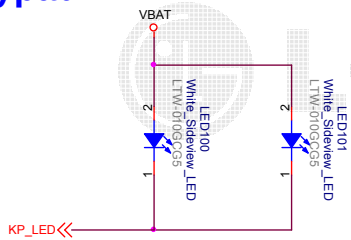


Circuit Diagram

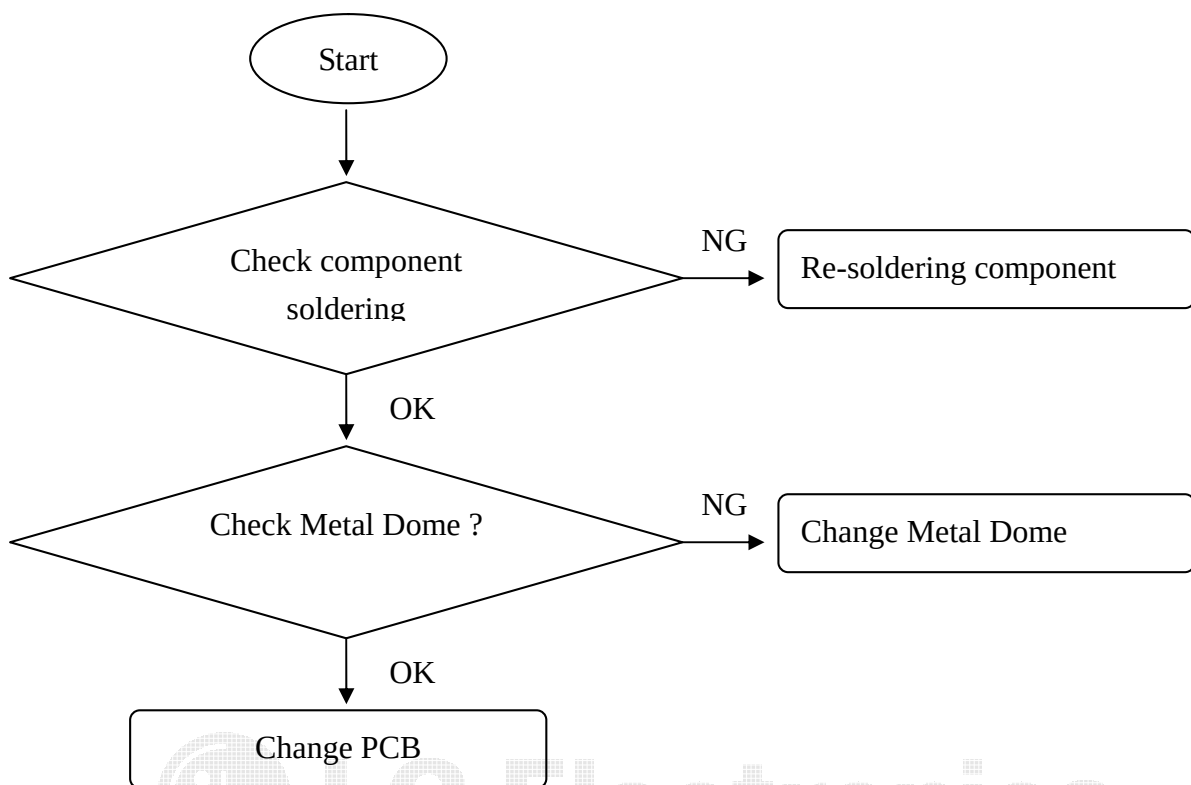
Key Matrix



Keypad LED

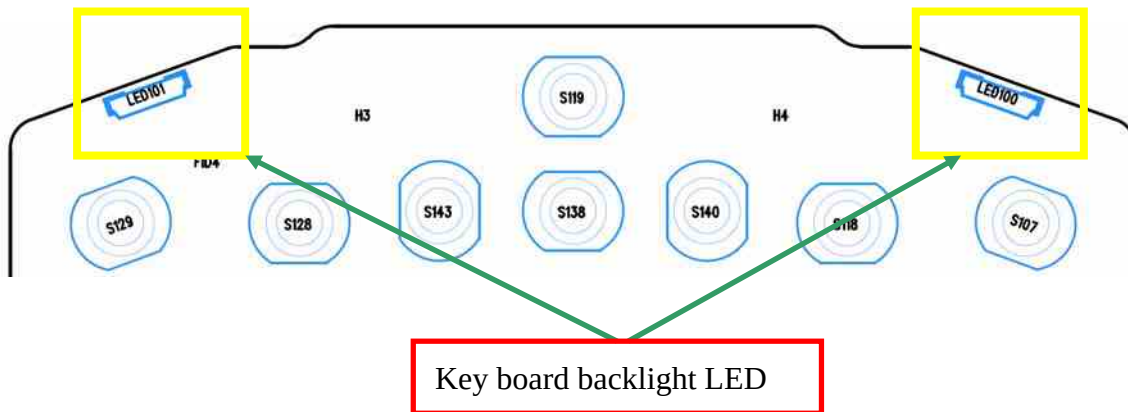


Checking Flow



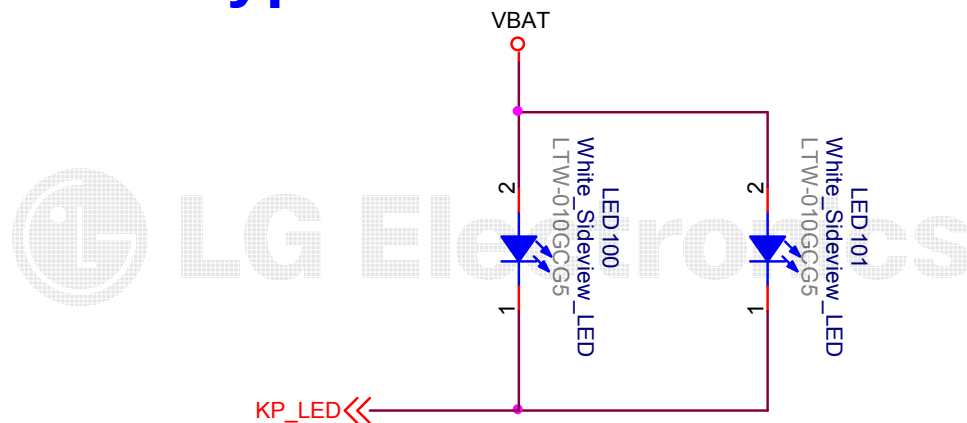
4.5 Key Backlight Trouble

Test Point

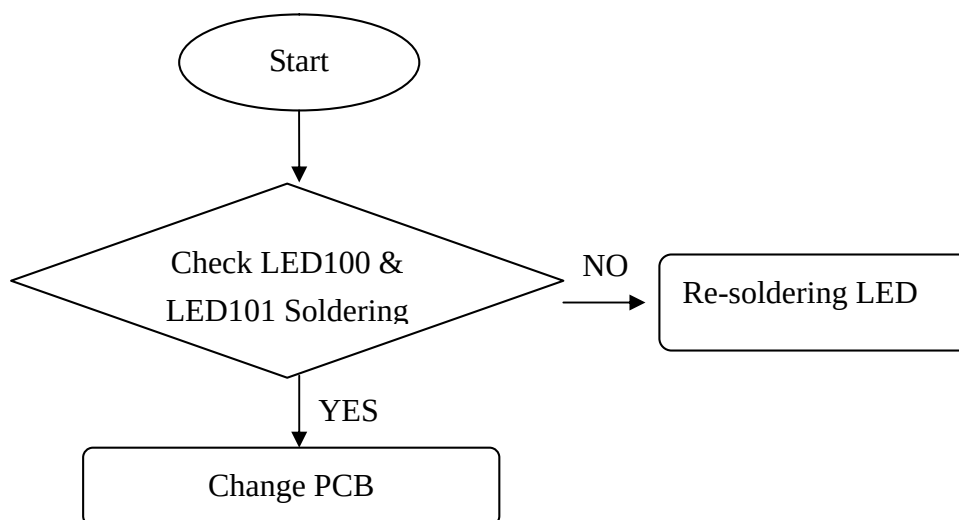


Circuit Diagram

Keypad LED

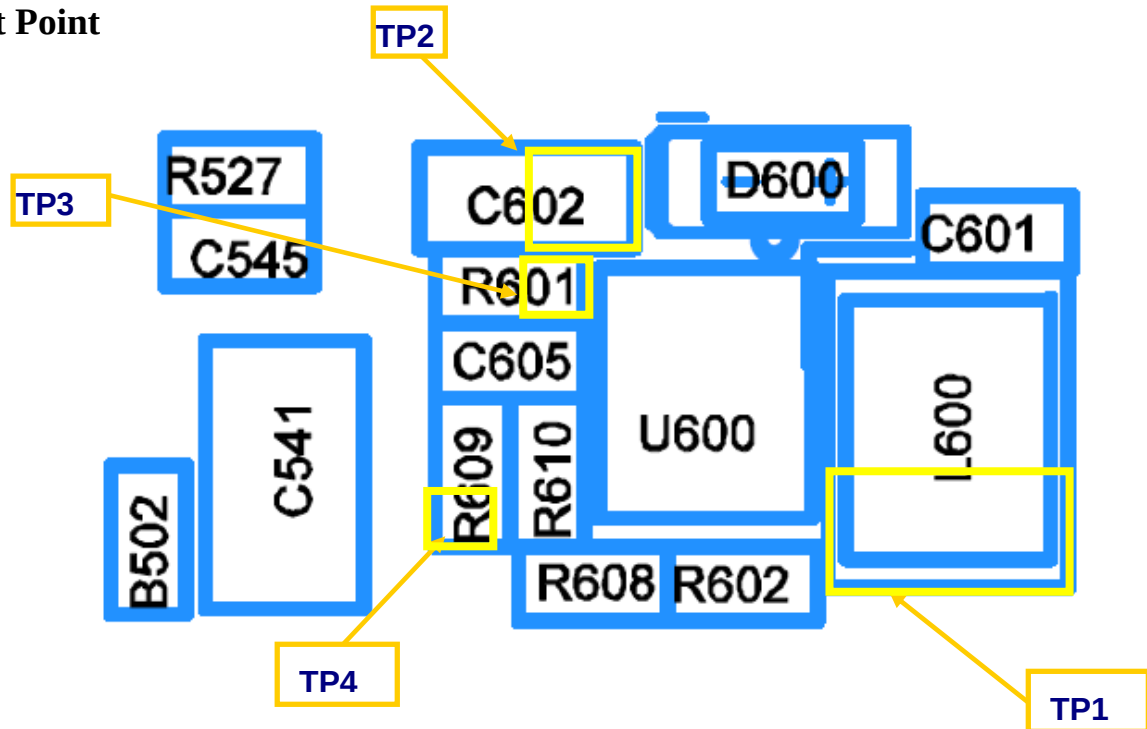


Checking Flow

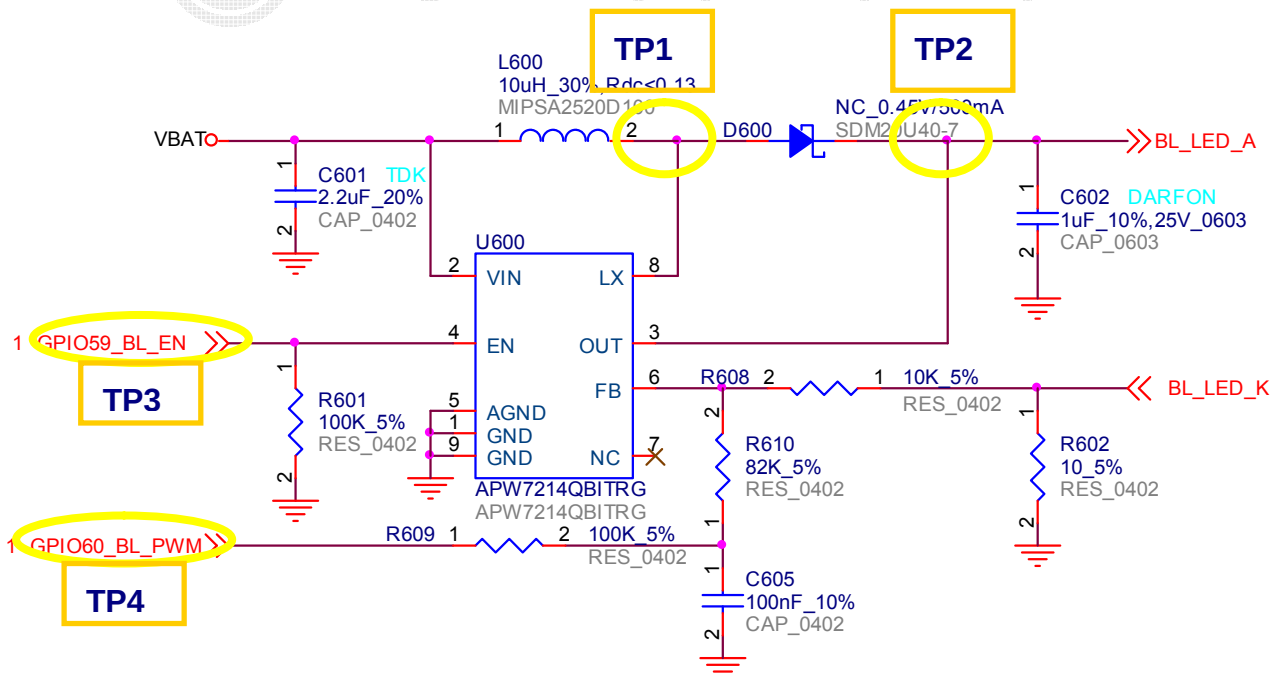


4.6 LCM Backlight Trouble

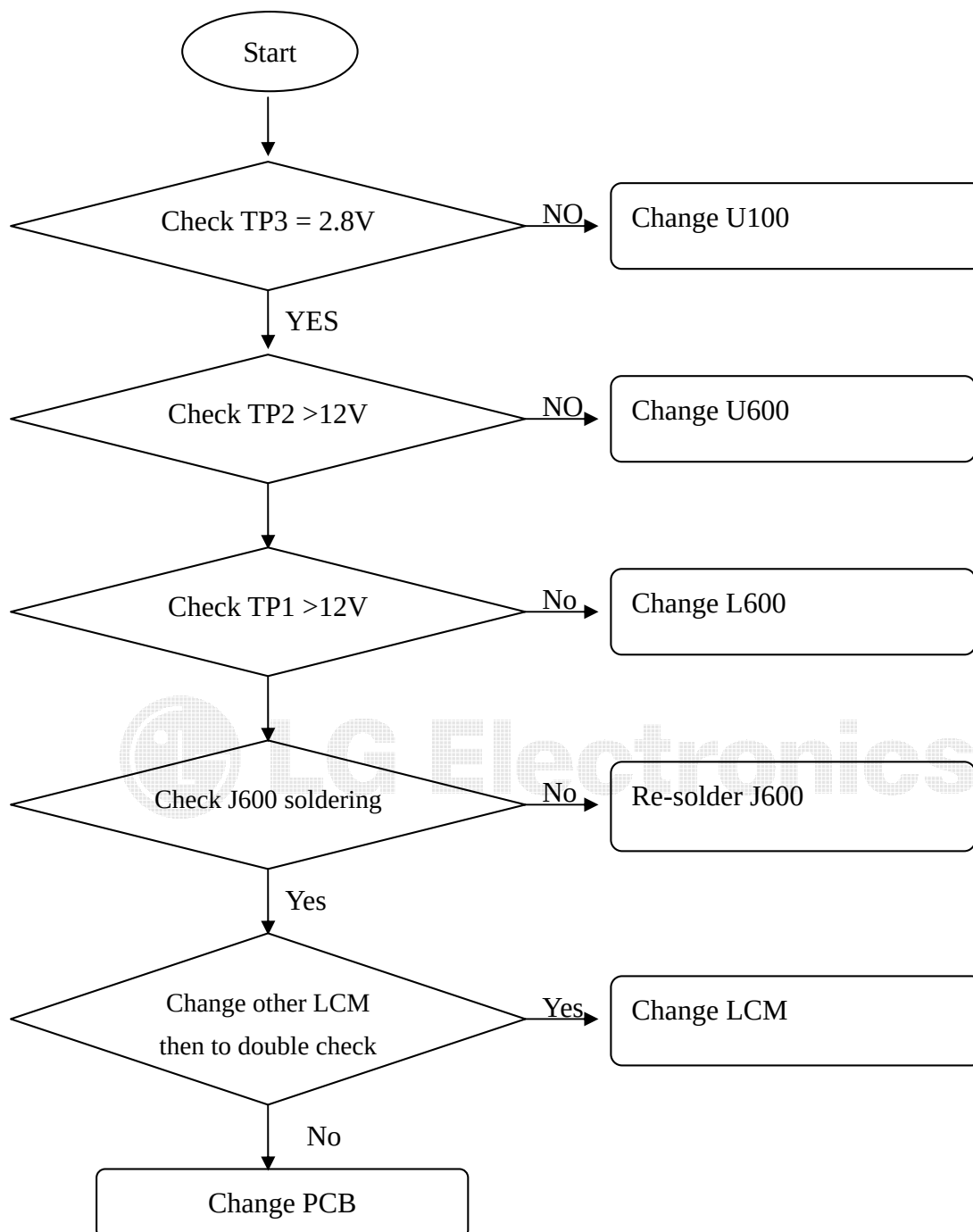
Test Point



Circuit Diagram

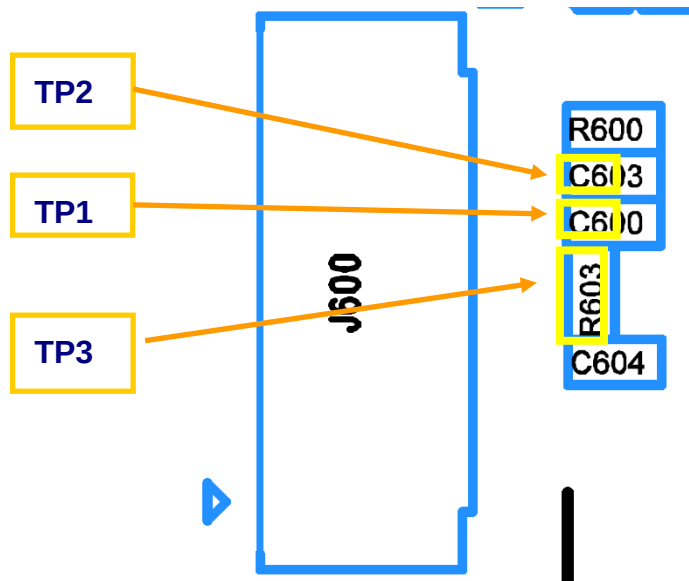


Checking Flow

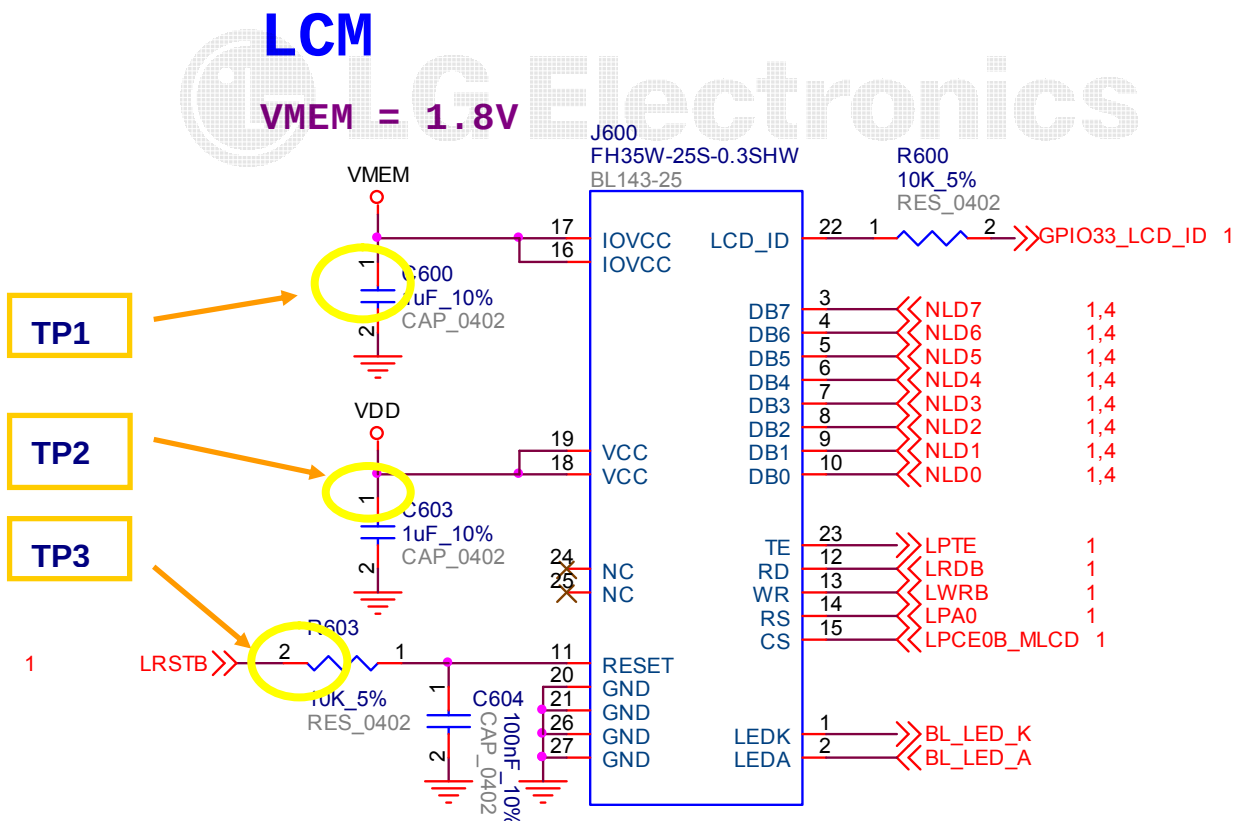


4.7 LCM Trouble

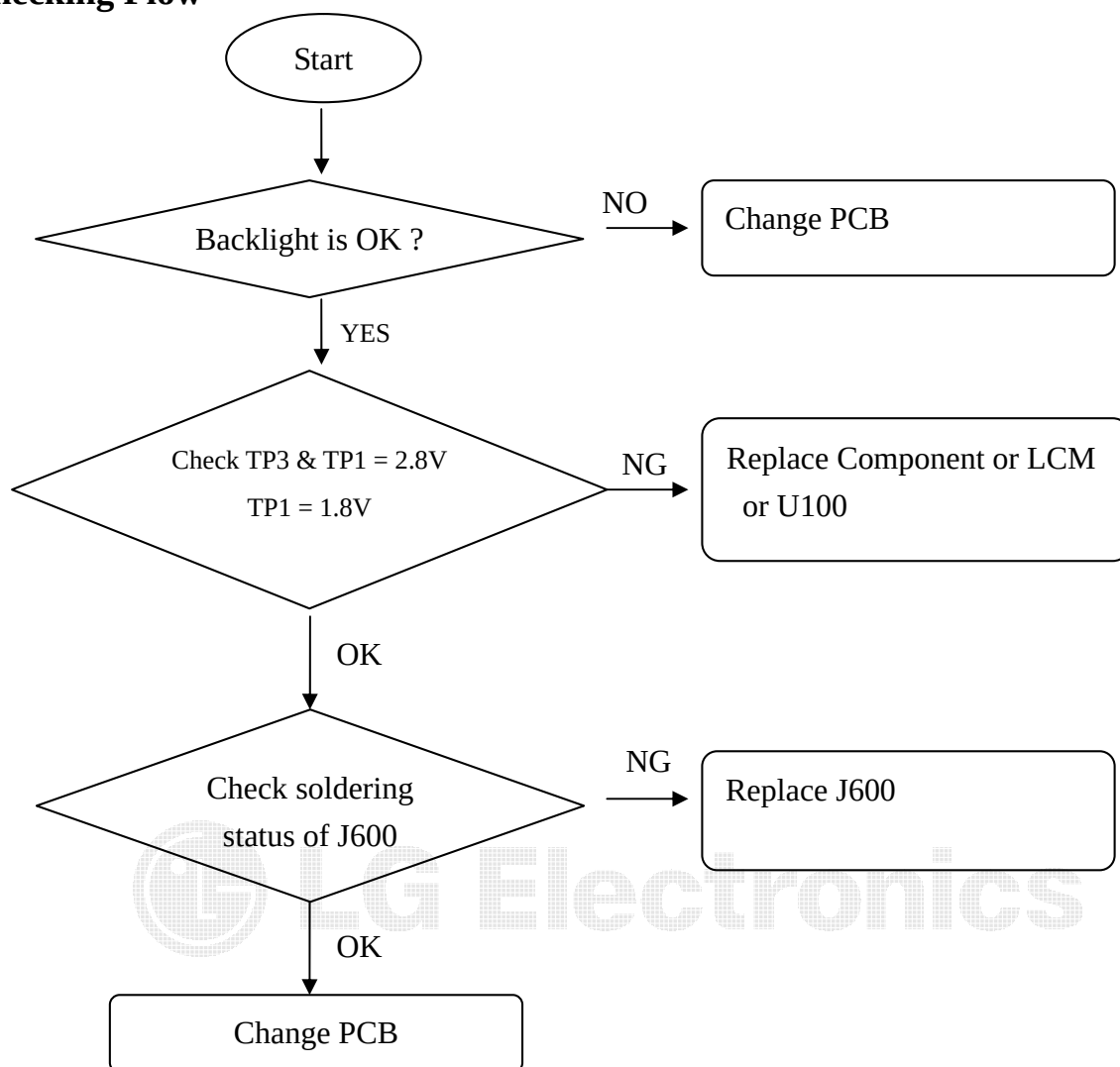
Test Point



Circuit Diagram

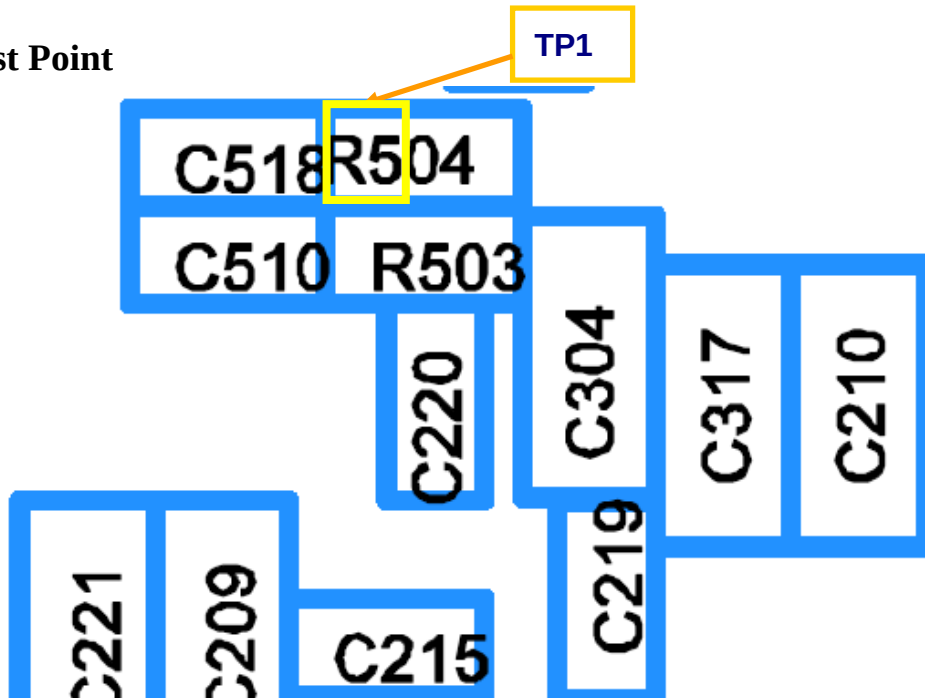


Checking Flow



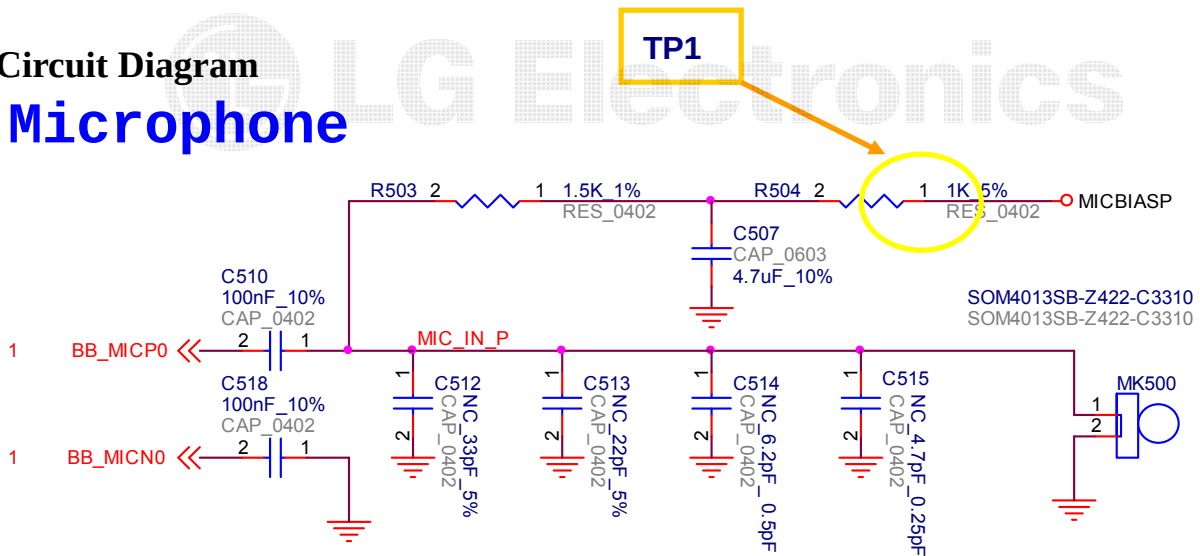
4.8 Microphone Trouble

Test Point

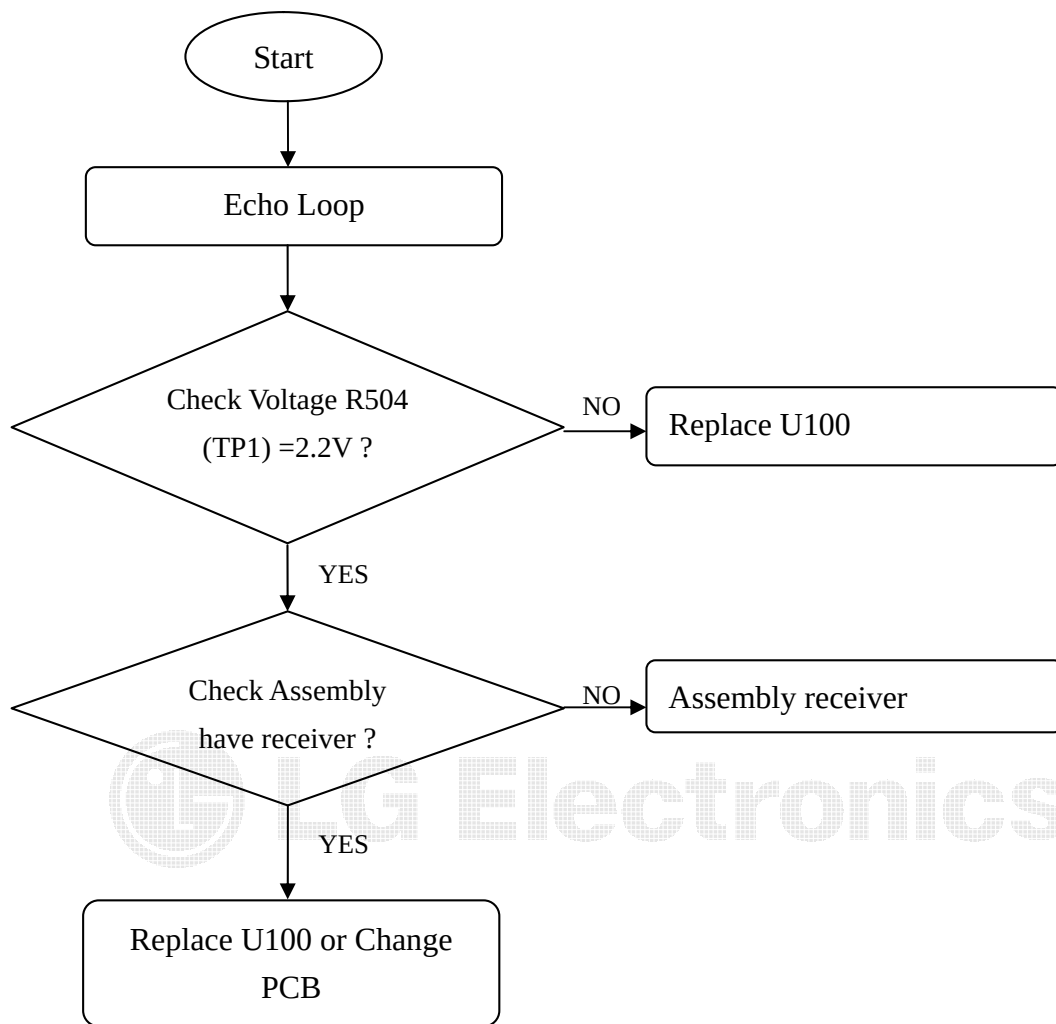


Circuit Diagram

Microphone

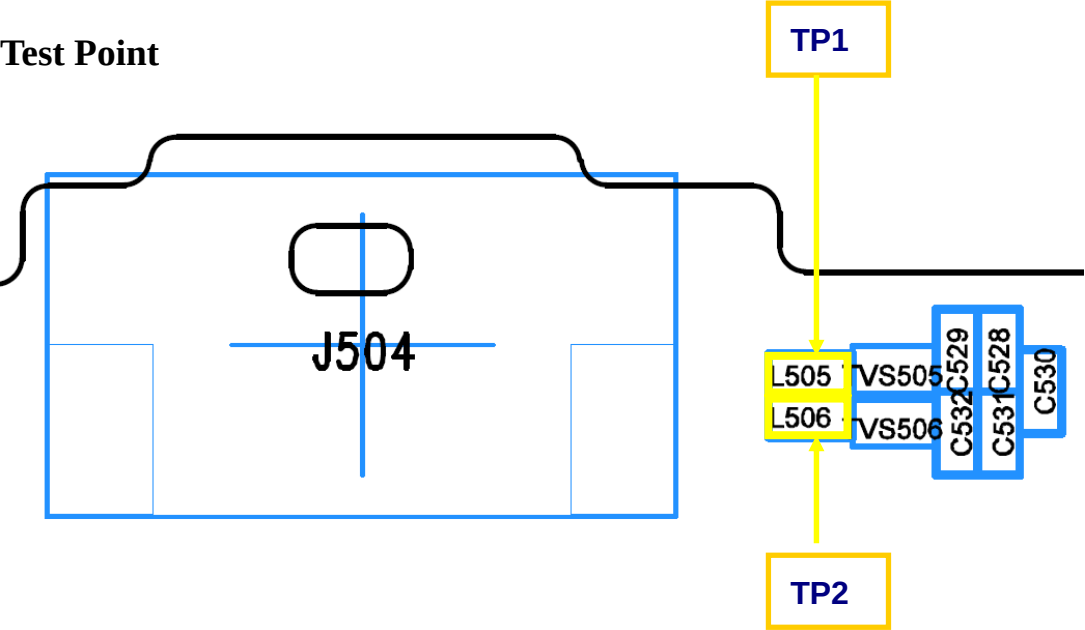


Checking Flow

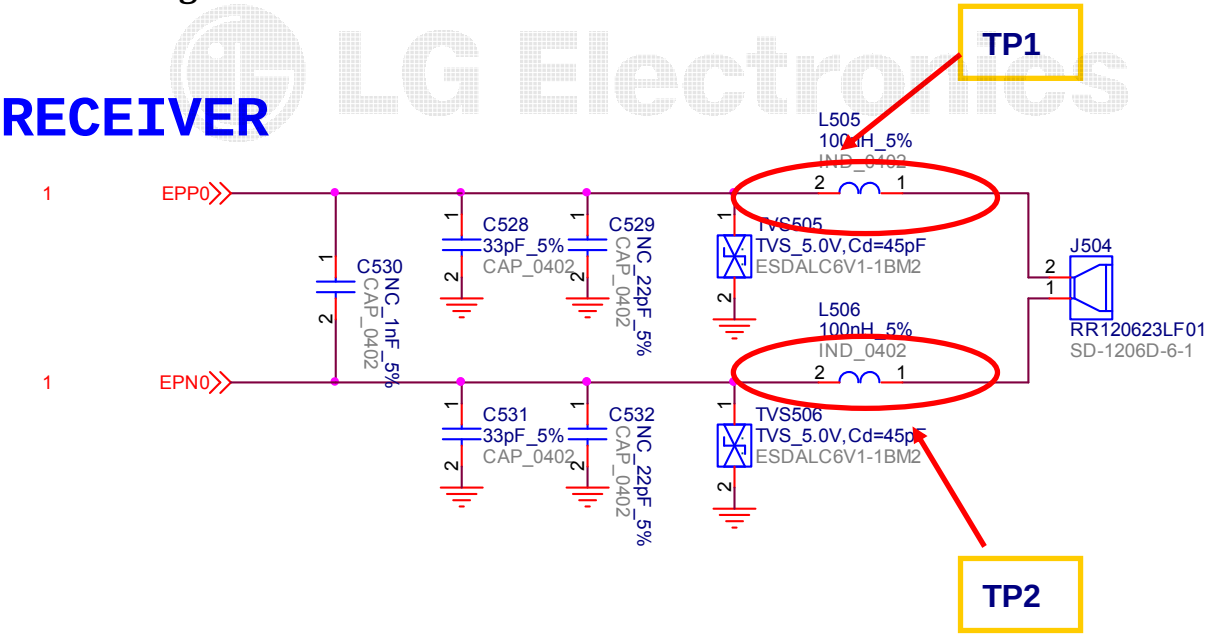


4.9 Receiver Trouble

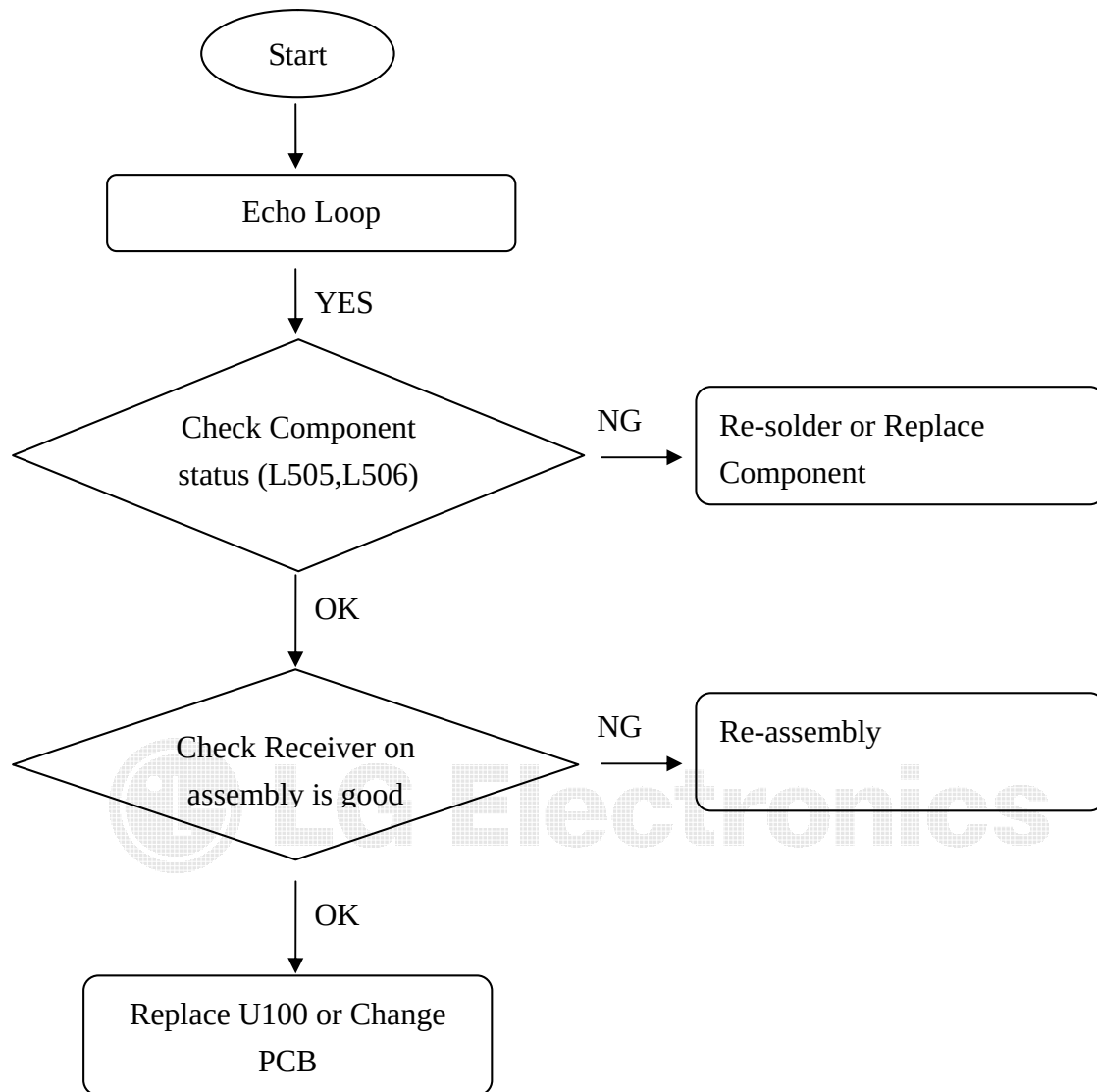
Test Point



Circuit Diagram

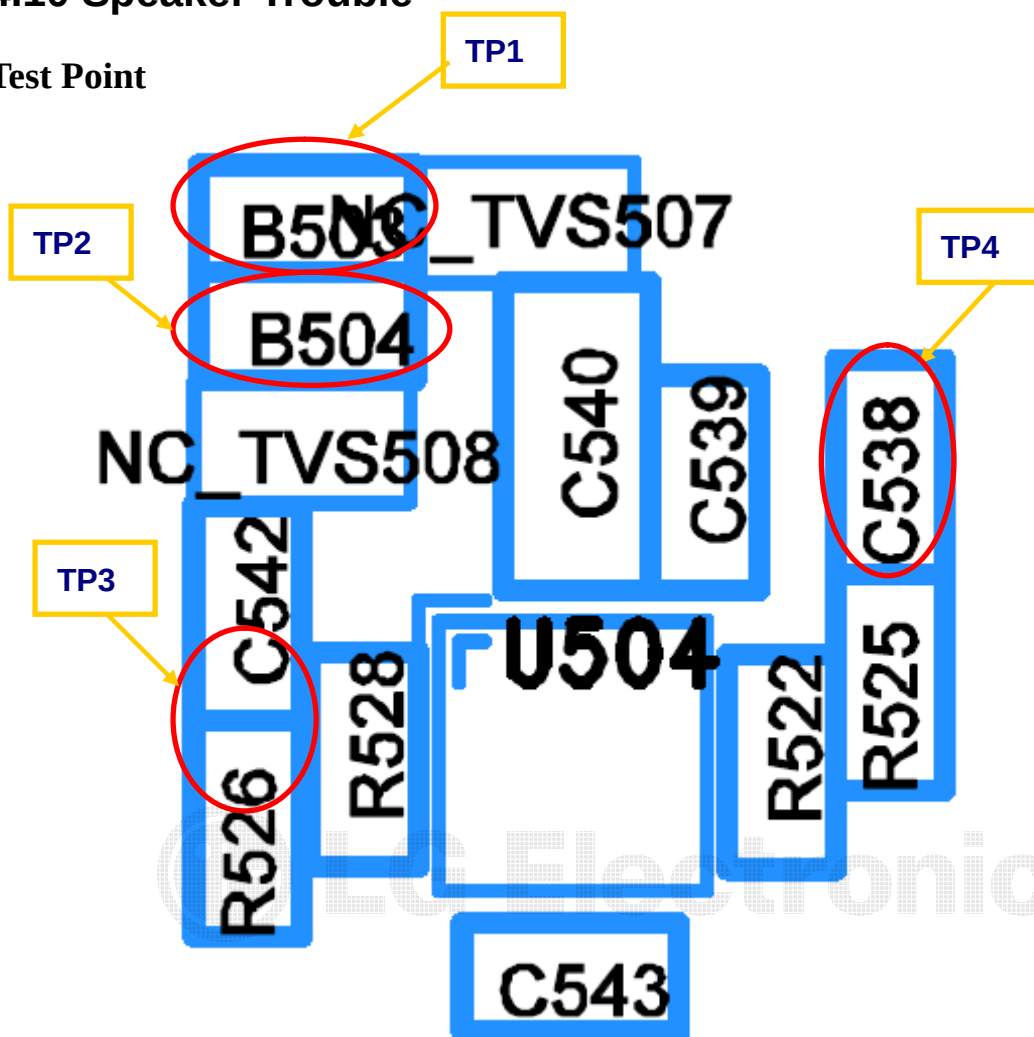


Checking Flow

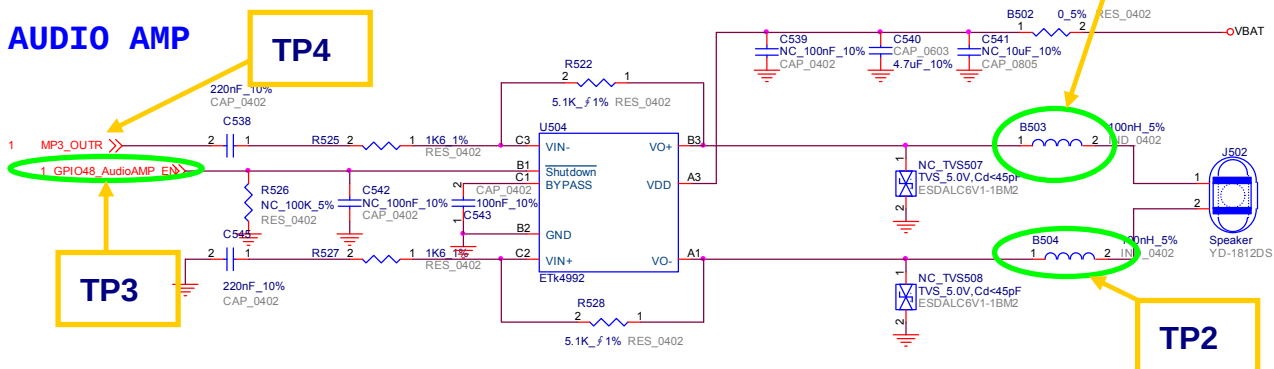


4.10 Speaker Trouble

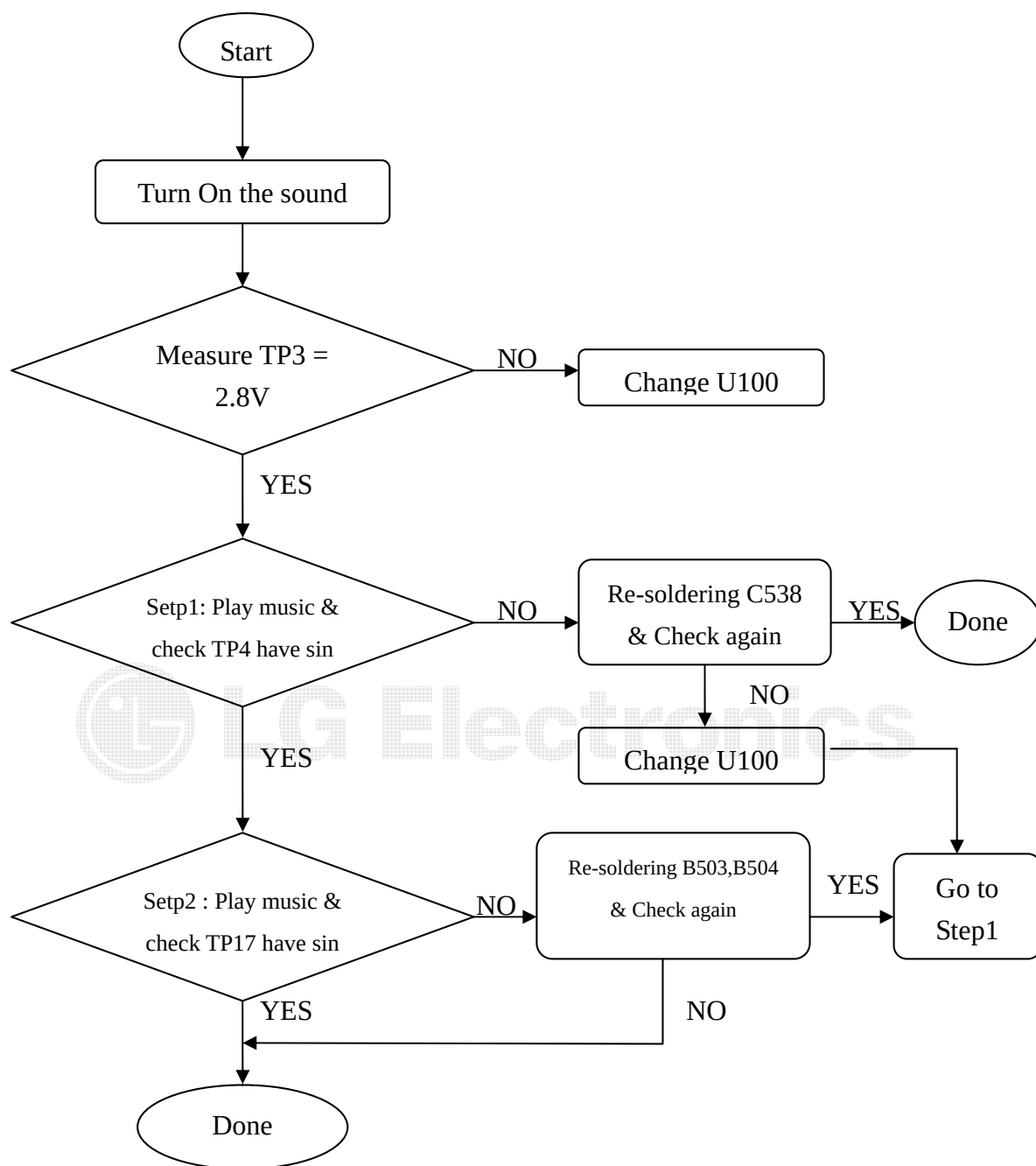
Test Point



Circuit Diagram

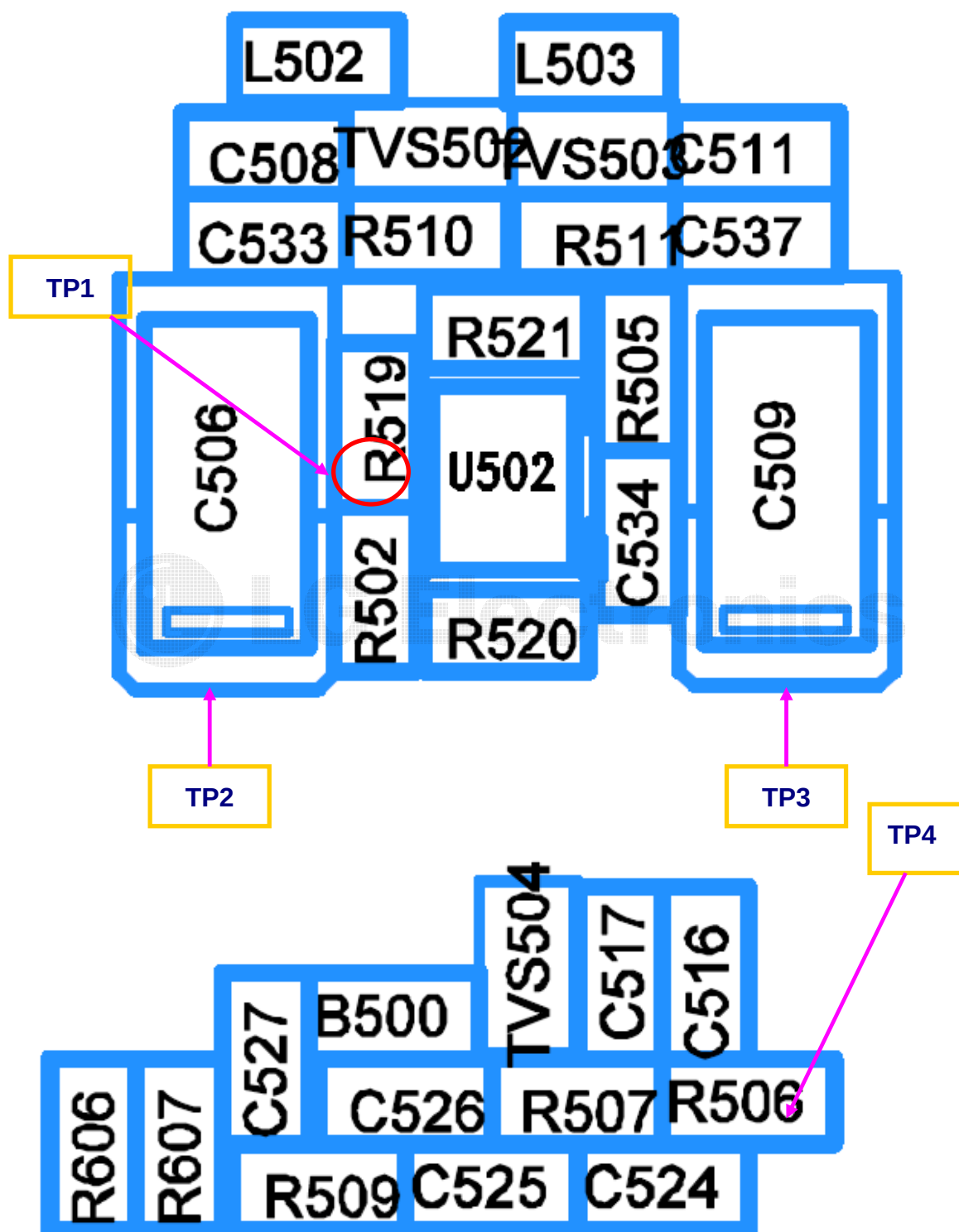


Checking Flow



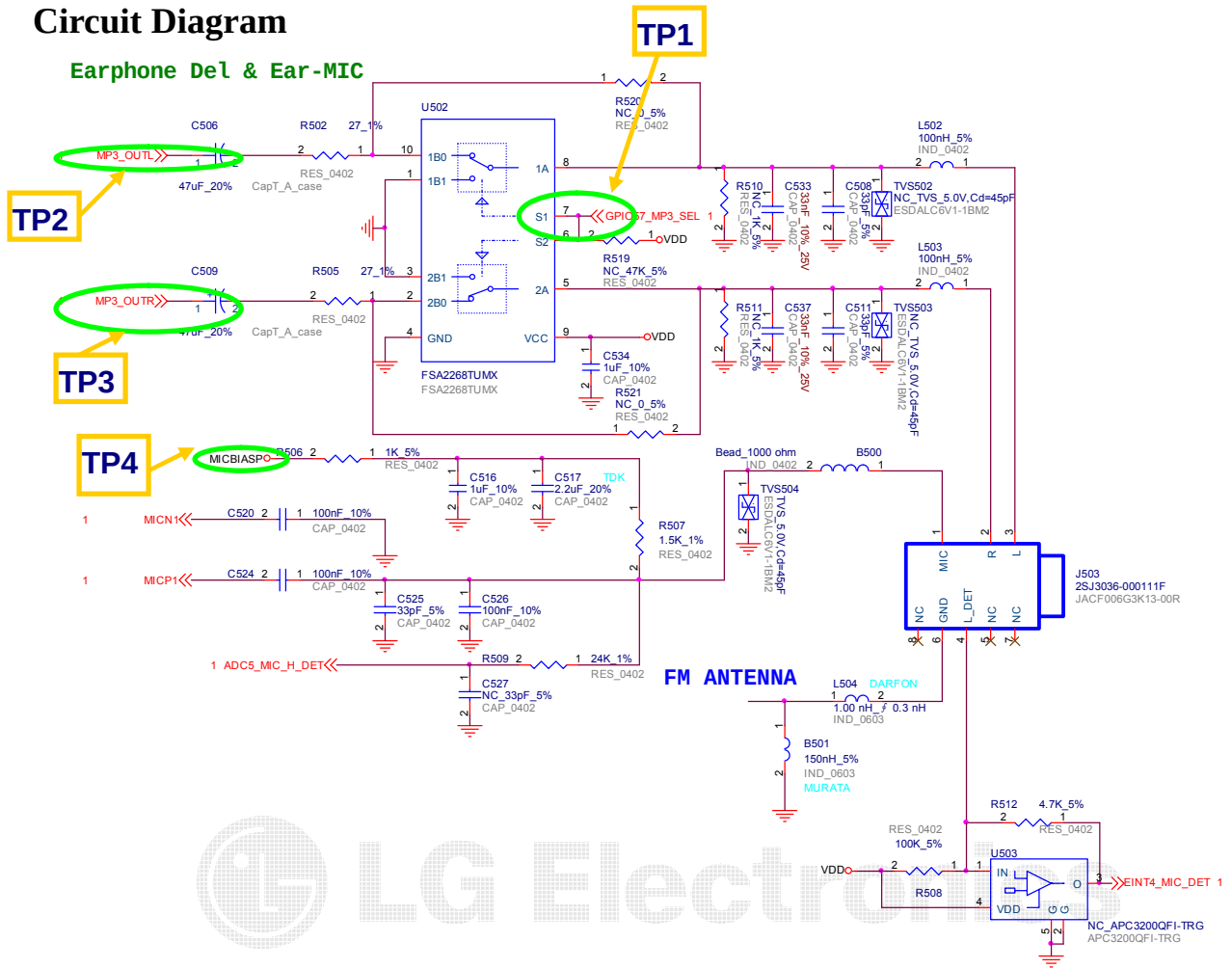
4.11 Headphone Trouble

Test Point

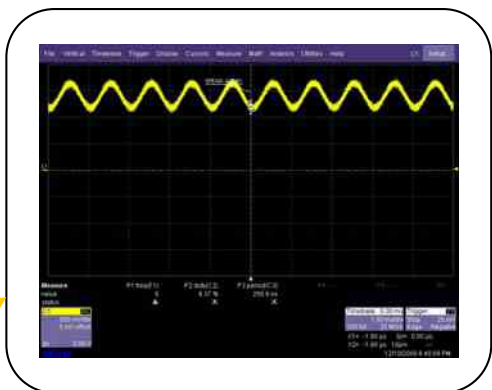
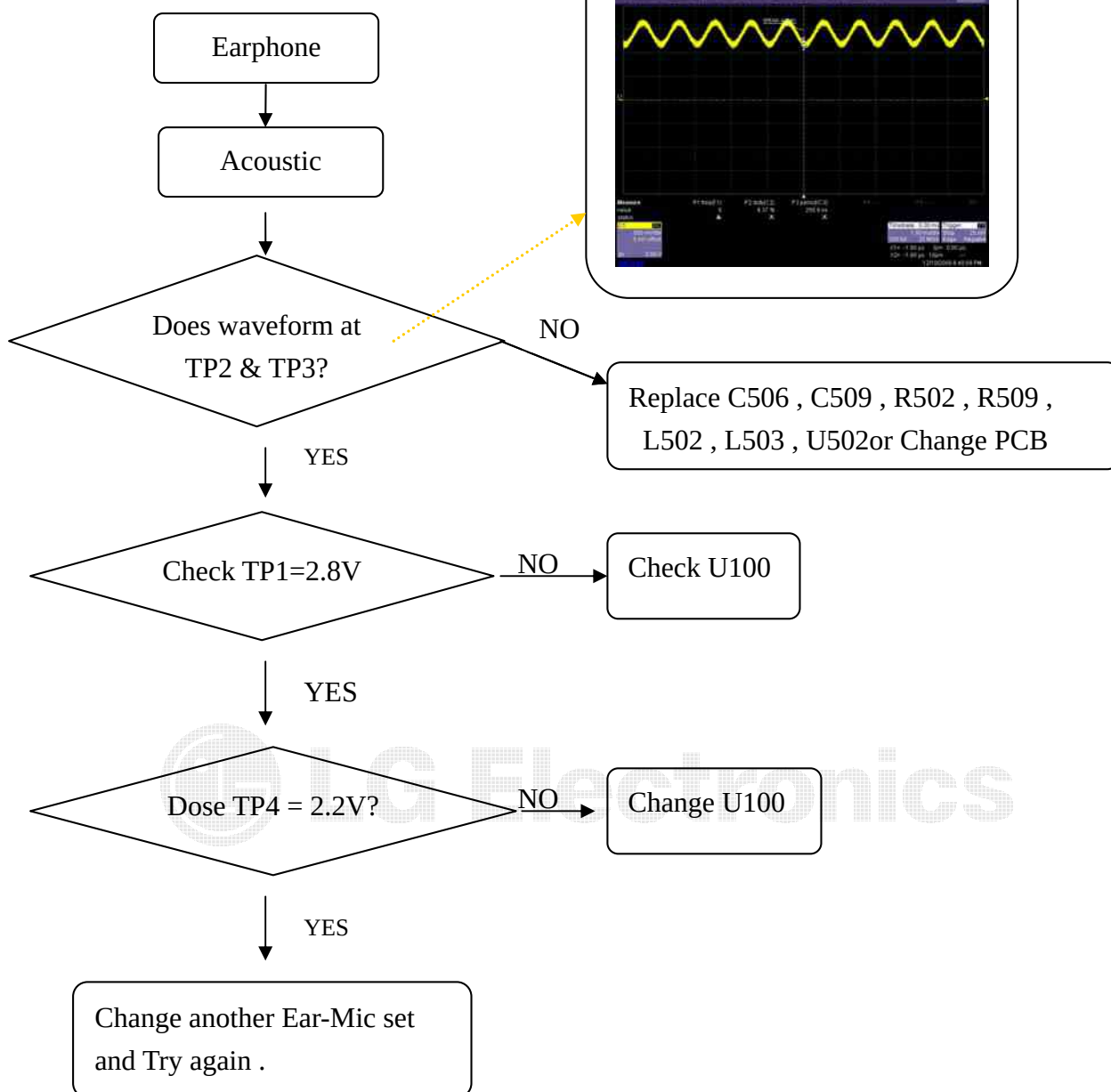


Circuit Diagram

Earphone Del & Ear-MIC

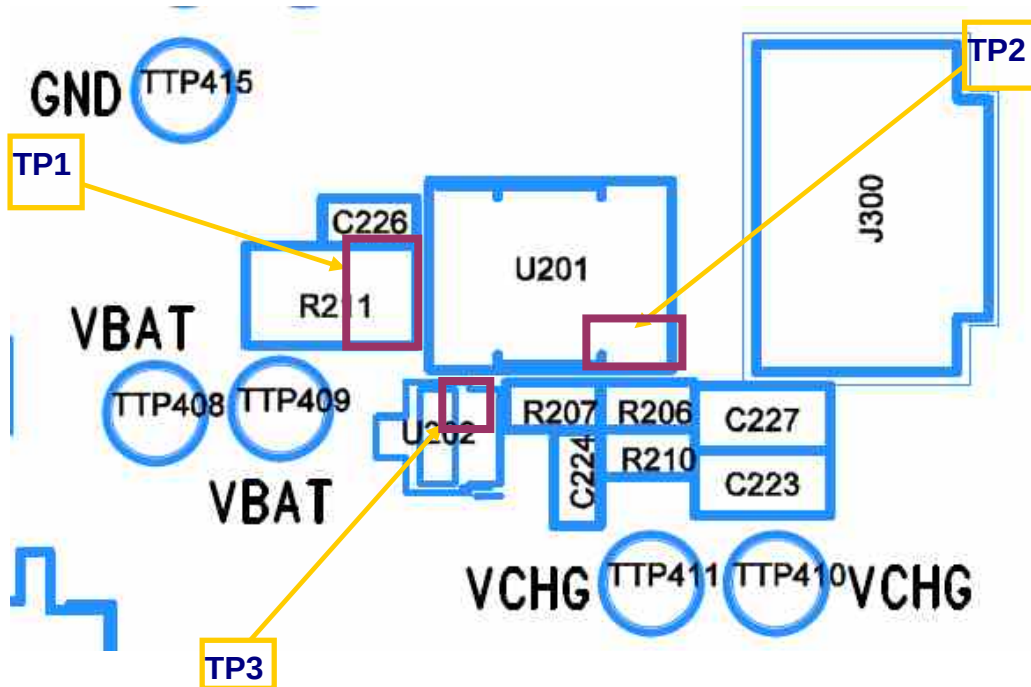


Checking Flow

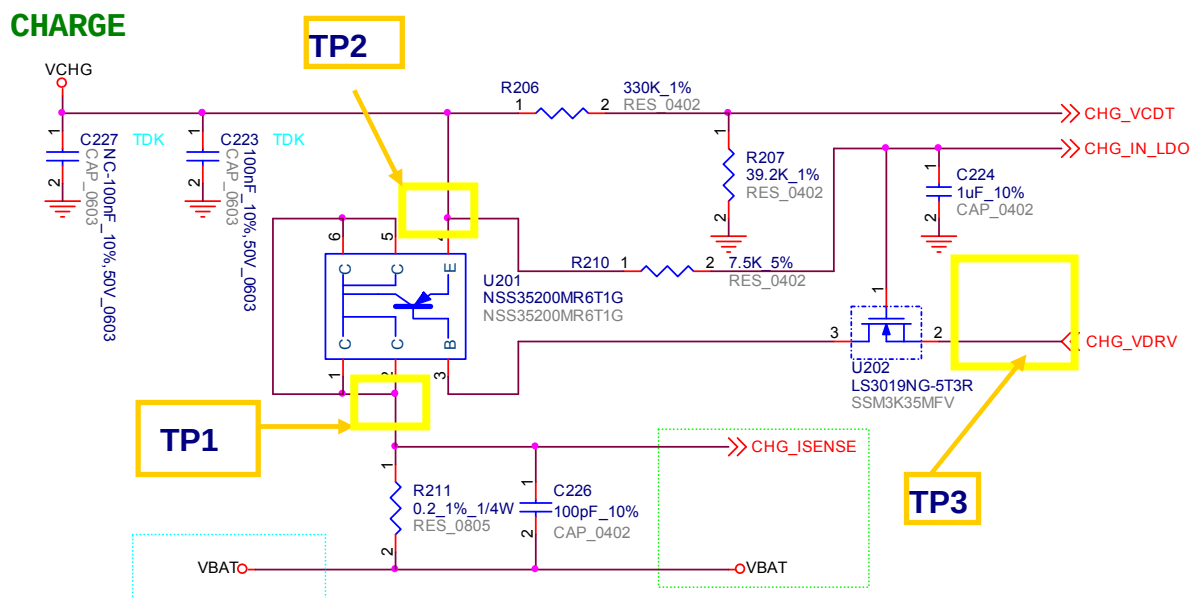


4.12 Charging Trouble

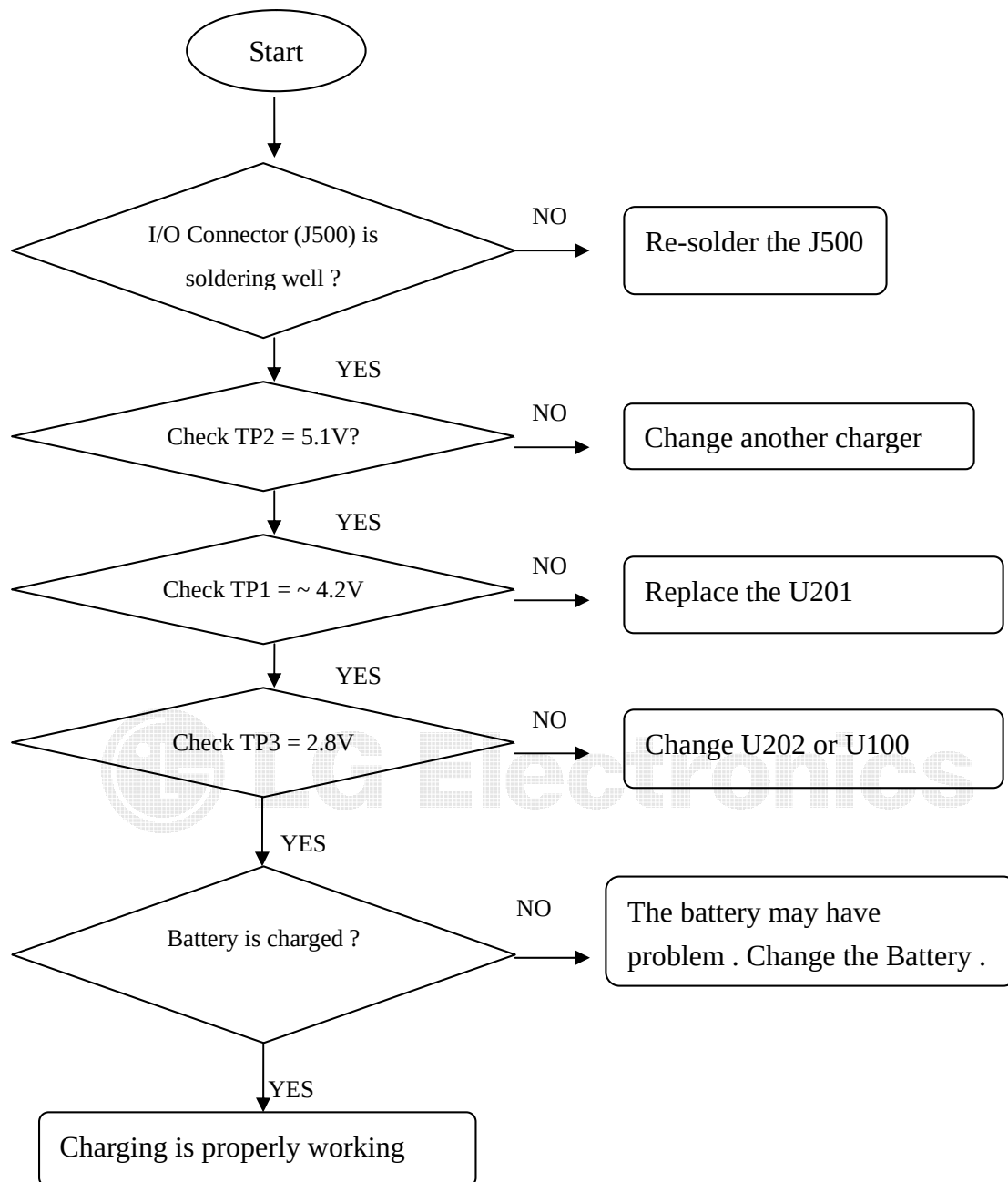
Test Point



Circuit Diagram

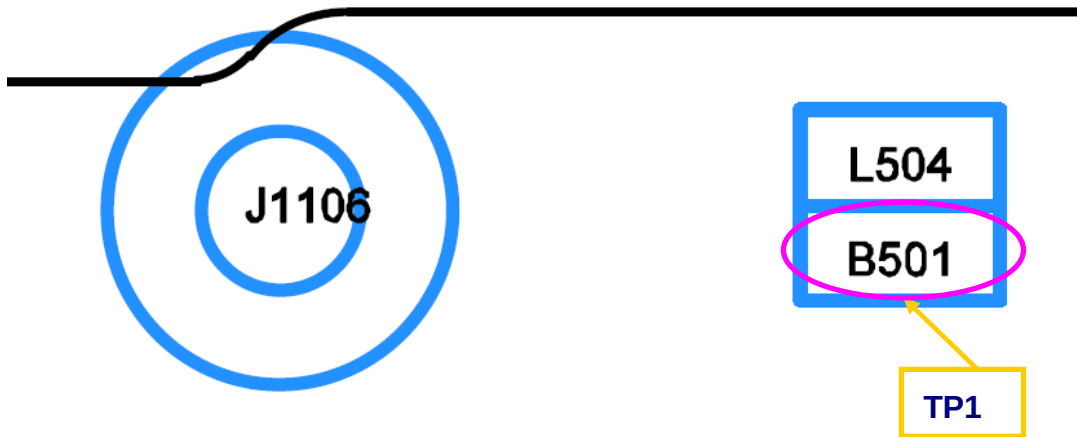


Checking Flow

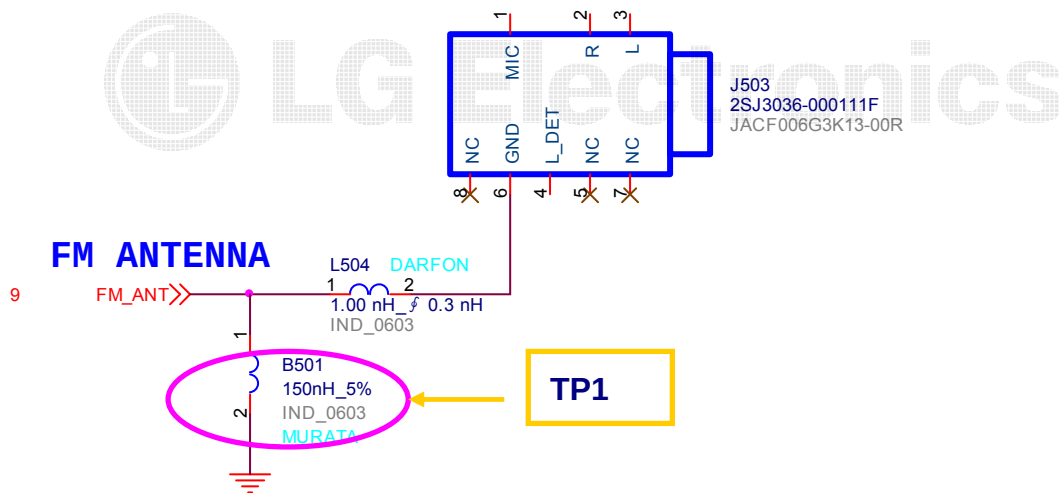


4.13 FM Radio Trouble

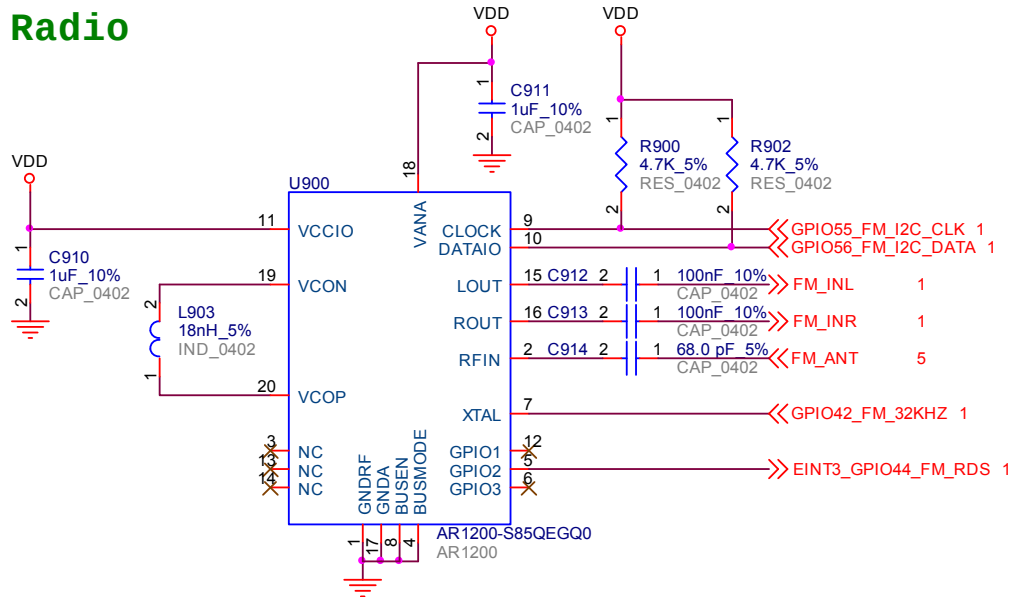
Test Point



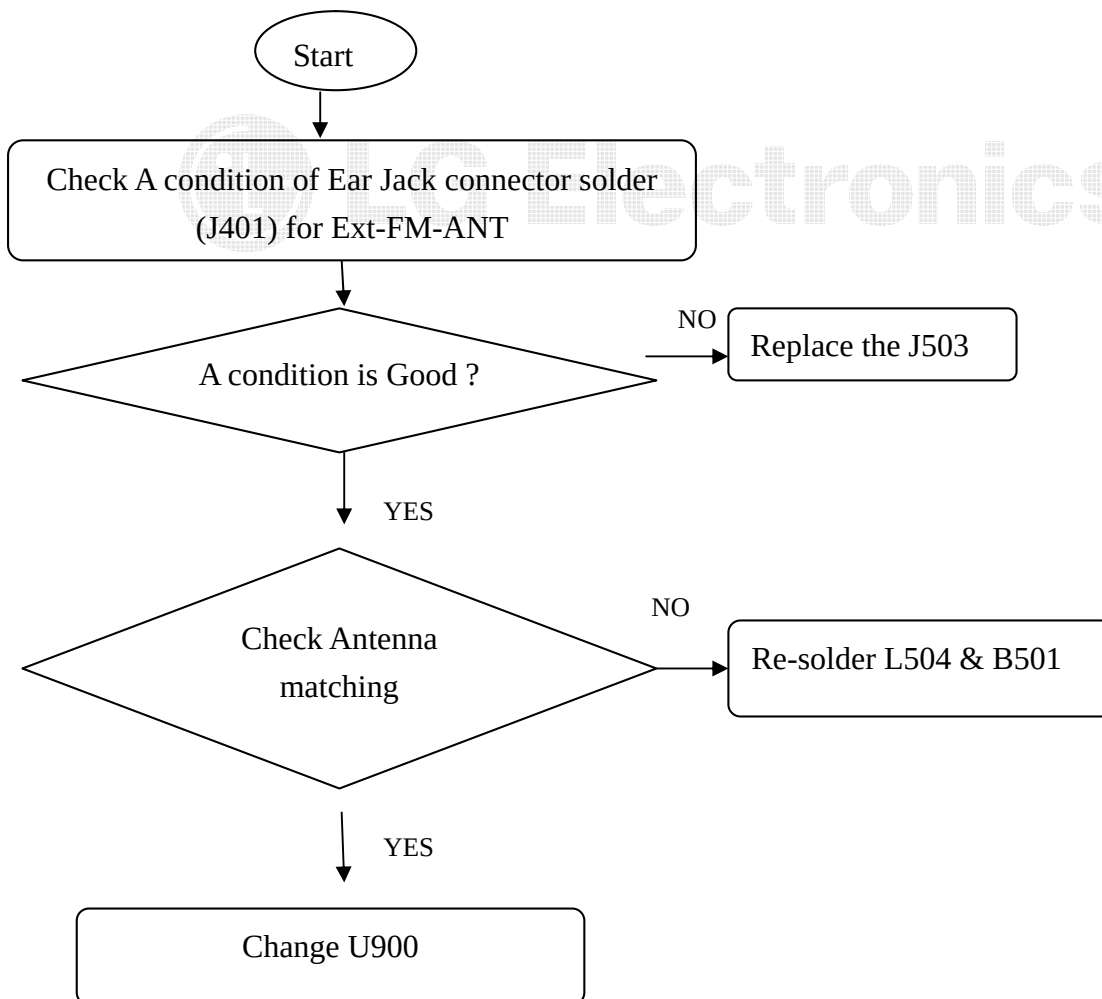
Circuit Diagram



FM Radio

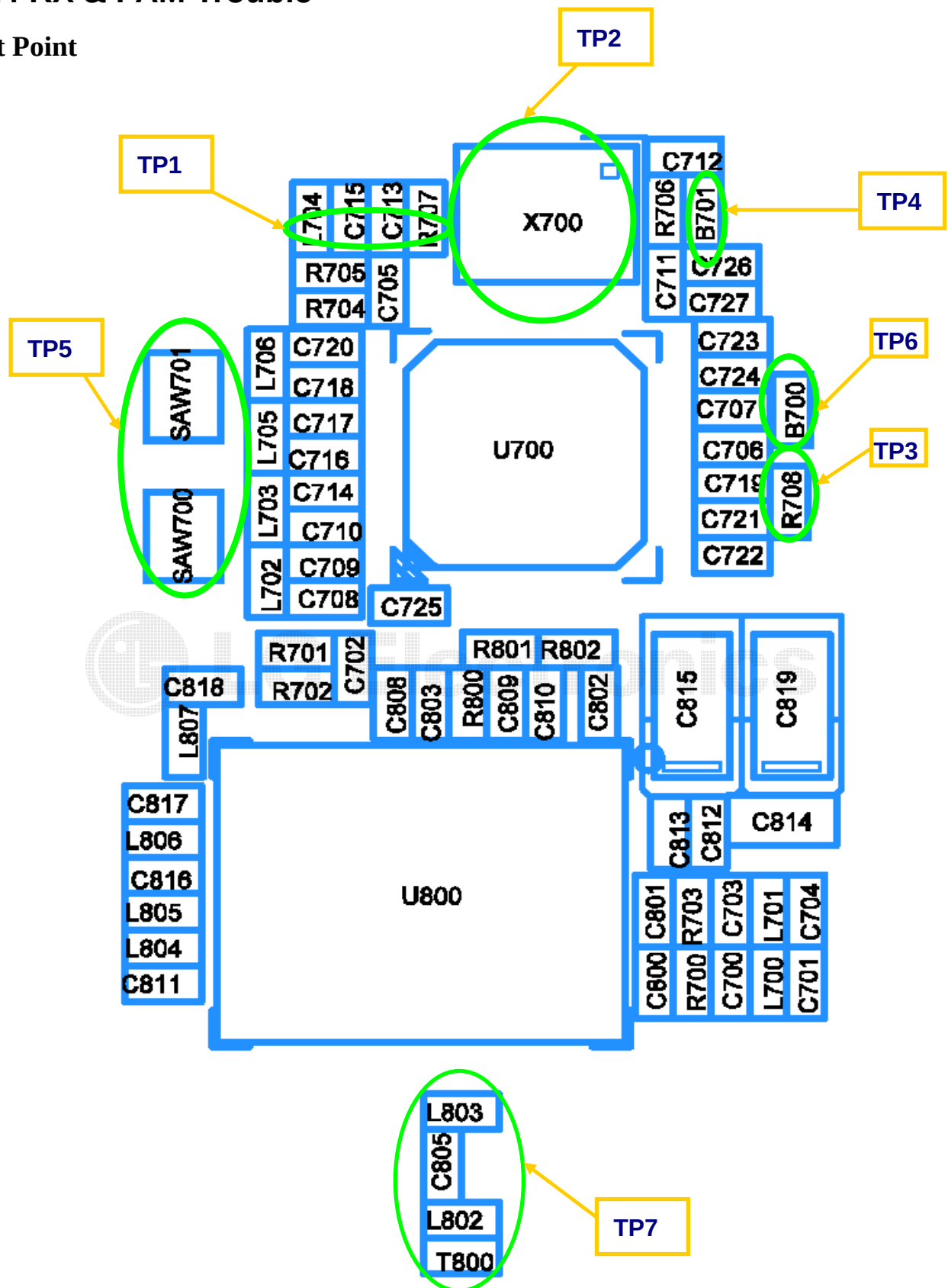


Checking FM

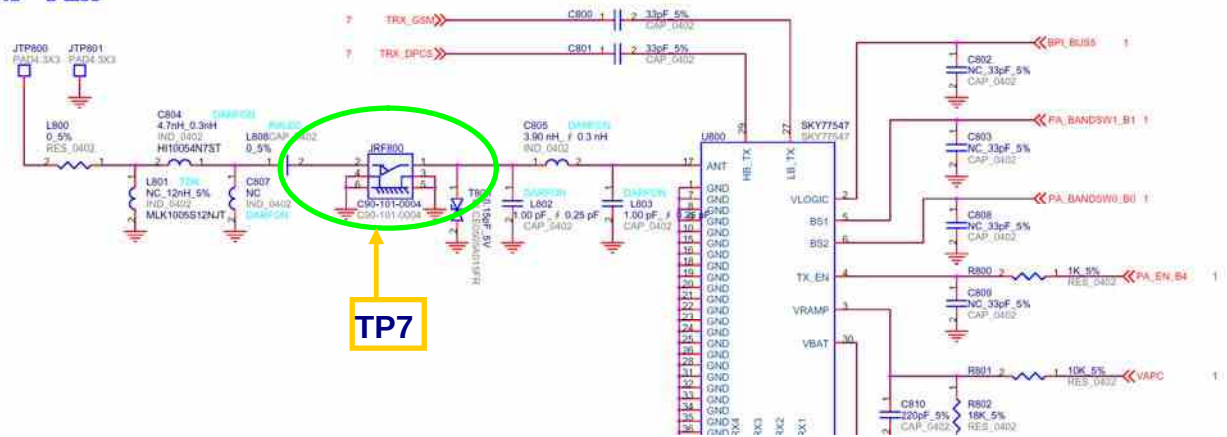


4.14 RX & PAM Trouble

Test Point



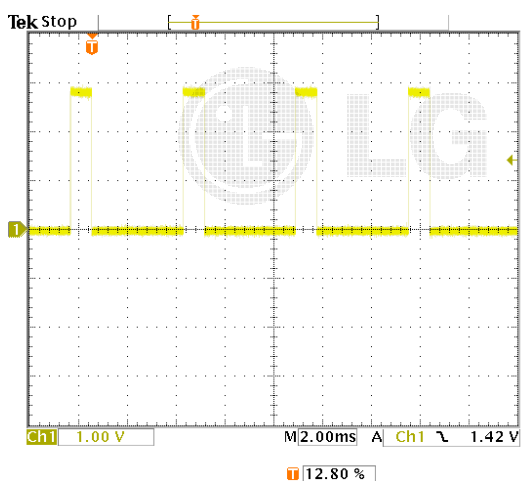
RF FEM



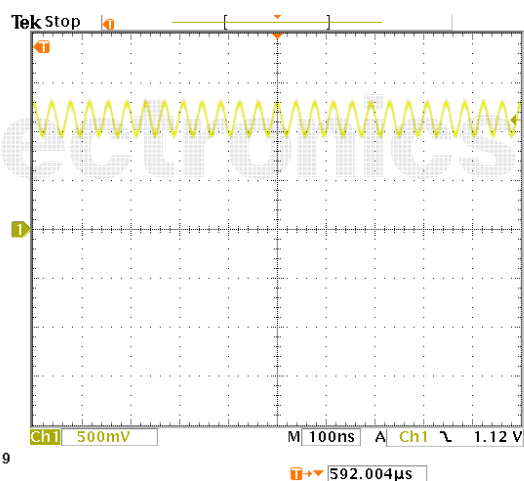
B701 & B700: $2.8 \pm 0.2V$

2.8±0.2V(frequency=216.7Hz)

X700: 26MHz



19 Feb 2009
02:59:09



18 Feb 2009
18:59:56

TX signal (GSM850/GSM900)

(1) Use Meta tool connect PCBA

(2) Choose TX level and profile

(3) Band: GSM 900 ARFCN : 37 TSC : 5 PCL : 5 AFC:4100

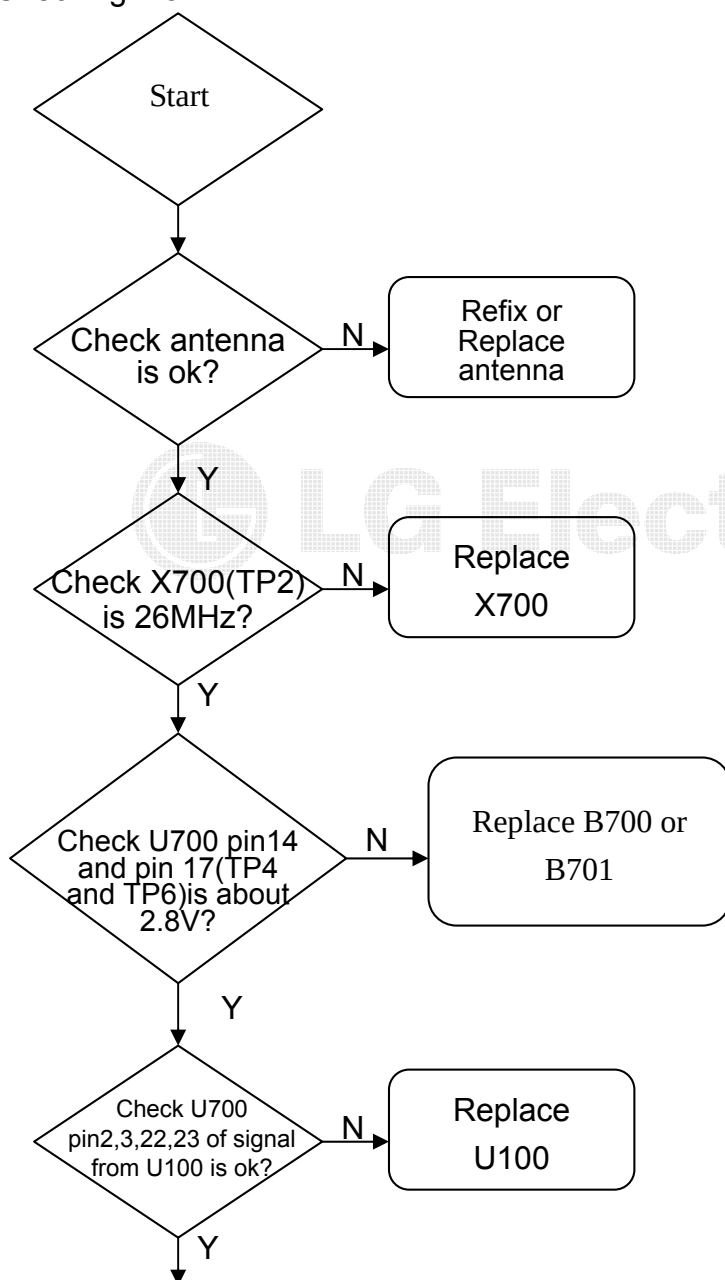
GSM850 ARFCN:190 TSC:5 PCL:5 AFC:4100

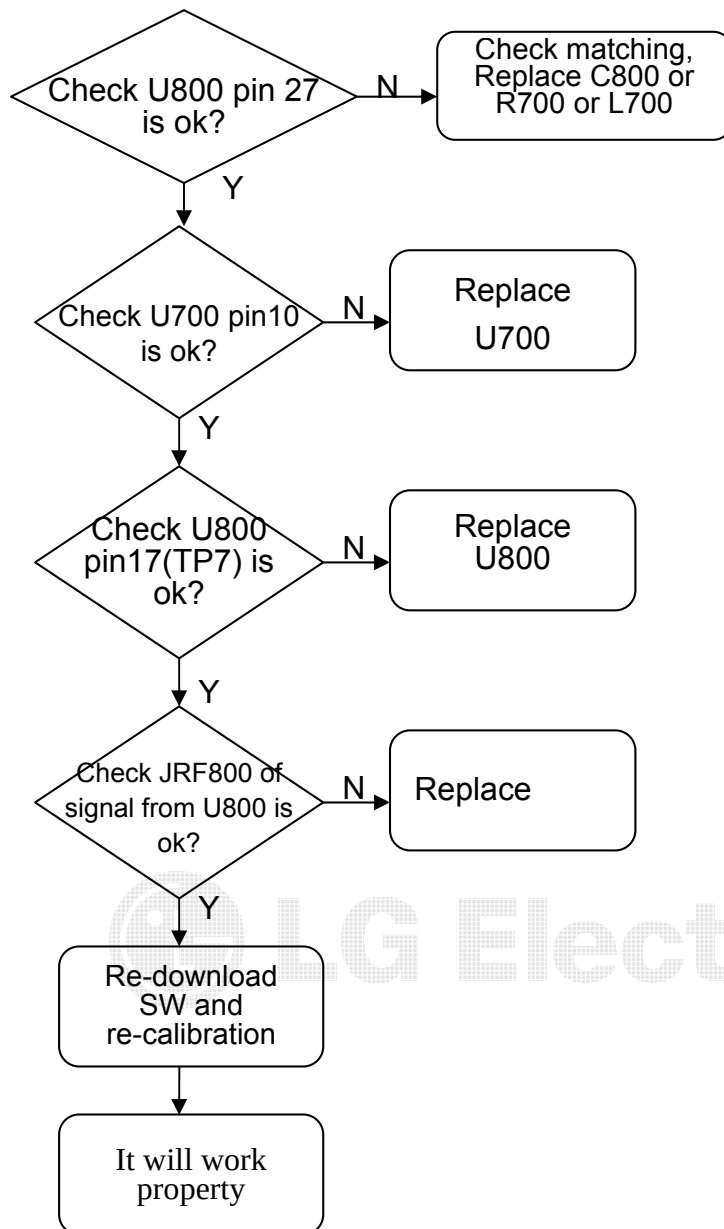
(4) Spectrum key in FREQ: 897.4 MHz(GSM900)

836.6MHz(GSM850) SPAN: 20 MHz

Comment : GSM850 and GSM900 have a same TX path

Checking Flow





RX signal (GSM900/GSM850)

(1) Use Meta tool connect PCBA

(2) Choose PM

(3) Band: GSM 900 ARFCN : 37 TSC : 5 PCL : 5 AFC:4100

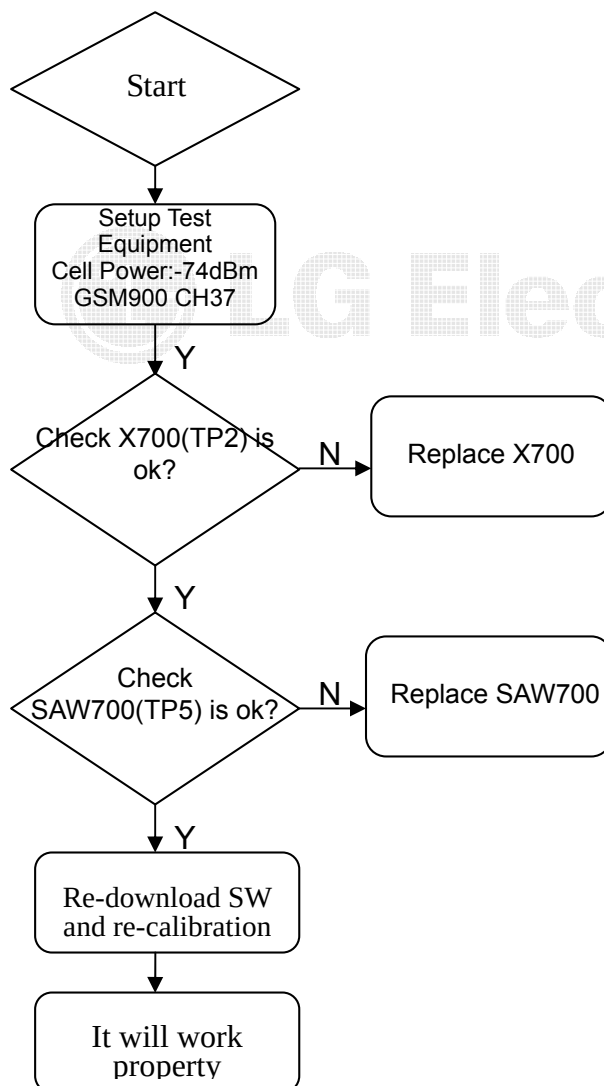
GSM850 ARFCN:190 TSC:5 PCL:5 AFC:4100

(4) Spectrum key in FREQ: 942.4 MHz(GSM900) 881.6MHz(GSM850)

SPAN: 20 MHz

Comment : GSM850 and GSM900 have a same saw filter .

Checking Flow



TX signal (DCS1800/PCS1900)

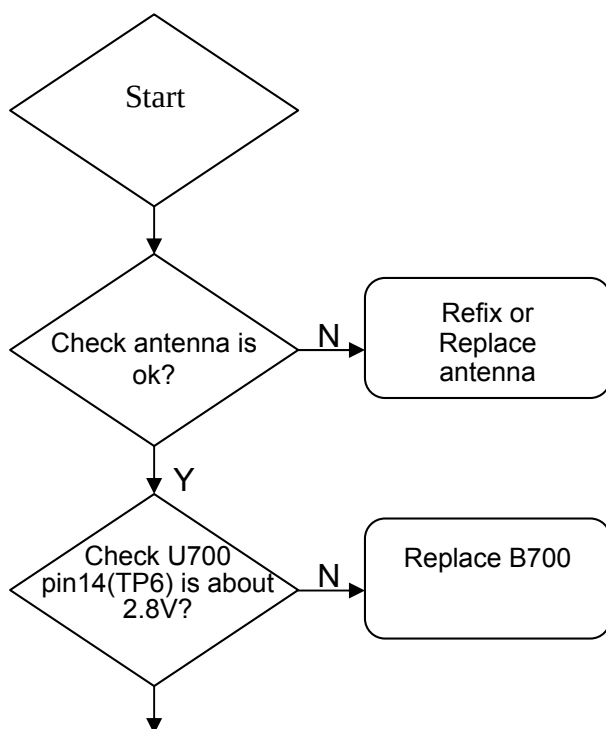
- (1) Use Meta tool connect PCBA
- (2) Choose TX level and profile
- (3) Band : DCS 1800 ARFCN : 700 TSC : 5 PCL : 5 AFC:4100
- (4) Spectrum key in FREQ: 1.7478GHz

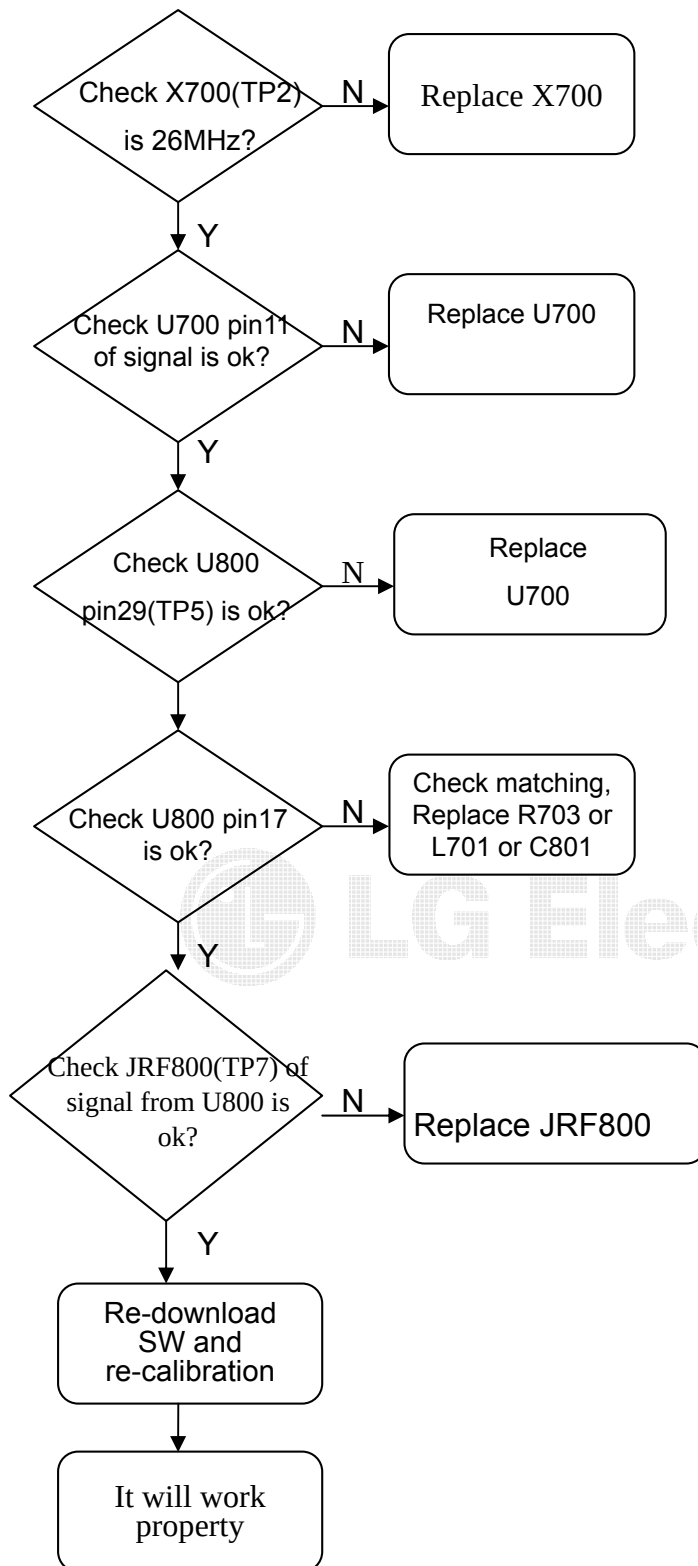
SPAN: 20 MHz

Comment : DCS1800 and PCS1900 have a same TX path



Checking Flow:





RX signal (DCS1800/PCS1900)

(1) Use Meta tool connect PCBA

(2) Choose PM

(3) Band : DCS 1800 ARFCN : 698 TSC : 0 PCL : 5 AFC:4100

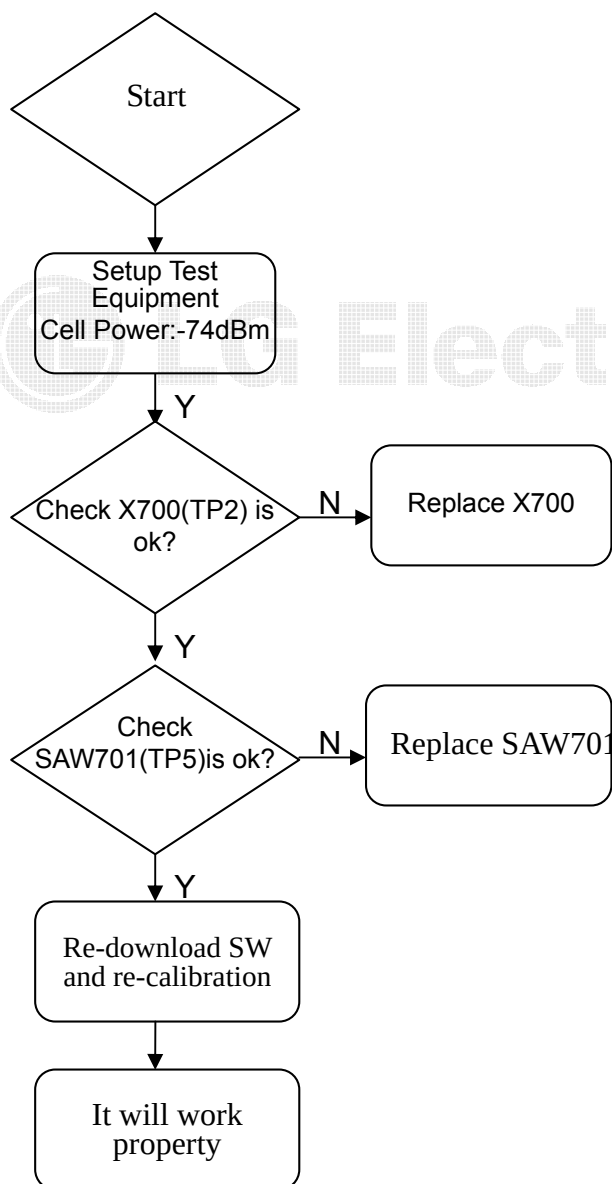
PCS 1900 ARFCN : 661 TSC : 0 PCL : 5 AFC:4100

(4) Spectrum key in FREQ: 1.8424GHz (DCS1800), 1.9600GHz (PCS1900)

SPAN: 20 MHz

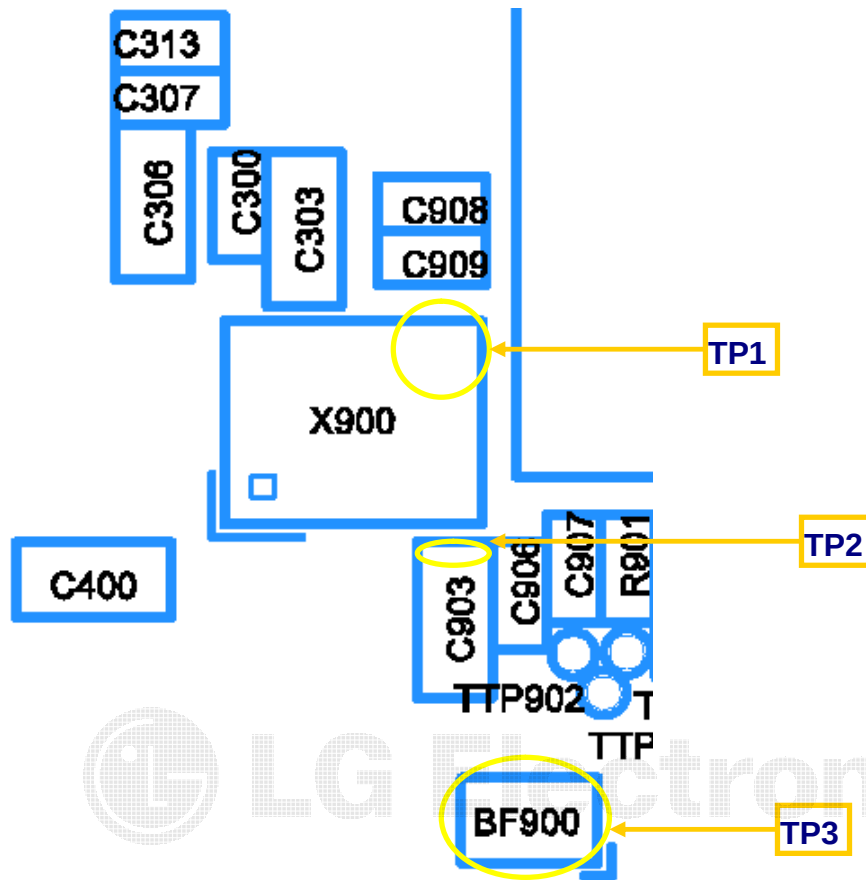
Comment : DCS1800 and PCS1900 have a same saw filter .

Checking Flow:

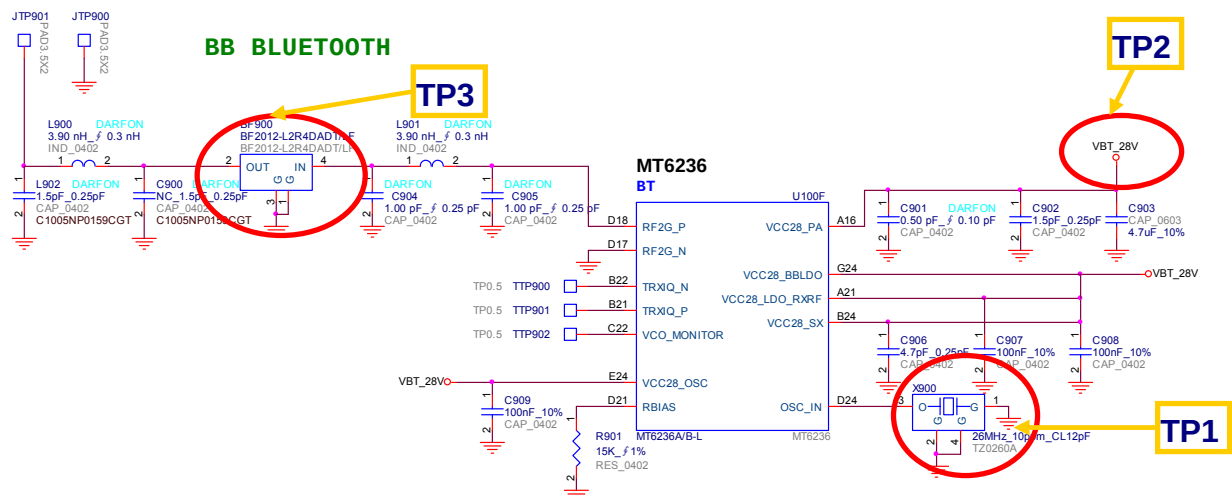


4.15 Bluetooth Trouble

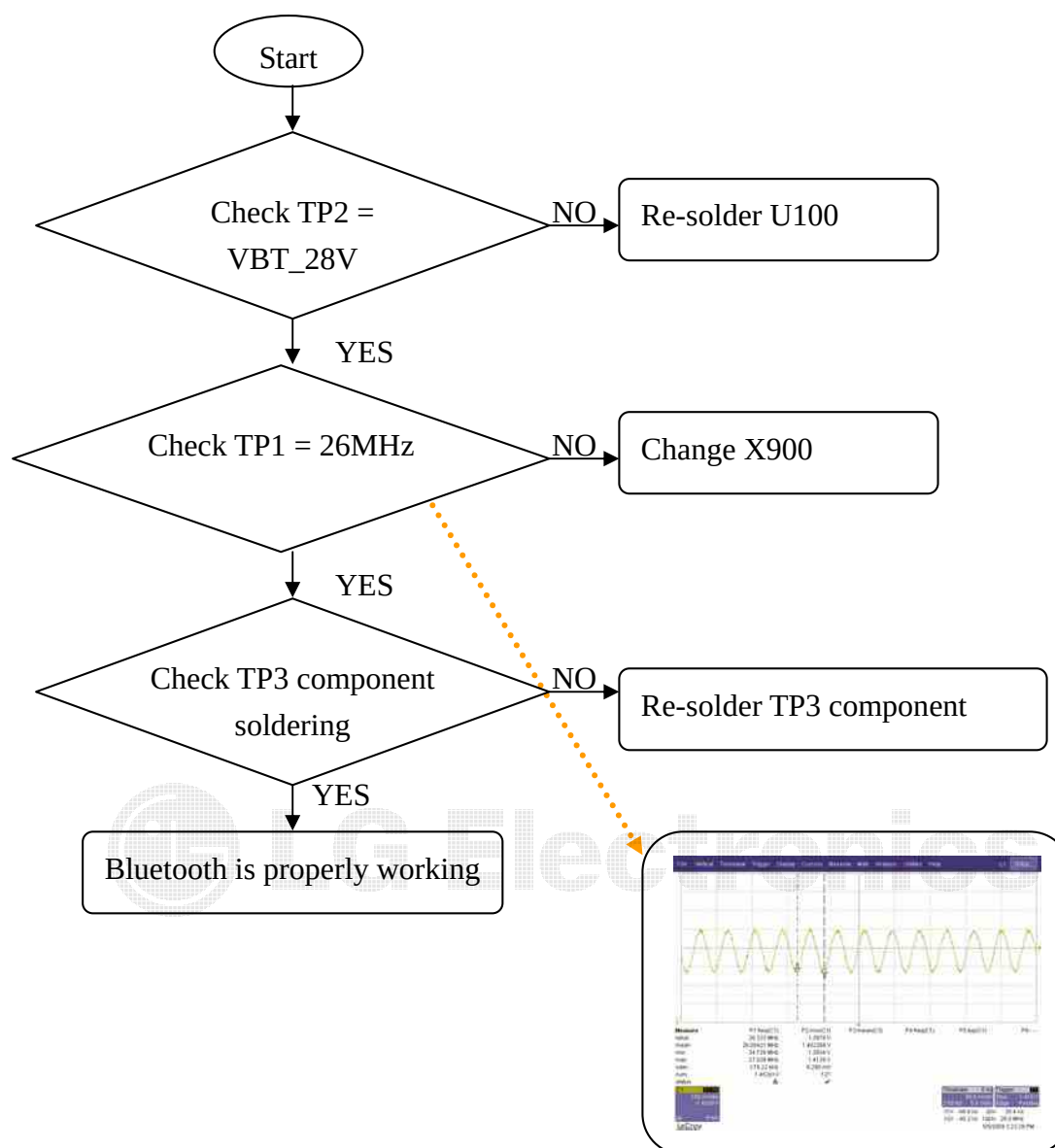
Test Point



Circuit Diagram

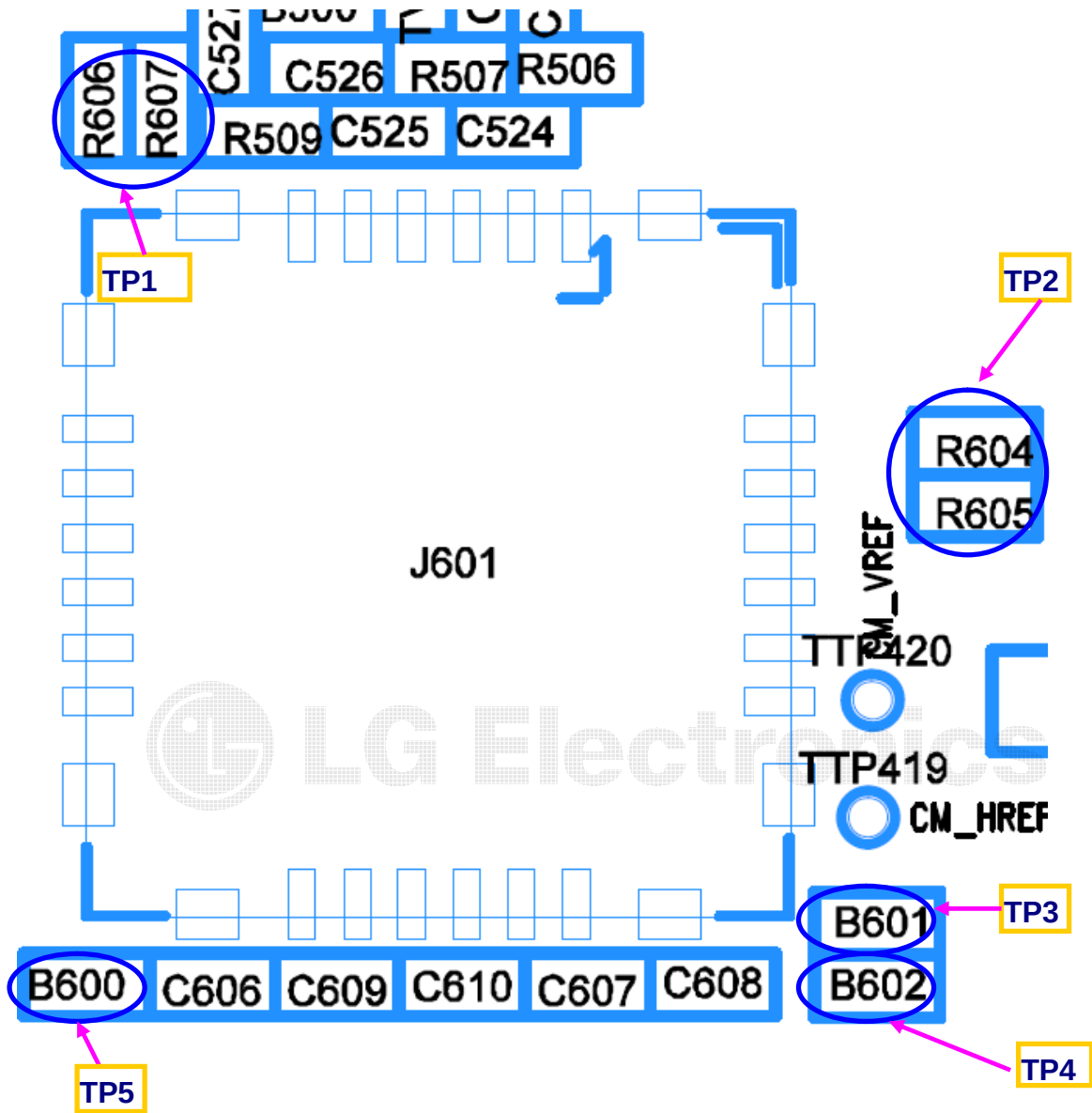


Checking Bluetooth

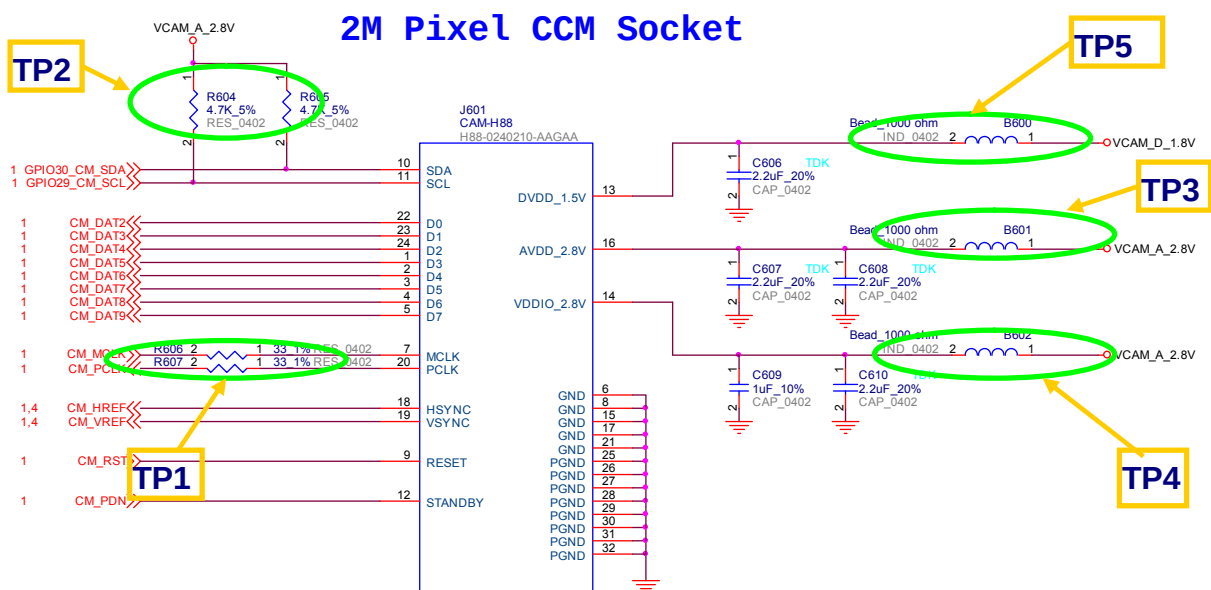


4.16 Camera Trouble

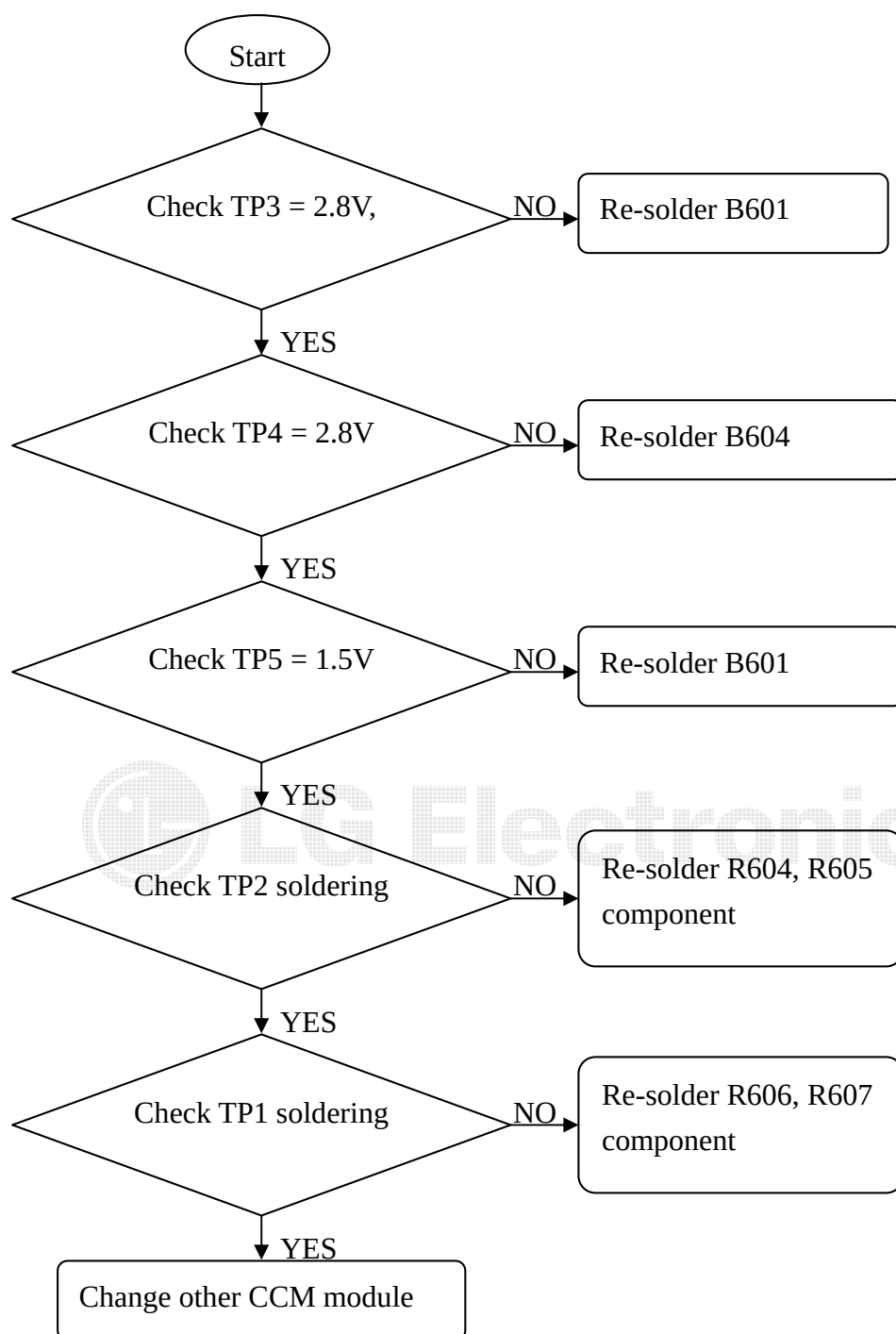
Test Point



Circuit Diagram

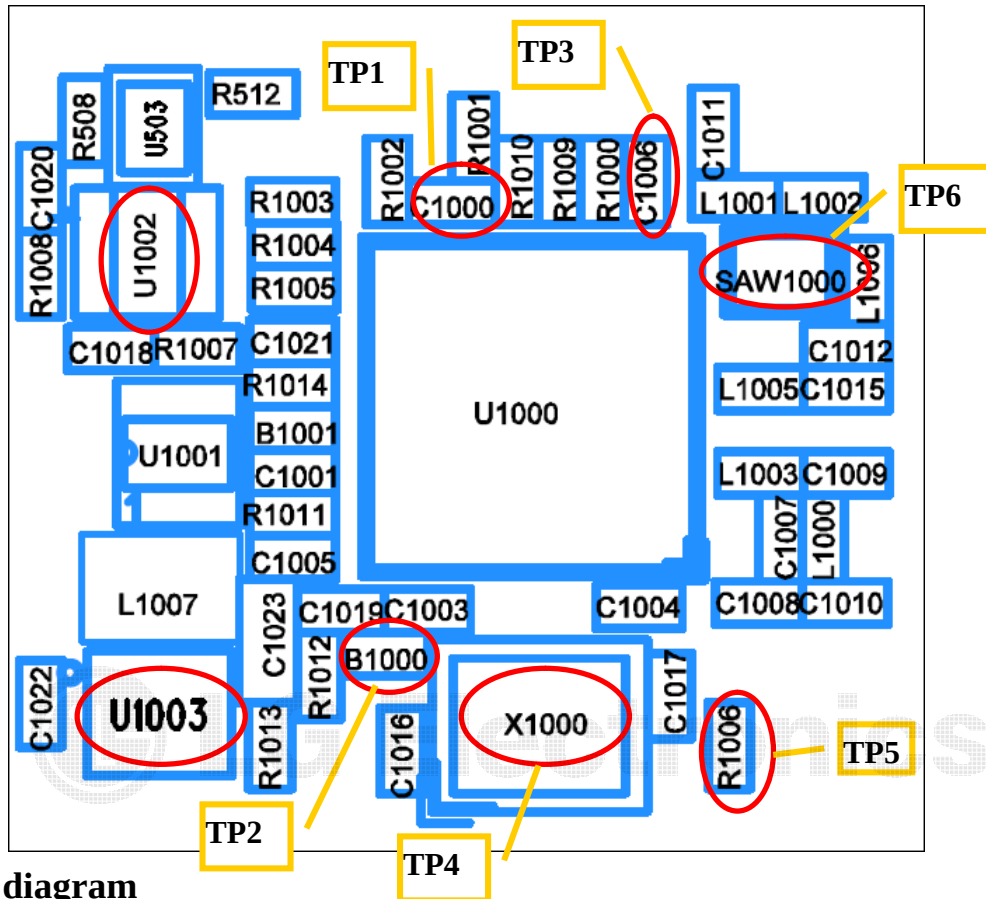


Checking Camera

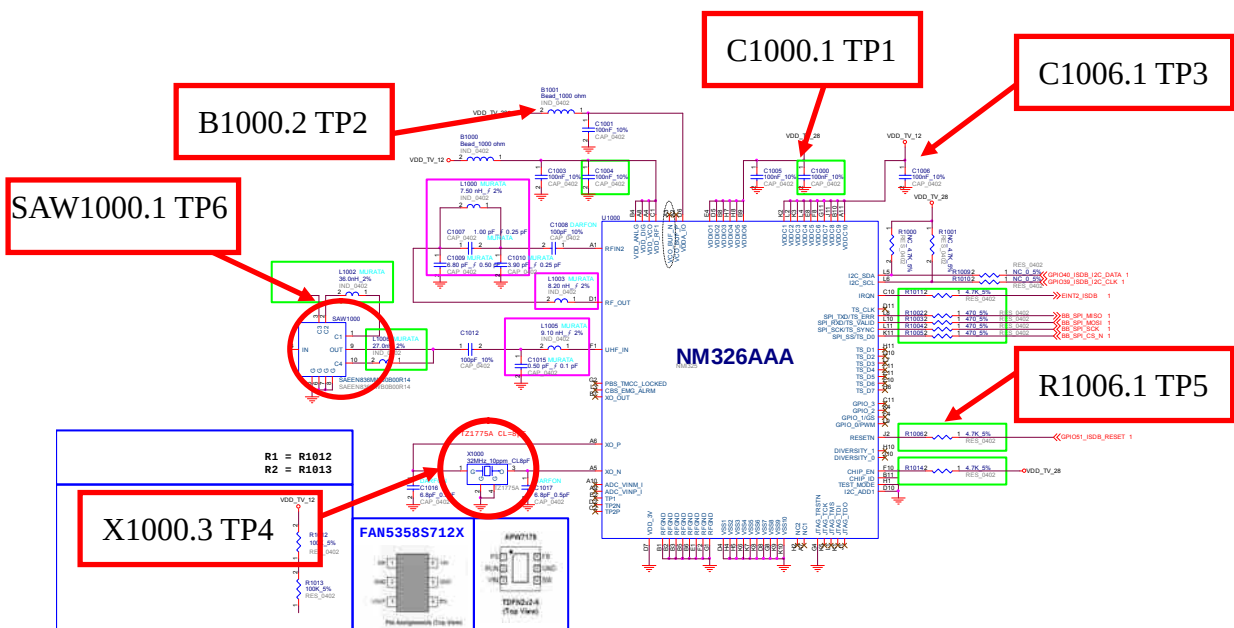


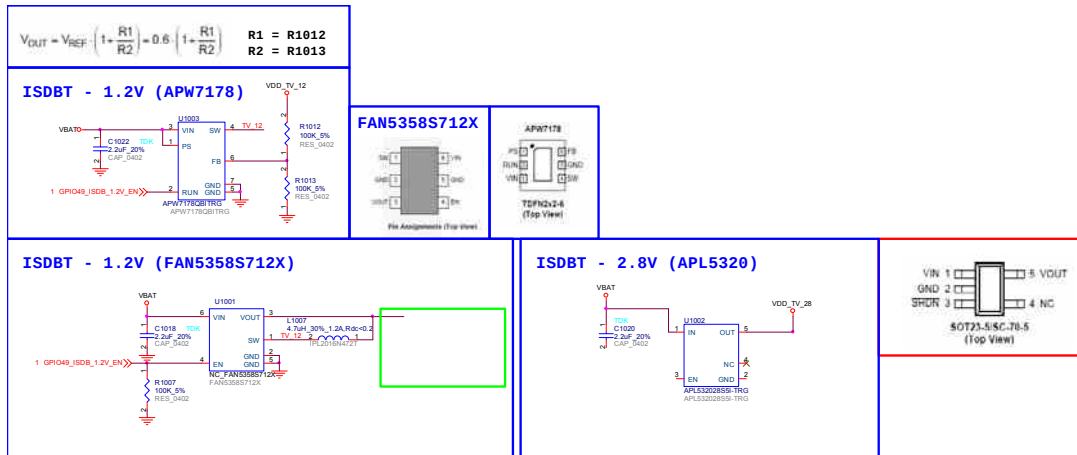
4.17 ISDB-T(NMI326) Trouble

Test point

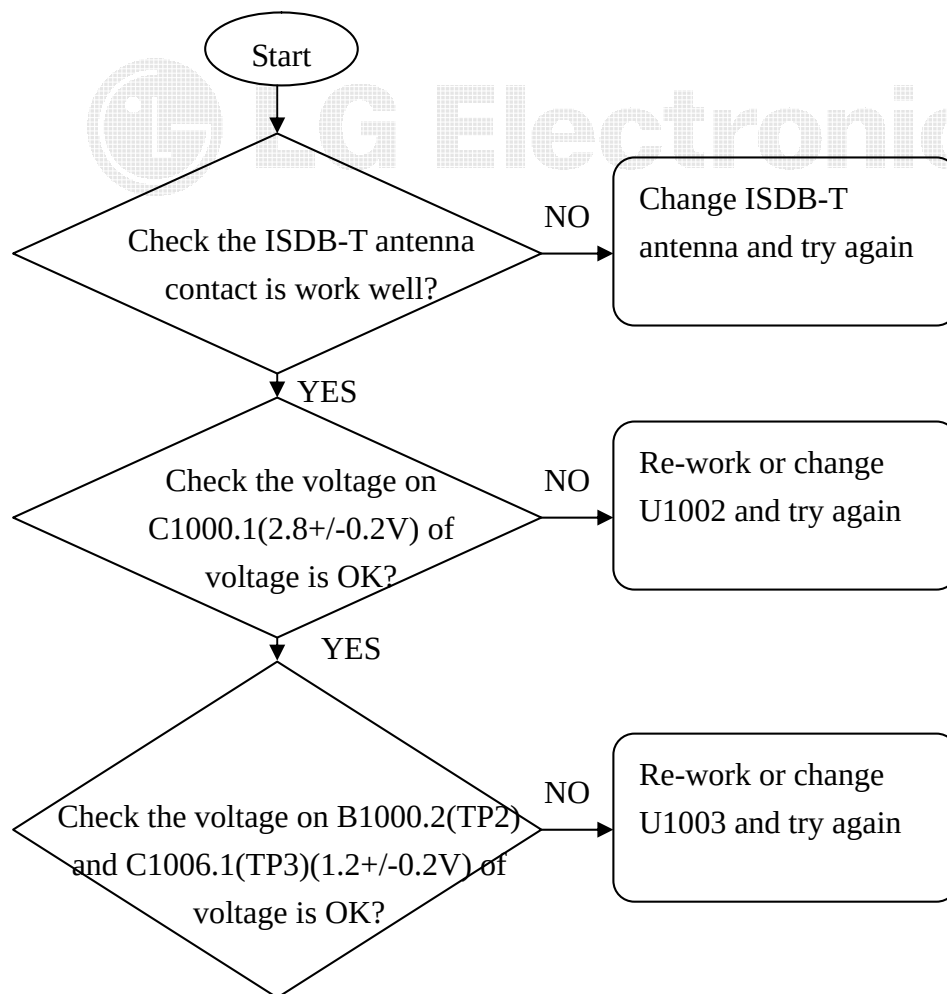


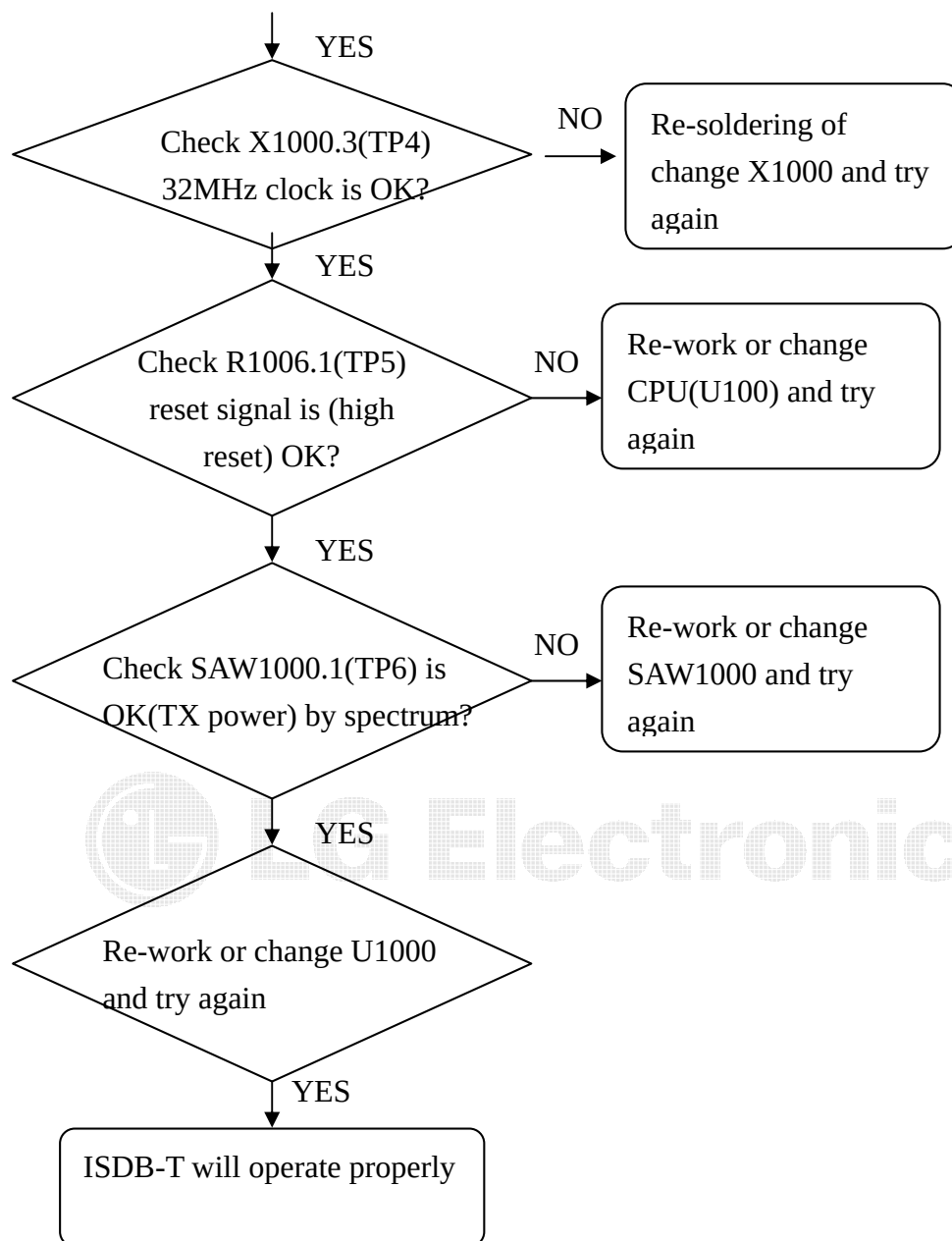
Circuit diagram





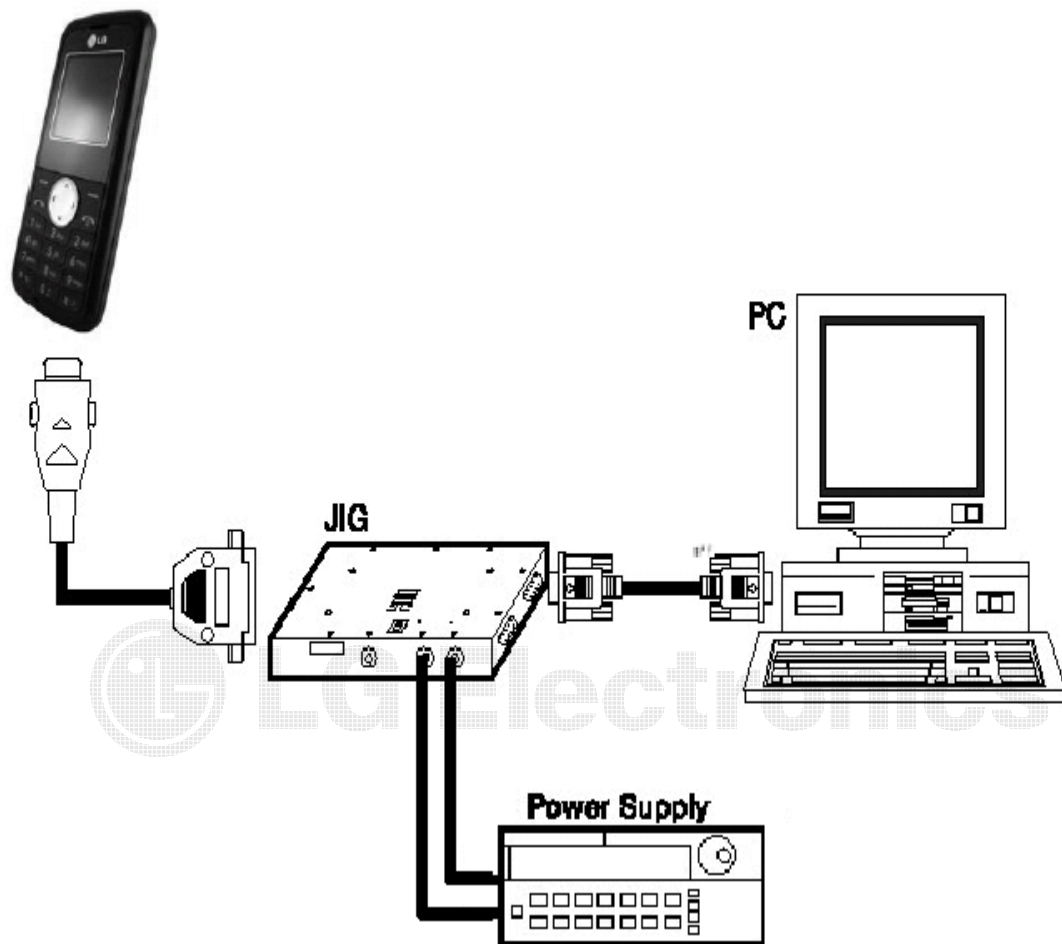
Checking flow





5.DOWNLOAD

5.1 Download setup



5.2 Download Process

LEO Download Tool

■ Tools

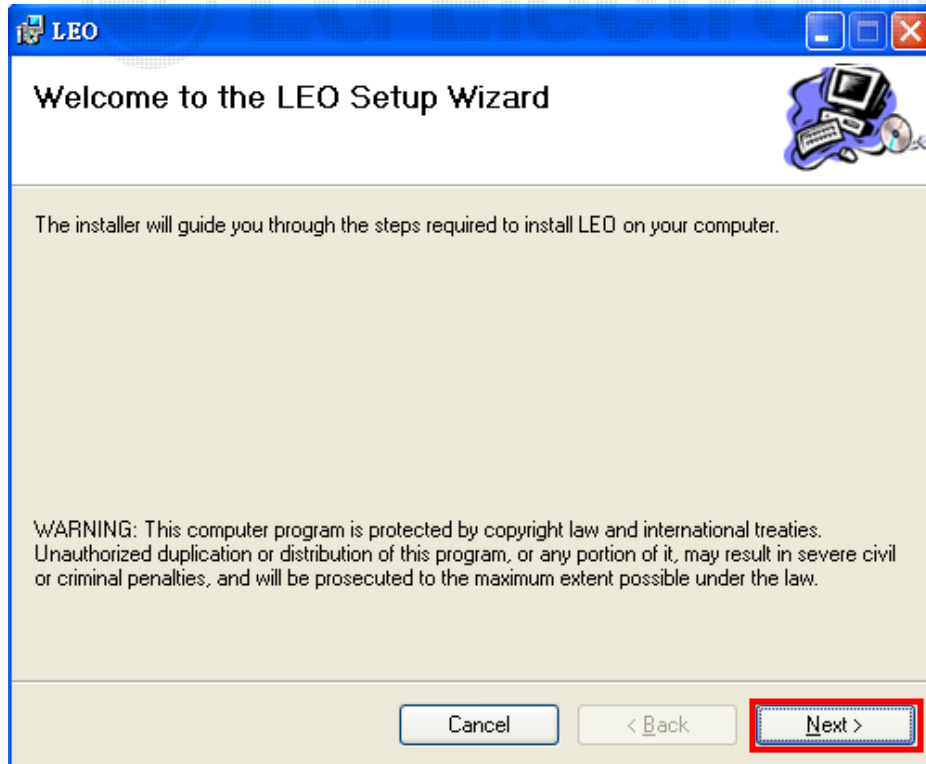
1. Download cable(**Prolific USB-to-Serial**)
2. PC
3. Battery (3.8 V Li-ion Battery)

■ How to install Leo download tool

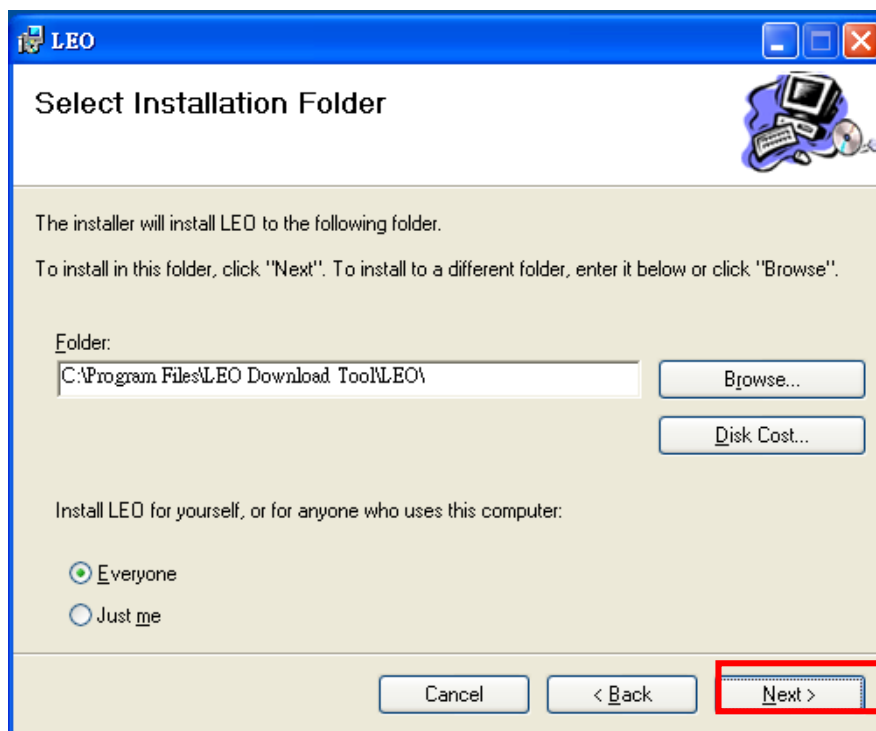
1. You must install “Prolific USB-to-Serial Comm Port” driver first before installing this program, and then double click the “Setup.msi” start installation.



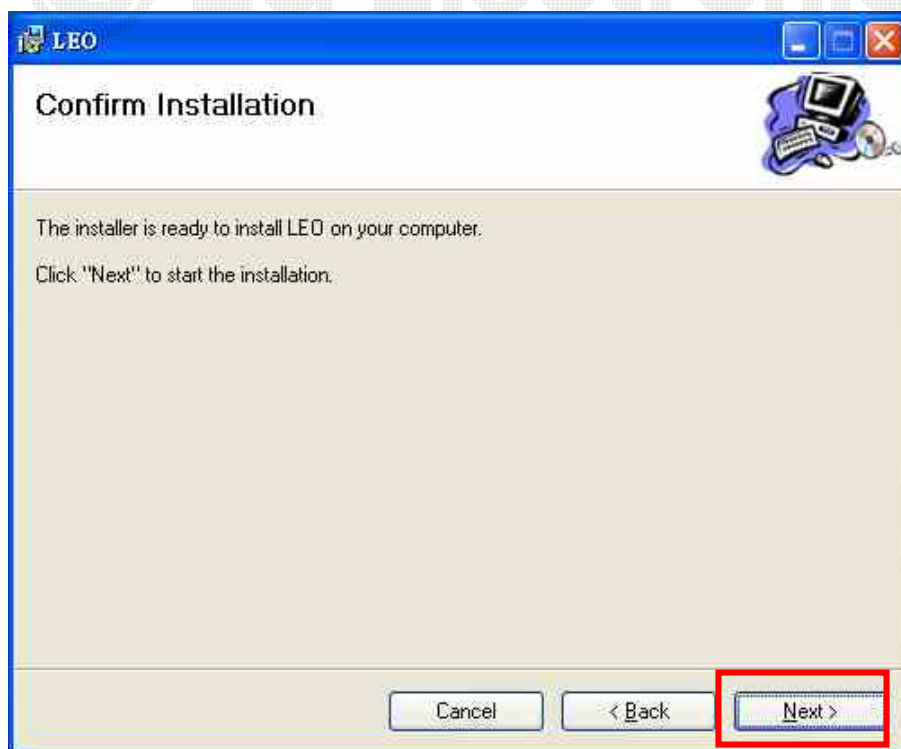
2. You can see the below picture, and then click the “Next” button.



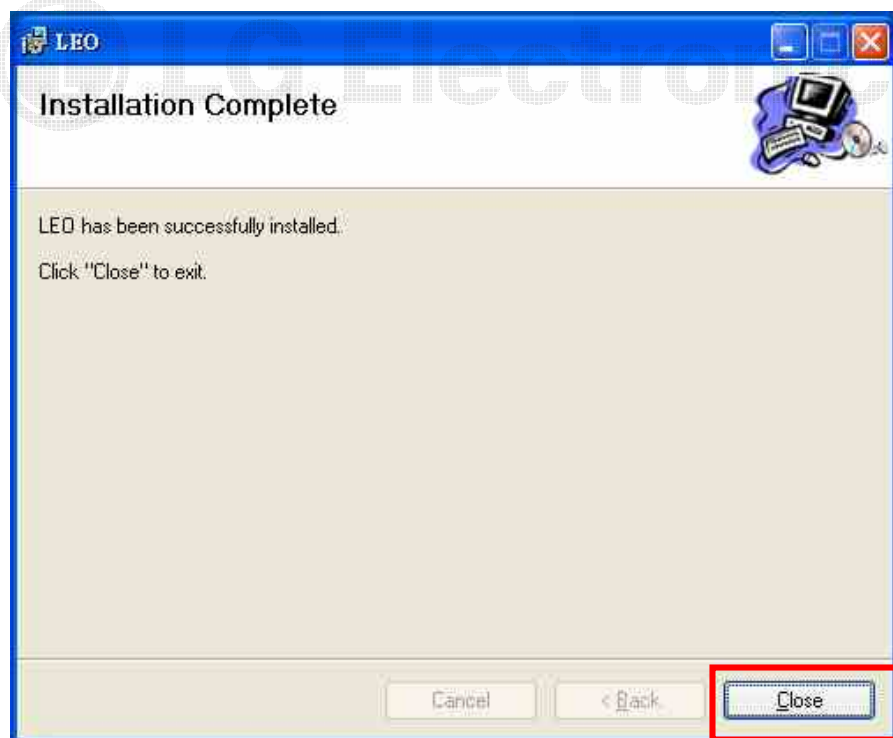
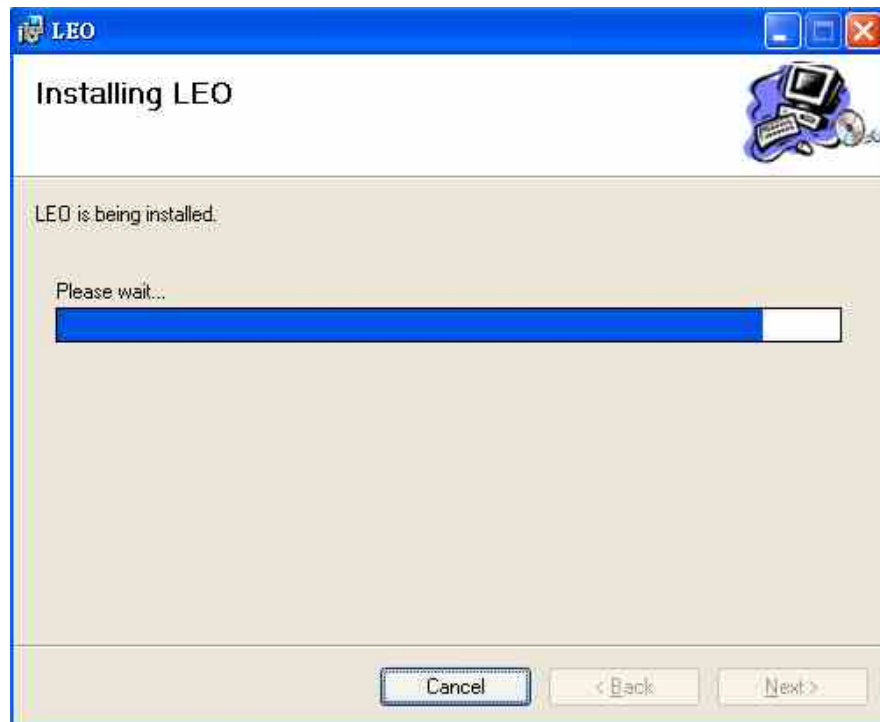
3. You can see the below picture, and then click the “Next” button.



4. You can see the below picture, and then click the “Next” button.



5. You can see the below Installing picture, and then click the “Close” button installation complete.



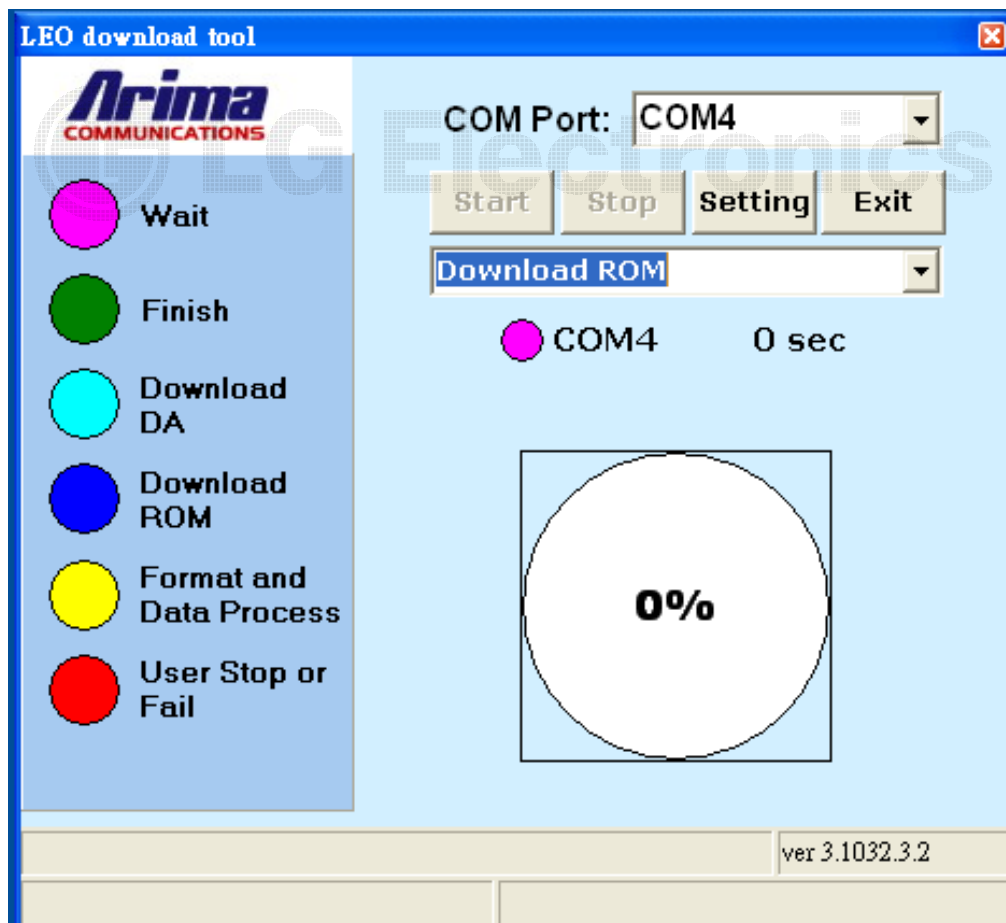
■ How to use Leo download tool

For example: LGX350-00-V09b-724-XX-JUN-30-2011

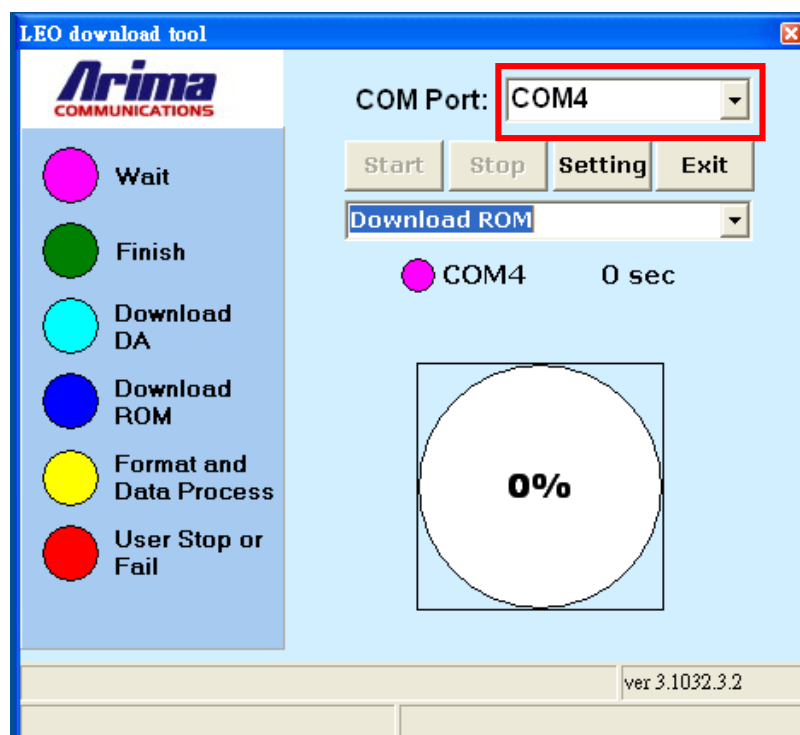
1.Connect Download cable with computer, and then double click the” **LEO Download Tool**”.



2.you can see the below picture.



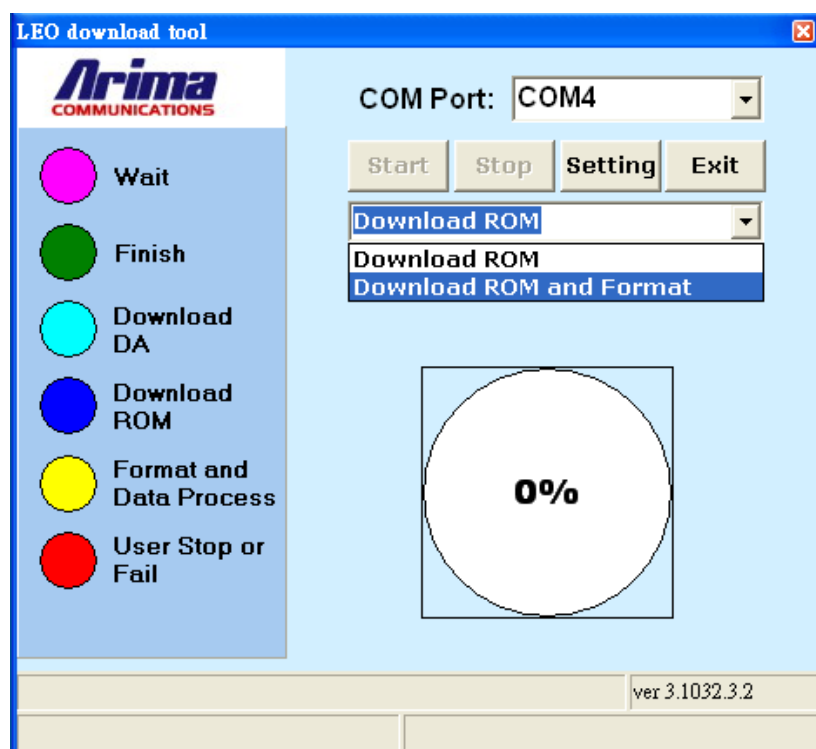
3. Select COM port (LEO will auto detect COM port)



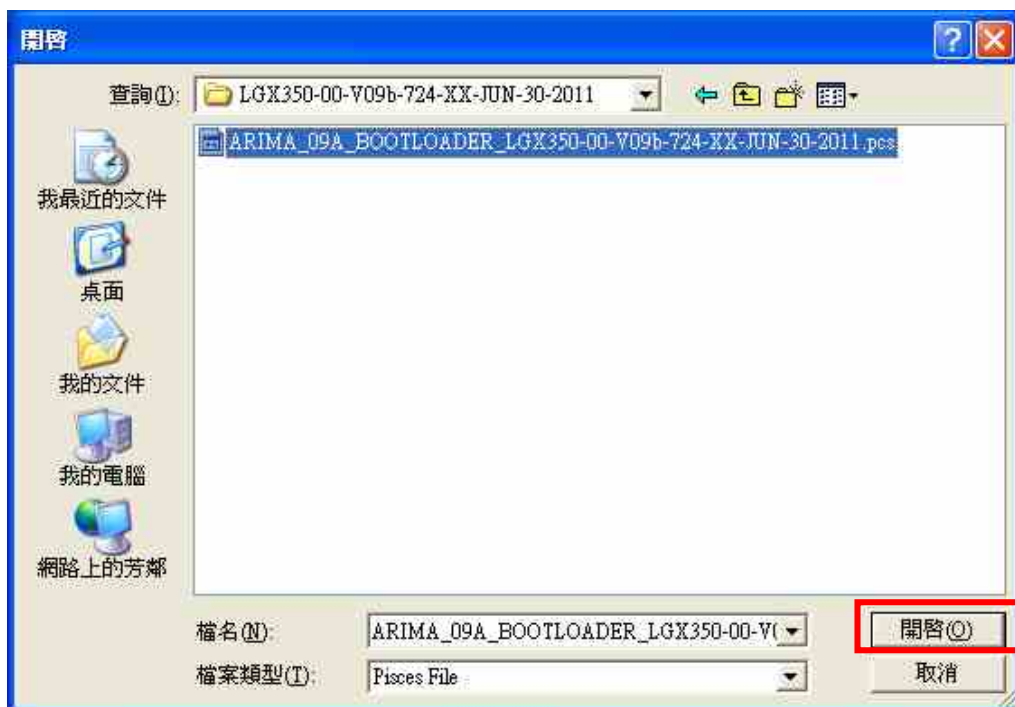
4. Select Download mode.

Note: ① If you select “Download ROM”, it will download software only.

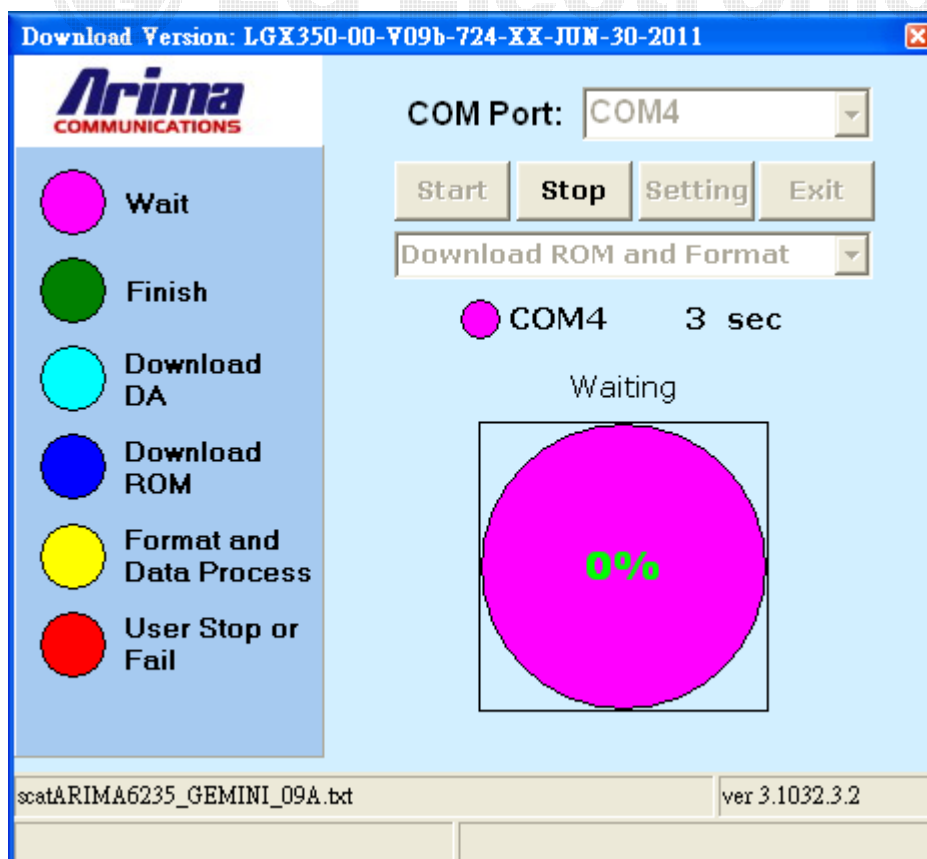
② If you select “Download ROM and Format”, it will download software and delete NVRAM all data except calibration data and IMEI number, and delete user disk data include contact information 、 message etc, also it still will reset META_NVRAM to factory default.



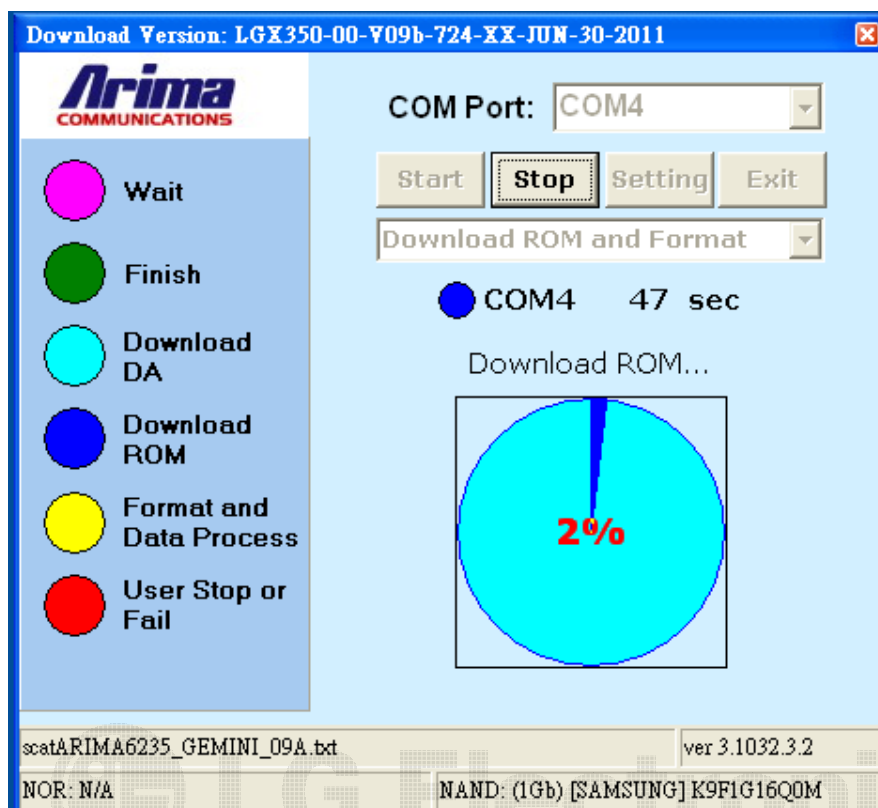
5. Click the “Setting” button and select a valid file. The file always be end of “.PCS” , reference below picture.



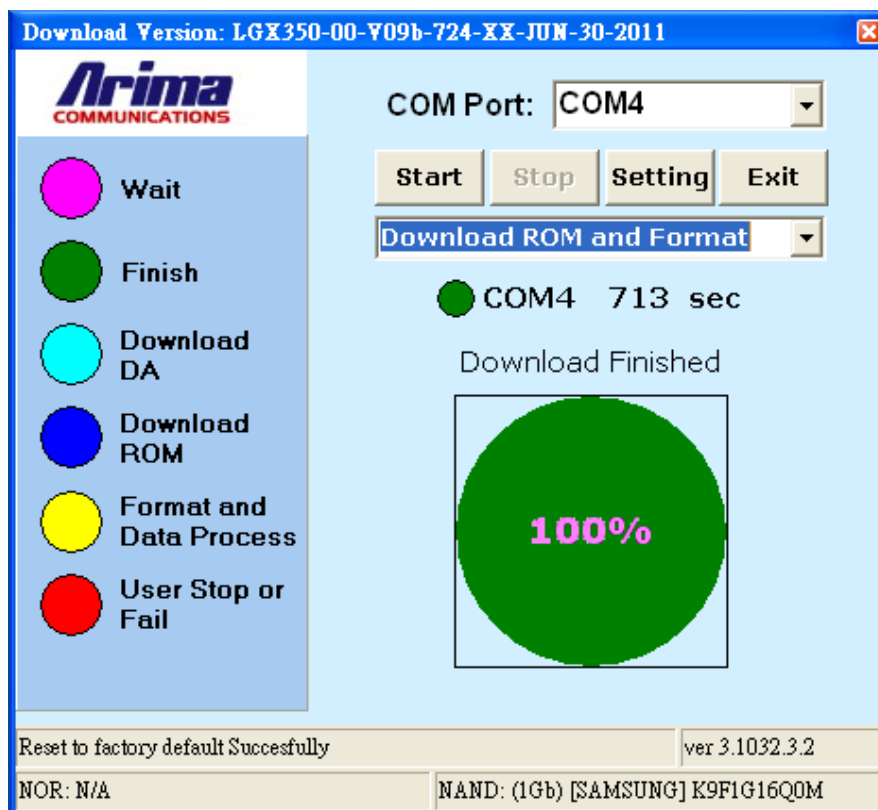
6. Select the “.PCS” file and press open, you can see following picture.



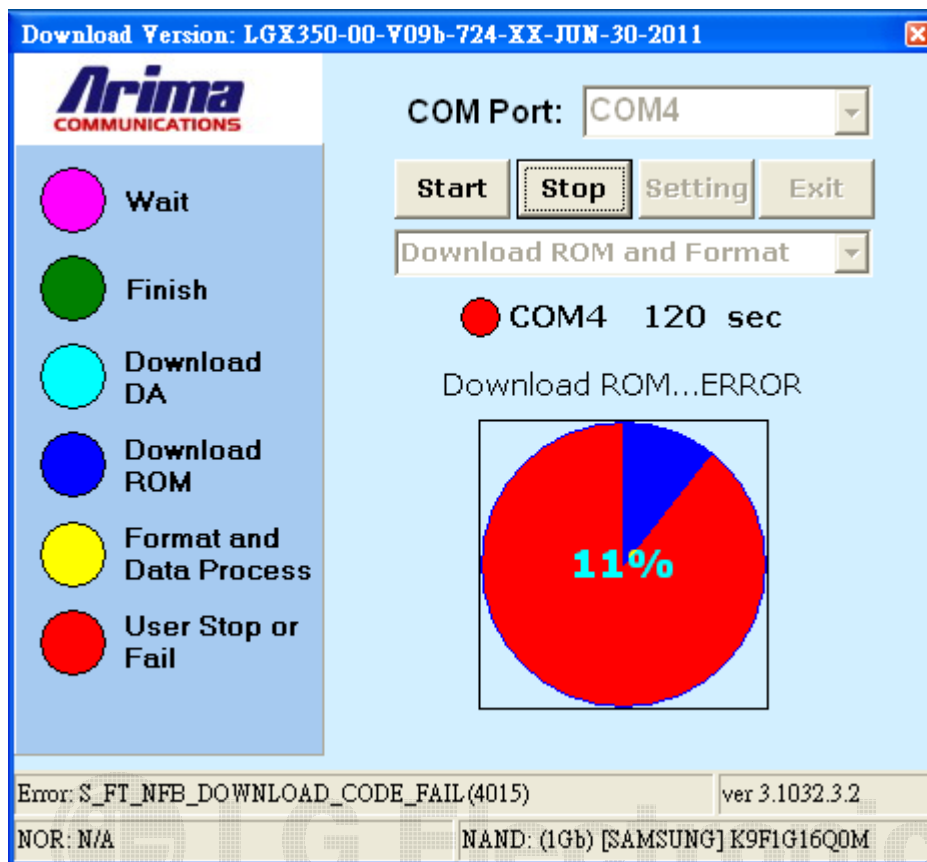
7. After you see the pink cycle, connect download cable with handset, and then press the power key, you will see below picture.



8. After reach to 100%, SW download finish.

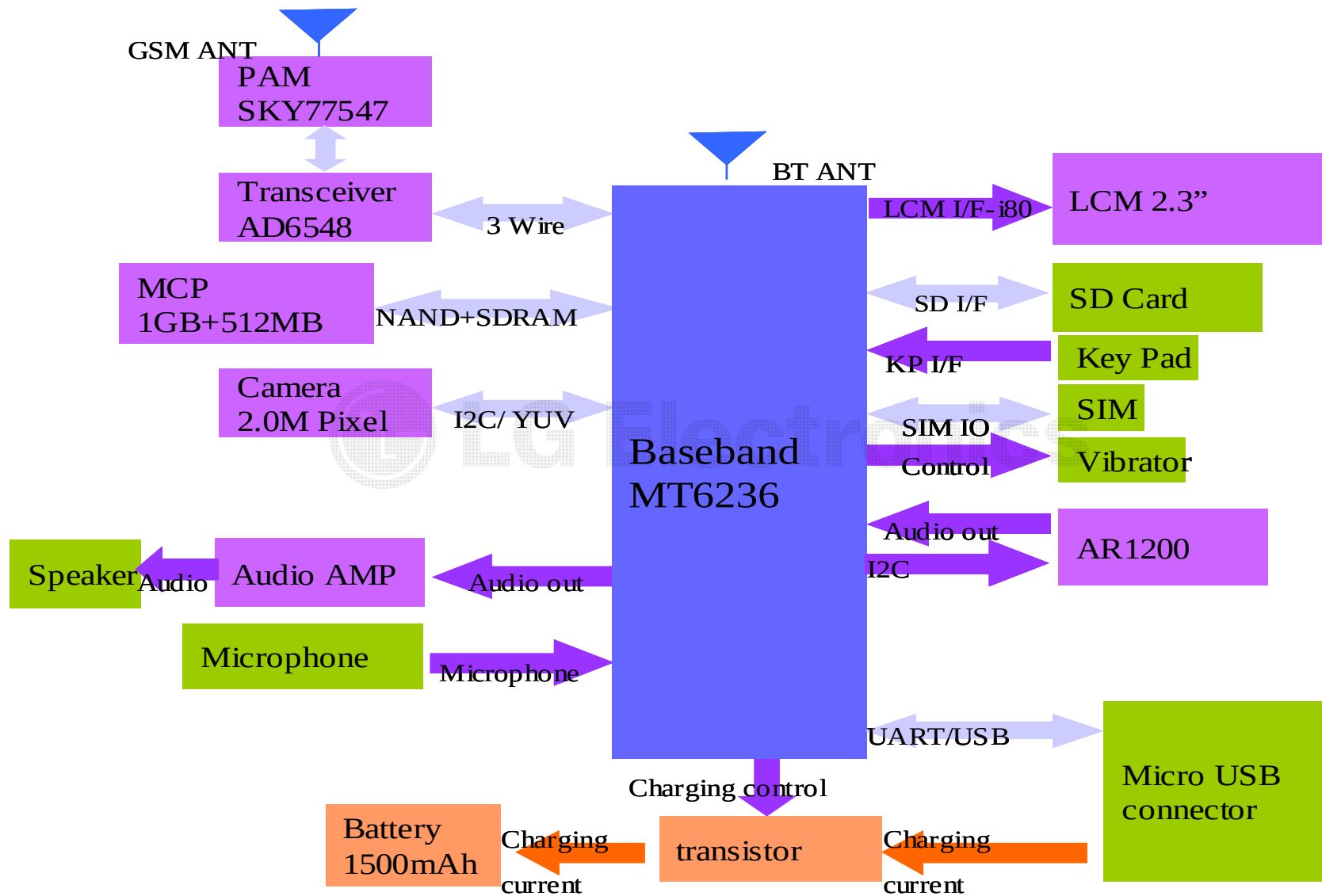


9.If download failed, you will see the below picture.

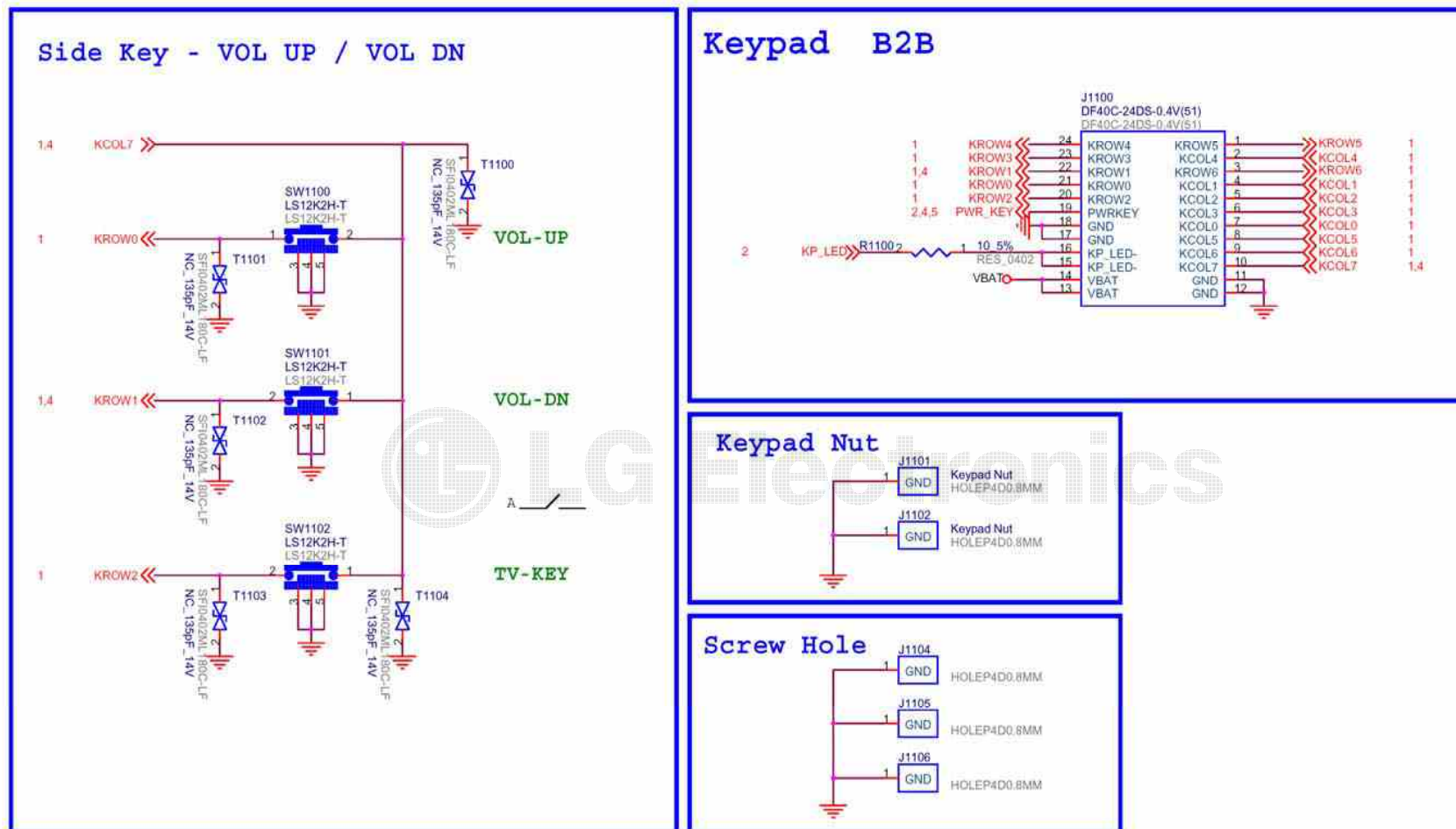


Attention: If appear failed image, Please try close LEO and try open again.

6.BLOCK DIAGRAM

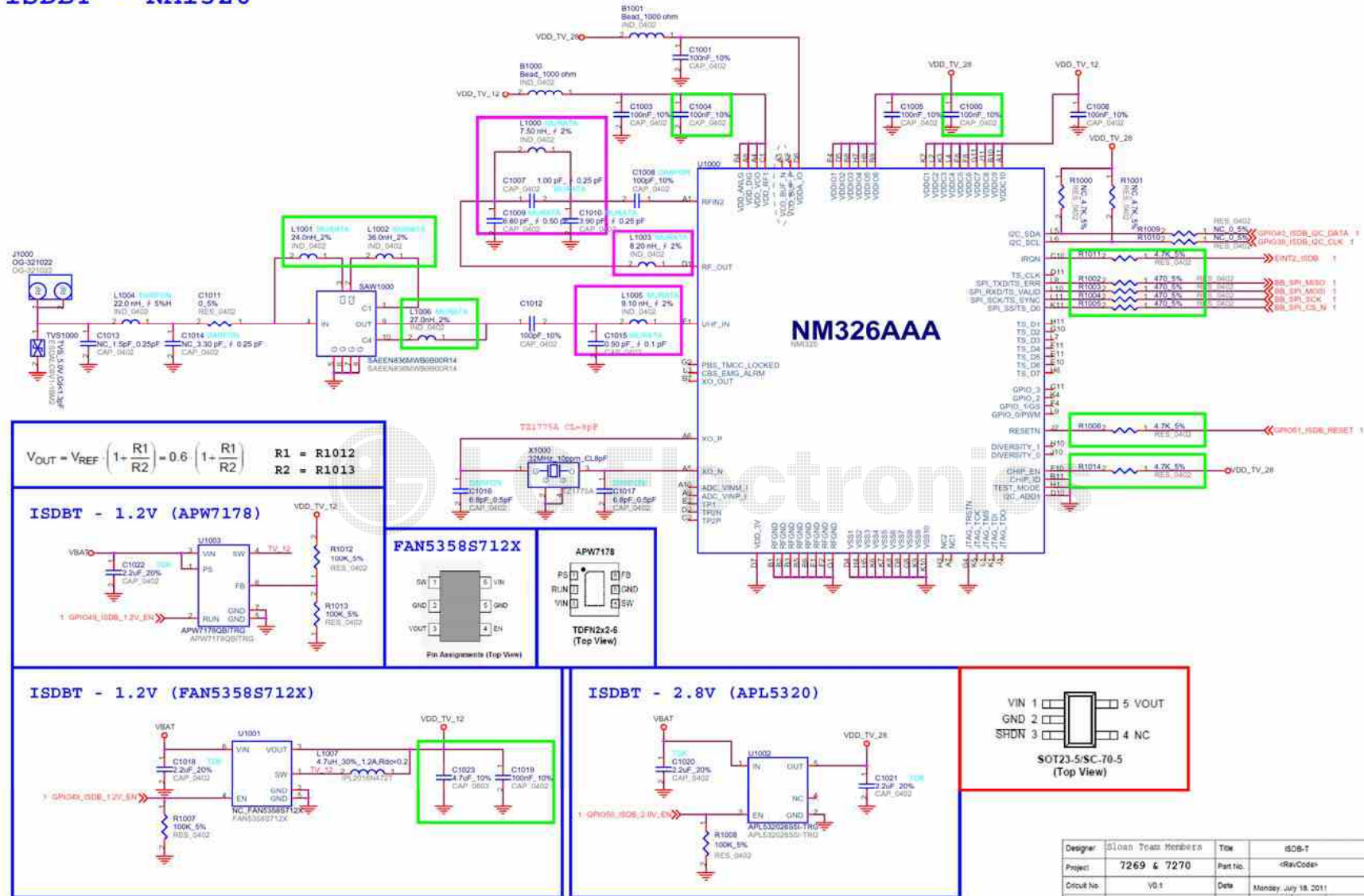


7. CIRCUIT DIAGRMA

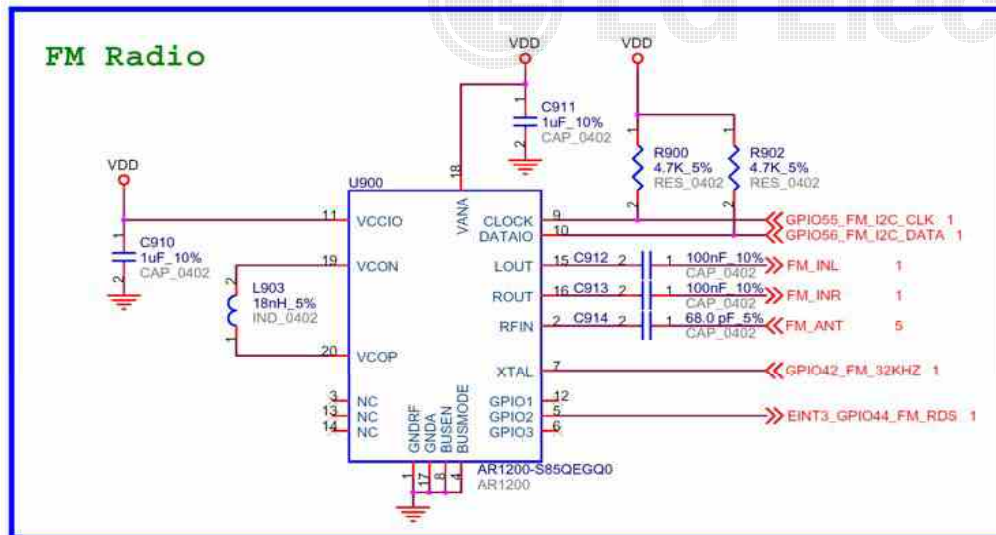
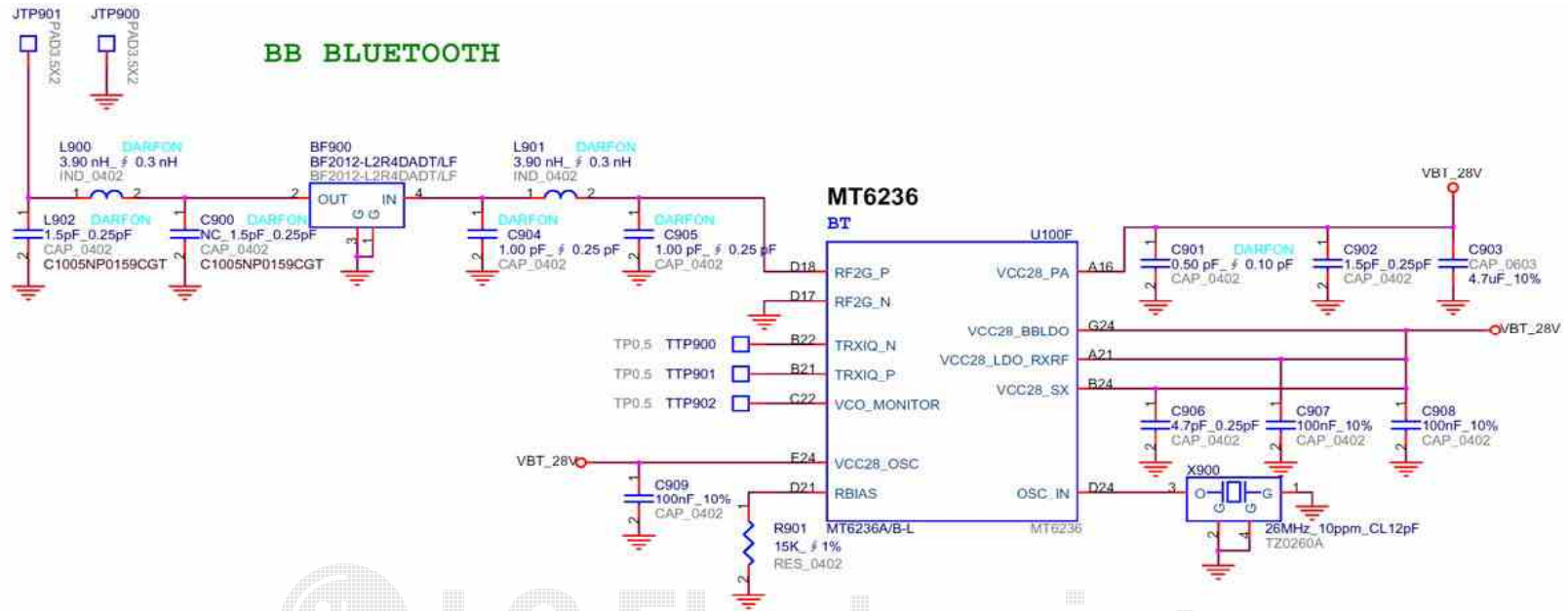


LGE Internal Use Only

ISDBT - NMI326



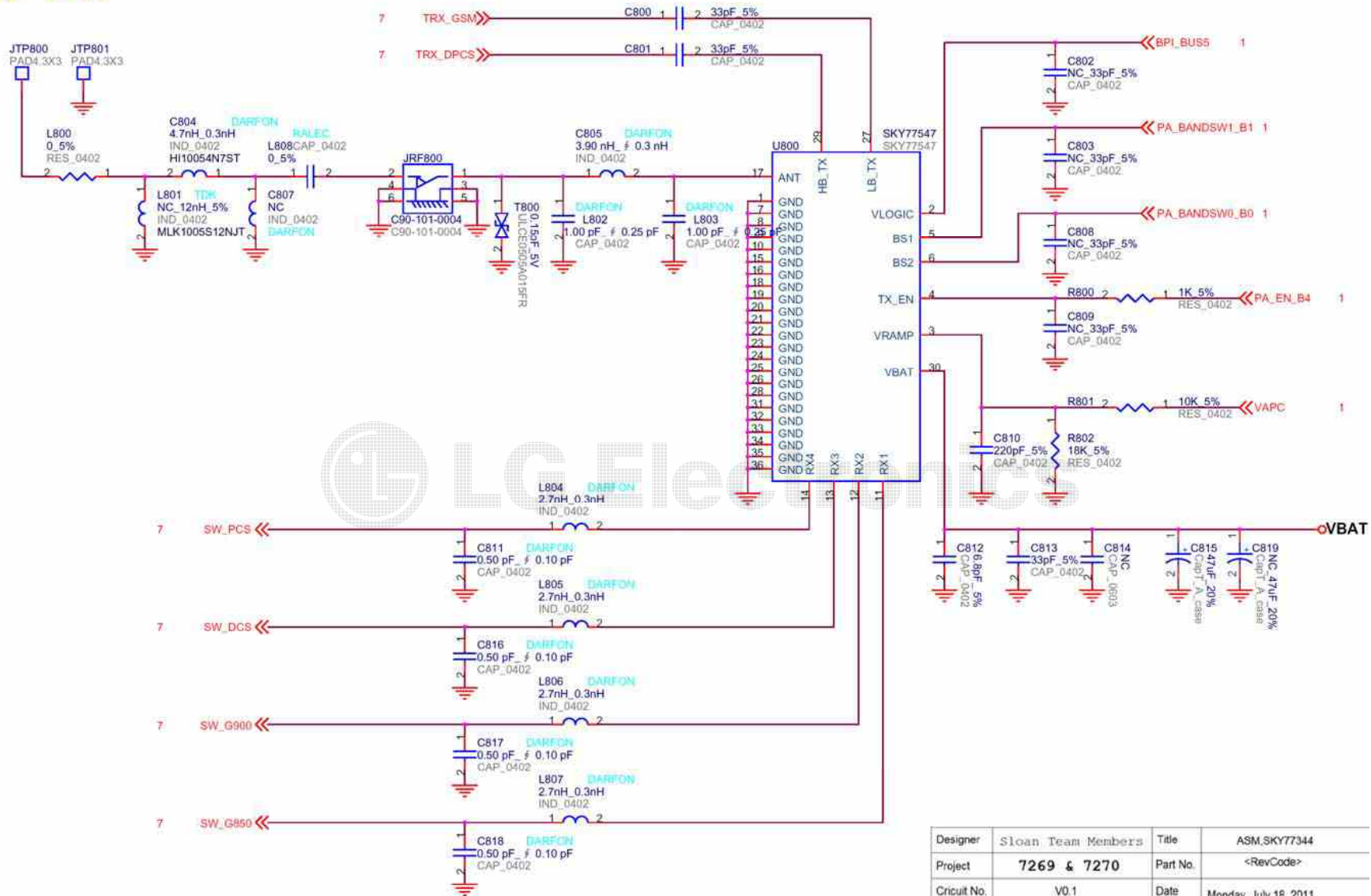
LGE Internal Use Only



Designer	Sloan Team Members	Title	MT6236_BT,FM
Project	7269 & 7270	Part No.	<RevCode>
Circuit No.	V0.1	Date	Monday, July 18, 2011
Sheet	9 of 12	DCC Ver	Approval

LGE Internal Use Only

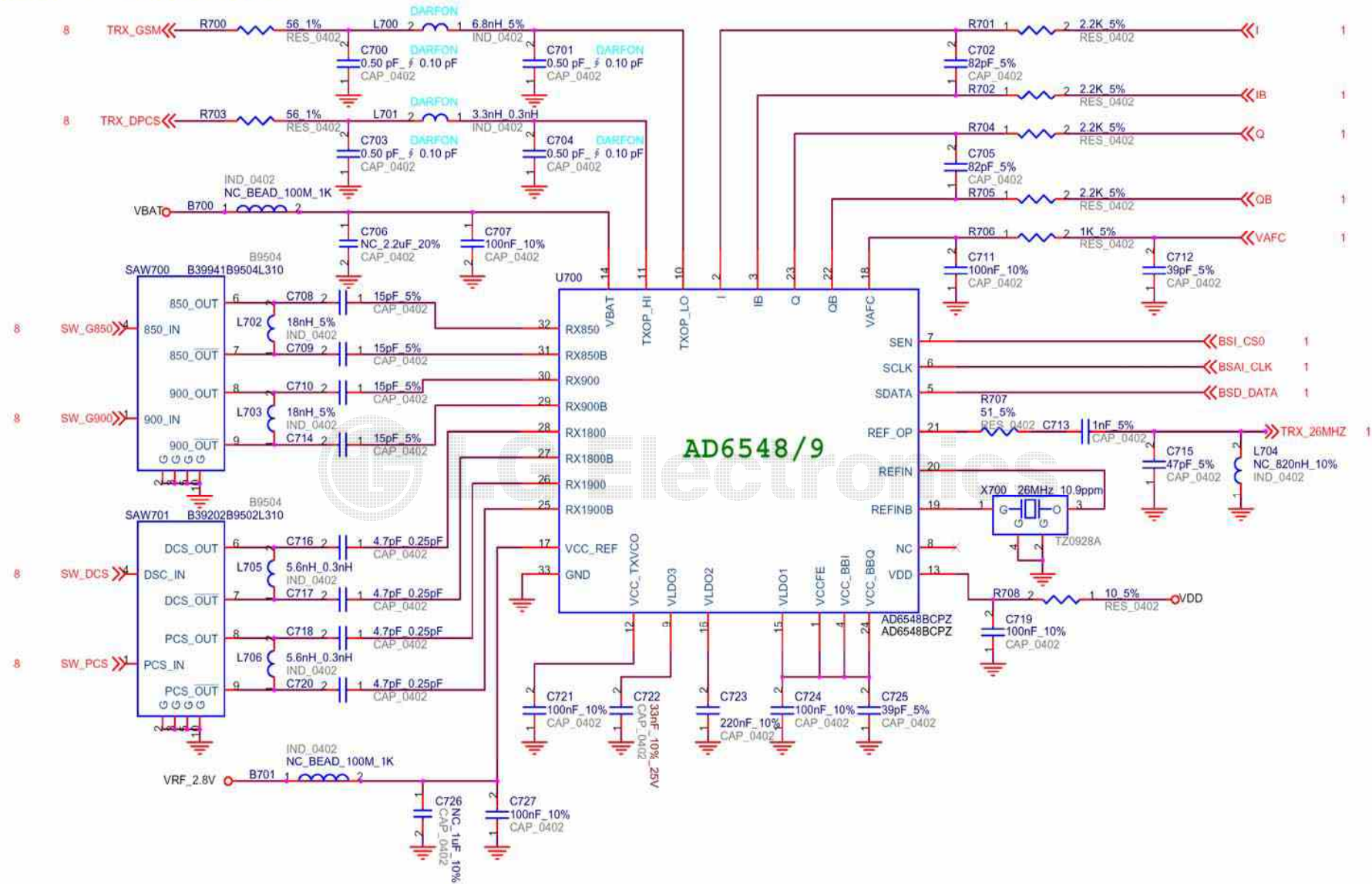
RF FEM



Designer	Sloan Team Members	Title	ASM.SKY77344
Project	7269 & 7270	Part No.	<RevCode>
Circuit No.	V0.1	Date	Monday, July 18, 2011

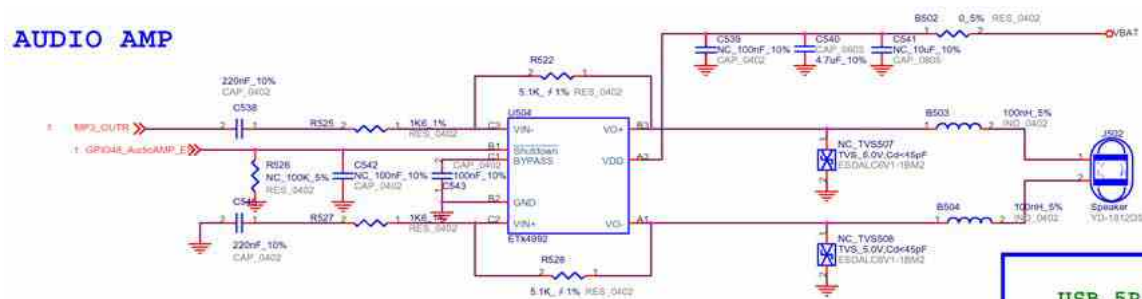
LGE Internal Use Only

TRX - AD6548/9

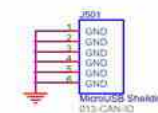


LGE Internal Use Only

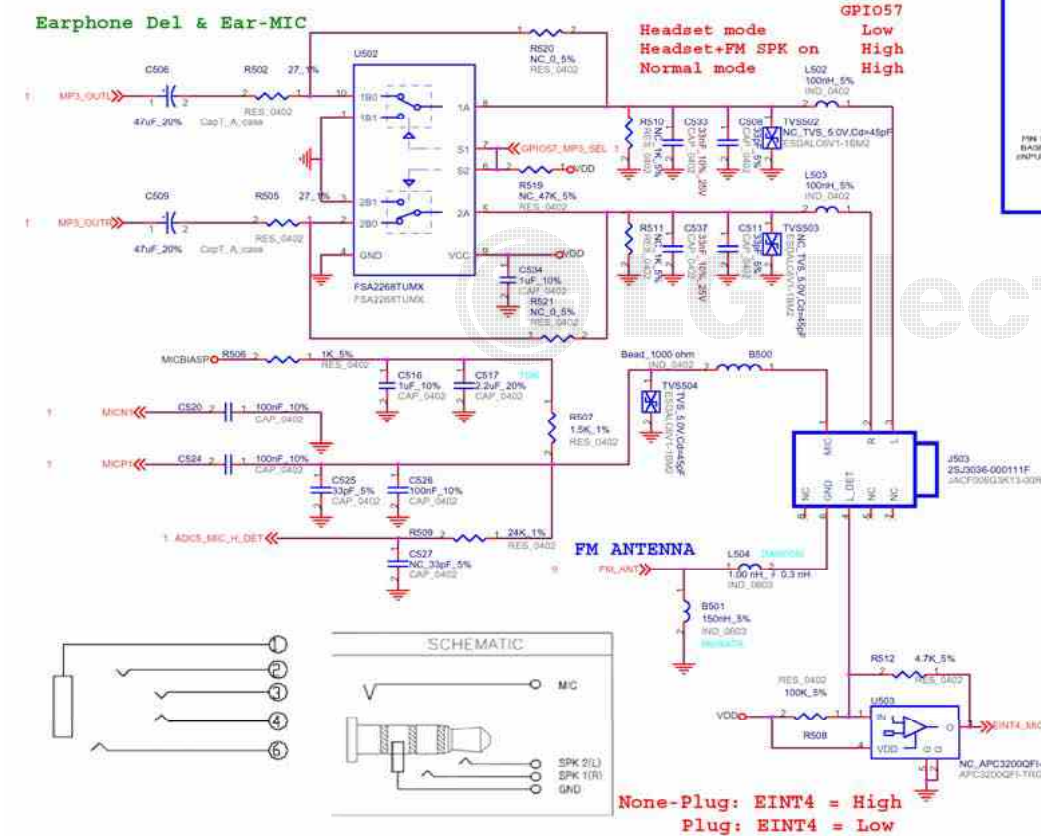
AUDIO AMP



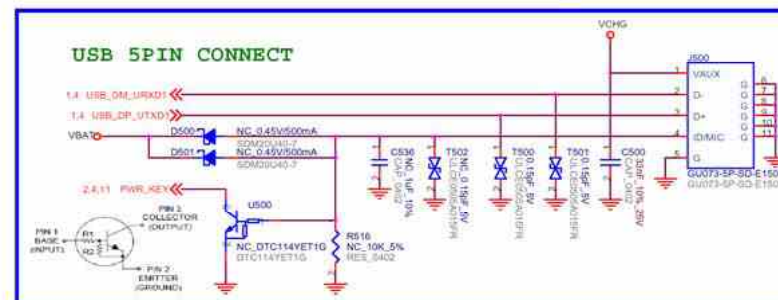
IO Shielding



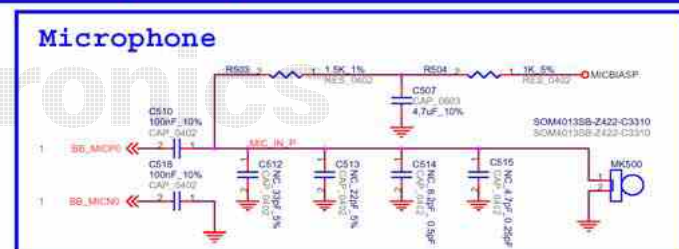
Earphone Del & Ear-MIC



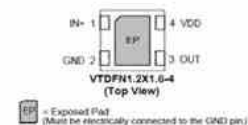
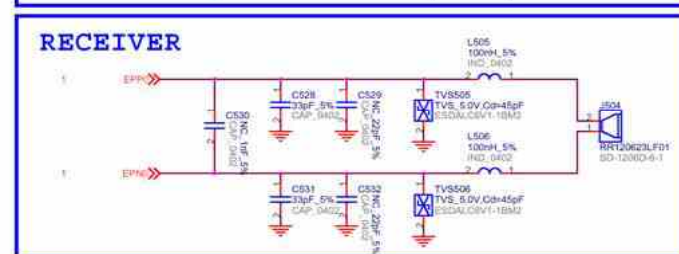
USB 5PIN CONNECT



Microphone



RECEIVER



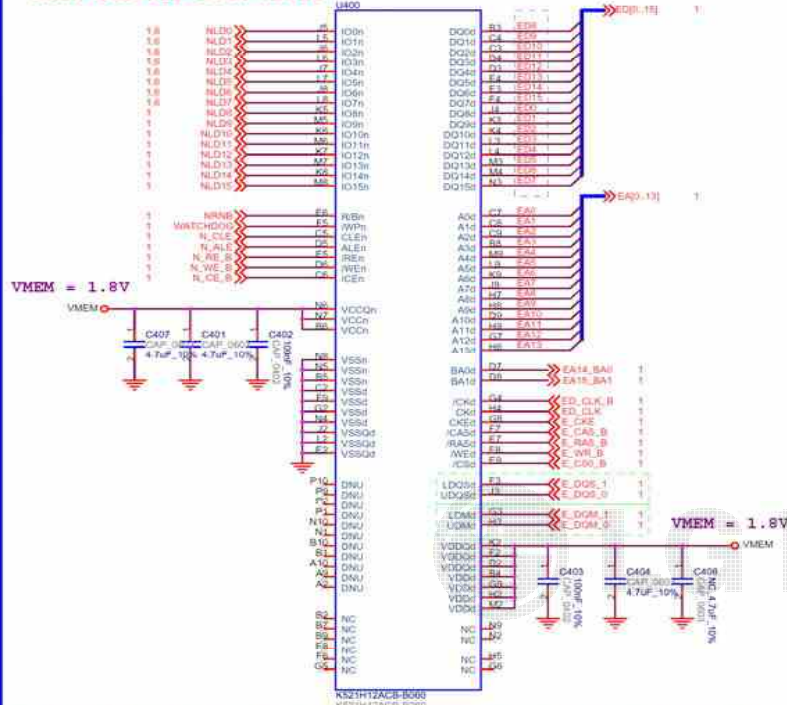
Designer	Sloan Team Members		Title	IO Audio Voice
Project	7269 & 7270		Part No.	«Rev Code»
Circuit No.	VG.1		Date	Monday, July 16, 2011
Sheet	5 of 12	DOC Ver	Approval	

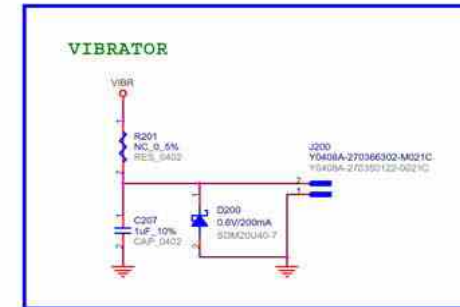
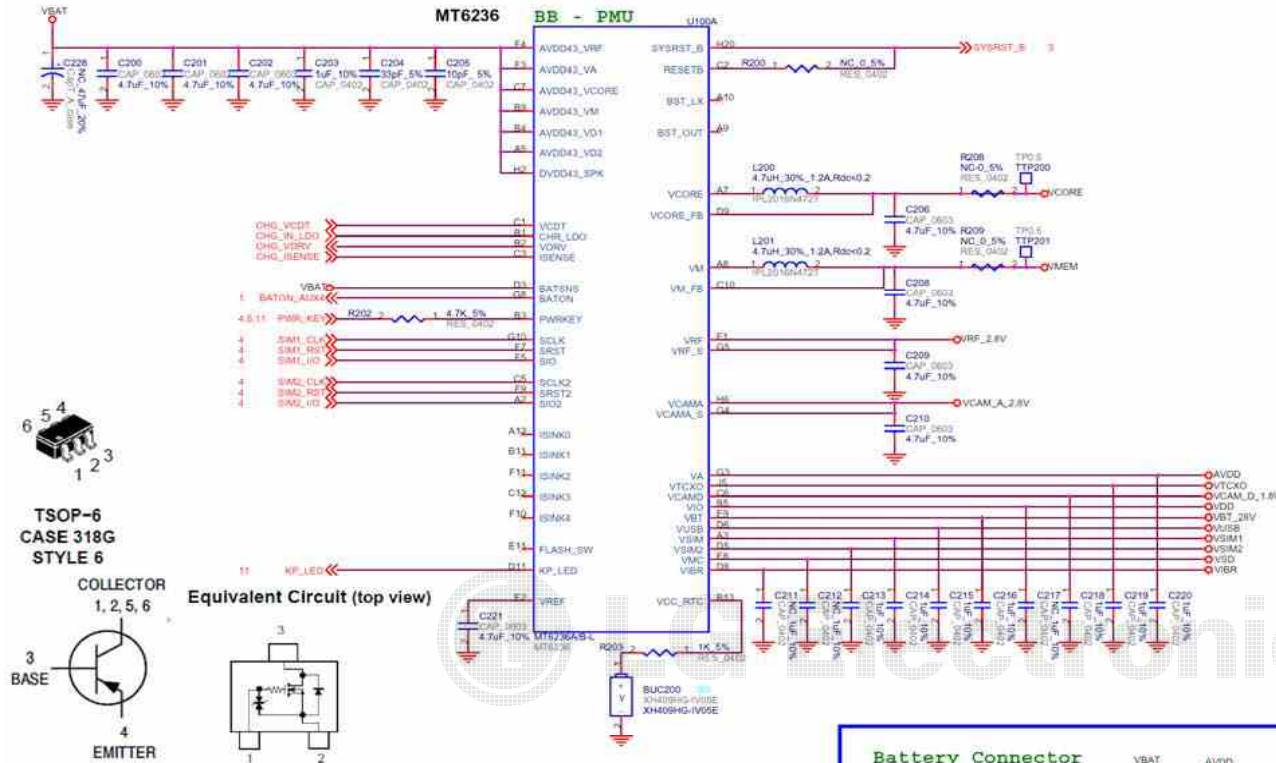
LGE Internal Use Only

SAMSUNG K521H12ACB-B060 NAND 1GNAND + 512MDDR 107BALLS-FGBA

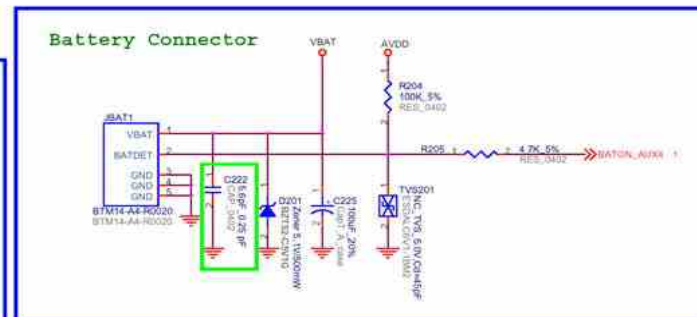
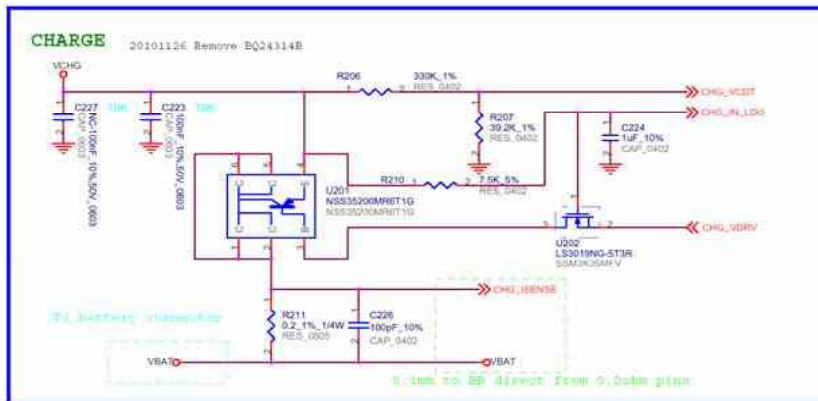
LGE=311-0000-01092

None-LGE=311-0000-01032





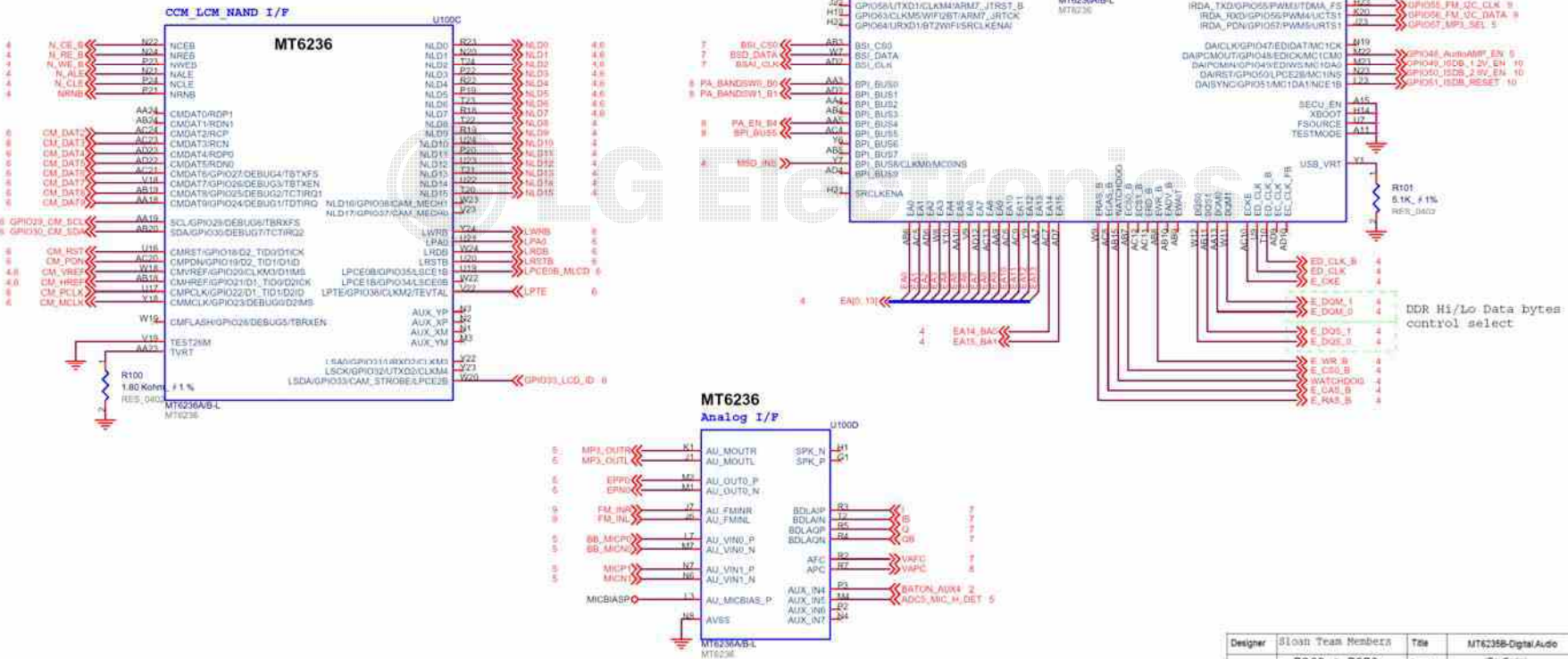
MT6236 PMU		
Power	Voltage(V)	Current(mA)
VRF	2.8	250
VCORE	0.8~1.3	350
VMEM	1.8	300
VDD	2.8	100
AVDD	2.8	125
VTCX0	2.8	40
VRTC	2.8	
VSIM1	1.8/3	100
VSIM2	2.8/3	100
VUSB	3.3	100
VBT	1.3/1.5/1.8/2.5/2.8/3.1/3.3	100
VCAMA	1.5/1.8/2.5/2.8	250
VCAMD	1.3/1.5/1.8/2.5/2.8/3.3	100



Designer	Sloan Team	Version	1	Title	MT6236 PMU CHG, VIB
Project	7269 & 7270	Part No.		Rev	RevC001
Circuit No.	V0.1	Date	Monday, July 18, 2011		
Sheet	2	of	12	DCC Ver	Approval

LGE Internal Use Only

Pin name	Pin out	Normal mode	Test mode	Enable security	Burn EPUSER
TESTMODE	A11	L	VDD	L	L
XBOOT	H14	L	L	L	L
SECURITY_EN	A15	L	L	VDD	L
FSOURCE	U7	L	L	L	VBAT
VDD	VDD	VDD	VDD	VDD	VBAT



LGE Internal Use Only

8. BGA IC PIN Check

8.1 BGA PIN Check of main chip (MT6236)

BB_MT6236 (U100)

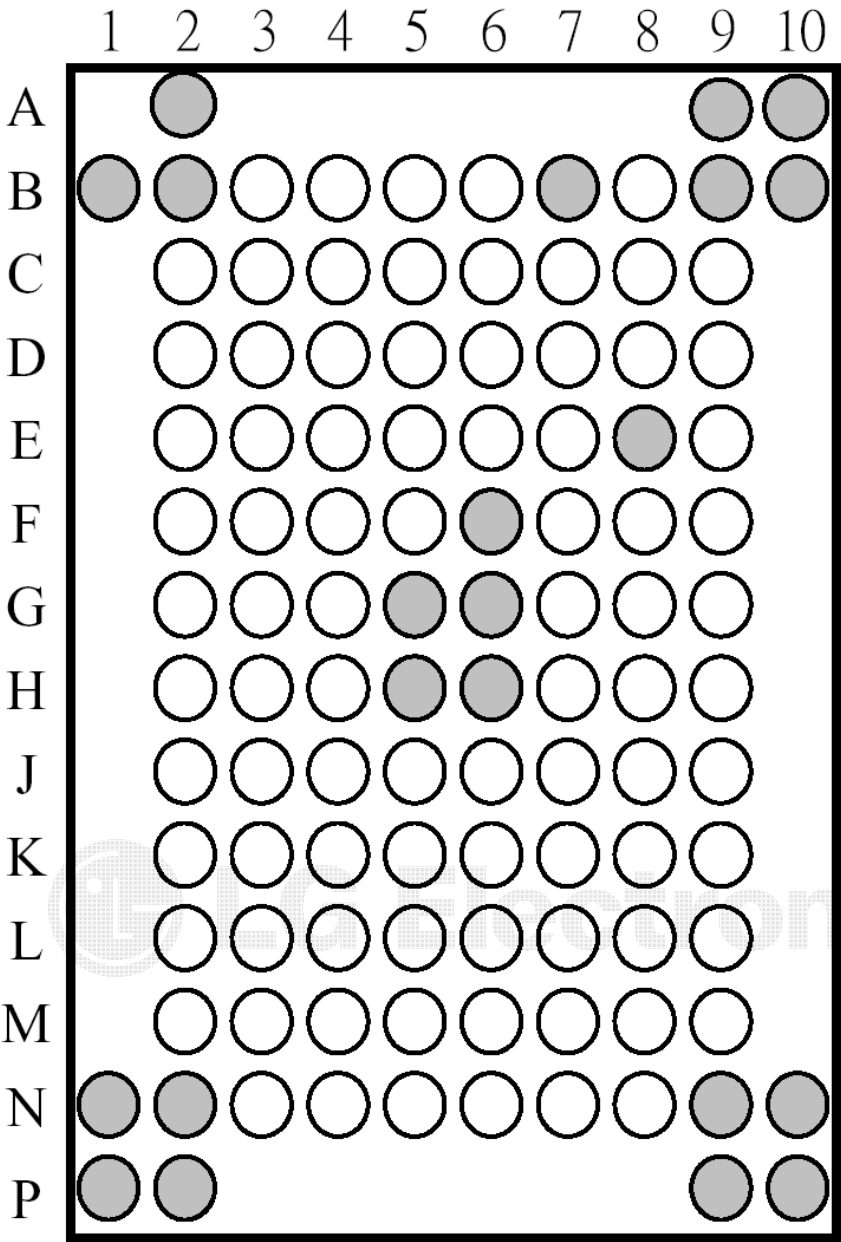
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
A		○	○		○		○	○	●	○	○	○	○	○	○	○		○	○	○	○	○	○	○
B	○	○	○	○	○	○			○	○	○	○	○	○	○	○		○		○	○	○	○	○
C	○	○	○	○	○	○	○		○	○	○	○	○	○	○	○		○			○	○	○	○
D		○		○	○	○		○	○		○		○		○		○	○	○		○	○	○	○
E	○	○						○	○		○			○	○		○	○					○	○
F	○		○	○	○		○		○	○	○	○	○						○	○	○	○	○	○
G	○		○	○	○		○	○		○			○	○		○	○	○	○	○	○	○	○	○
H	○	○				○		○	○					○	○			○	○	○	○	○	○	○
J	○		○	○	○	○	○		○						○			○	○		○	○	○	○
K	○	○	○	○				○		○		○	○		○			○	○	○	○	○	○	○
L			○				○		○		○			○		○	○	○	○	○	○	○	○	○
M	○	○	○	○			○						○		○	○		○	○		○	○	○	○
N	○	○	○	○		○	○	○	○		○		○	○	○	○		○	○	○	○	○	○	○
P		○	○									○				○		○	○	○	○	○	○	○
R	○	○	○	○	○	○	○		○									○	○			○	○	○
T		○		○		○			○	○	○	○	○	○	○			○	○	○	○	○	○	○
U	○	○	○		○		○		○		○		○	○	○	○	○	○	○	○	○	○	○	○
V	○		○		○	○			○			○	○					○	○		○	○	○	○
W	○	○	○	○	○		○	○	○		○	○	○	○	○	○	○	○	○	○	○	○	○	○
Y	○	○				○	○		○	○		○	○		○	○		○	○	○	○	○	○	○
AA		○	○	○	○		○		○	○		○	○		○	○		○	○		○	○	○	○
AB	○		○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○
AC	○	○		○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○
AD	○	○	○		○	○		○	○	○	○	○	○	○	○	○		○	○	○	○	○	○	○

○ BGA use

● BGA non-use

8.2 BGA PIN Check of Memory (K521H12ACB-B060)

K521H12ACB-B060 (U400)



○ BGA use

● BGA non-use

8.3 BGA PIN Check of TV module (NM326AAA)

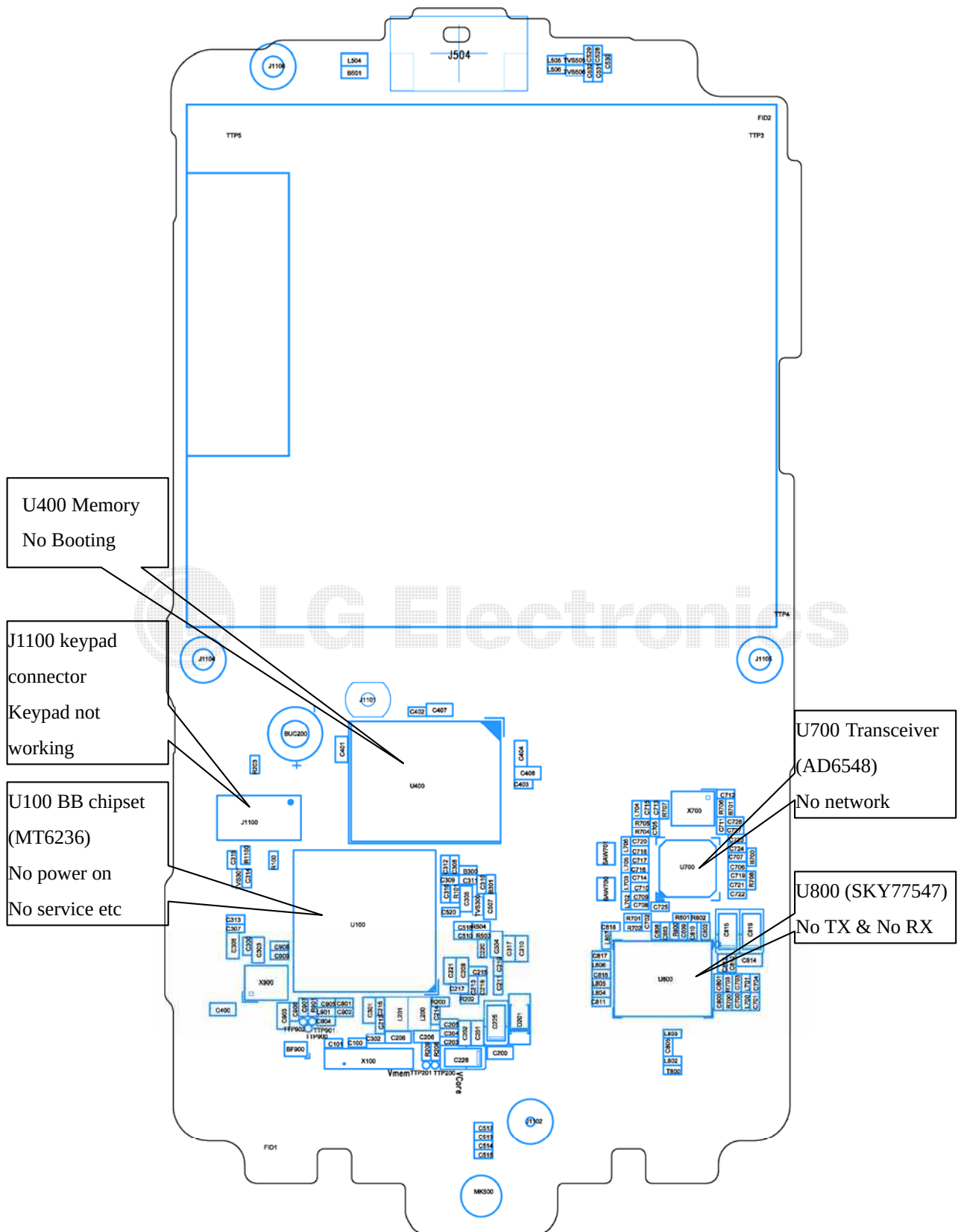
NM326AAA (U1000)

	1	2	3	4	5	6	7	8	9	10	11
A											
B											
C											
D											
E											
F											
G											
H											
J											
K											
L											

 BGA use

 BGA non-use

9. PCB LAYOUT



TOP

10. Engineering mode

Enter and Exit Engineering Mode

In idle screen, enter the Engineering Mode menu by typing “*#3646633#”.
Like normal menu operations, press Right-Soft-Key (RSK) “Back” to the previous screen or End key to go back to the idle screen.

1. Network Setting

1.1 Network Info

This function is for engineers to monitor the status of the protocol stack and interactions with the network. Use CSK ok to toggle the value of a selected item. After selection of items to monitor is complete, press RSK Back. If any settings are changed, a screen appears with “Update Parameter?”. Pressing CSK Yes saves the new values to the protocol stack; Pressing RSK NO discards all changes. These settings are not stored in NVRAM; thus the settings are lost if the handset is powered off.

Item	Function Selection	Value	Description
Network Info.	RR Cell Sel	On/Off	Radio Resource cell selection information
	RR Ch Dscr	On/Off	Radio Resource channel description information
	RR Ctrl Chan	On/Off	Radio Resource control channel information
	RR RACH Ctrl	On/Off	Radio RACH control channel information
	RR LAI Info	On/Off	Radio Resource LAI information
	RR Radio Link	On/Off	Radio Resource radio link information
	RR Meas Rep	On/Off	Radio Resource measurement report information
	CC Chan Info	On/Off	Call Control Channel information
	CC Call Info	On/Off	Call Control Call information
	CA List Info	On/Off	CA List information
	PLMN Info	On/Off	PLMN information
	GPRS Info	On/Off	GPRS information
	Si2Q/Mi Info	On/Off	Si2Q/Mi information
	TBF Status	On/Off	TBF Status
	Block Info	On/Off	Block information

1.2 Band Selection

Item	Function Selection	Value
Band	Auto	ON/OFF
	850 MHz	ON/OFF
	900 MHz	ON/OFF
	1800 MHz	ON/OFF
	1900 MHz	ON/OFF
	850/1800 MHz	ON/OFF
	900/1800 MHz	ON/OFF
	850/1900 MHz	ON/OFF

2. GPRS attach

This menu is for GPRS mode test only (if supported).

2.1 Power on

To start GPRS attach when MS powers on.

2.2 When needed

To start GPRS attach only when MS needs to attach GPRS.

3. SET GPRS Attach Value

To set GPRS attach value

4. Device

4.1 LCD

Item	Function Selection	Value	Description
Version	Set Contrast	0-255	To set the Contrast of Main LCD.
	Set Bias Ratio	0-255	To set the Bias Ratio of Main LCD.
	Set Line rate	0-255	To set the Line Rate of Main LCD.
	Set Temperature	0-255	To set the Temperature of Main LCD.
	Set Color		To set the value of RGB.
	Display Demo Pic.		To display the picture.

The string to enter Engineering Mode can be changed in the source code.

The larger the value set in the input box, the brighter the screen is.

4.2 GPIO

The GPIO includes **List GPIO**, **GPIO Editor** and **GPO Editor**.

Item	Function Selection	Value	Description
List GPO/GPIO	WIFI_32K	On/Off	To switch the GPO On or Off
	...	...	...
GPIO Editor	View/Edit the state of the GPIO entered		Enter the GPIO to view and edit the state: Current Level: High/Low Mode: [0~3] Direction: IN/OUT
GPO Editor	View/Edit the state of the GPO entered		Enter the GPO to view and edit the state: Current Level: High/Low Mode: [0~3] Direction: IN/OUT

List GPO/GPIO is for predefined functional GPIOs. The setting depends on the project specification.

GPIO Editor and **GPO Editor** are extensive functionalities that allow engineers to change the state of any GPIO/GPO.



4.3 PWM

There are 3 PWMs.

Item	Function Selection	Value	Description
PWM	PWM1	Freq = 0-255, Duty = 0- 100	PWM Editor shows the following info: Level: [1~5] Frequency: [0~255] Duty: [0~100]
	PWM2	Freq = 0-255, Duty = 0-100	PWM Editor shows the following info: Level: [1~5] Frequency: [0~255] Duty: [0~100]
	PWM3	Freq = 0-255, Duty = 0-100	PWM Editor shows the following info: Level: [1~5] Frequency: [0~255] Duty: [0~100]

1. On entry of the PWM Editor, the setting of the current level is shown. Settings of all levels can be adjusted, stored (if the ok key is pressed) and applied.

4.4 EINT

Display External Interrupt Status. 1 means ON, 0 means OFF.

Item	Function Selection	Value	Description
EINT	ISDB 0	0/1	ISDB interrupt
	FM 0	0/1	FM interrupt

4.5 ADC

Display the ADC value. The ADC value is displayed and updated every 1 second on this screen.

Item	Function Selection	Value	Description
ADC	VBAT	XX.XX (V)	Battery voltage
	ISENSE	XX.XX (A)	Charging current
	CHARGER	XX.XX (V)	Charger voltage
	TMP	XX.XX (C)	Battery Temperature
	ACCESSARY DET	XX.XX	

4.6 Set Default Level

Item	Function Selection	Value	Description
Main LCD Contrast	LEVEL1-15	0-255	To set the contrast of 15 levels of Main LCD.
Battery	LEVEL 1	0-9999999	Voltage below LEVEL 1 powers off automatically.
	LEVEL 2	0-9999999	Voltage below LEVEL 2 prohibits MO calls.
	LEVEL 3	0-9999999	Voltage below LEVEL 3 pops up a warning screen.
	LEVEL 4	0-9999999	Voltage below LEVEL 4 displays battery status icon with 0 level (empty).
	LEVEL 5	0-9999999	Voltage below LEVEL 5 displays battery status icon with 1 level.
	LEVEL 6	0-9999999	Voltage below LEVEL 5 displays battery status icon with 2 level.
	LEVEL 7	0-9999999	Voltage above LEVEL 6 displays battery status icon with 3 level (full).
	LEVEL 8-10	9999999	Reserved, should be 9999999.
PWM 1	Freq 1-5	0-255	Frequency of PWM for 5 levels.
	Duty 1-5	0-100	Duty Cycle of PWM for 5 levels.
PWM 2	Freq 1-5	0-255	Frequency of PWM for 5 levels.
	Duty 1-5	0-100	Duty Cycle of PWM for 5 levels.
PWM 3	Freq 1-5	0-255	Frequency of PWM for 5 levels.
	Duty 1-5	0-100	Duty Cycle of PWM for 5 levels.

1. All settings are saved in NVRAM when the user presses CSK ok on the confirmation screen "Update Parameters?"

4.7 Set UART

Change the UART setting for AT Command. If AT Command is set to UART1, then TST² is set to UART2. If AT Command is set to UART2, then TST is set to UART1.

The UART POWER ON/OFF is for selection of UART1/2/3 on or off.

4.8 Sleep Mode

Enable or disable sleep mode.

4.9 DCM Mode

Enable or disable sleep mode.

4.10 Memory info

Item	Description
EMI register settings	To see EMI parameters
Custom settings	To see FS parameters
Flash physical info	To see flash physical information

4.11 FM Radio

Item	Function Selection	Value	Description
MONO	Disable/Enable		To Power on/off MONO
STEREO	Disable/Enable		To Power on/off STEREO
RSSI		Level1~6	To set the signal level
If Count Delta		10,15,20,25,30 KHz	To set the frequency increment value
RSSI Info.			To display the signal information

5. Audio

5.1 Set Mode.

Set Normal Mode, LoudSp Mode or Headset Mode.

5.2 Normal Mode, LoudSp Mode, Headset Mode

Five types of audio settings can be set.

Item	Function Selection	Value	Description
FIR		0-5	Set the 6 levels for FIR
Speech	0-6	0-255	Set the 7 volume levels for Speech
Key tone	0-6	0-255	Set the 7 volume levels for Key tone
Melody	0-6	0-255	Set the 7 volume levels for Melody
FM Radio	0-6	0-255	Set the 7 volume levels for FM Radio
Sound	0-6	0-255	Set the 7 volume levels for Sound
Microphone	0-6	0-255	Set the 7 volume levels for Microphone
Side tone	Only one level	0-255	Set the gain of the Side tone

1. The preset value is read from NVRAM when entering Normal Mode.
2. To change the volume, use the up and down keys. Use CSK set to set the value (not yet stored to NVRAM).
3. On exiting Normal Mode, a confirmation screen "Update Parameter?" appears. Pressing CSK yes stores the setting to NVRAM. Pressing RSK no discards all changes and returns the user to the Audio screen.
4. Only the active volume level gain of the active mode is set to the hardware register. For example: If the MS is in Normal Mode and the current speech volume level is LEVEL4, updating the parameter sets the gain value of LEVEL4 of Speech to the hardware register.
5. To change the current volume level of Speech, use side key when in call.
6. To change the current volume level of Key tone, use side key when in idle screen.
7. To change the current level of Ring tone, enter User Profiles->[Active Mode]->Customize->Volume->Ring tone to change volume.
8. The parameters of Headset Mode are applied only when the headset is plugged in.
9. The parameters of LoudSp Mode are applied only when handsfree is turned on during a call.

5.3 Ring Tone

This audio setting is used to demonstrate all melodies on the MS, including iMelodies, Midis and Sounds (MIDI format). Use the up and down keys to scroll over the ring tones: highlighting a selection longer than one second plays the ring tone.

5.4 Speech Enhancement

You can set the Common Parameters 0~7.

There are 9 Mode for selection: Common Parameters, Normal Mode, Headset

Mode, LoudSp Mode, BT Earphone Mode, BT Cordless Mode, AUX1 Mode, AUX2 Mode, AUX3 Mode. The 8 Mode have 0~15 parameter to set.

5.5 Max Swing

This setting sets the maximum swing value.

5.6 Debug Info

The setting sets the Debug information parameter :0~15.

5.7 Auto Record Setting

The Setting includes: VM Support, Auto Speech Record.

You can turn on/off them.

6. GPRS Act

This menu is for GPRS mode test only (if supported).

6.1 Attach

To start GPRS attach.

6.2 1st PDP

To activate GPRS PDP context1-15.

6.3 2nd PDP

To deactivate GPRS PDP context.

7. Misc.

This screen contains other miscellaneous settings.

Item	Function Selection	Description
Auto Answer	ON/OFF	To toggle on or off the Auto Answer function.
RAM Test		
Memory Dump	ON/OFF	
MMI Debug	ON/OFF	To switch between ON and OFF for MMI debug level.
AMR	ON/OFF	To enable or disable AMR.
Internet Application		
J2ME TCK	J2ME TCK testing	To choose among the listed TCKs.
Fixed Gain For Helios2	ON/OFF	
Fake language		To show string identifier for the specified string
Call timer		Show the call time.
Auto Test List		To see the auto test items in factory mode
Java Performance		To test Java performance
MED Memory Setting	Debug Info ON/OFF	To enable or disable Debug Info display

8. Auto Test List

To see the auto test items in factory mode and

8.1 Add From List

Add item to auto test list.

8.2 Change Priority

Change test item order in auto test list

8.3 Remove

Remove item in auto test list.

9. Debug Info

Item	Function Selection	Description
Last Exception	Read	To read and display information about the last exception.



10. Bluetooth

To test the Bluetooth Module

Item	Function Selection	Value	Description
Bluetooth	General Test	Press OK to Start testing	To do the General Test
	Bluetooth RF Test	Press OK to Start testing	To test the Bluetooth RF
	Get Chip Version	Press OK	To get the Chip Version
	Bluetooth UPF Test	Press OK to Start testing	To test Bluetooth UPF
	Enable item	Enable/Disable	Enable/Disable item for PTS test

11. Profiling

Item	Function Selection	Value	Description
Profiling	Frame Profiling		
	Memory Monitor	Press OK	To monitor control buffer (OSL) and APPMEN (ASM), and after setting will display result at left screen.
	Memory Statistic	Press OK	Statistics of ASM to display Pool size and max of screen id and app id.
	Fs Test		

12. RF Test Tool

Test the RF. Press CSK OK and begin to test the RF.

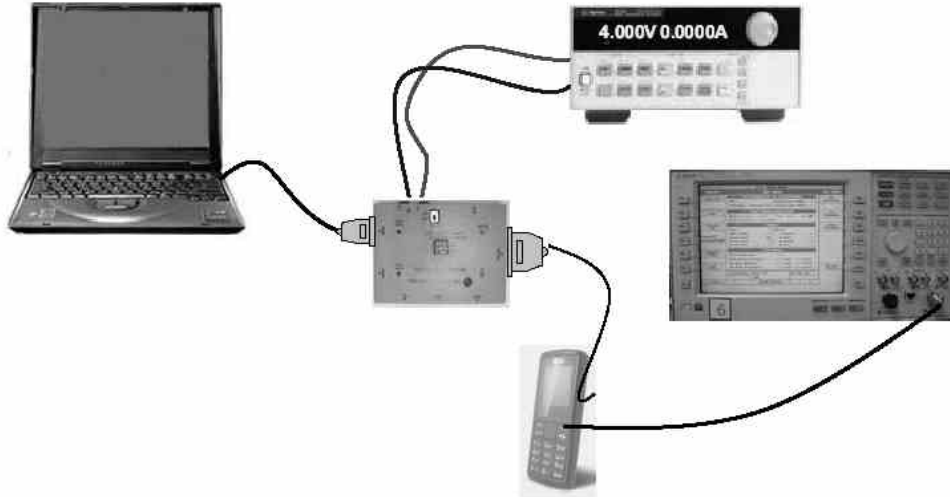
13. Format NVRAM but keep K

To Erase NVRAM Data but will keep calibration data and IMEI, BT Address, SN number.



11. CALIBRATION

11.1 Test Equipment set up



11.2 Calibration Steps

Environment Requirement:

OS:

MS Windows 2000 or XP

Hardware:

Generic Pentium III or above PC (256M RAM or above)

GPIO Card

- National Instruments GPIO device and driver
- Agilent GPIO card and driver
- KEITHLEY GPIO card and driver

Radio Communication Tester

- Rohde & Schwarz CMU 200
- Agilent 8960
- Anritsu MT8820
- Rohde & Schwarz CMD55
- Willtek WT4400
- Agilent N4010A (for Bluetooth test)
- Rohde & Schwarz CBT (for Bluetooth test)
- Anritsu MT88852 (for Bluetooth test)

DC Power Supply

- Agilent 661x or Agilent 663x2 series power supply
- R&S NGSM Power Supply
- KEITHLEY 2303, 2304, 2306
- Agilent 3631A power supply
- Willtek WT4400 power supply option

Others

USB download cable

Dummy battery

RF cable

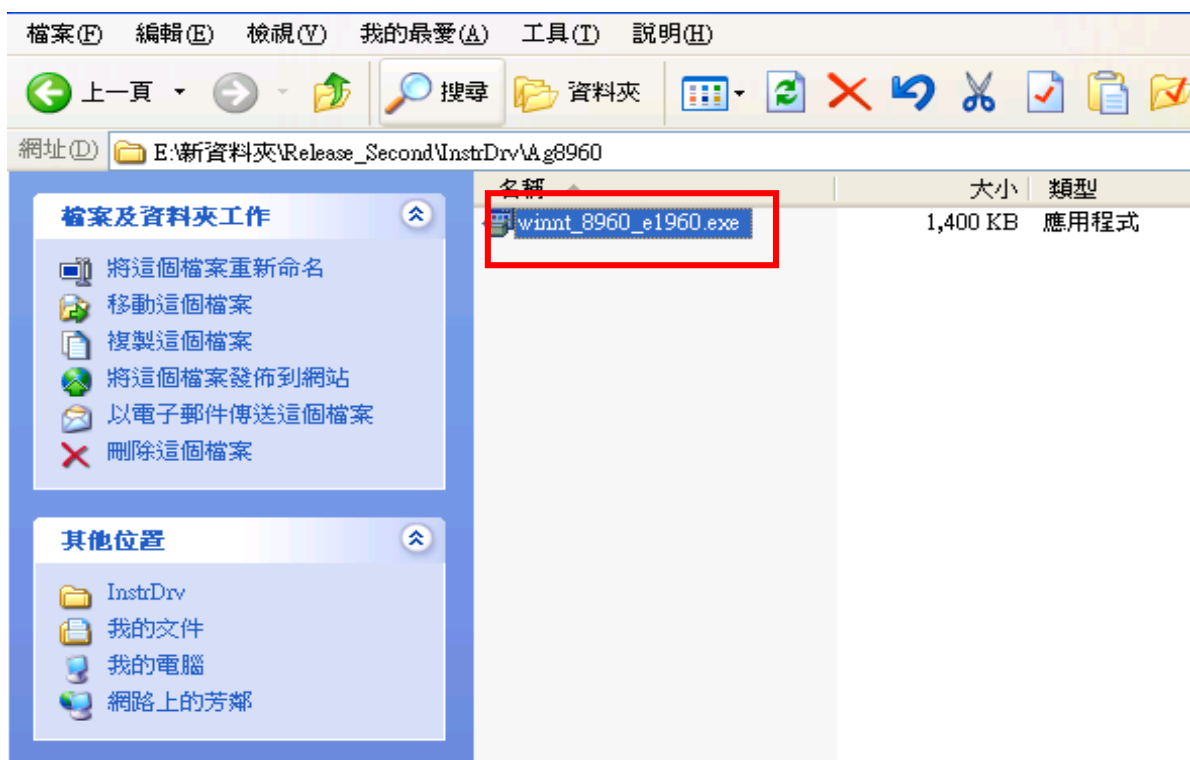
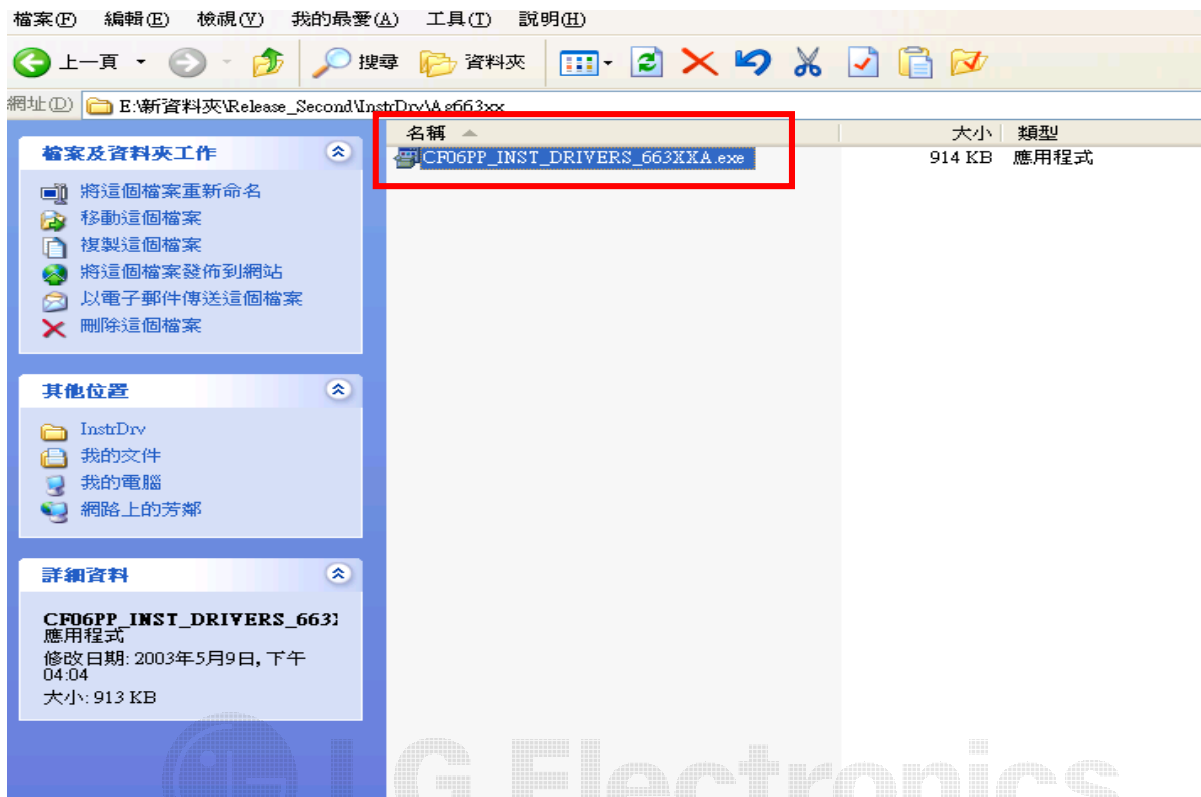
The following diagrams depict the system setups when using the Agilent test platform.

Connect 8960, power supply , computer ,phone



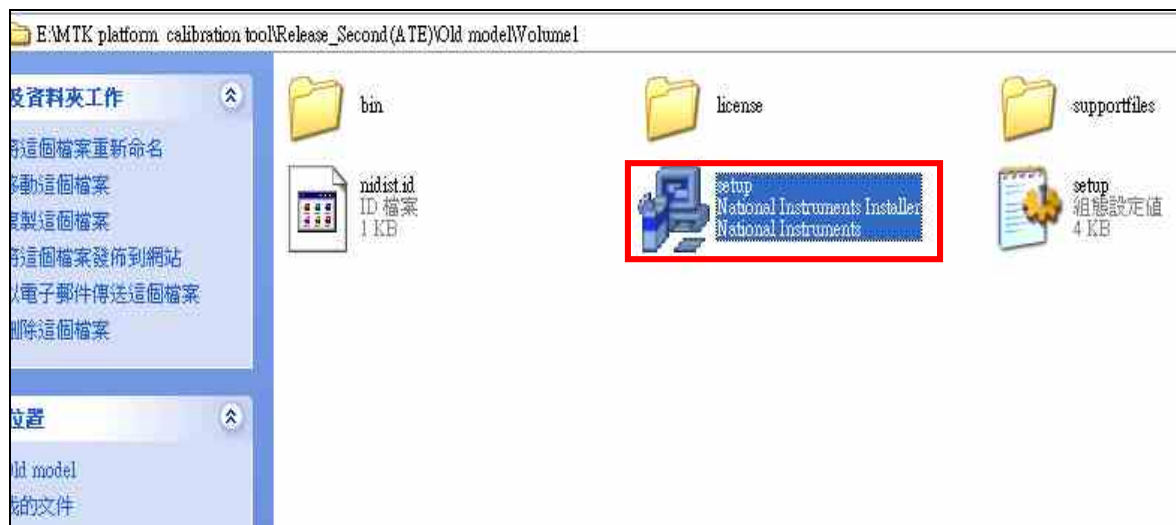
When install the MTK ATE tool, first install driver.

In turn execute [CF06PP_INST_DRIVERS_663XXA.exe](#), [winnt_8960_e1960.exe](#), [230x-850a01.exe](#).

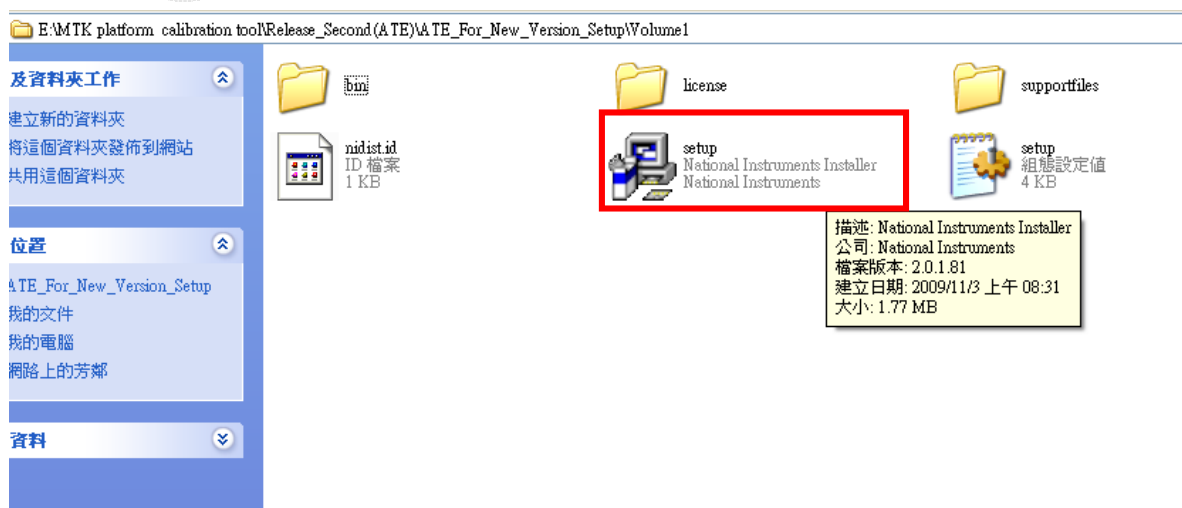




Second, to install the MTK ATE tool, execute the [Old model \ Volume1 \setup.exe](#) file. The Installation Wizard guides the user through the installation process step by step, up to Installation finish.



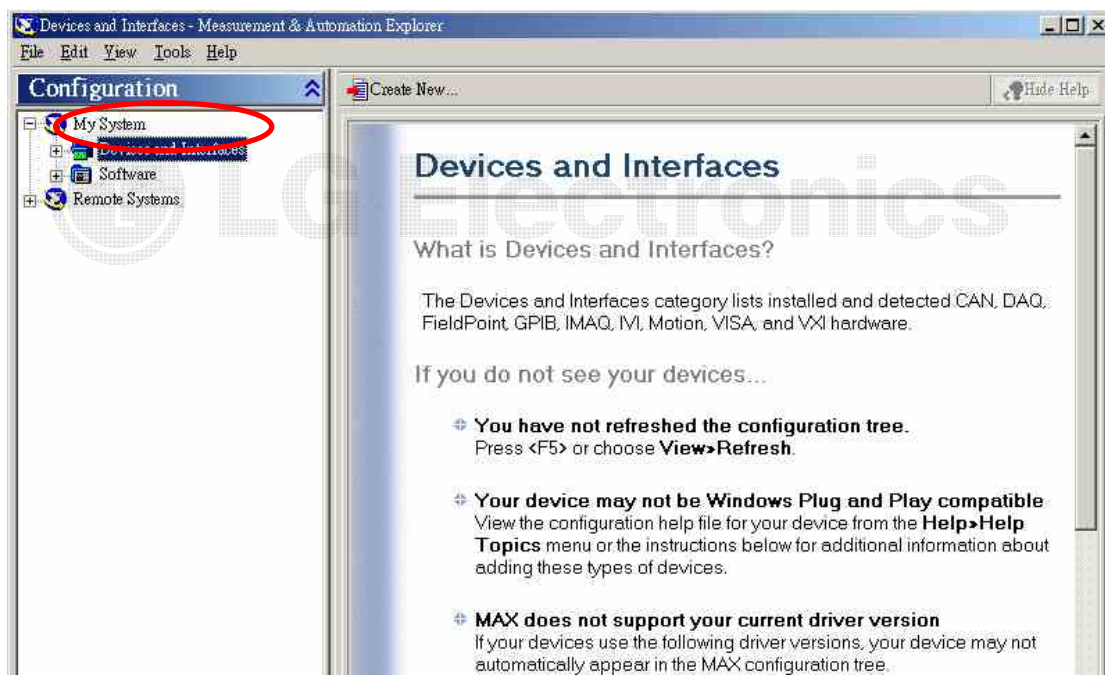
Third, to install the MTK ATE tool, execute the [ATE_For_New_Version_Setup \ Volume1 \setup.exe](#) file. The Installation Wizard guides the user through the installation process step by step, up to Installation finish.



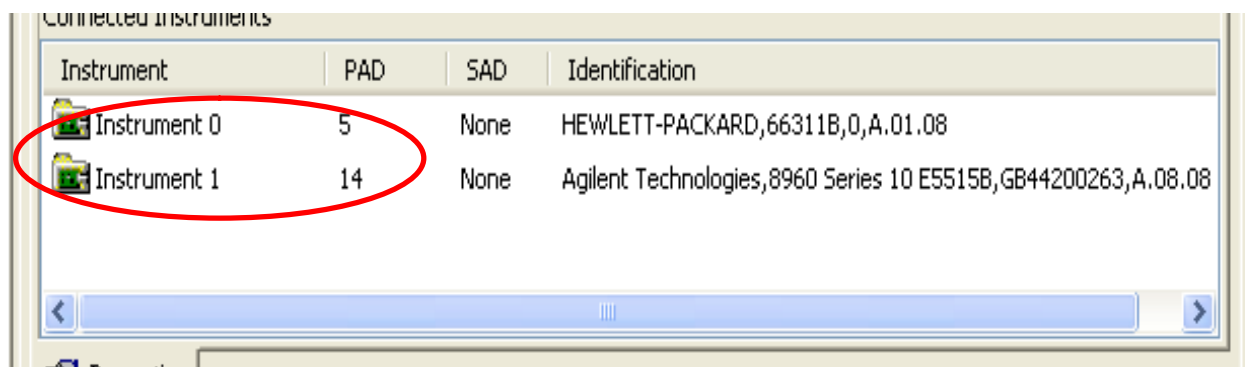
Execute Measurement & Automation to check equipment address



Choose Devices and Interfaces



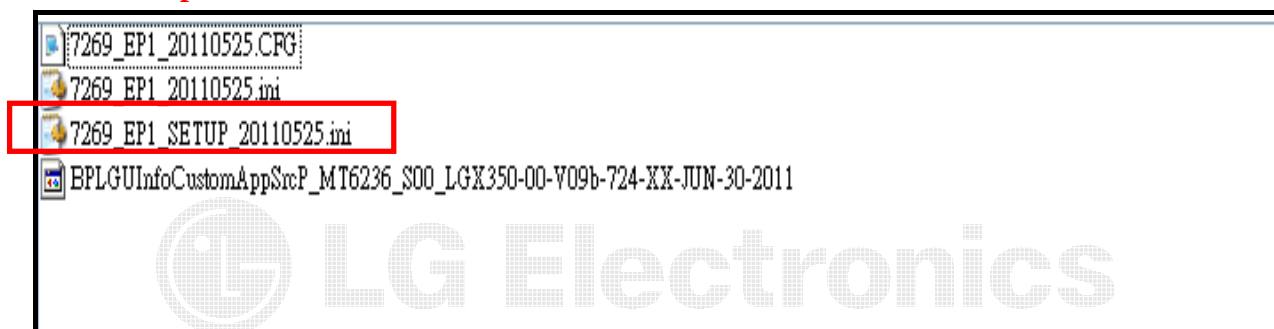
You can see your equipment address



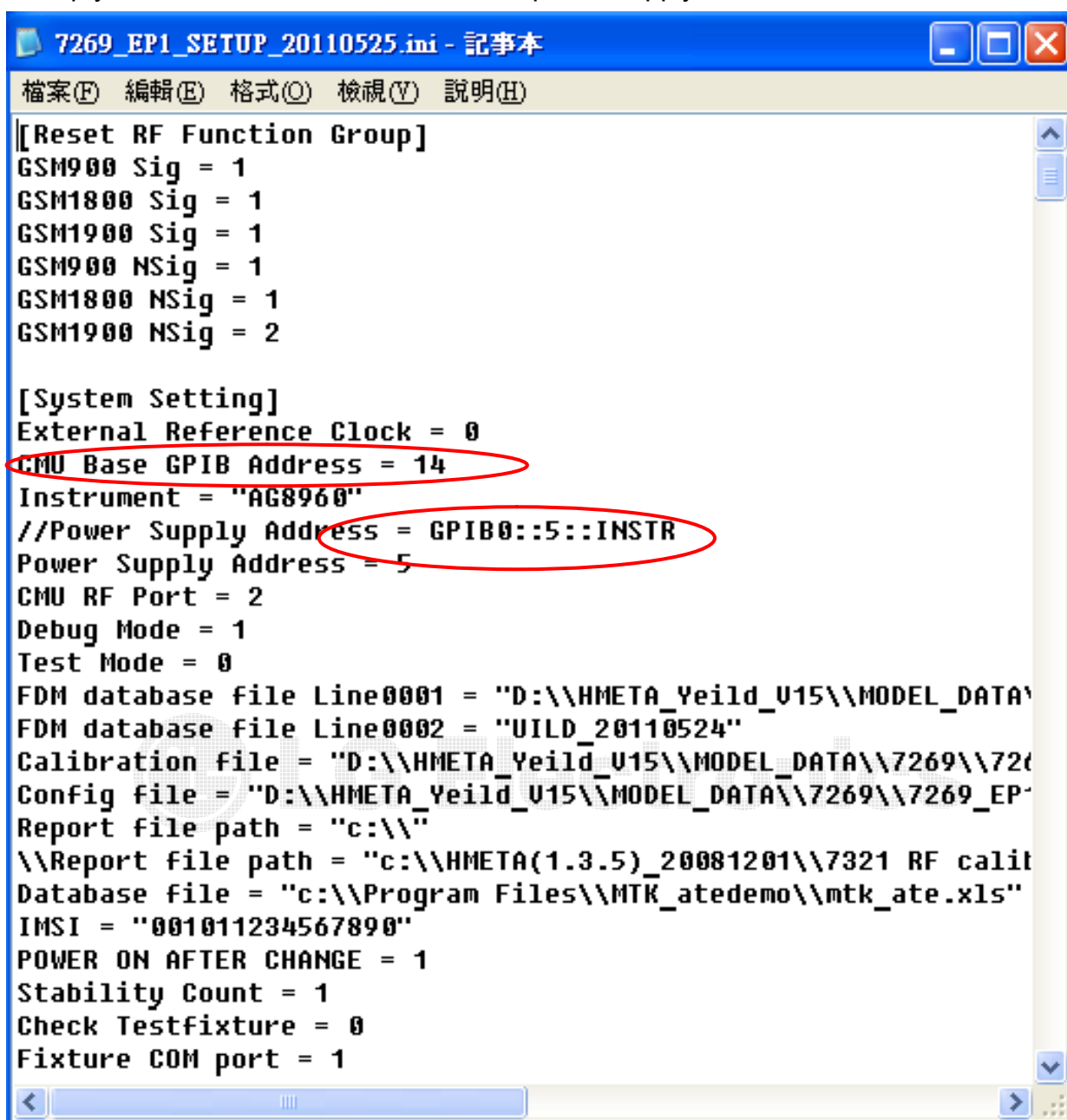
Instrument	PAD	SAD	Identification
Instrument 0	5	None	HEWLETT-PACKARD,66311B,0,A.01.08
Instrument 1	14	None	Agilent Technologies,8960 Series 10 E5515B,GB44200263,A.08.08

Choose [7269_EP1_SETUP_20110525.ini](#) and open the file to setup from data files .

(For example: X350)



Setup your CMU Base GPIB address and power supply address

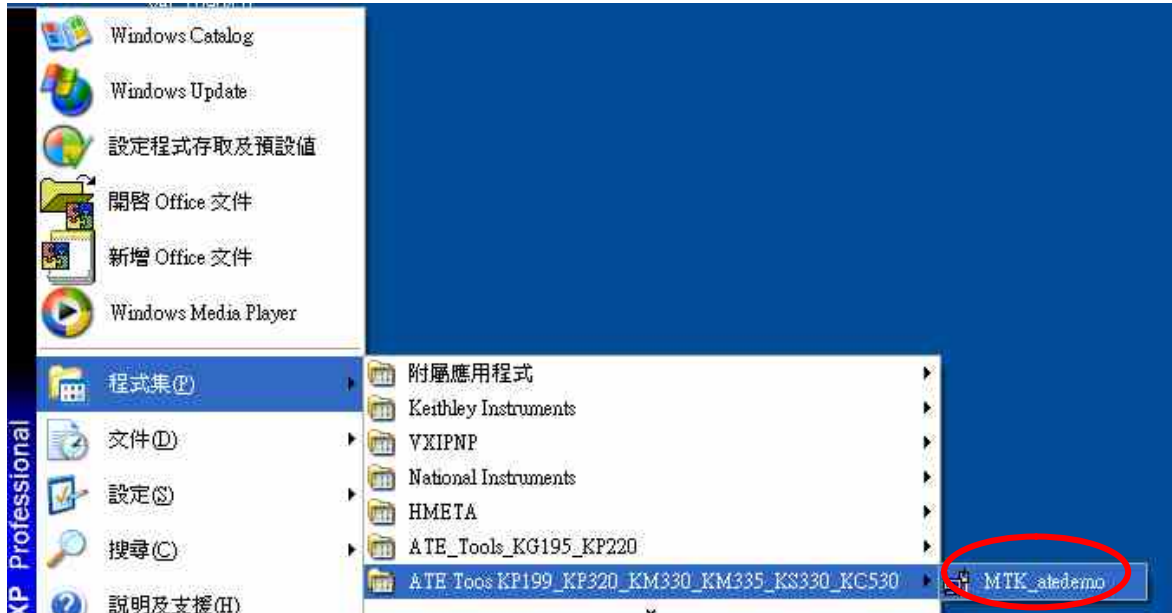


```
[Reset RF Function Group]
GSM900 Sig = 1
GSM1800 Sig = 1
GSM1900 Sig = 1
GSM900 NSig = 1
GSM1800 NSig = 1
GSM1900 NSig = 2

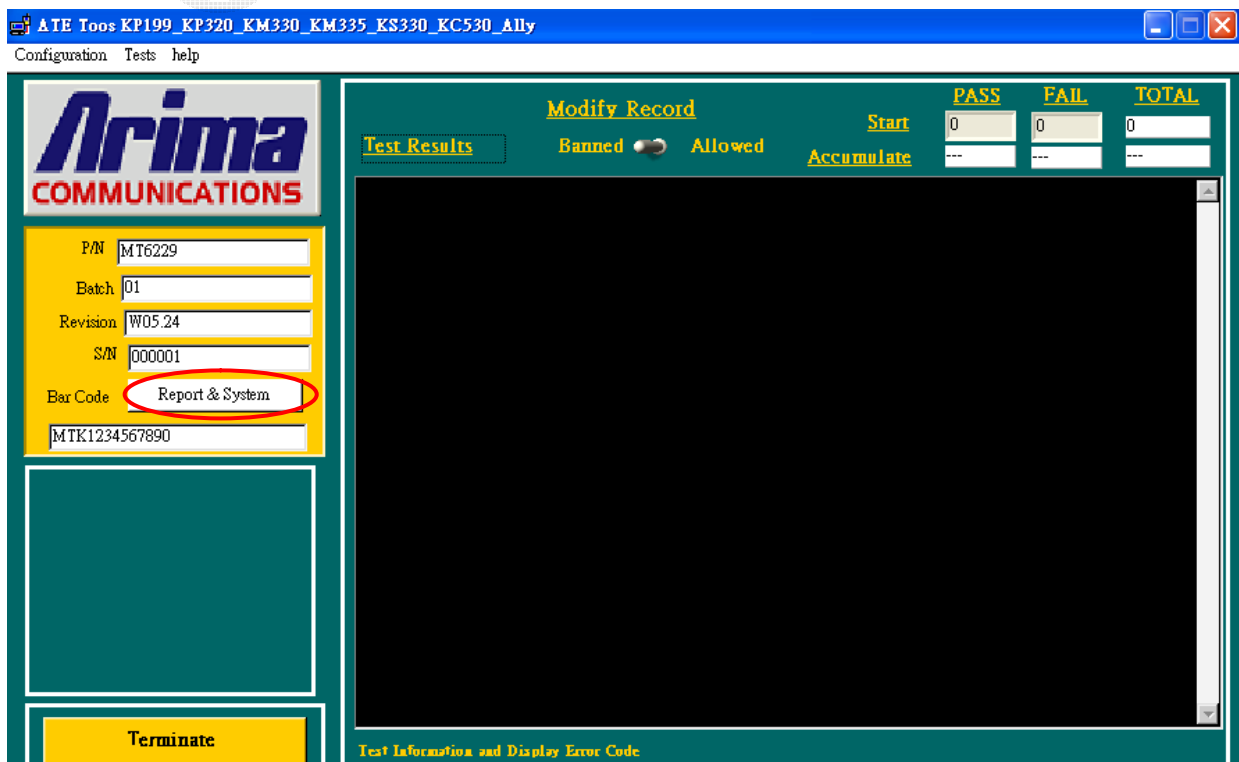
[System Setting]
External Reference Clock = 0
CMU Base GPIB Address = 14
Instrument = "AG8960"
//Power Supply Address = GPIB0::5::INSTR
Power Supply Address = 5
CMU RF Port = 2
Debug Mode = 1
Test Mode = 0
FDM database file Line0001 = "D:\\\\HMETA_Yeild_U15\\\\MODEL_DATA\\
FDM database file Line0002 = "UILD_20110524"
Calibration file = "D:\\\\HMETA_Yeild_U15\\\\MODEL_DATA\\\\7269\\\\726
Config file = "D:\\\\HMETA_Yeild_U15\\\\MODEL_DATA\\\\7269\\\\7269_EP
Report file path = "c:\\\\"
\\\\Report file path = "c:\\\\HMETA(1.3.5)_20081201\\\\7321 RF calit
Database file = "c:\\\\Program Files\\\\MTK_atedemo\\\\mtk_ate.xls"
IMSI = "001011234567890"
POWER ON AFTER CHANGE = 1
Stability Count = 1
Check Testfixture = 0
Fixture COM port = 1
```

ATE Tool system setting

Execute MTK _ ate demo



Press Report & System button



Setting your equipment

The screenshot shows a software interface for configuring equipment. It includes several sections:

- Part Information:** Fields for Part Number (MT6235), Batch (01), Revision (W08.20), Serial Number (000001), and Bar Code (MTK1234567890).
- GSM/EDGE Cal Setting:** Includes Band selection (GSM850, GSM900, DCS, PCS), RX (Xtal Tx) settings, TX GSM/EDGE settings, and Battery/ADC settings.
- System Setting:** Contains TEST MODE SELECT (Manual Initial), Bar Code Get Type When Calibration (Scan Barcode), Power Supply Type (Agilent E3631/2), PSU GPIB Address (GPIB0:5:INSTR), GSM/EDGE Tester (Agilent 8960), CMU RF Port (RF2), WiFi Tester (N4010A), BT Tester (MT8852B), WCDMA Tester (AG8960), Baseband Chip Type (AutoDetect), COM Port Select (COM 4), and checkboxes for Cal INP LOSS and Cal OUP LOSS.
- File Locations:** Fields for NVRAM Database file, Config File Location (CFG file), Calibration File Location (ini file), Battery DFI file, Test Setup File Location (Setup file), Test Report Location, and Report Database Location, each with a corresponding 'Select' button.

Setting your power supply type

This close-up view of the 'System Setting' tab highlights the 'Power Supply Type' dropdown menu. The menu is currently set to 'KEITHLEY230'. A red circle is drawn around the dropdown, and a blue arrow points from a text box to it.

Choose your Power Supply Type

Setting your GSM/EDGE Tester

System Setting

TEST MODE SELECT
Manual Initial

Bar Code Get Type When Calibration
Scan Barcode

Power Supply Type PSU GPIB Address
KEITHLEY230. 7

GSM/EDGE Tester CMU RF Port
Agilent 8960 RF2

WiFi Tester
IQVIEW

BT Tester
CMU200

Baseband Chip Type COM Port Select
6226 COM 6

☐ Cal INP LOSS ☐ Cal OUP LOSS

Choose your Tester



Choose your download com port

System Setting

TEST MODE SELECT
Manual Initial

Bar Code Get Type When Calibration
Scan Barcode

Power Supply Type PSU GPIB Address
KEITHLEY230. 7

GSM/EDGE Tester CMU RF Port
Agilent 8960 RF2

WiFi Tester
IQVIEW

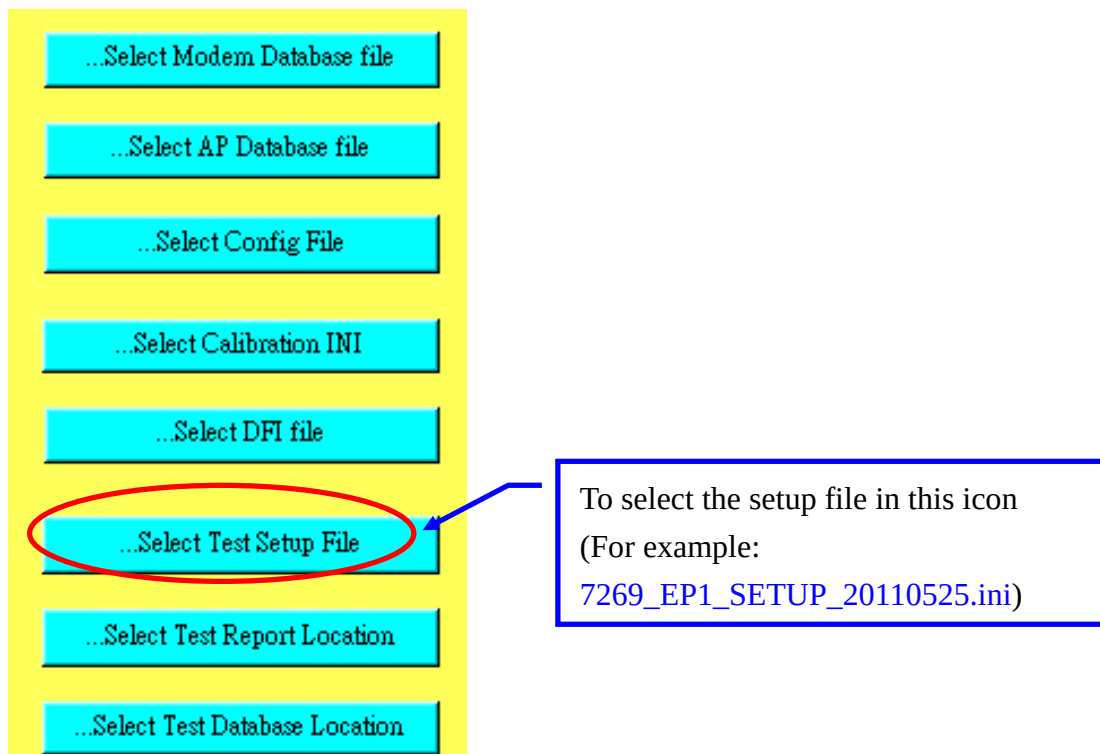
BT Tester
CMU200

Baseband Chip Type COM Port Select
6226 COM 6

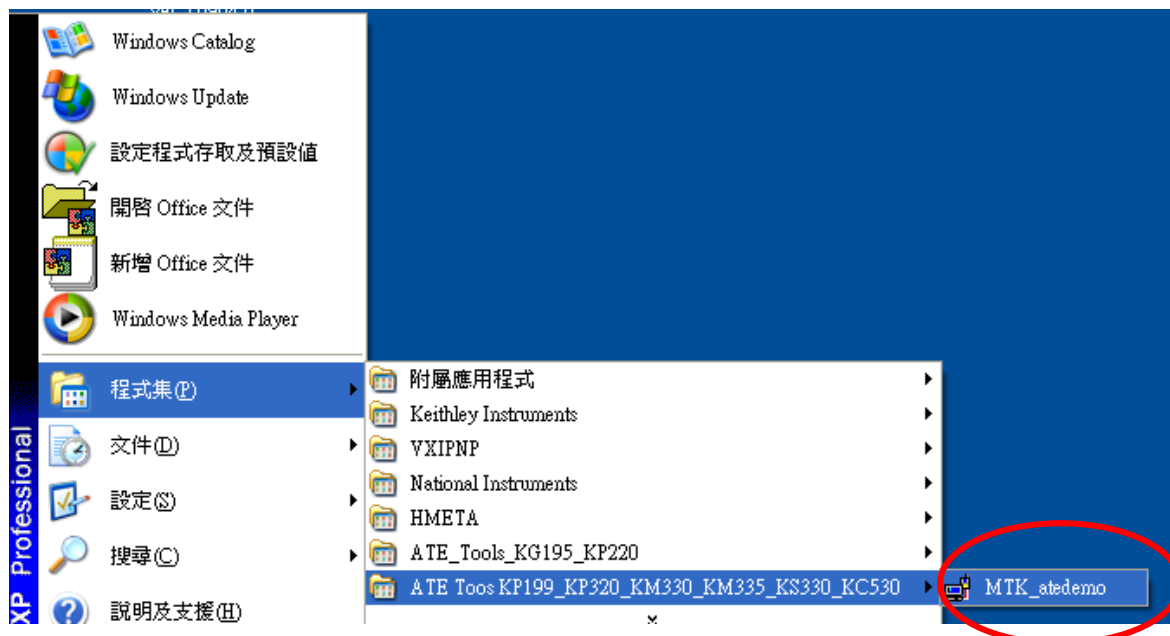
☐ Cal INP LOSS ☐ Cal OUP LOSS

Choose your download
cable connect COM port

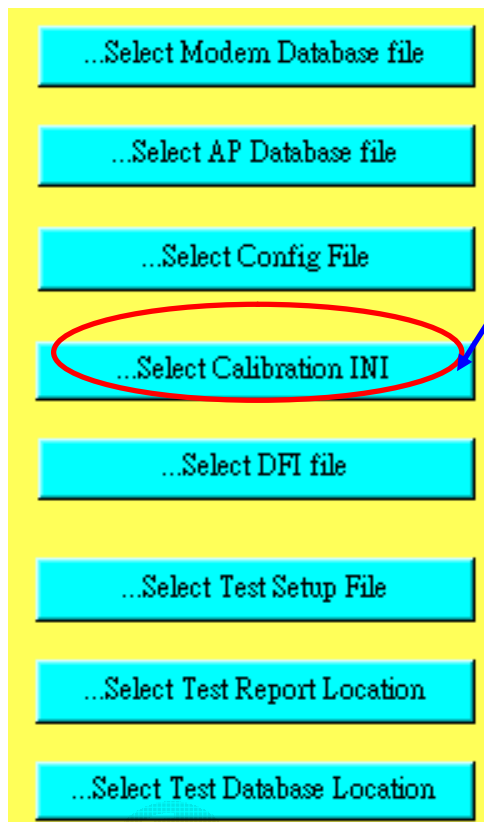
Choose “select test setup file”



Execute MTK_ate demo again

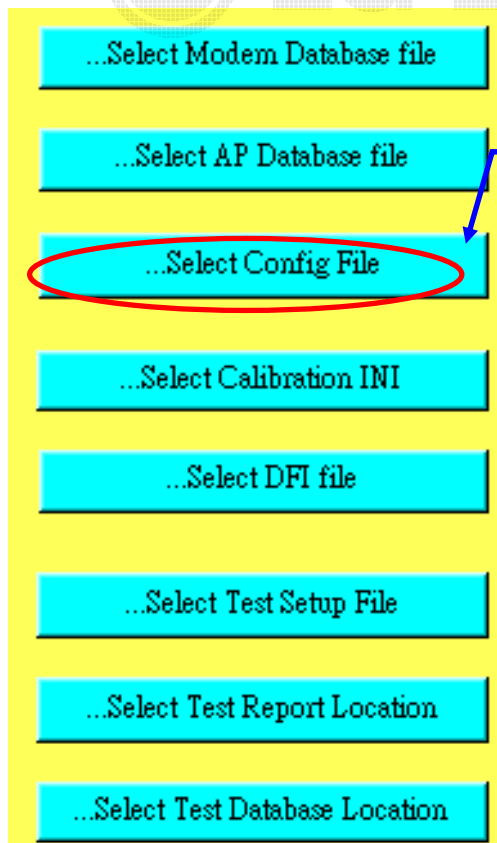


Choose Calibration INI



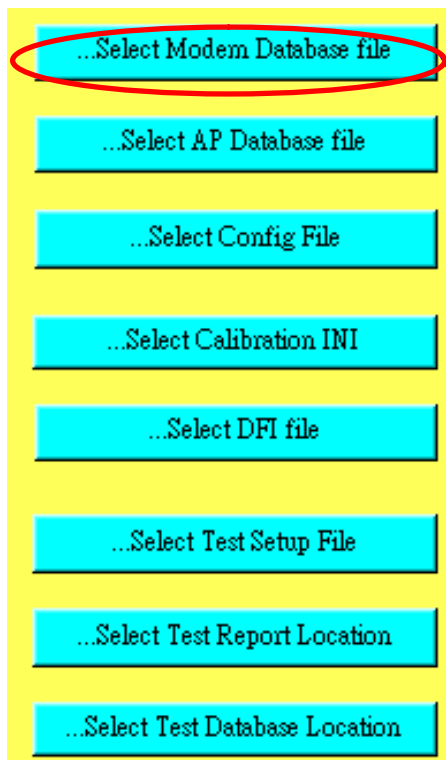
To select the ini file in this icon
(For example: 7269_EP1_20110525.ini)

Choose Con fig File



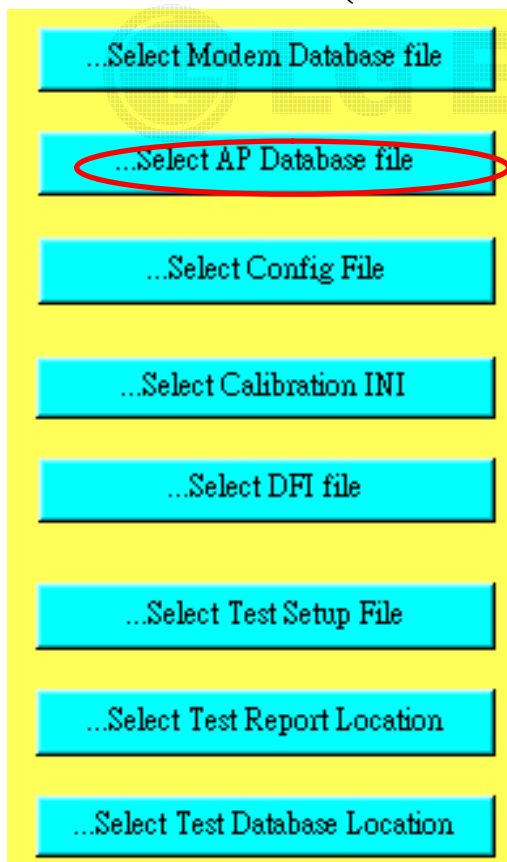
To select the CFG file in this icon
(For example: 7269_EP1_20110525.CFG)

Choose NVRAM Database file



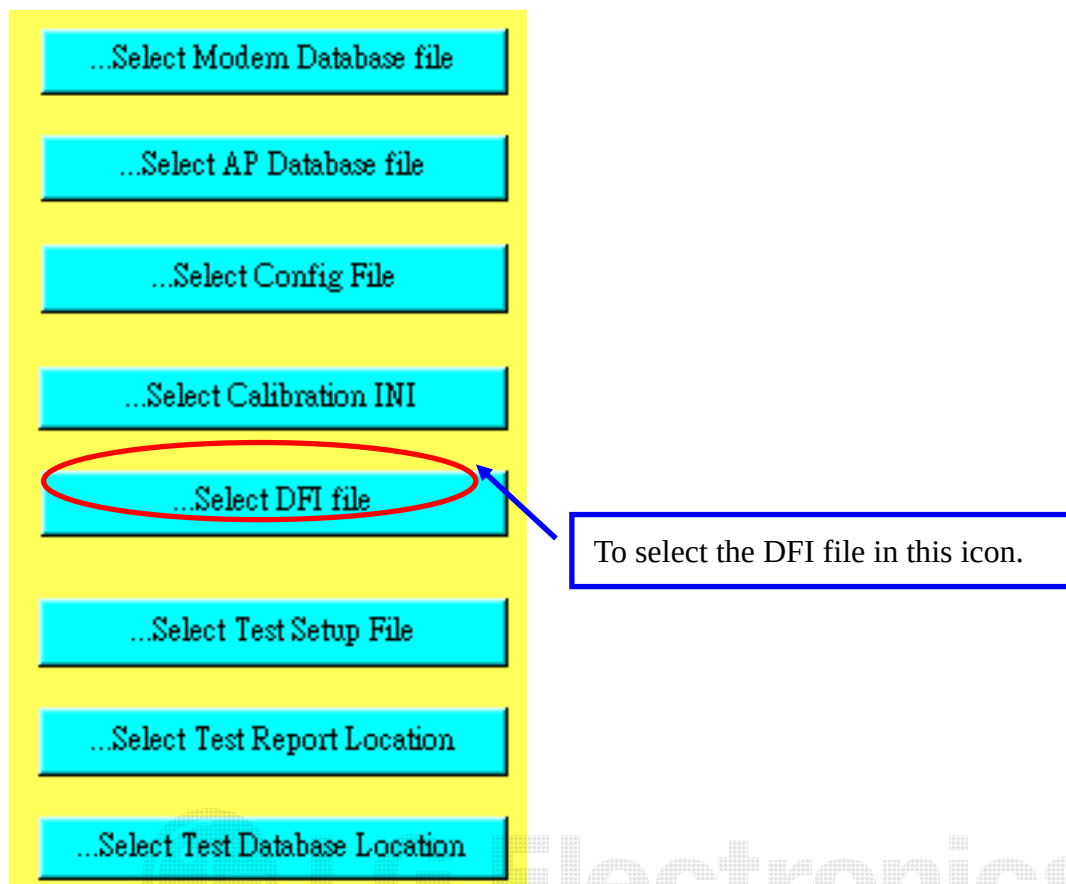
To select the SW database in this icon.

Choose AP database file (Caution: ONLY Smart phone need choose it)



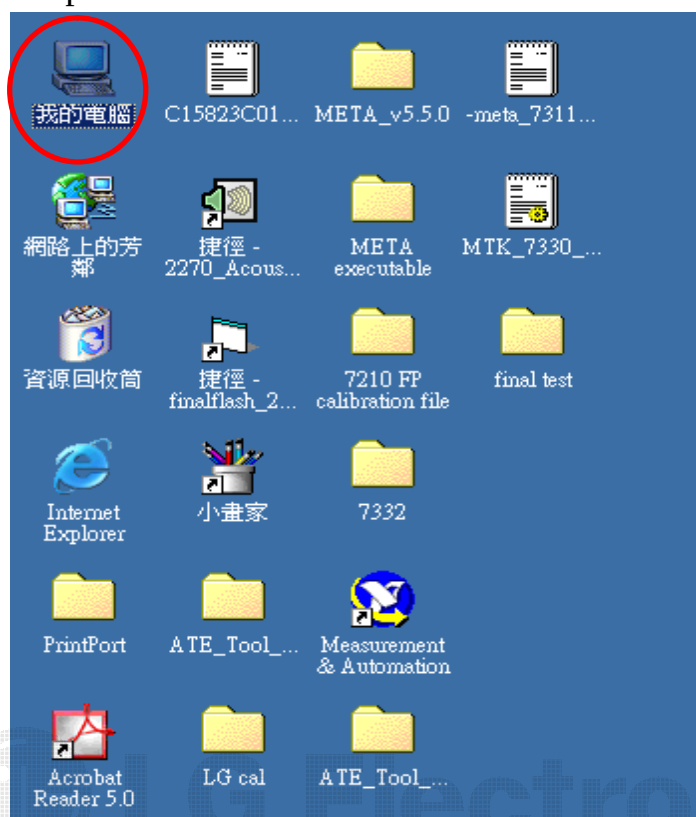
To select the AP database in this icon.

Choose Battery DFI file (**Caution: ONLY Smart battery need choose it**)



How to setup your test report location

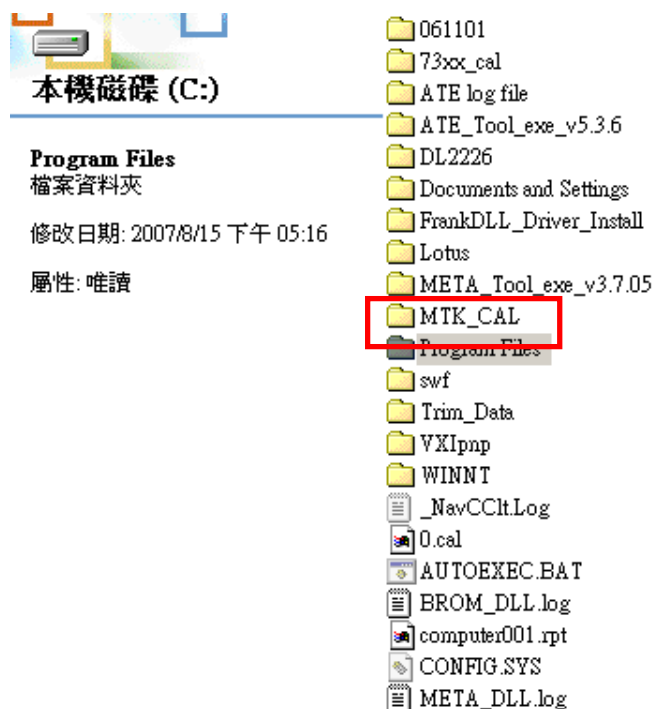
Choose my computer



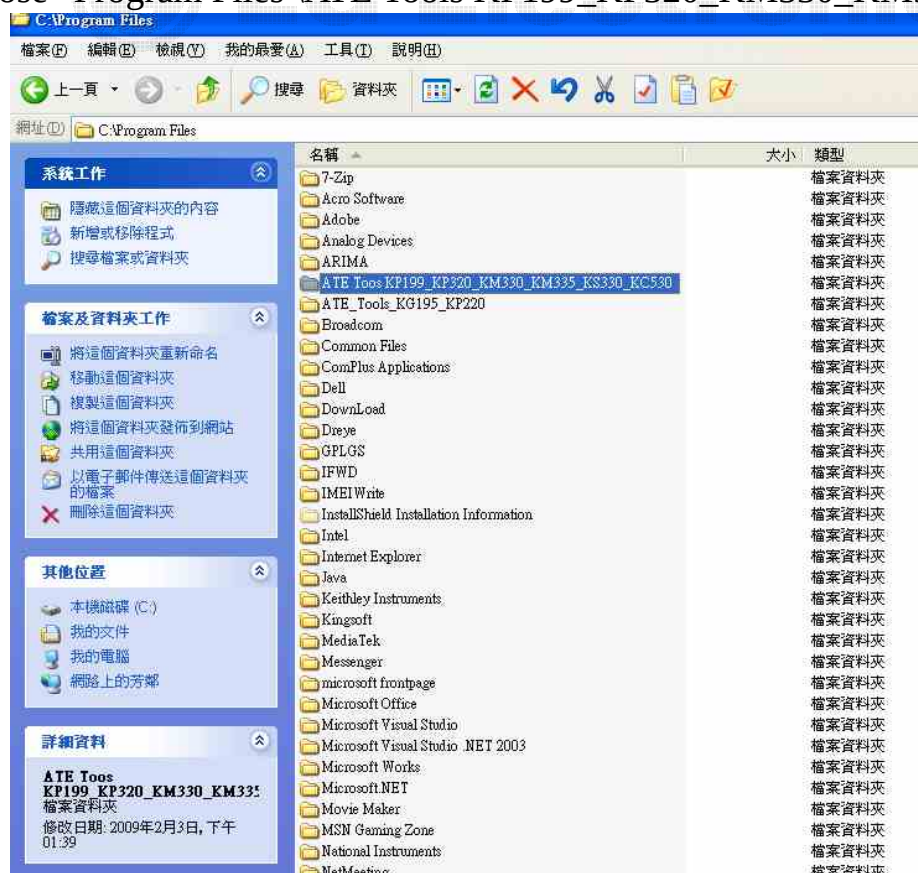
Choose “C” disk

名稱 ▲	類型	大小總計	可用空間
3.5 軟碟機 (A:)	3.5 吋軟式磁碟機		
本機磁碟 (C:)	本機磁碟	18.6 GB	15.6 GB
新增磁碟區 (D:)	本機磁碟	18.6 GB	16.0 GB
控制台	可用空間: 15.6 GB, 容量: 18.6 GB		

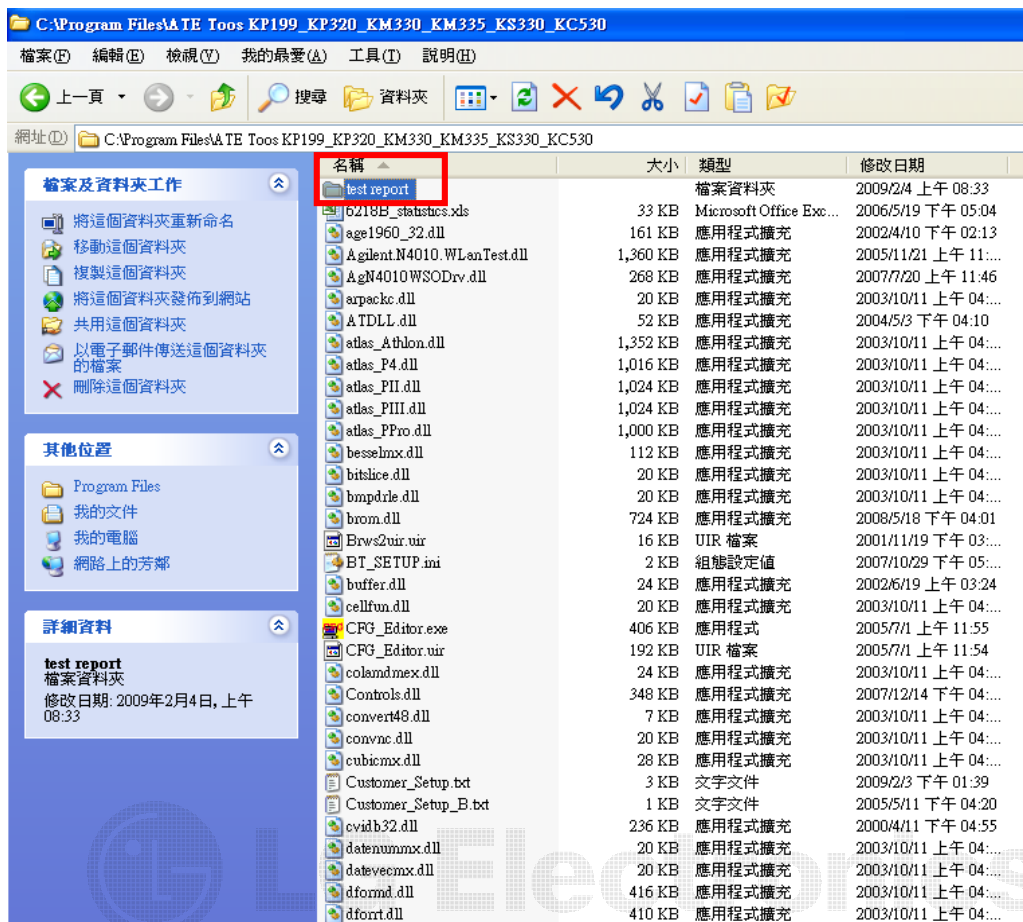
Choose “program files”



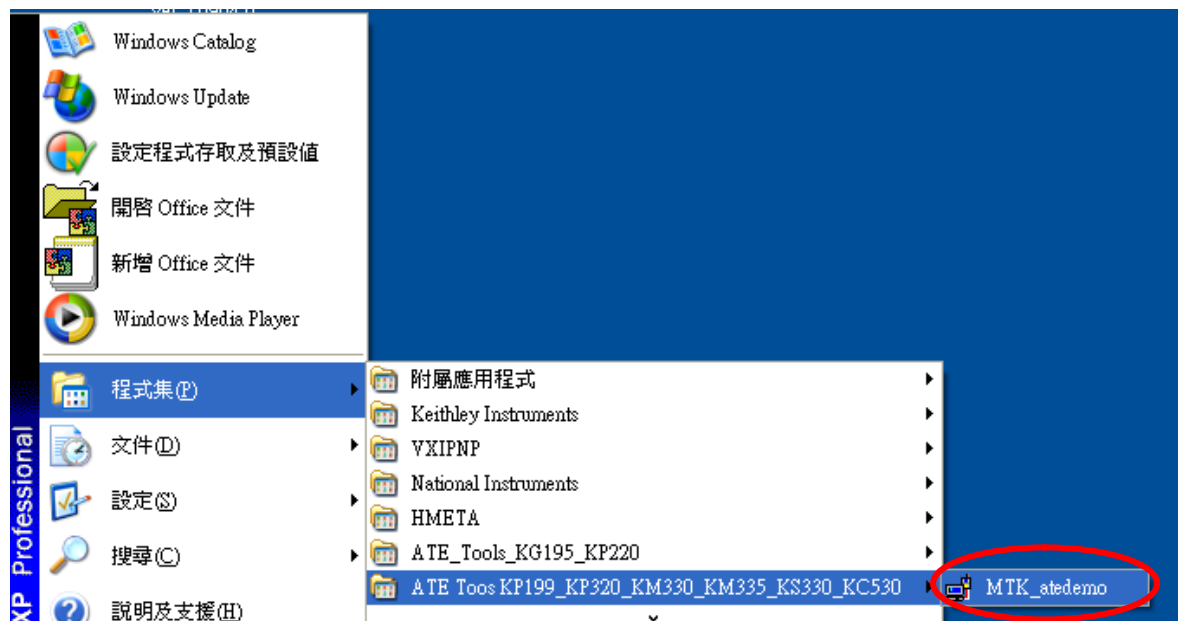
Choose “Program Files \ATE Tools KP199_KP320_KM330_KM335_KC530”file



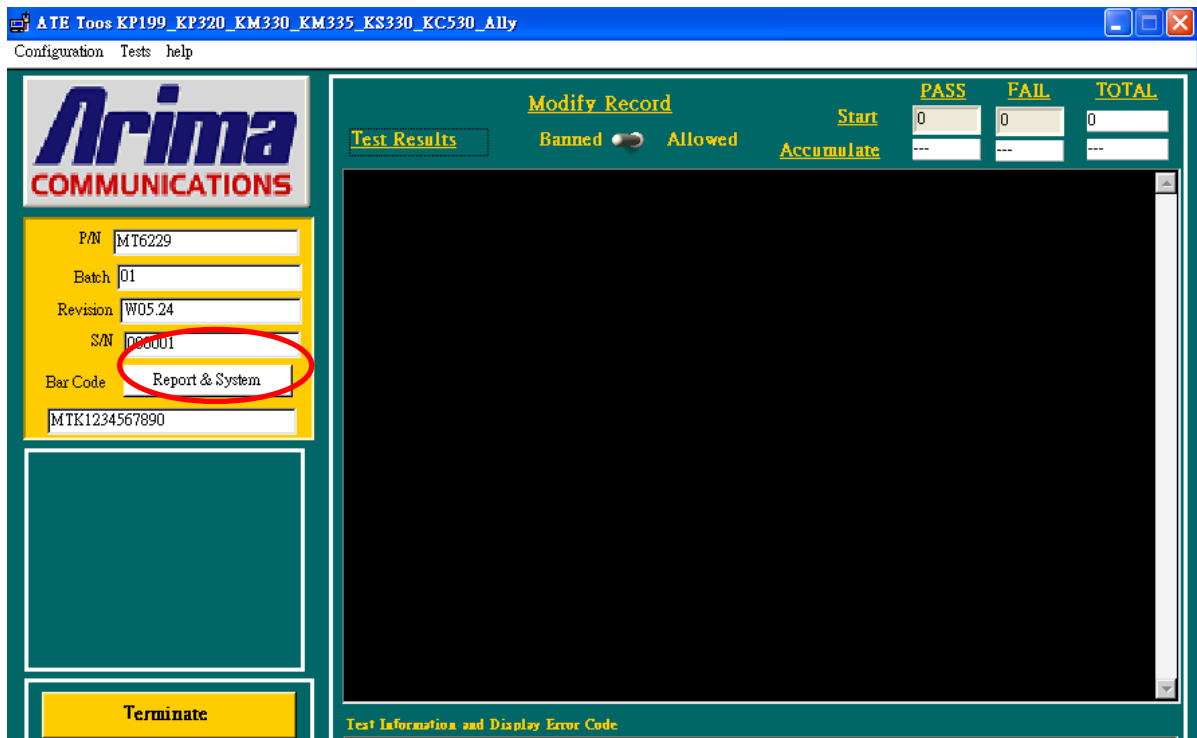
Setup new file and leave the window



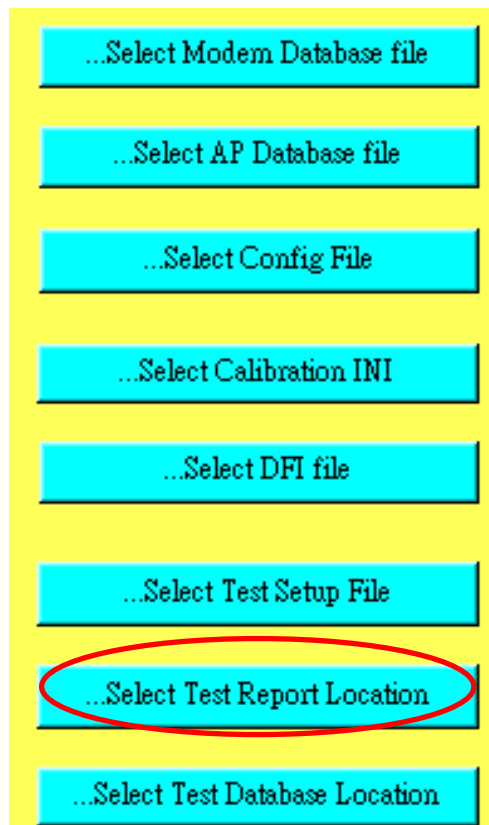
Execute MTK_ate demo



Press Report & System button



Press “select test report location”



Choose your setup report



Press “Done”



Setup finish

Batch [01]

Revision [W05.24]

Serial Number [000001]

Bar Code [MTK1234567890]

☒ Fast Power Measurement (CMU 3.50)

☐ Wireless test ☐ Fast Handset Calibration

☒ GSM Default Items

☐ Stop Condition

☐ Add Final Status

☐ RF Final Test with Check Bar Code

☐ Final Test with IMEI Write

☐ Add Cal Status

☐ Multi MS MS # [2 Handsets]

AFC Type [Crystal AFC] ☒ AFC Cal

☒ PathLoss Calibration ☒ AFC T/R Cal

☒ AFC CapId Cal

TX GSM/EDGE

☐ APCDC Cal(Skyworks only) ☒ Slope Skew ☒ FB dac

TXIQ [GMSK] ☒ TXIQ PCL Check [None]

PA [GSM Full PCL] ☒ TXP Cal

Battery/ADC: ☒ ADC Cal/PSU Ctrl

WiFi Cal:

☐ TxDeOffset ☐ EEPROM Copy ☐ TXP CAL ☐ RF Check

☐ Cap Id ☐ Internal Sensor

BT Cal:

☐ BT CapId ☐ wo Tester

GSM/EDGE Final Setting

☒ GSM850 ☒ GSM900 ☒ DCS ☒ PCS ☐ GPRS Test

Auto Bar Code With Date & Batch

Power Supply Type PSU GPIB Address

[Agilent 663xx] [5]

GSM/EDGE Tester CMU RF Port

[Agilent 8960] [RF2]

WiFi Tester

[N4010A]

BT Tester WCDMA Tester

[N4010A] [AG8960]

Baseband Chip Type COM Port Select

[AutoDetect] [COM 4]

☐ Cal INP LOSS ☐ Cal OUP LOSS

Save Change

NVRAM Database file (For Modem and feature phone)

[e:\calibration data\7269 calibration data\EPLGUIInfoCustomAppSrcP_MT6236_S00_LGX350-00-V09b-724-XX-JUN-30-2

...Select Modem Database file

NVRAM Database file (For AP, Smart phone only)

...Select AP Database file

Config File Location (CFG file)

[e:\calibration data\7269 calibration data\7269_EP1_20110525.CFG

...Select Config File

Calibration File Location (.ini file)

[e:\calibration data\7269 calibration data\7269_EP1_20110525.ini

...Select Calibration INI

Battery DFI file (For smart battery)

...Select DFI file

Test Setup File Location (Setup file)

[e:\calibration data\7269 calibration data\7269_EP1_SETUP_20110525.ini

...Select Test Setup File

Test Report Location

[e:\calibration data\report

...Select Test Report Location

Report Database Location

[e:\calibration data\report

...Select Test Database Location

When you finish the setup then you press **save change** icon.

Batch [01]

Revision [W05.24]

Serial Number [000001]

Bar Code [MTK1234567890]

☒ Fast Power Measurement (CMU 3.50)

☐ Wireless test ☐ Fast Handset Calibration

☒ GSM Default Items

☐ Stop Condition

☐ Add Final Status

☐ RF Final Test with Check Bar Code

☐ Final Test with IMEI Write

☐ Add Cal Status

☐ Multi MS MS # [2 Handsets]

AFC Type [Crystal AFC] ☒ AFC Cal

☒ PathLoss Calibration ☒ AFC T/R Cal

☒ AFC CapId Cal

TX GSM/EDGE

☐ APCDC Cal(Skyworks only) ☒ Slope Skew ☒ FB dac

TXIQ [GMSK] ☒ TXIQ PCL Check [None]

PA [GSM Full PCL] ☒ TXP Cal

Battery/ADC: ☒ ADC Cal/PSU Ctrl

WiFi Cal:

☐ TxDeOffset ☐ EEPROM Copy ☐ TXP CAL ☐ RF Check

☐ Cap Id ☐ Internal Sensor

BT Cal:

☐ BT CapId ☐ wo Tester

GSM/EDGE Final Setting

☒ GSM850 ☒ GSM900 ☒ DCS ☒ PCS ☐ GPRS Test

Auto Bar Code With Date & Batch

Power Supply Type PSU GPIB Address

[Agilent 663xx] [5]

GSM/EDGE Tester CMU RF Port

[Agilent 8960] [RF2]

WiFi Tester

[N4010A]

BT Tester WCDMA Tester

[N4010A] [AG8960]

Baseband Chip Type COM Port Select

[AutoDetect] [COM 4]

☐ Cal INP LOSS ☐ Cal OUP LOSS

Save Change

NVRAM Database file (For Modem and feature phone)

[e:\calibration data\7269 calibration data\EPLGUIInfoCustomAppSrcP_MT6236_S00_LGX350-00-V09b-724-XX-JUN-30-2

...Select Modem Database file

NVRAM Database file (For AP, Smart phone only)

...Select AP Database file

Config File Location (CFG file)

[e:\calibration data\7269 calibration data\7269_EP1_20110525.CFG

...Select Config File

Calibration File Location (.ini file)

[e:\calibration data\7269 calibration data\7269_EP1_20110525.ini

...Select Calibration INI

Battery DFI file (For smart battery)

...Select DFI file

Test Setup File Location (Setup file)

[e:\calibration data\7269 calibration data\7269_EP1_SETUP_20110525.ini

...Select Test Setup File

Test Report Location

[e:\calibration data\report

...Select Test Report Location

Report Database Location

[c:\Program Files\MTK_atedemo\mtk_ate.xls

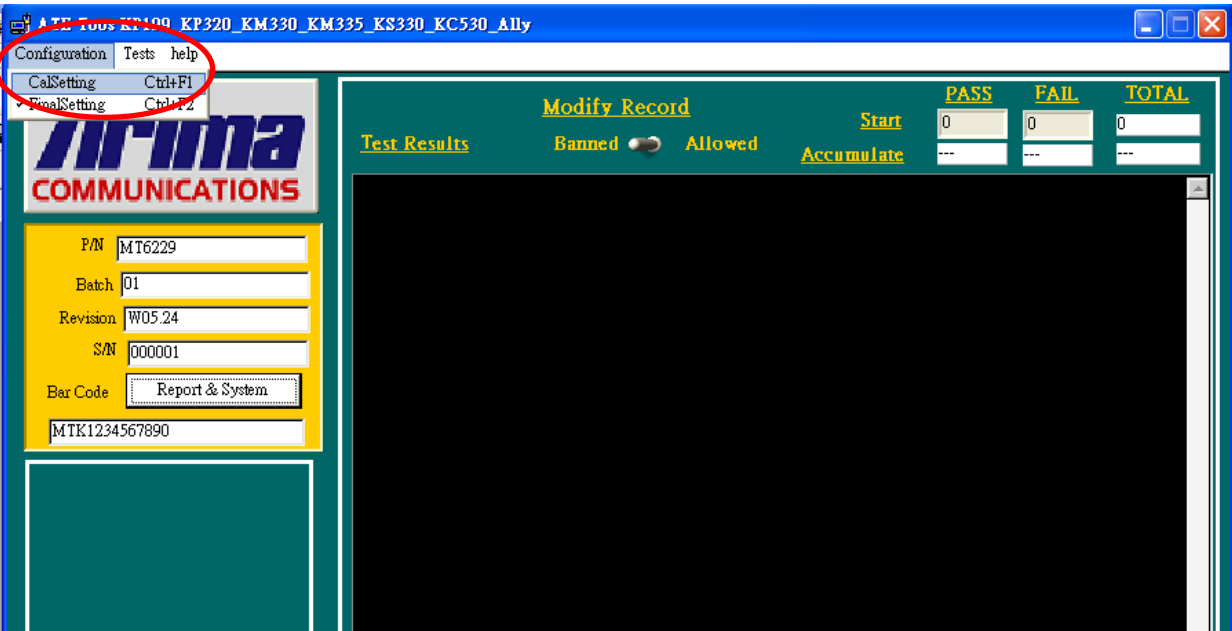
...Select Test Database Location

ATE INFORMATION

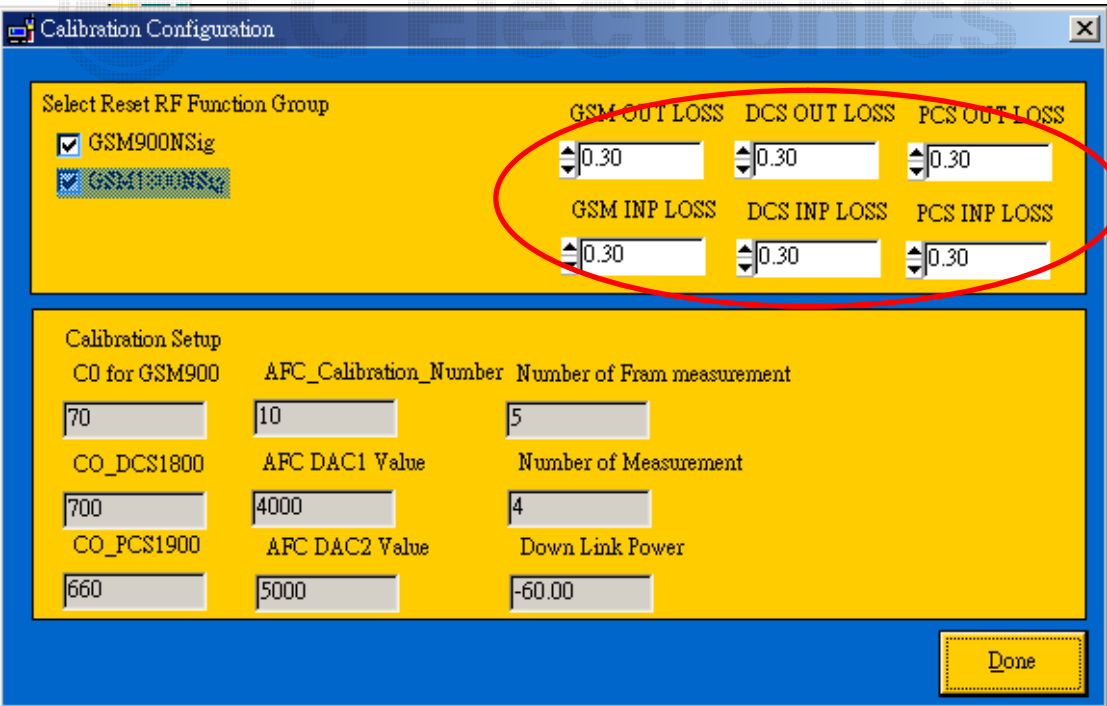
Setting change, Please restart program!

OK

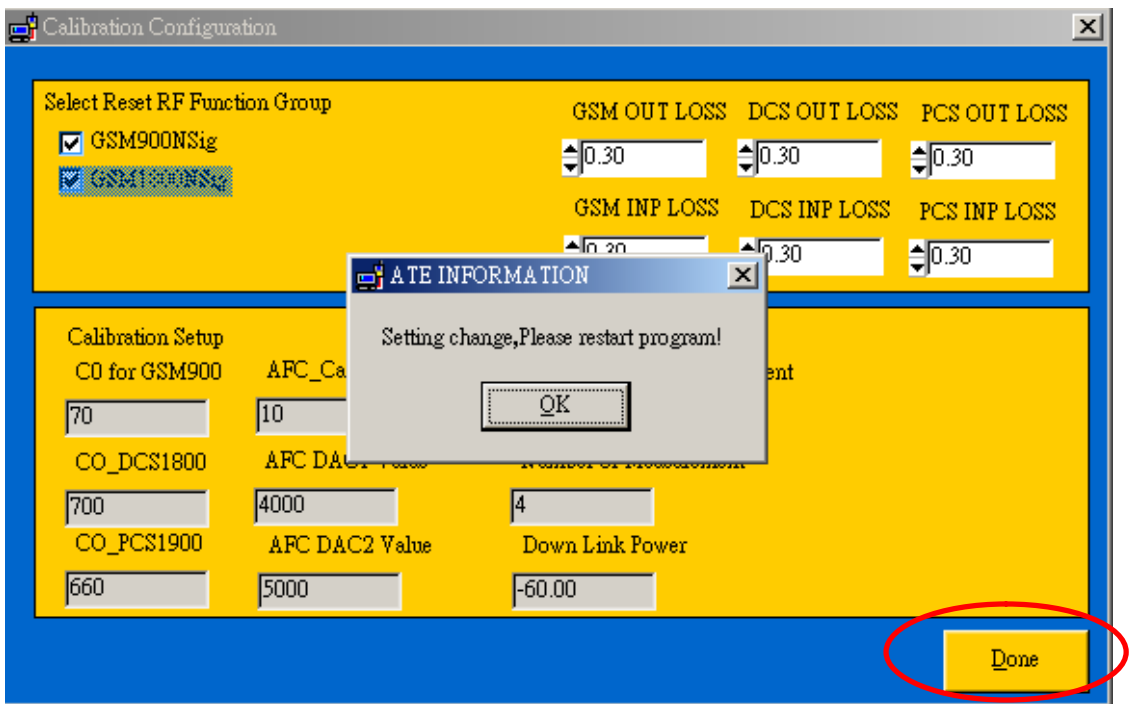
Press Configuration choose Cal Setting



Setting your cable loss



Press Done to save



Press Configuration choose Final setting



Choose “MT Call” from Establish Call Type

Select Reset RF Function Group

- ☒ GSM850 Sig
- ☐ GSM 900 Sig
- ☒ GSM 1800 Sig
- ☒ GSM 1900 Sig

Establish Call Type ☒ MT Call ☐ MO Call

IMSI NUMBER
001011234567890

☐ External 10MHz Reference Clock

Call Setup Configuration

GSM850
Call Setup Channel: 128, BCCH Channel: 128

GSM
Call Setup Channel: 1, BCCH Channel: 32, Call Setup Network: GSM900

DCS
Call Setup Channel: 512, BCCH Channel: 700, BCCH RF LEVEL: -60.00

PCS
Call Setup Channel: 512, BCCH Channel: 700, BS TCH LEVEL: -81.00

GPRS
GPRS Test Mode: AG USFER, MCS9, GPRS ACK ON/OFF: ☐

Key in your test SIM card number form IMSI NUMBER

Select Reset RF Function Group

- ☒ GSM850 Sig
- ☐ GSM 900 Sig
- ☒ GSM 1800 Sig
- ☒ GSM 1900 Sig

Establish Call Type ☒ MT Call ☐ MO Call

IMSI NUMBER
001011234567890

☐ External 10MHz Reference Clock

Call Setup Configuration

GSM850
Call Setup Channel: 128, BCCH Channel: 128

GSM
Call Setup Channel: 1, BCCH Channel: 32, Call Setup Network: GSM900

DCS
Call Setup Channel: 512, BCCH Channel: 700, BCCH RF LEVEL: -60.00

PCS
Call Setup Channel: 512, BCCH Channel: 700, BS TCH LEVEL: -81.00

GPRS
GPRS Test Mode: AG USFER, MCS9, GPRS ACK ON/OFF: ☐

Press “Done” and save your setting

Final Test Configurations

Select Reset RF Function Group

- ☒ GSM850 Sig
- ☐ GSM 900 Sig
- ☒ GSM 1800 Sig
- ☒ GSM 1900 Sig

Establish Call Type ☒ MT Call ☐ MO Call

IMSI NUMBER
001011234567890

☐ External 10MHz Reference Clock

Call Setup Configuration

GSM850
Call Setup Channel: 128, BCCH Channel: 128

GSM
Call Setup Channel: 1, BCCH Channel: 32, Call Setup Network: GSM900

DCS
Call Setup Channel: 512, BCCH Channel: 700, BCCH RF LEVEL: -60.00

PCS
Call Setup Channel: 512, BCCH Channel: 700, BS TCH LEVEL: -81.00

GPRS
GPRS Test Mode: AG USFER, MCS9, GPRS ACK ON/OFF: ☐

Measurement Configurations

Power
Power Measurement Burst: 10, Average Burst Power: ☒, Peak Burst Power: ☒, PVT Match: ☒, Timing Error: ☒

Modulation
Modulation: 10, Phase Error: ☒, Phase Error RMS: ☒, Frequency Error: ☒

Bluetooth Loss
BT OUT LOSS: 0.50, BT INP LOSS: 0.50

Rx Report
RX Quality: ☒, RX Level: ☒, RFER: ☐, BBB: ☒, Manual BER Limit Check: ☒

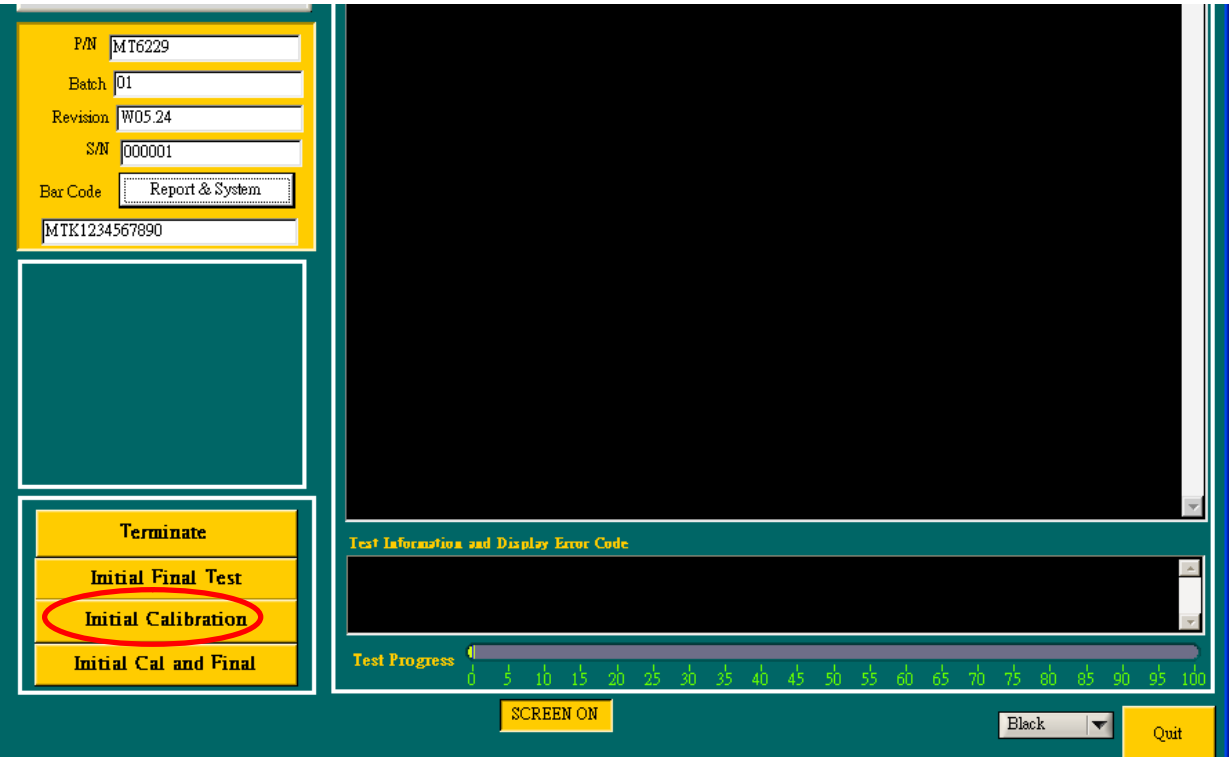
Rx Quality
GSM Rx Meas Level: -100.00, PCS Rx Meas Level: -100.00, DCS Rx Meas Level: -100.00, GSM850 Rx Meas Level: -100.00, Rx RFER Burst: 128, Rx BBB Burst: 88

Bluetooth Loss
GSM850 LOSS / GSM OUT LOSS: 0.30, DCS OUT LOSS: 0.30, PCS OUT LOSS: 0.30, GSM INP LOSS: 0.30, DCS INP LOSS: 0.30, PCS INP LOSS: 0.30

Setting change, Please restart program!
OK

Done

If you want calibration , you can press “initial calibration”



 Press Calibration Test



Key-in your phone bar Code

Arima
COMMUNICATIONS

P/N

MT6229

Batch

01

Revision

W05.24

S/N

000001

Bar Code

Report & System

MTK1234567890

Terminate

Calibration Test

Terminate

Initial Final Test

Initial Calibration

Initial Cal and Final

Modify Record

Test Results

Banned

Allowed

Accumulate

Start

0

0

0

FAIL

0

TOTAL

0

Please Scan bar Code & Press OK to continue...

OK

Test Information and Display Error Code

Test Progress

0

5

10

15

20

25

30

35

40

45

50

55

60

65

70

75

80

85

90

95

100

Press your phone of power on key and Start calibration

Configuration Tests help

Arima
COMMUNICATIONS

P/N

MT6229

Batch

01

Revision

W05.24

S/N

000001

Bar Code

Report & System

111

Terminate

Calibration Test

Terminate

Modify Record

Test Results

Banned

Allowed

Accumulate

Start

0

0

0

FAIL

0

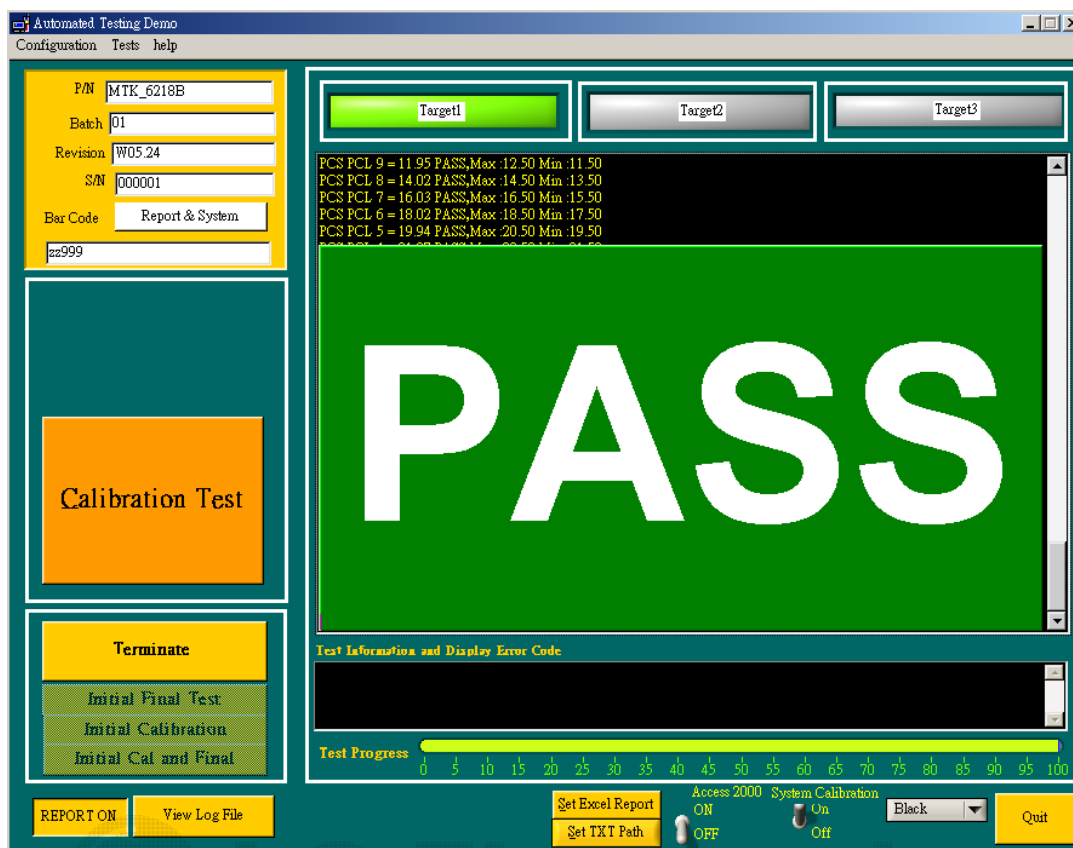
TOTAL

0

Booting Into META mode, Please Wait.

Test Information and Display Error Code

Calibration is ok and will show “PASS”



You can see the test report

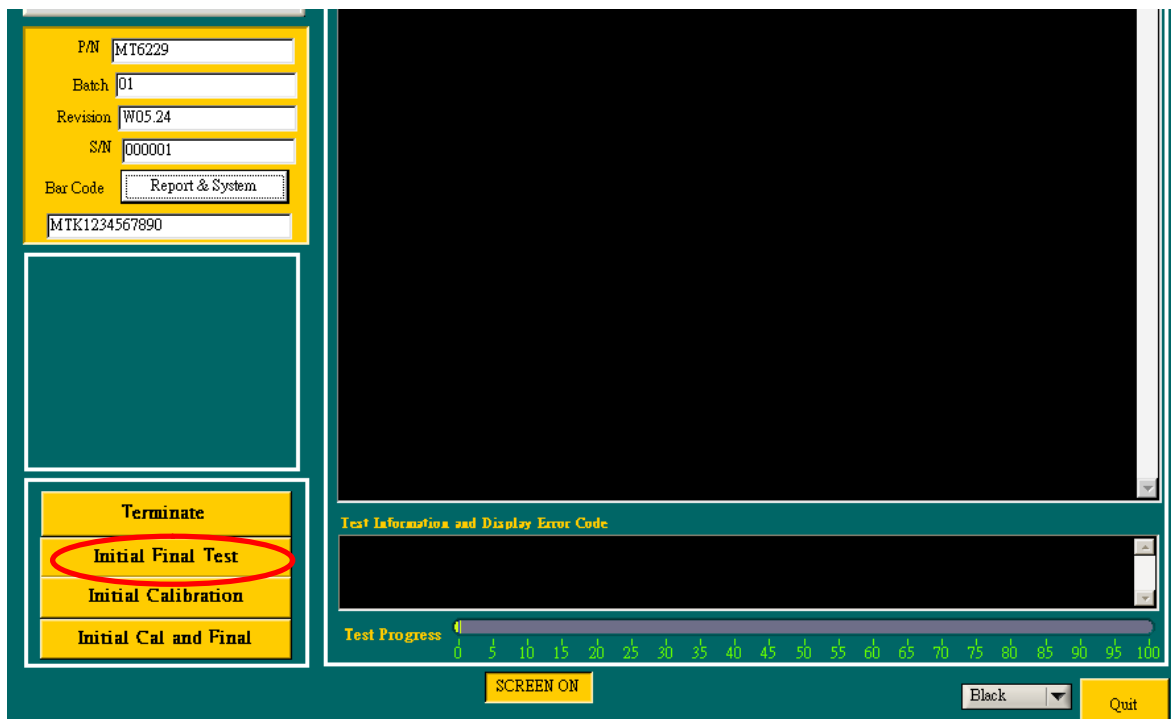
```

-----
ATE Tool Version:5.0.3
Part Number: MTK_6218B
Serial Number: 000001
Revision: W05.24
Batch: 01
Bar Code: qqq
Error Code: 000
-----

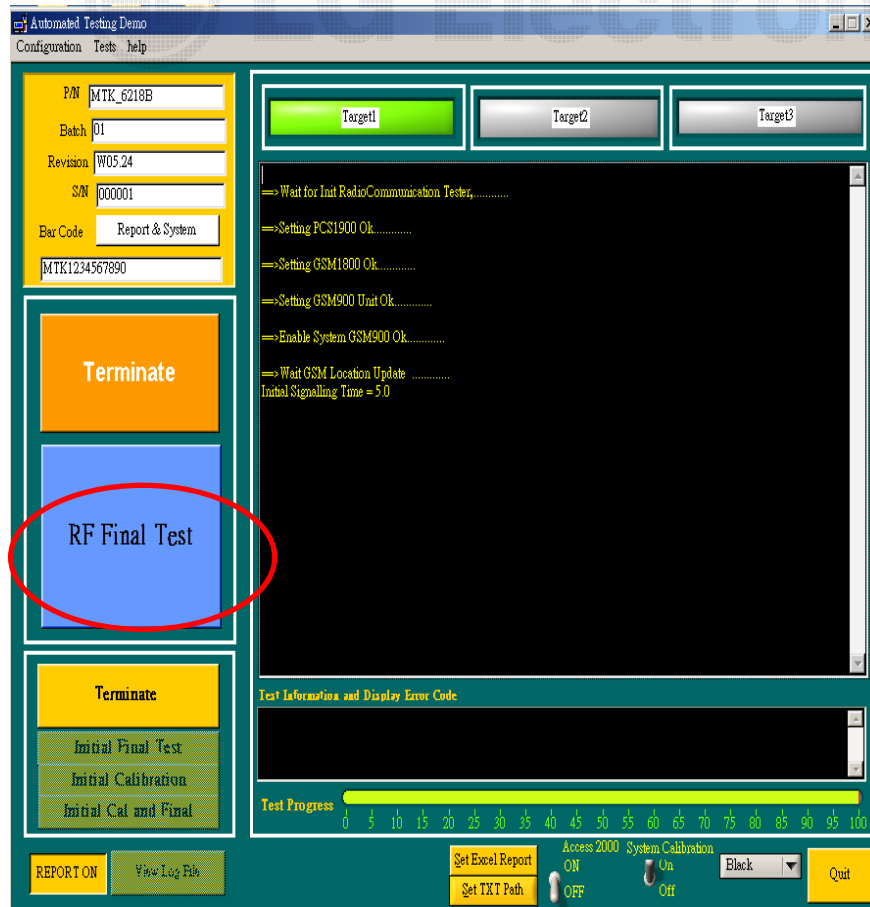
==>Wait GSM Location Update .....
Enter into META Mode OK
AFC Calibration OK
Slope=2.824,min:1.000,max:10.000
Use Default Value=3836
AFC Calibration time=1.64(sec)
PL GSM TCH 15 = 1.25 Pass MAX:3.00 MIN:-3.00
PL GSM TCH 30 = 1.00 Pass MAX:3.00 MIN:-3.00
PL GSM TCH 45 = 0.88 Pass MAX:3.00 MIN:-3.00
PL GSM TCH 60 = 1.25 Pass MAX:3.00 MIN:-3.00
PL GSM TCH 75 = 1.38 Pass MAX:3.00 MIN:-3.00
PL GSM TCH 80 = 1.50 Pass MAX:3.00 MIN:-3.00
PL GSM TCH 100 = 1.25 Pass MAX:3.00 MIN:-3.00
PL GSM TCH 124 = 1.25 Pass MAX:3.00 MIN:-3.00
PL GSM TCH 975 = 1.50 Pass MAX:3.00 MIN:-3.00
PL GSM TCH 1000 = 1.38 Pass MAX:3.00 MIN:-3.00
PL GSM TCH 1023 = 1.00 Pass MAX:3.00 MIN:-3.00
PL DCS TCH 550 = 0.50 Pass MAX:3.00 MIN:-3.00
PL DCS TCH 590 = 1.00 Pass MAX:3.00 MIN:-3.00

```

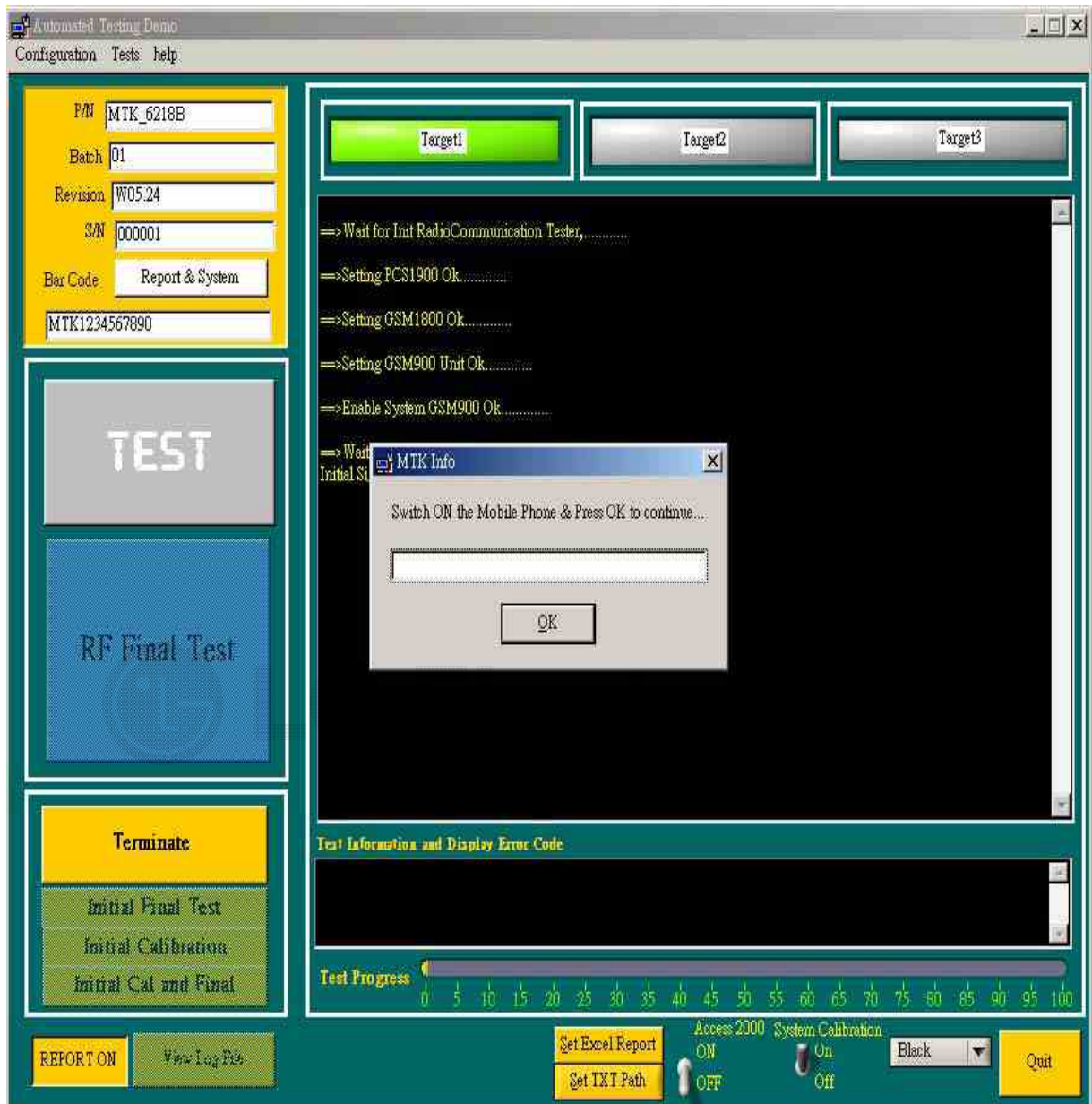
If you want final test , you can press “initial final test “



Press “RF Final test”



1. Handset to insert SIM card
2. Key-in bar code or IMEI number
3. Power on handset



ATE start final test

Automated Testing Demo
Configuration Tests help

P/N: MTK_6218B
Batch: 01
Revision: W05.24
S/N: 000001
Bar Code: Report & System
zz999

TEST

RF Final Test

Terminate

Initial Final Test
Initial Calibration
Initial Cal and Final

REPORT ON View Log File

Set Excel Report
Set TXT Path

Access 2000 ON
System Calibration On
Black

Quit

Target1 Target2 Target3

GSM Band TCH 124, PCL 5

Avg. Burst Power (Avg.) [dBm] = 32.228660 Pass
Peak Burst Power [dBm] = 32.228660 Pass
Burst Power Matching = 0 Pass
Maximum phase error peak [deg] = 5.259489 Pass
Maximum phase error RMS [deg] = 2.343793 Pass
Maximum frequency error [Hz] = -20.146050 Pass
Timing Advance error = 0.000000 Pass
Rx Level = 29 Pass
Rx Quality = 0 Pass
Class II = 0.029904 Pass
Class Ib = 0.000000 Pass
Modulation +400kHz = -66.839870
Modulation -400kHz = -65.896740 PASS
Modulation +600kHz = -68.604150
Modulation -600kHz = -69.584160 PASS
Modulation +1.2MHz = -71.047200
Modulation -1.2MHz = -70.857630 PASS
Modulation +1.8MHz = -78.453870
Modulation -1.8MHz = -79.476660 PASS
Modulation = 0 Pass
Switching +400kHz = -31.559020
Switching -400kHz = -30.744700 PASS
Switching +1.8MHz = -43.066520
Switching -1.8MHz = -40.874710 PASS
Switching = 0 Pass

Test Information and Display Error Code

Test Progress 0 5 10 15 20 25 30 35 40 45 50 55 60 65 70 75 80 85 90 95 100

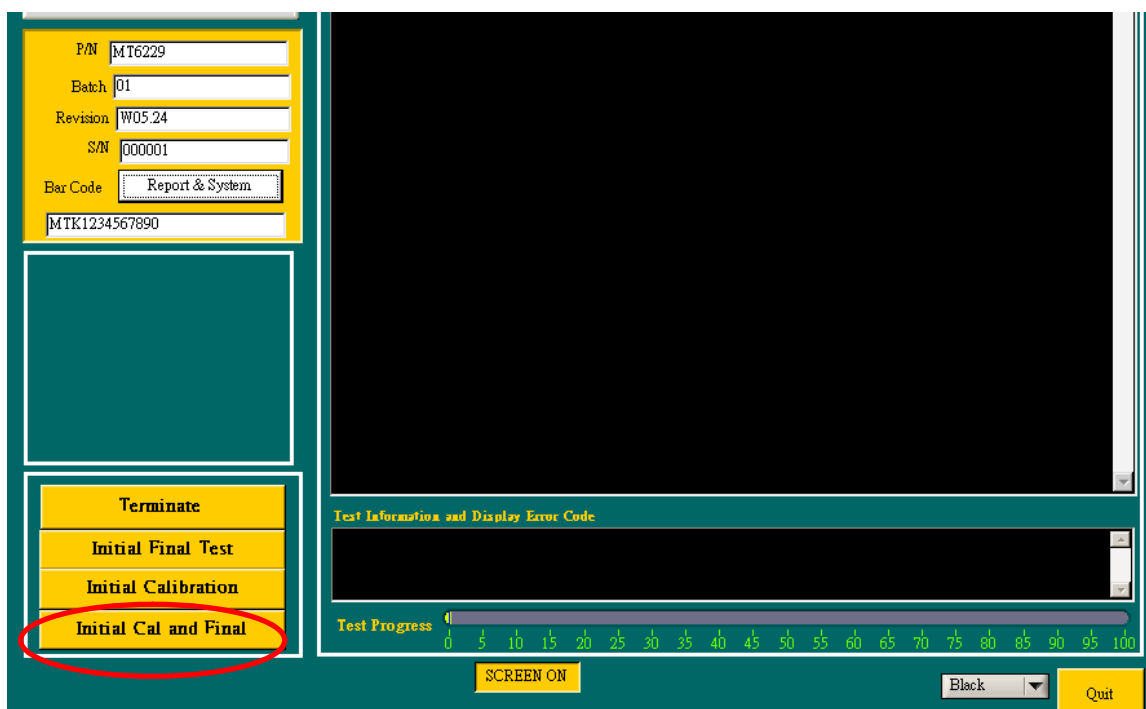
If ATE test finish , ATE will show pass



You can see the test report

```
-----  
ATE Tool Version:5.0.3  
Part Number: MTK_6218B  
Serial Number: 000001  
Revision: W05.24  
Batch: 01  
Bar Code: qqg  
Error Code: 000  
-----  
  
==>Wait GSM Location Update .....  
Enter into META Mode OK  
AFC Calibration OK  
Slope=2.824,min:1.000,max:10.000  
Use Default Value=3836  
AFC Calibration time=1.64(sec)  
PL GSM TCH 15 = 1.25 Pass MAX:3.00 MIN:-3.00  
PL GSM TCH 30 = 1.00 Pass MAX:3.00 MIN:-3.00  
PL GSM TCH 45 = 0.88 Pass MAX:3.00 MIN:-3.00  
PL GSM TCH 60 = 1.25 Pass MAX:3.00 MIN:-3.00  
PL GSM TCH 75 = 1.38 Pass MAX:3.00 MIN:-3.00  
PL GSM TCH 80 = 1.50 Pass MAX:3.00 MIN:-3.00  
PL GSM TCH 100 = 1.25 Pass MAX:3.00 MIN:-3.00  
PL GSM TCH 124 = 1.25 Pass MAX:3.00 MIN:-3.00  
PL GSM TCH 975 = 1.50 Pass MAX:3.00 MIN:-3.00  
PL GSM TCH 1000 = 1.38 Pass MAX:3.00 MIN:-3.00  
PL GSM TCH 1023 = 1.00 Pass MAX:3.00 MIN:-3.00  
PL DCS TCH 550 = 0.50 Pass MAX:3.00 MIN:-3.00  
PL DCS TCH 590 = 1.00 Pass MAX:3.00 MIN:-3.00
```

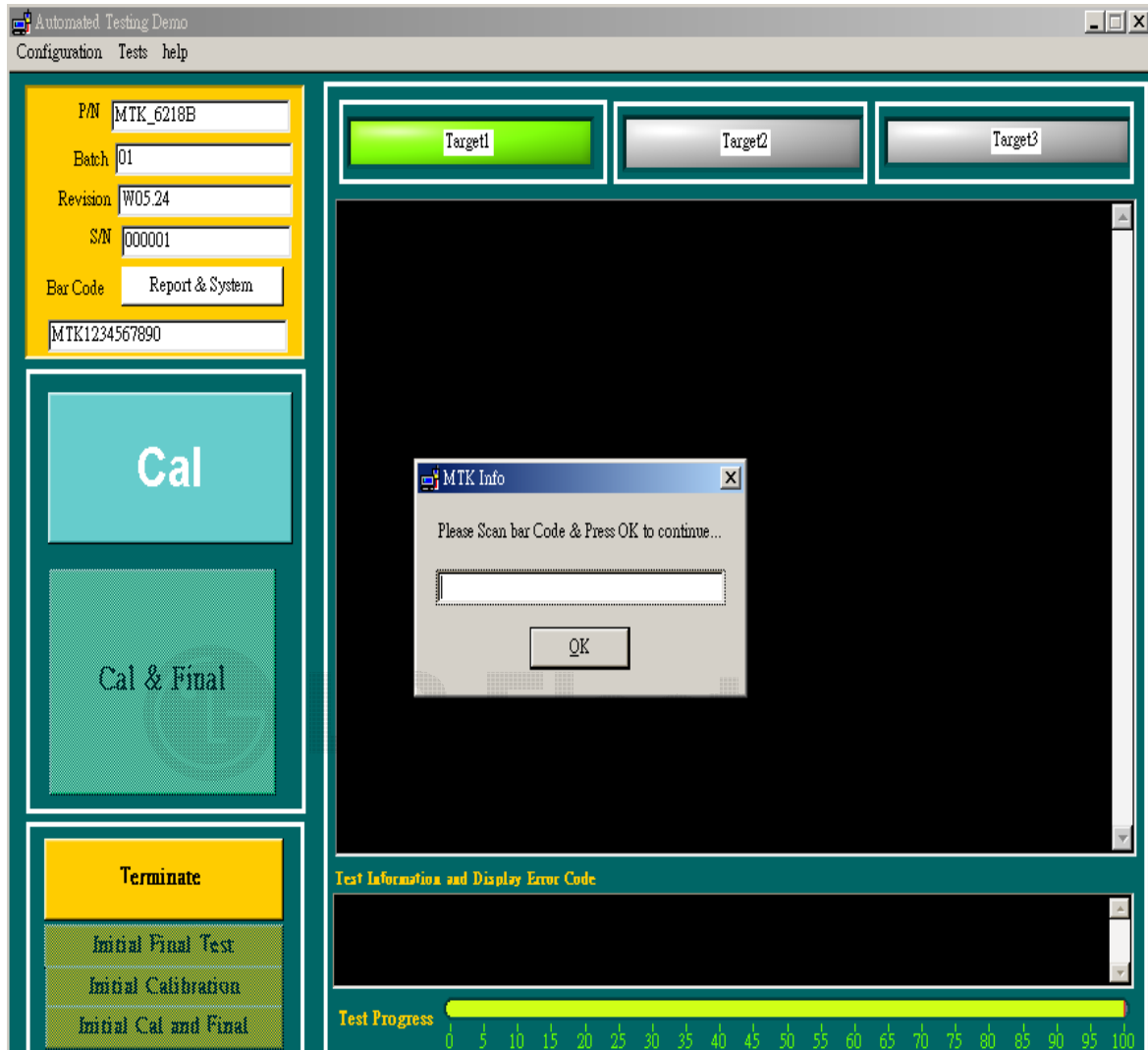
If you want initial cal and final test , you can press “initial cal and final test”



Press “Cal & Final”



- 1.Handset to insert SIM card
- 2.Key-in bar code or IMEI number
- 3.Power on handset



Start calibration

Automated Testing Demo
Configuration Tests help

P/N MTK_6218B

Batch 01

Revision W05.24

S/N 000001

Bar Code Report & System

zz999

Cal

Cal & Final

Terminate

Initial Final Test

Initial Calibration

Initial Cal and Final

Target1

Target2

Target3

```

PL DCS TCH 710 = 0.00 Pass MAX:3.00 MIN:-3.00
PL DCS TCH 740 = 0.00 Pass MAX:3.00 MIN:-3.00
PL DCS TCH 770 = 0.25 Pass MAX:3.00 MIN:-3.00
PL DCS TCH 810 = 0.50 Pass MAX:3.00 MIN:-3.00
PL DCS TCH 850 = 0.38 Pass MAX:3.00 MIN:-3.00
PL DCS TCH 885 = 0.50 Pass MAX:3.00 MIN:-3.00
PL PCS TCH 550 = 1.25 Pass MAX:3.00 MIN:-3.00
PL PCS TCH 590 = 1.12 Pass MAX:3.00 MIN:-3.00
PL PCS TCH 620 = 1.12 Pass MAX:3.00 MIN:-3.00
PL PCS TCH 650 = 1.12 Pass MAX:3.00 MIN:-3.00
PL PCS TCH 680 = 1.25 Pass MAX:3.00 MIN:-3.00
PL PCS TCH 710 = 1.25 Pass MAX:3.00 MIN:-3.00
PL PCS TCH 740 = 1.50 Pass MAX:3.00 MIN:-3.00
PL PCS TCH 770 = 1.38 Pass MAX:3.00 MIN:-3.00
PL PCS TCH 810 = 1.62 Pass MAX:3.00 MIN:-3.00
Path Loss Calibration OK
Pathloss Calibration time=8.31(sec)
-----GSM900 APC Cal-----
delta s = 0
Cal APC Power:19.03
delta s = 0
Cal APC Power:32.24
-----DCS1800 APC Cal-----
delta s = 0
Cal APC Power:13.96
delta s = 0
Cal APC Power:29.20
-----PCS1900 APC Cal-----
  
```

Test Information and Display Error Code

Test Progress 0 5 10 15 20 25 30 35 40 45 50 55 60 65 70 75 80 85 90 95 100

Calibration finish and power on handset again

Automated Testing Demo
Configuration Tests help

P/N MTK_6218B
Batch 01
Revision W05.24
S/N 000001
Bar Code Report & System
999

Target1

Target2

Target3

Enter into META Mode OK
AFC Calibration OK
Slope=2.816,min:1.000,max:10.000
Use Default Value=3803
AFC Calibration time=1.67(sec)
PL GSM TCH 15 = 1.25 Pass MAX:3.00 MIN:-3.00
PL GSM TCH 30 = 1.12 Pass MAX:3.00 MIN:-3.00
PL GSM TCH 45 = 0.88 Pass MAX:3.00 MIN:-3.00
PL GSM TCH 60 = 1.12 Pass MAX:3.00 MIN:-3.00
PL GSM TCH 75 = 1.50 Pass MAX:3.00 MIN:-3.00
PL GSM TCH 80 = 1.50 Pass MAX:3.00 MIN:-3.00
PL GSM TCH 100 = 1.38 Pass MAX:3.00 MIN:-3.00
PL GSM TCH 124 = 1.38 Pass MAX:3.00 MIN:-3.00
PL GSM TCH 975 = 1.50 Pass MAX:3.00 MIN:-3.00
PL GSM TCH 1000 = 1.38 Pass MAX:3.00 MIN:-3.00
PL GSM TCH 1023 = 1.12 Pass MAX:3.00 MIN:-3.00
PL DCS TCH 550 = 0.62 Pass MAX:3.00 MIN:-3.00
PL DCS TCH 590 = 1.12 Pass MAX:3.00 MIN:-3.00
PL DCS TCH 620 = 1.00 Pass MAX:3.00 MIN:-3.00
PL DCS TCH 650 = 0.62 Pass MAX:3.00 MIN:-3.00
PL DCS TCH 680 = 0.25 Pass MAX:3.00 MIN:-3.00
PL DCS TCH 710 = 0.12 Pass MAX:3.00 MIN:-3.00
PL DCS TCH 740 = 0.12 Pass MAX:3.00 MIN:-3.00
PL DCS TCH 770 = 0.25 Pass MAX:3.00 MIN:-3.00
PL DCS TCH 810 = 0.62 Pass MAX:3.00 MIN:-3.00
PL DCS TCH 850 = 0.38 Pass MAX:3.00 MIN:-3.00
PL DCS TCH 885 = 0.62 Pass MAX:3.00 MIN:-3.00

TEST

Cal & Final

Terminate

Initial Final Test

Initial Calibration

Initial Cal and Final

Test Information and Display Error Code

Test Progress
0 5 10 15 20 25 30 35 40 45 50 55 60 65 70 75 80 85 90 95 100

REPORT ON
View Log File

Set Excel Report
Set TXT Path

Access 2000 ON
System Calibration On
Black

Quit

Start final test

Automated Testing Demo
Configuration Tests help

P/N MTK_6218B
Batch 01
Revision W05.24
S/N 000001
Bar Code Report & System
zz999

Cal

Cal & Final

Terminate

Initial Final Test
Initial Calibration
Initial Cal and Final

Target1 Target2 Target3

PL DCS TCH 710 = 0.00 Pass MAX:3.00 MIN:-3.00
PL DCS TCH 740 = 0.00 Pass MAX:3.00 MIN:-3.00
PL DCS TCH 770 = 0.25 Pass MAX:3.00 MIN:-3.00
PL DCS TCH 810 = 0.50 Pass MAX:3.00 MIN:-3.00
PL DCS TCH 850 = 0.38 Pass MAX:3.00 MIN:-3.00
PL DCS TCH 885 = 0.50 Pass MAX:3.00 MIN:-3.00
PL PCS TCH 550 = 1.25 Pass MAX:3.00 MIN:-3.00
PL PCS TCH 590 = 1.12 Pass MAX:3.00 MIN:-3.00
PL PCS TCH 620 = 1.12 Pass MAX:3.00 MIN:-3.00
PL PCS TCH 650 = 1.12 Pass MAX:3.00 MIN:-3.00
PL PCS TCH 680 = 1.25 Pass MAX:3.00 MIN:-3.00
PL PCS TCH 710 = 1.25 Pass MAX:3.00 MIN:-3.00
PL PCS TCH 740 = 1.50 Pass MAX:3.00 MIN:-3.00
PL PCS TCH 770 = 1.38 Pass MAX:3.00 MIN:-3.00
PL PCS TCH 810 = 1.62 Pass MAX:3.00 MIN:-3.00
Path Loss Calibration OK
Pathloss Calibration time=8.31(sec)
-----GSM900 APC Cal-----
delta s = 0
Cal APC Power:19.03
delta s = 0
Cal APC Power:32.24
-----DCS1800 APC Cal-----
delta s = 0
Cal APC Power:13.96
delta s = 0
Cal APC Power:29.20
-----PCS1900 APC Cal-----

Test Information and Display Error Code

Test Progress 0 5 10 15 20 25 30 35 40 45 50 55 60 65 70 75 80 85 90 95 100

Finish “Cal & Final test”

Automated Testing Demo
Configuration Tests help

P/N MTK_6218B
Batch 01
Revision W05.24
S/N 000001
Bar Code Report & System
qq88

Target1 Target2 Target3

PCS Band TCH 810, PCL 0
Avg. Burst Power (Avg.) [dBm] = 29.143900 Pass
Peak Burst Power [dBm] = 29.143900 Pass

PASS

Cal & Final

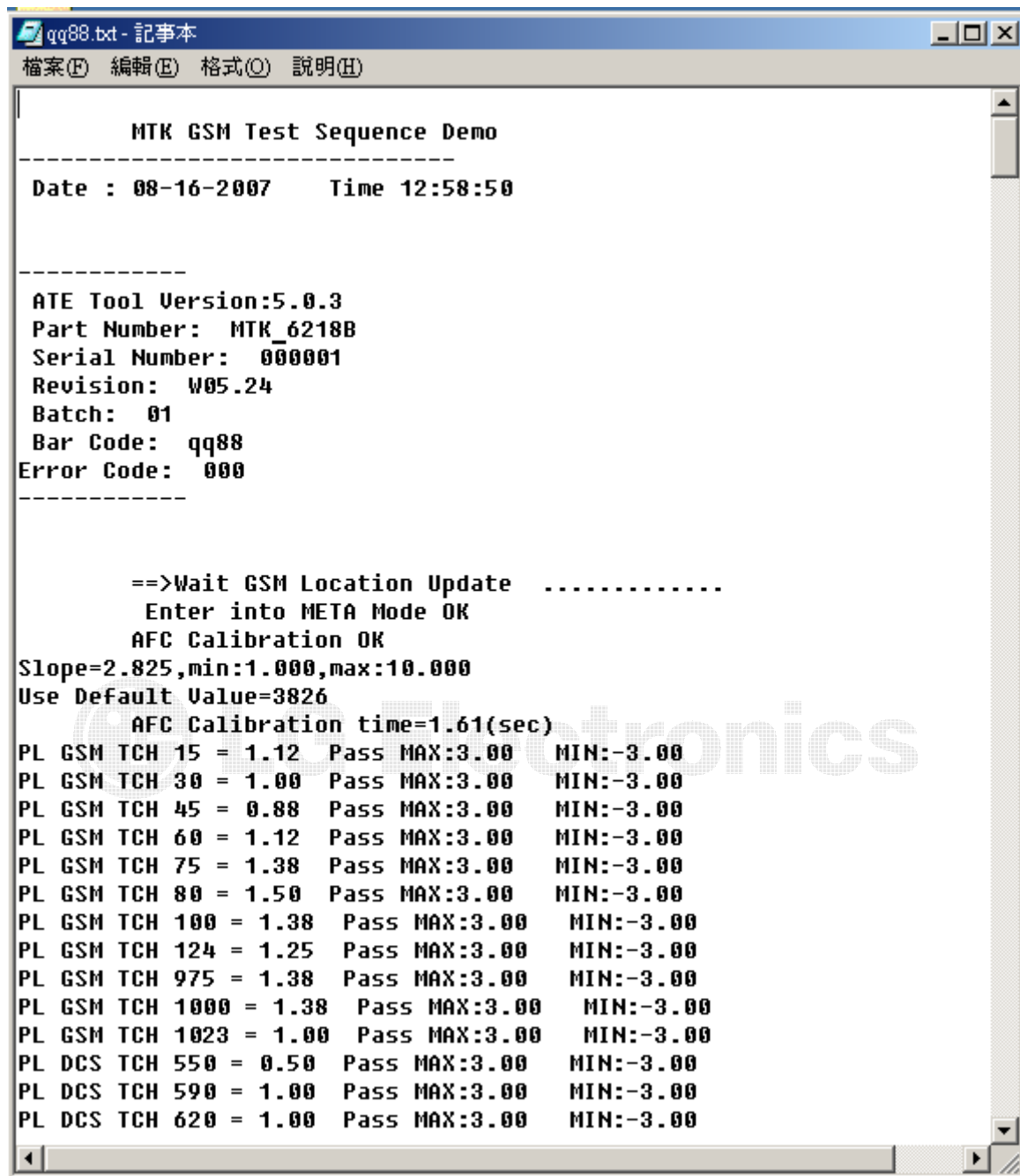
Terminate
Initial Final Test
Initial Calibration
Initial Cal and Final

Test Information and Display Error Code

Test Progress 0 5 10 15 20 25 30 35 40 45 50 55 60 65 70 75 80 85 90 95 100

REPORT ON View Log File Set Excel Report Set TXT Path Access 2000 ON OFF System Calibration On Off Black Quit

Ate show the test report



```
qq88.txt - 記事本
檔案(F) 編輯(E) 格式(O) 說明(H)

MTK GSM Test Sequence Demo
-----
Date : 08-16-2007    Time 12:58:50
-----

ATE Tool Version:5.0.3
Part Number:  MTK_6218B
Serial Number:  000001
Revision:  W05.24
Batch:  01
Bar Code:  qq88
Error Code:  000
-----

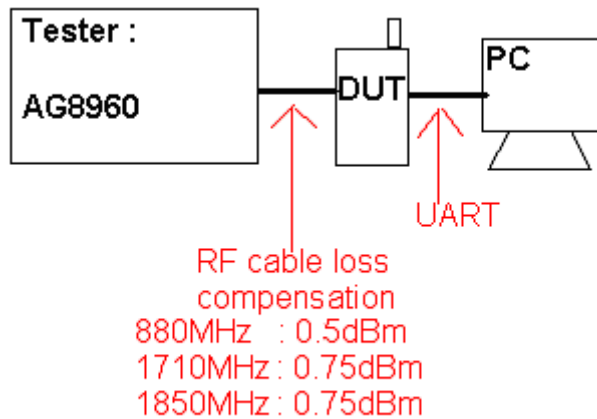
==>Wait GSM Location Update .....
Enter into META Mode OK
AFC Calibration OK
Slope=2.825,min:1.000,max:10.000
Use Default Value=3826
AFC Calibration time=1.61(sec)
PL GSM TCH 15 = 1.12  Pass MAX:3.00  MIN:-3.00
PL GSM TCH 30 = 1.00  Pass MAX:3.00  MIN:-3.00
PL GSM TCH 45 = 0.88  Pass MAX:3.00  MIN:-3.00
PL GSM TCH 60 = 1.12  Pass MAX:3.00  MIN:-3.00
PL GSM TCH 75 = 1.38  Pass MAX:3.00  MIN:-3.00
PL GSM TCH 80 = 1.50  Pass MAX:3.00  MIN:-3.00
PL GSM TCH 100 = 1.38  Pass MAX:3.00  MIN:-3.00
PL GSM TCH 124 = 1.25  Pass MAX:3.00  MIN:-3.00
PL GSM TCH 975 = 1.38  Pass MAX:3.00  MIN:-3.00
PL GSM TCH 1000 = 1.38  Pass MAX:3.00  MIN:-3.00
PL GSM TCH 1023 = 1.00  Pass MAX:3.00  MIN:-3.00
PL DCS TCH 550 = 0.50  Pass MAX:3.00  MIN:-3.00
PL DCS TCH 590 = 1.00  Pass MAX:3.00  MIN:-3.00
PL DCS TCH 620 = 1.00  Pass MAX:3.00  MIN:-3.00
```

12. STAND ALONE TEST

12.1 RF TX & RX Test :

Test Configuration & Expected Outcome

Test Configuration :



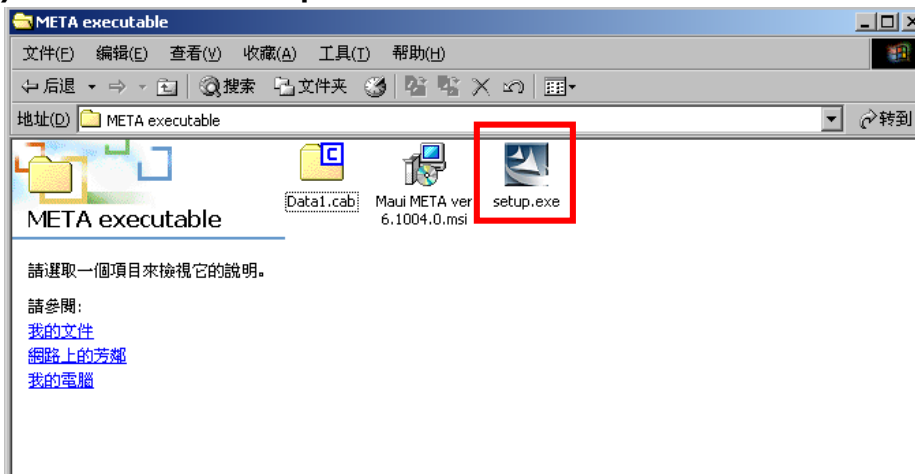
Expected Outcome :

TX	power	: 32.5	+/-	1.5 dBm	for GSM900
TX	power	: 29.5	+/-	1.5 dBm	for DCS1800
RX	power	: -85	+/-	4 dBm	for GSM900, DCS1800

12.2 META Install & RF TX & RX Check

META Tool Install process :

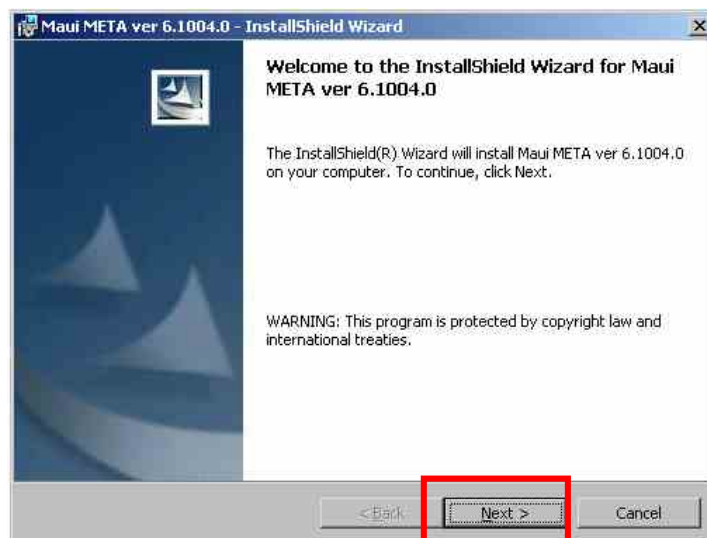
(1) Double click “setup.exe”



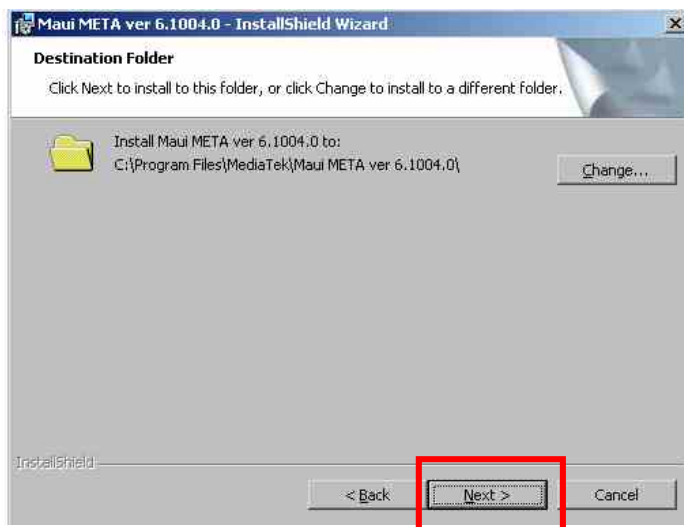
(2) Install Process



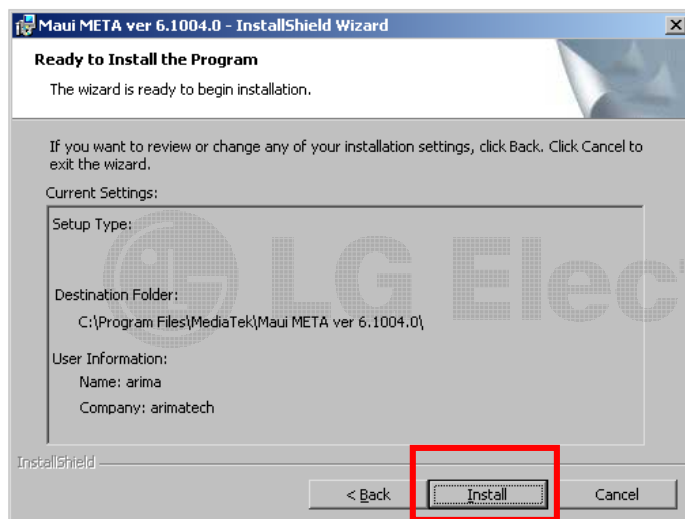
(3) Install Process – press “Next”



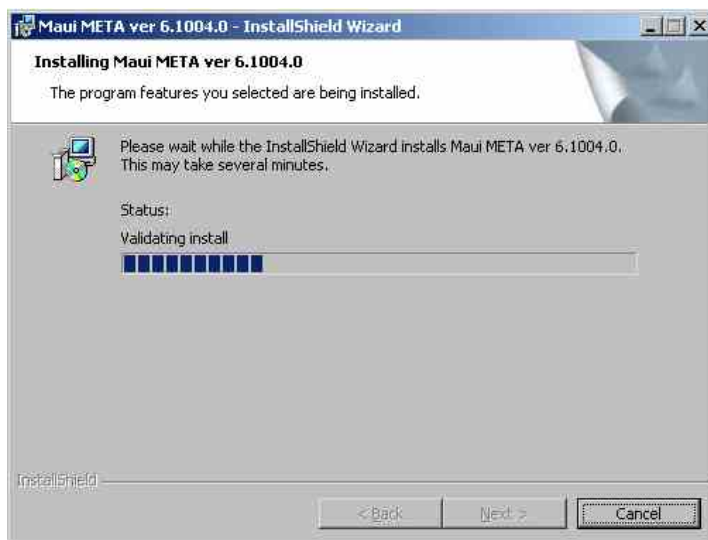
(4) Install Process – press “Next”



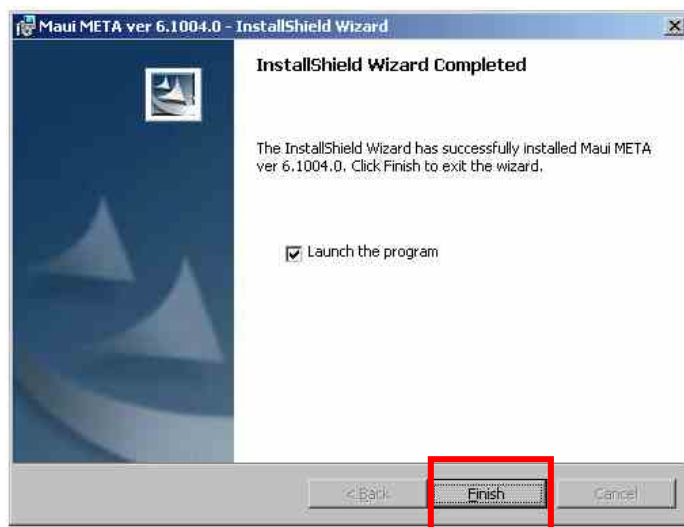
(5) Install Process – press “Install”



(6) Install Process



(7) Install Process – press “Finish”



12.3 RF RX Check:

(1) Open “ Meta_RF_Tool ”



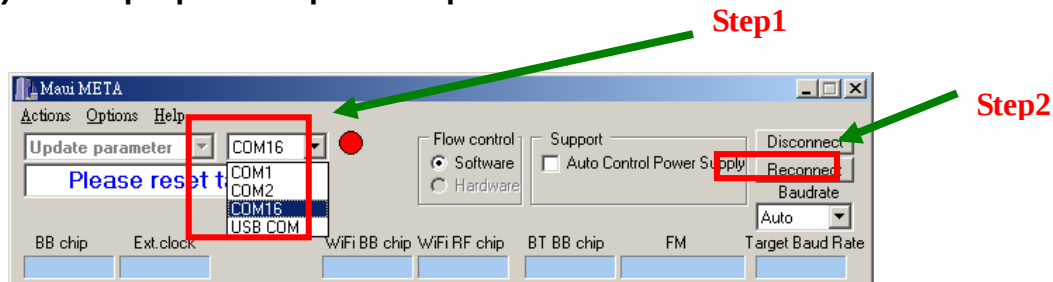
(2) Pull in UART cable



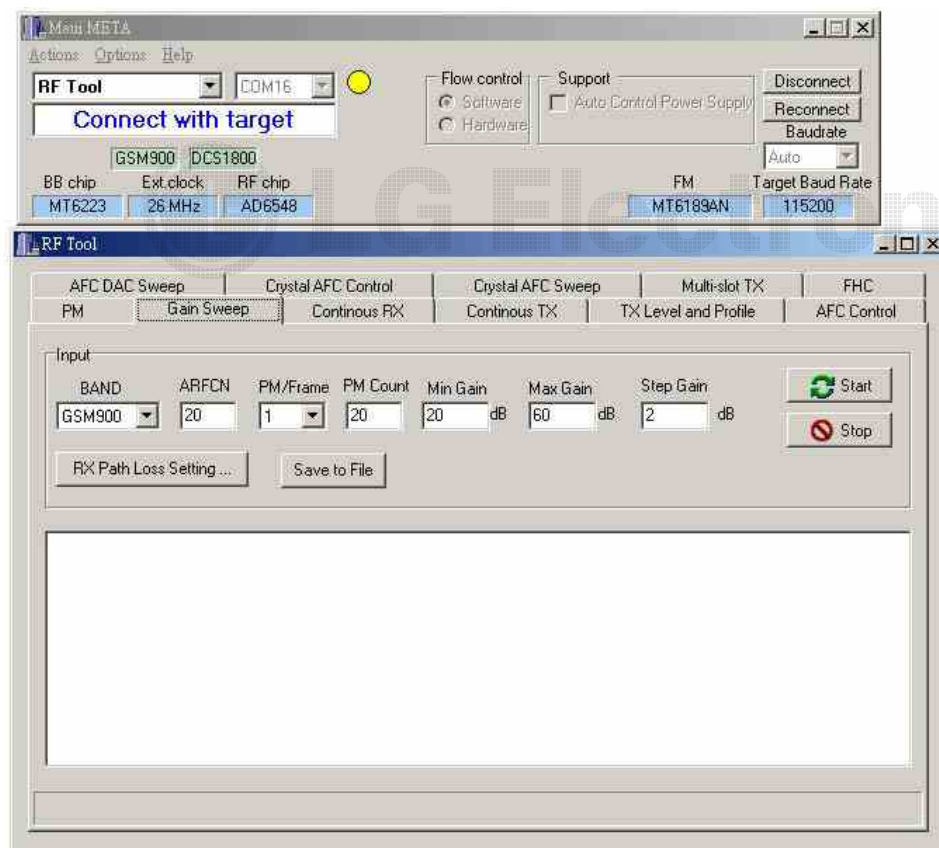
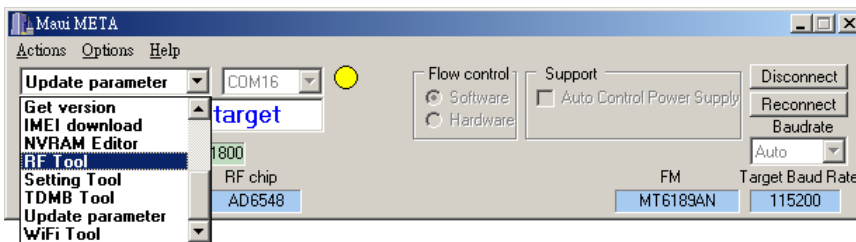
(3) Inset RF-Cable (AG8960)



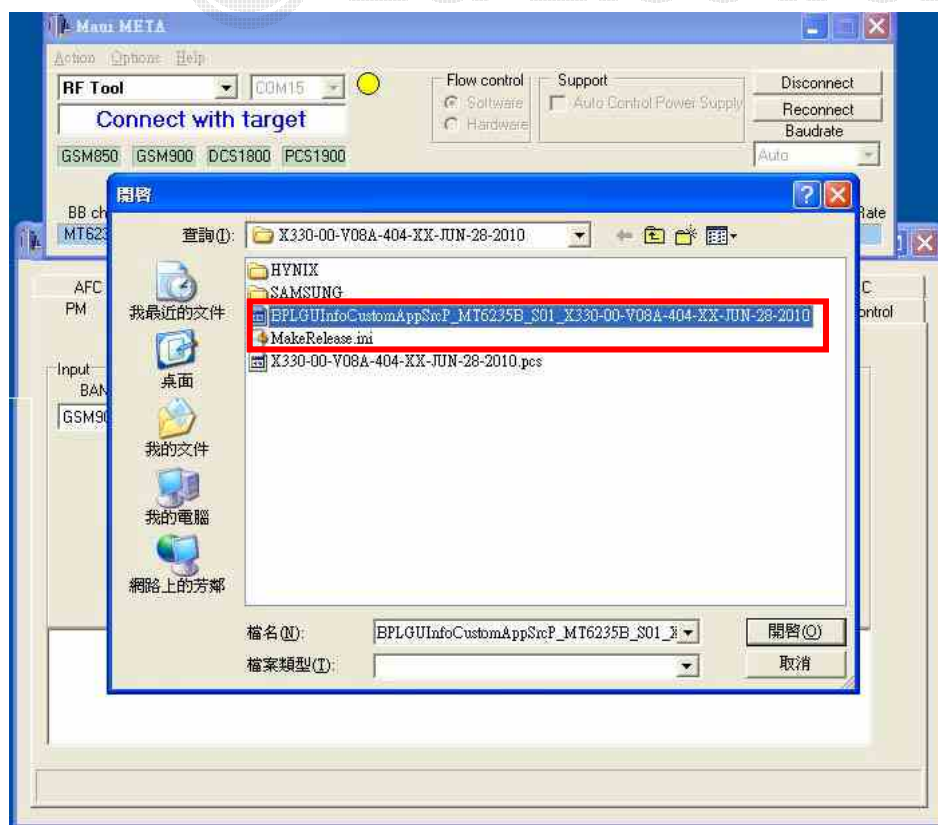
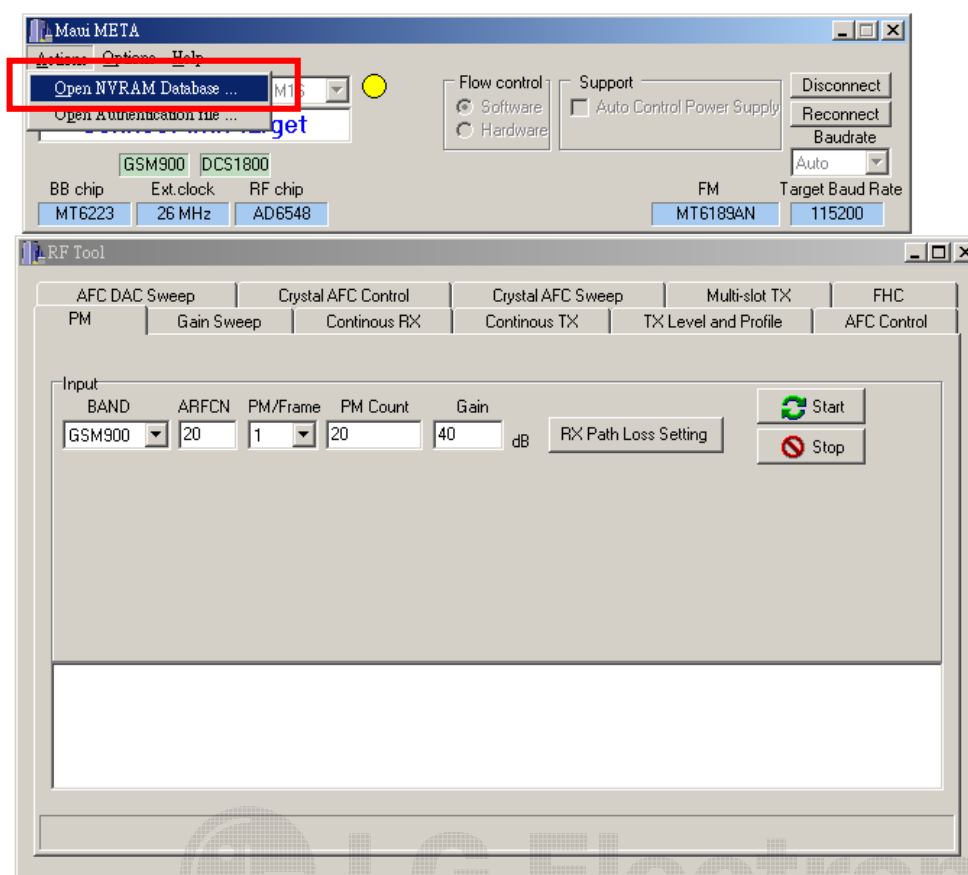
(4) Select proper com port and press “Reconnect” .



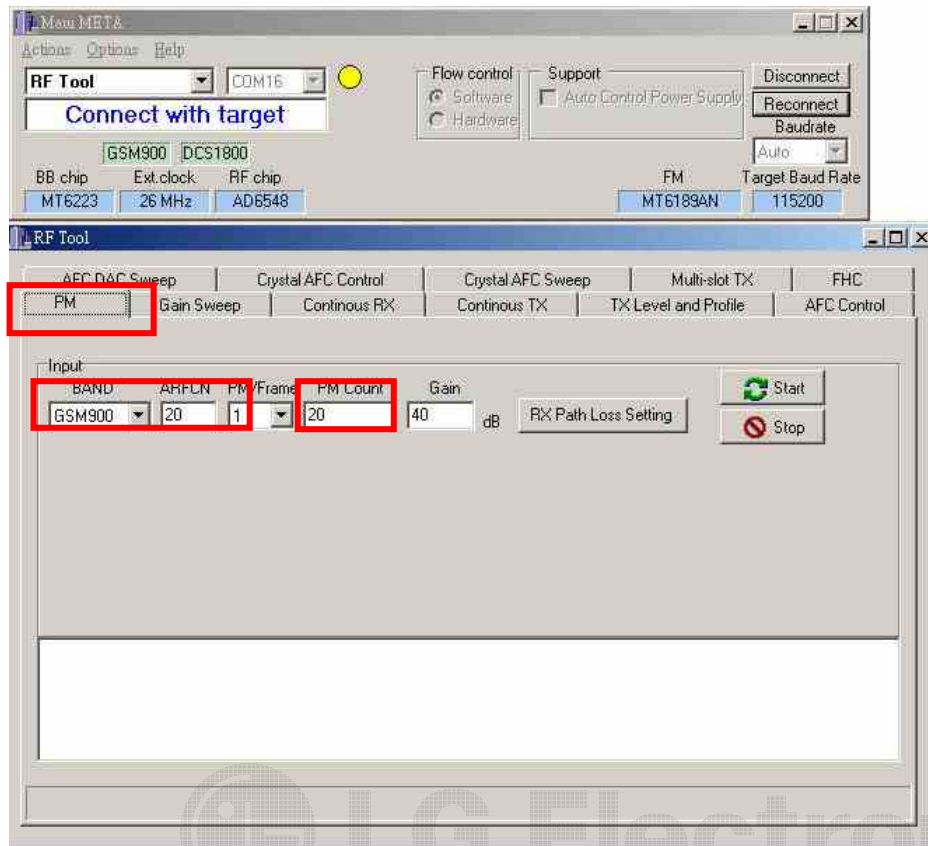
(5) Press handset's power key and it will show LG logo. Then appear the following picture. Select RF Tool.



(6) Loading database Make sure the same to handset.

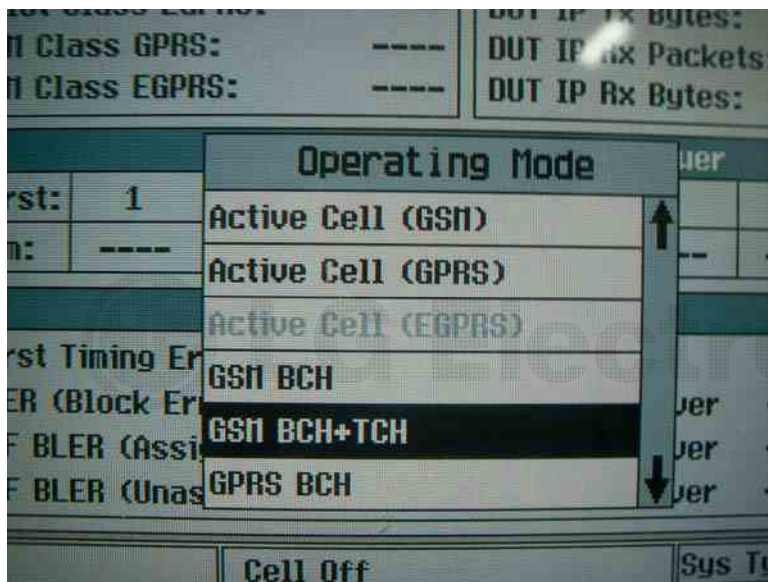
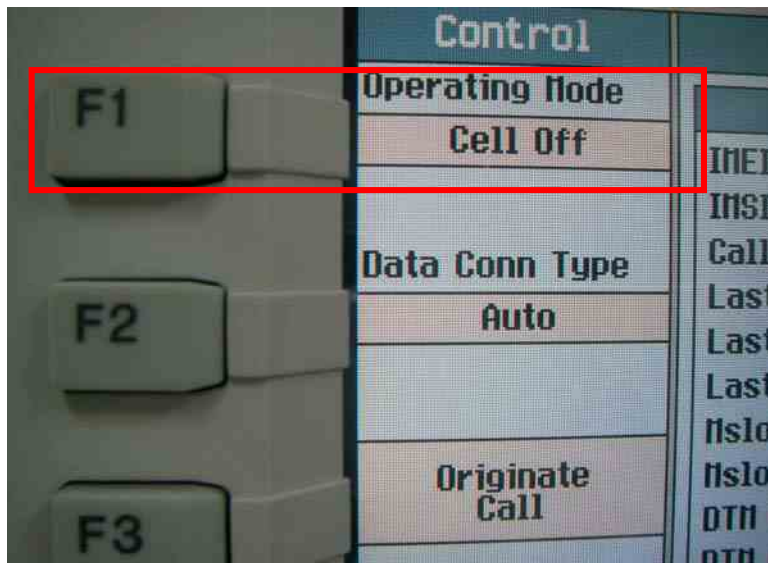


(7) Press “PM” to RX check. Select proper Band, ARFCN and PM Count.



(8) Setup AG8960: Press CALL SETUP, than press “F1”, and select “GSM BCH+TCH”.

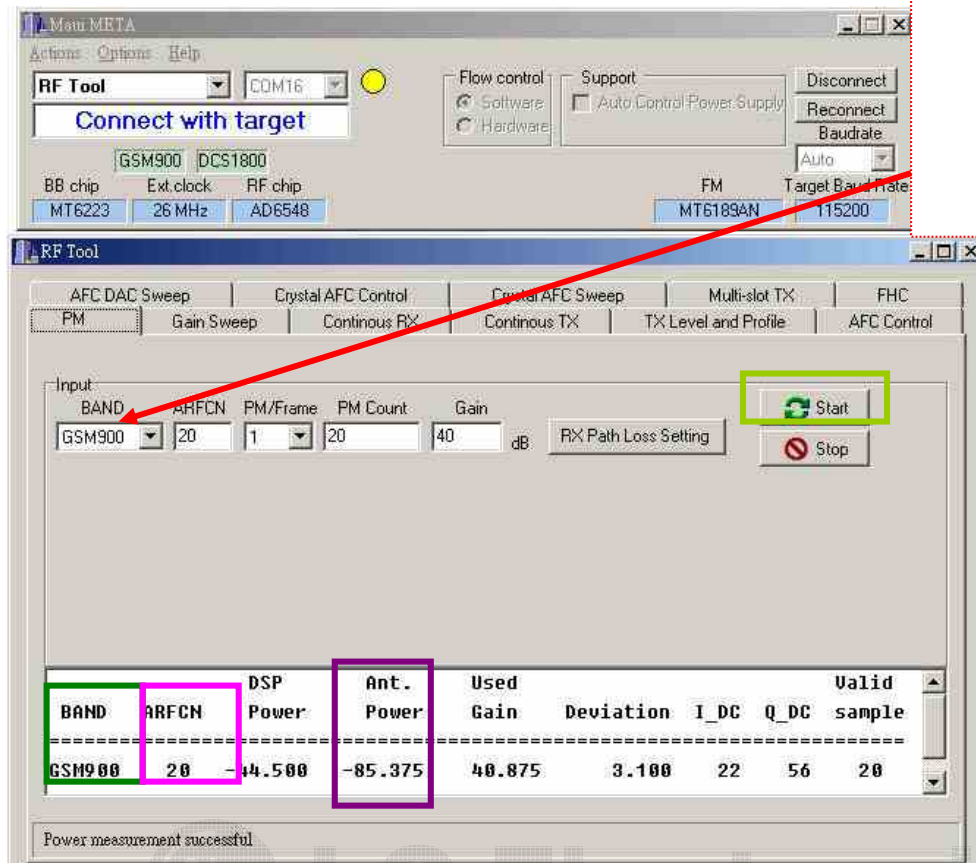




(9) RX Test (AG8960)--- Set "BCH Params" ,



(10) RX Test --- Press Start



1. Select Band (GSM/DCS)
2. Set channel number
3. Press "Start"

12.4 RF TX Check :

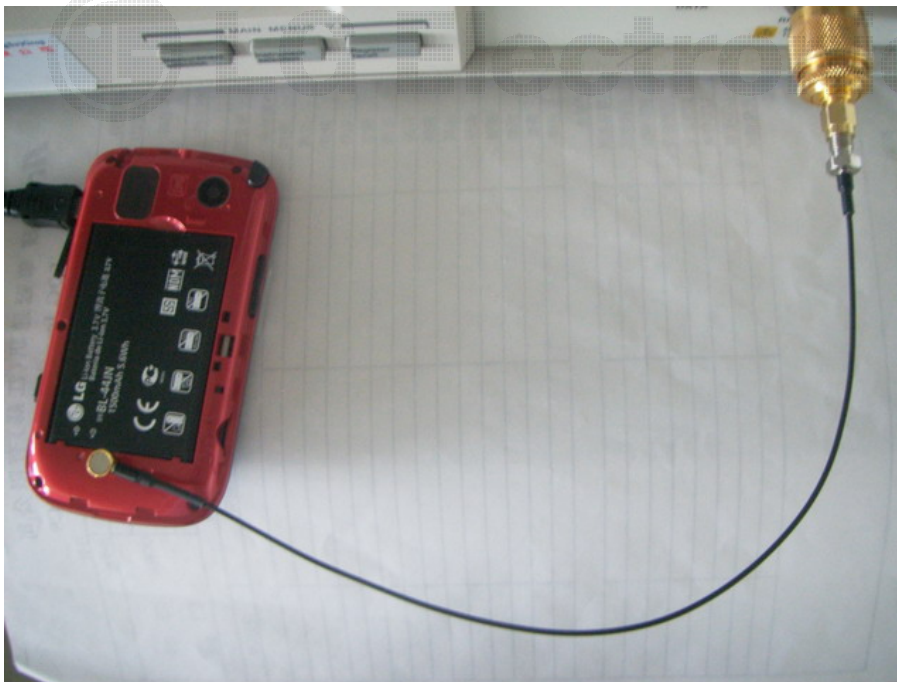
(1) Open “ Meta_RF_Tool ”.



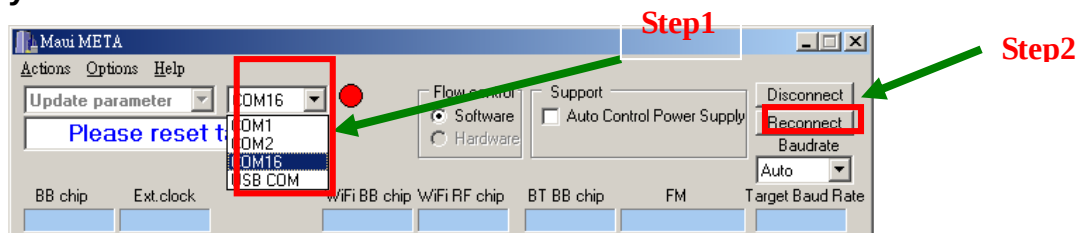
(2) Pull in UART cable.



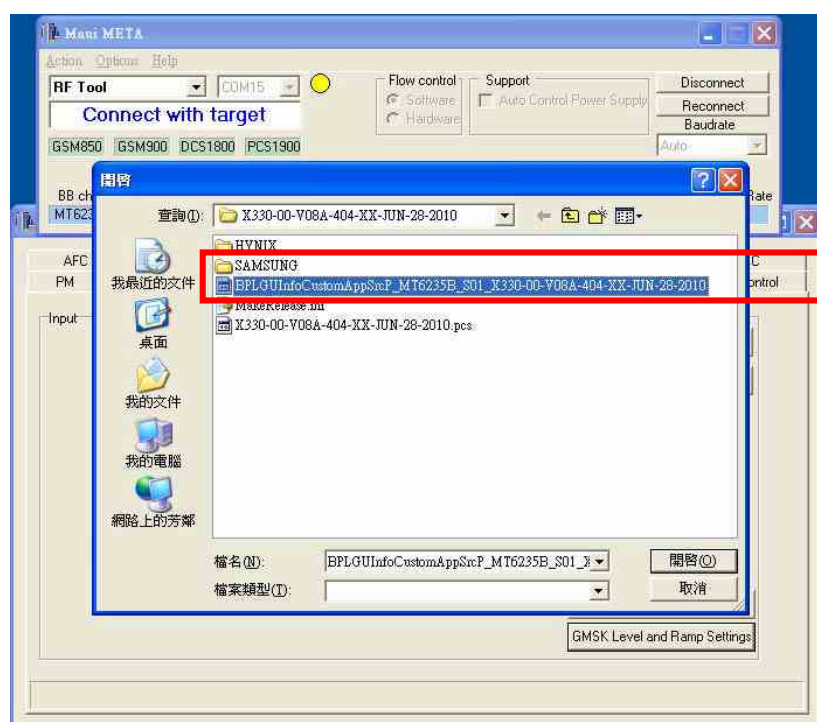
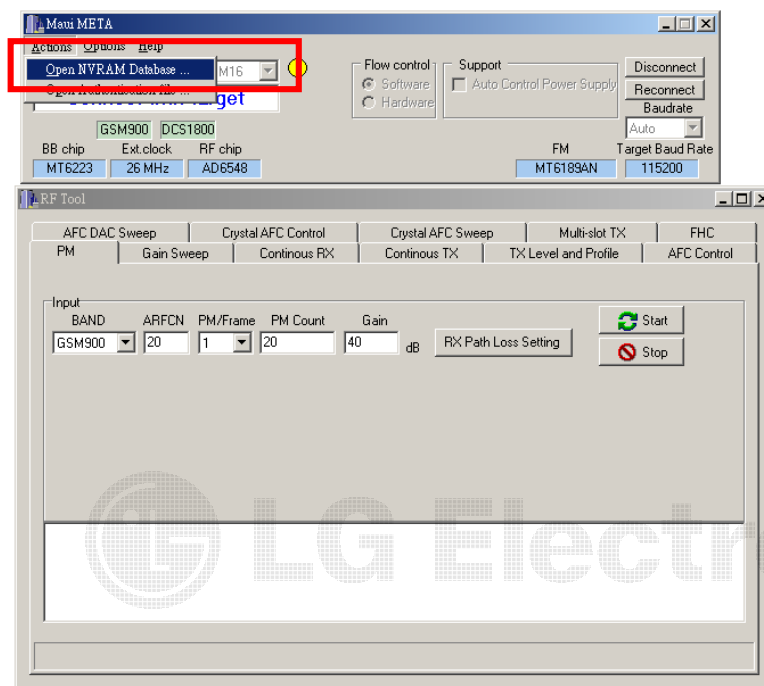
(3) Inset RF-Cable (AG8960).



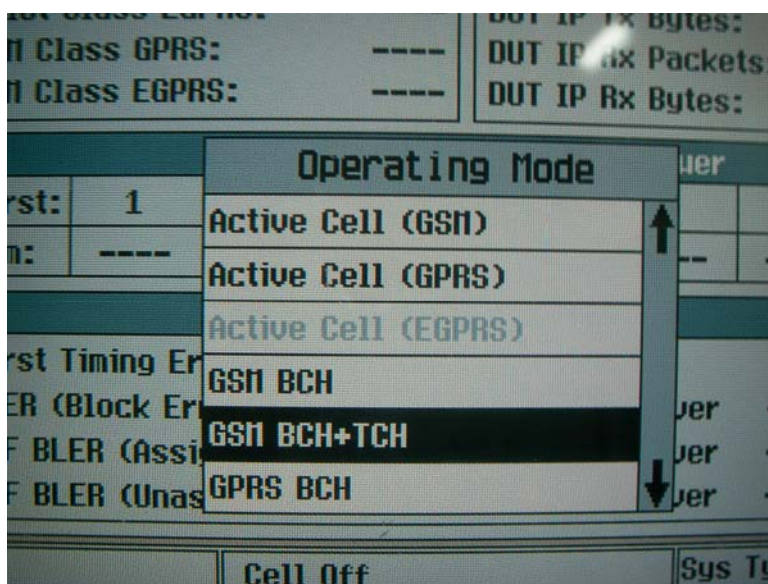
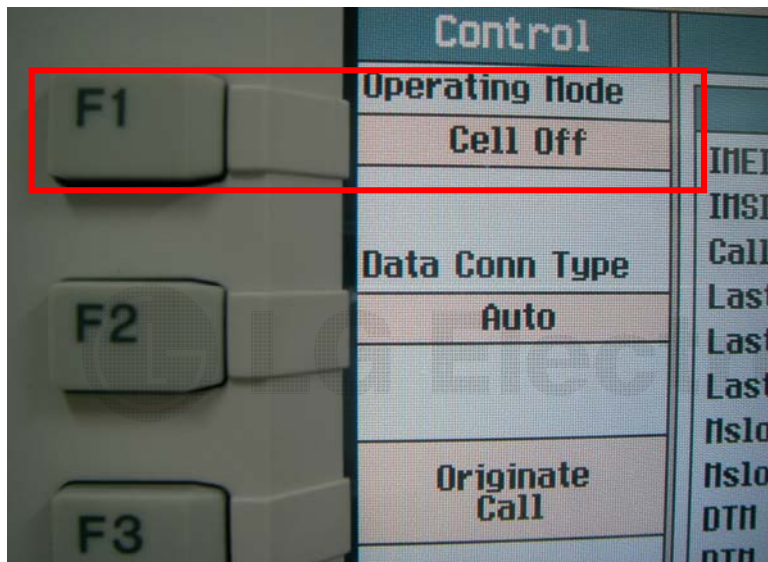
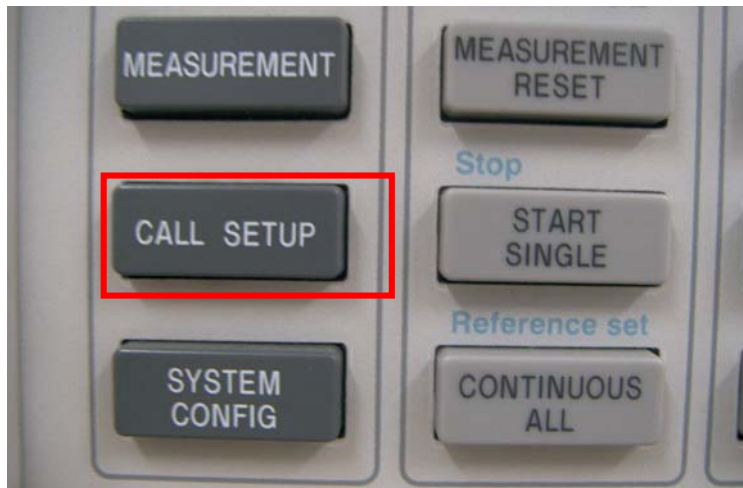
(4) Select proper com port and press “Reconnect” and then press handset’s power key.



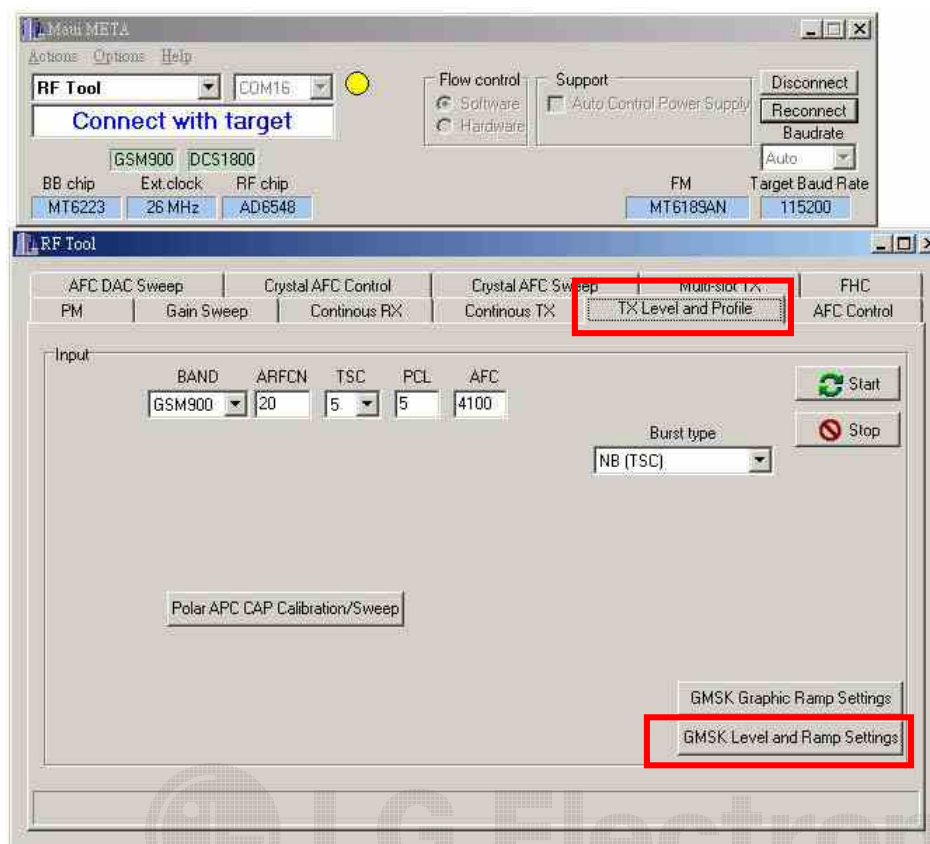
(5) Loading database Make sure the same to handset.



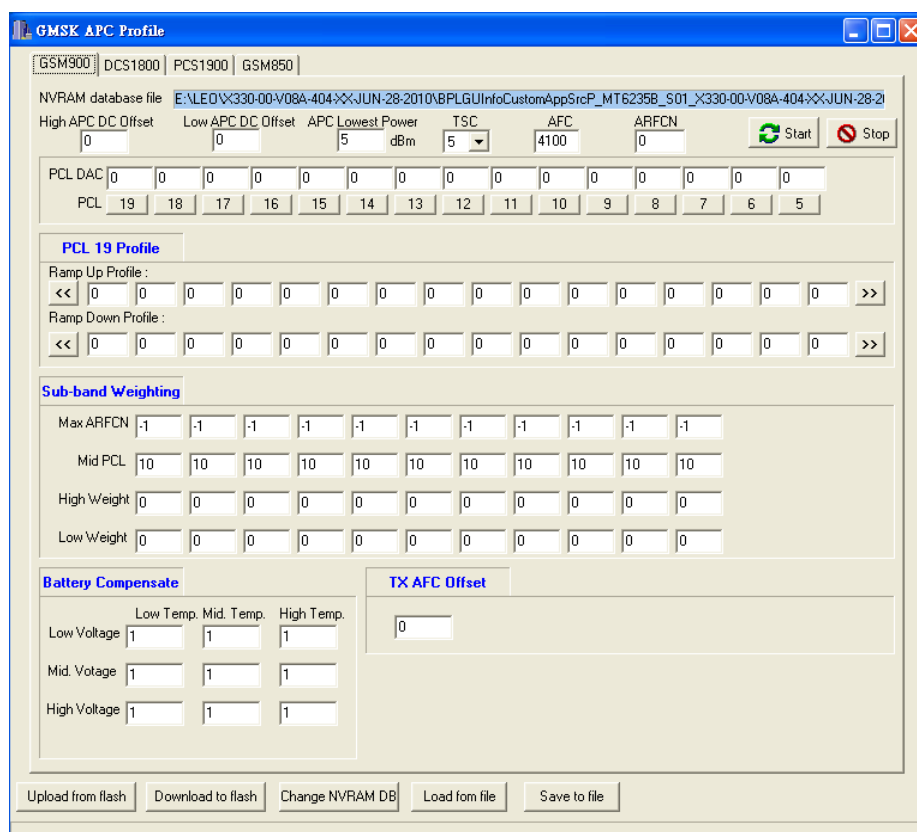
(6) AG8960 need to set TCH Params.

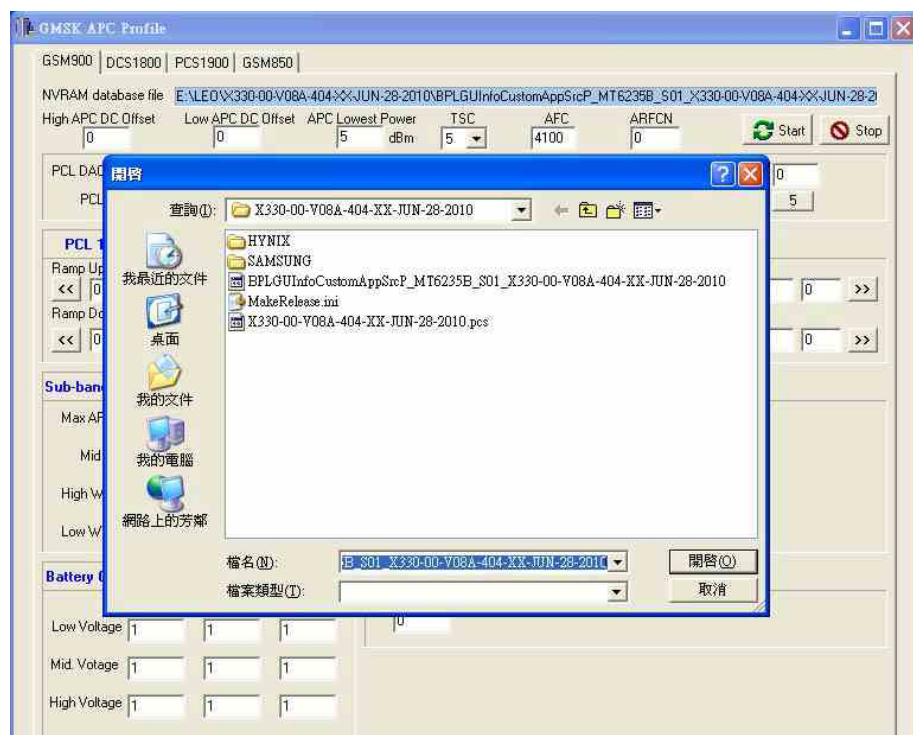


(7) Press “TX level and profile” to TX Test, and click “GMSK Level and Ramp settings”.

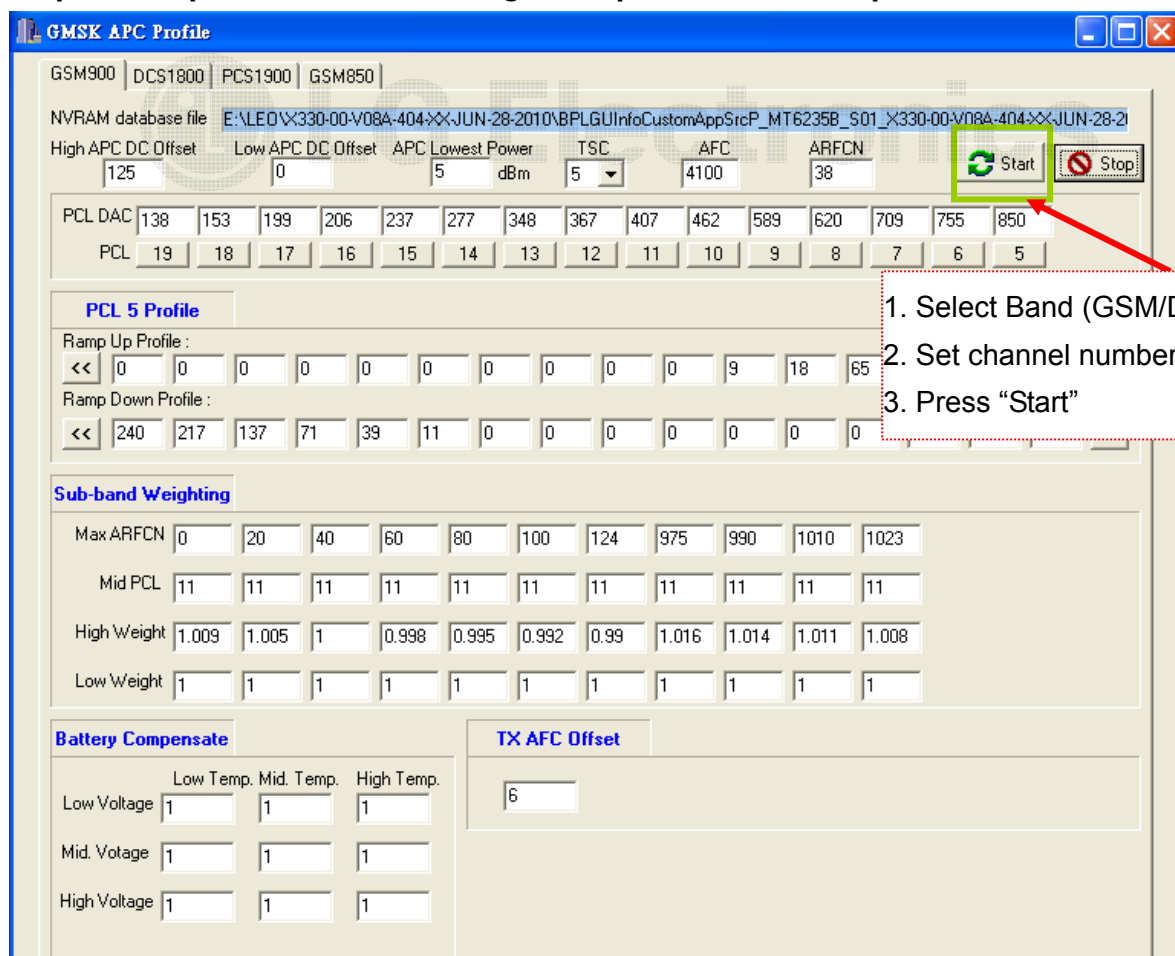


(8) Select “Change NVRAM DB” to choose the right database of the mobile.





Then press “Upload from flash” to get the parameters, and press “Start”



(9) TX Test (AG8960) --- Need set “Band”, “Channel” & “Power Level”

GSMK APC Profile

GSM900 | CS1800 | PCS1900 | GSM850

NVRAM database file: E:\LED\X330-00-V08A-404\XJUN-28-2010\BPLGUIInfoCustomAppSrcP_MT6235B_S01_X330-00-V08A-404\XJUN-28-2

High APC DC Offset: 125 Low APC DC Offset: 0 APC Lowest Power: 5 dBm TSC: 5 AFC: 4100 ARFCN: 38 Start Stop

PCL DAC	138	153	199	206	237	277	348	367	407	462	589	620	709	755	858
PCL	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5

PCL 5 Profile

Ramp Up Profile: << 0 0 0 0 0 0 0 0 0 0 0 9 18 65 116 168 245 >>

Ramp Down Profile: << 240 217 137 71 39 11 0 0 0 0 0 0 0 0 0 0 >>

Sub-band Weighting

Max ARFCN	0	20	40	60	80	100	124	975	990	1010	1023
Mid PCL	11	11	11	11	11	11	11	11	11	11	11
High Weight	1.009	1.005	1	0.998	0.995	0.992	0.99	1.016	1.014	1.011	1.008
Low Weight	1	1	1	1	1	1	1	1	1	1	1

Battery Compensate

	Low Temp.	Mid. Temp.	High Temp.
Low Voltage	1	1	1
Mid. Voltage	1	1	1
High Voltage	1	1	1

TX AFC Offset

6

COMMUNICATIONS TEST SET

Measurement / Instrument Screen

GSM/GPRS Transmit Power

Minimum	Maximum
32.55 dBm	32.55 dBm
Average	Std. Dev.
32.55 dBm	0.00 dB

10 / 10 Continuous

TCH Params

Downlink Traffic Power

Traffic Band: EGSM

Traffic Channel: 38

MS TX Level: 5

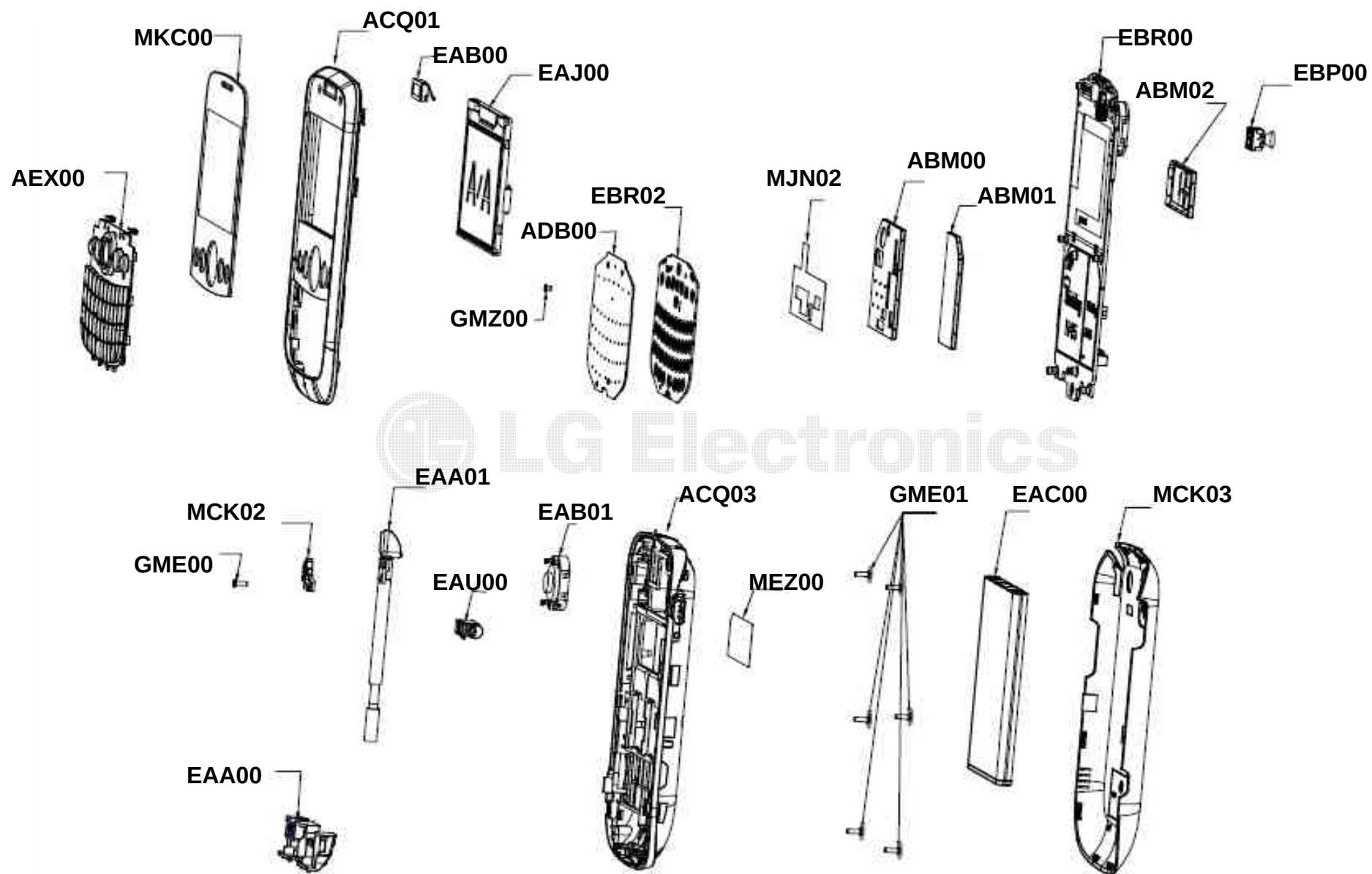
Channel Mode Setup

Return

TX Output power

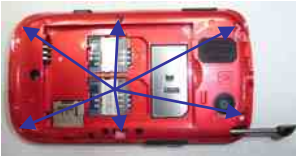
Background GSM BCH+TCH Sys Type: GSM

13. #EV#



Location No	Description	Location No	Description
MCK03	Cover, Battery	GMZ00	Machine Screw
ACQ01	Cover Assembly, Front	MEZ00	Label, Approval
EAB00	Receiver	EBR00	PCB Assembly, Main
MKC00	Window, LCD	ADB00	Dome Assembly, Metal
ACQ03	Cover Assembly, Rear	EBR02	PCB Assembly, Keypad
MCK02	Cover	EBP00	Camera Module
EAA01	Retractable Antenna, DMB	EAJ00	LCD, Module
EAA00	Retractable Antenna, Multiple	ABM00	Can Assembly, Shield
EAU00	Motor, DC	ABM01	Can Assembly, Shield
EAB01	Speaker Module	ABM02	Can Assembly, Shield
AEX00	Keypad Assembly, Main	MJN02	Tape
GME01	Machine Screw	EAC00	Battery, Lithium Ion
GME00	Machine Screw		

14. Rear Cover Disassemble SOP

Process	Detail Guide	Description
Disassemble Rear Cabinet Ass'y	 Disassemble battery cover  Pull out TV antenna and unlock 6 screws  Push TV antenna into rear cabinet ass'y and disassemble the rear cabinet	1. Disassemble the battery cover. 2. Disassemble 6 screws on rear cabinet after pulling out TV antenna. 3. Push TV antenna into rear cabinet ass'y and disassemble the rear cabinet



X350 SBOM (BK)

Exploded	NO	Location No	Description	LG P/N	ODM P/N
	1		LG-X350 BRABK	LGX350.ABRABK@WWZ.EKHQ 3	#N/A
	2	AAD00	Addition Assembly	AAD85829801 7	#N/A
Exploded	3	EAC00	Rechargeable Battery,Lithium Ion	EAC61679611	306-0000-00116
	4	MFL00	Manual,Operation	MFL66985014	No part number in system
	5	SGD00	Accessory,Data Cable	EBX61368601	410-7221000006
	6	SGE00	Earphone,Stereo	SGEY0003744	333-0000-00080
	7	SSA00	Adapters	SSAD0036101	331-0000-00150
	8	*S*00	Adapters	*S*SSAD0036102	#N/A
	9	*S*00	Adapters	*S*SSAD0036001	#N/A
	10	AGF00	Package Assembly	AGF75963712	#N/A
	11	AGQ00	Phone Assembly	AGQ86627401 8	#N/A
	12	ACQ00	Cover Assembly,Front	ACQ85688901 3	#N/A
Exploded	13	ACQ01	Cover Assembly,Front	ACQ85627301 5	401-72690-0001
	14	MCK00	Cover,Front	MCK66846401	#N/A
	15	MCQ00	Damper,LCD	MCQ66724601	#N/A
	16	MDJ00	Filter	MDJ63207101	#N/A
	17	MDJ01	Filter	MDJ63207201	#N/A
	18	MJN00	Tape,Window	MJN67902701	#N/A
Exploded	19	EAB00	Receiver	EAB62448401	313-0000-00208
Exploded	20	MKC00	Window,LCD	MKC64121901	403-72690-0001
	21	ACQ02	Cover Assembly,Rear	ACQ85689001 7	402-72690-0001
Exploded	22	ACQ03	Cover Assembly,Rear	ACQ85635601 11	402-72690-0002
	23	MBG00	Button	MBG64326501	#N/A

	24	MBG01	Button,Side	MBG64349801	#N/A
	25	MBL00	Cap,Receptacle	MBL65025401	#N/A
	26	MCK01	Cover,Rear	MCK66854901	#N/A
	27	MCQ01	Damper,Speaker	MCQ66726001	#N/A
	28	MCQ02	Damper,Connector	MCQ66726101	#N/A
	29	MCQ03	Damper,Camera	MCQ66726201	#N/A
	30	MDJ02	Filter	MDJ63227501	#N/A
	31	MGJ00	Plate	MGJ63141401	#N/A
	32	MJN01	Tape	MJN67878201	#N/A
	33	MKC01	Window,Camera	MKC64122301	#N/A
Exploded	34	EAA00	Retractable Antenna,Multiple	EAA62645301	330-0000-00230
Exploded	35	EAA01	Retractable Antenna,DMB	EAA62645701	330-0000-00229
Exploded	36	EAB01	Speaker Module	EAB62510301	313-0000-00211
Exploded	37	EAU00	Motor,DC	EAU61446401	320-0000-00071
Exploded	38	GME00	Screw,Machine	GMEY0014301	409-00000-0059
Exploded	39	MCK02	Cover	MCK66960101	415-72690-0016
Exploded	40	AEX00	Keypad Assembly,Main	AEX73818401	404-72690-0001
Exploded	41	EBR00	PCB Assembly,Main	EBR74004501 3	Under registration
	42	EBR01	PCB Assembly,Main,Insert	EBR74004601 7	#N/A
Exploded	43	ABM00	Can Assembly,Shield	ABM73636001	415-72690-0014
Exploded	44	ABM01	Can Assembly,Shield	ABM73636101	415-72690-0013
Exploded	45	ABM02	Can Assembly,Shield	ABM73636201	415-72690-0015
Exploded	46	EAJ00	LCD Module	EAJ61947901	327-0000-00129
Exploded	47	EBP00	Camera Module	EBP61361801	335-0000-00110
Exploded	48	GMZ00	Machine Screw	GMZZ0031401	409-00000-0101
Exploded	49	MJN02	Tape	MJN67905601	415-72690-0012

Exploded	50	EBR02	PCB Assembly,Keypad	EBR74004701 3	Under registration
Exploded	51	ADB00	Dome Assembly,Metal	ADB73718901	415-72690-0002
	52	EDL00	LED,Chip	EDLH0015501	309-0000-00132
	53	ENB00	Connector,BtoB	ENBY0060901	314-0000-00381
	54	EBR03	PCB Assembly,Main,SMT	EBR74004801 35	#N/A
	55	ACK00	Can Assembly,Shield	ACKA0028101	415-72620-0023
	56	EAE00	Capacitor,Ceramic,Chip	EAE62561901	306-0000-00084
	57	EAG00	Connector,FFC/FPC/PIC	EAG63091501	314-0000-00501
	58	EAN00	IC,Digital Baseband Processor,GSM	EAN61828501	311-0000-01026
	59	EAN01	IC,LDO Voltage Regulator	EAN62166301	311-0000-01082
	60	EAN02	IC,Other Mobile DTV	EAN62166501	311-0000-00947
	61	EAN03	IC,FM	EAN62166601	311-0000-00982
	62	EAN04	IC,DC,DC Converter	EAN62221901	311-0000-01081
	63	EAN05	IC,DC,DC Converter	EAN62239501	311-0000-01080
	64	EAW00	Oscillator,Crystal	EAW61483501	305-0000-00124
	65	EAW01	Oscillator,Crystal	EAW61663401	305-0000-00142
	66	EAW02	Oscillator,Crystal	EAW61663501	305-0000-00133
	67	EBF00	Switch,Tact	EBF61534401	314-0000-00551
	68	EBF01	Switch,Tact	EBF61534501	310-0000-00134
	69	EBF02	Switch,Tact	EBF61574801	310-0000-00143
	70	ENB01	Connector,BtoB	ENBY0060601	314-0000-00430
	71	ENB02	Connector,BtoB	ENBY0060801	314-0000-00374
	72	ENR00	Connector,I/O	ENRY0009101	314-0000-00229
	73	ENS00	Socket,Camera	ENSY0022201	314-0000-00568
	74	ENS01	Card Socket	ENSY0025501	314-0000-00449
	75	ENW00	Connector,RF	ENWY0006901	314-0000-00016

	76	ENZ00	Connector,Terminal Block	ENZY0028301	314-0000-00441
	77	ESC00	Switch,Tact	ESCY0007001	315-0000-00047
	78	EUS00	IC,MCP,NAND	EUSY0335808	311-0000-01092
	79	EUS01	IC,Audio Amplifier	EUSY0399601	#N/A
	80	EUS02	IC,Transceiver	EUSY0399701	311-0000-00740
	81	EUS03	IC,Analog Switch	EUSY0403501	311-0000-00856
	82	EUS04	IC,Audio Amplifier	EUSY0420601	311-0000-00895
	83	EXS00	Oscillator,Crystal	EXSY0024801	305-0000-00026
	84	MIC00	Insert,Nut	MICE0017501	415-73210-0007
	85	SAD00	Software,Mobile	SAD33124301	#N/A
	86	SFS00	Filter,Saw	SFSY0038201	326-0000-00184
	87	SFS01	Filter,Saw	SFSY0041601	326-0000-00150
	88	SFS02	Filter,Saw	SFSY0041701	326-0000-00171
	89	SUM00	Microphone,Condenser	SUMY0012401	312-0000-00040
Exploded	90	GME01	Screw,Machine	GMEY0013901	409-00000-0014
Exploded	91	MCK03	Cover,Battery	MCK66866501	405-72690-0001
Exploded	92	MEZ00	Label,Approval	MEZ64148501	478-726500-001
	93	SMZ00	IC,Memory Card,MINI SD	SMZY0027202	#N/A